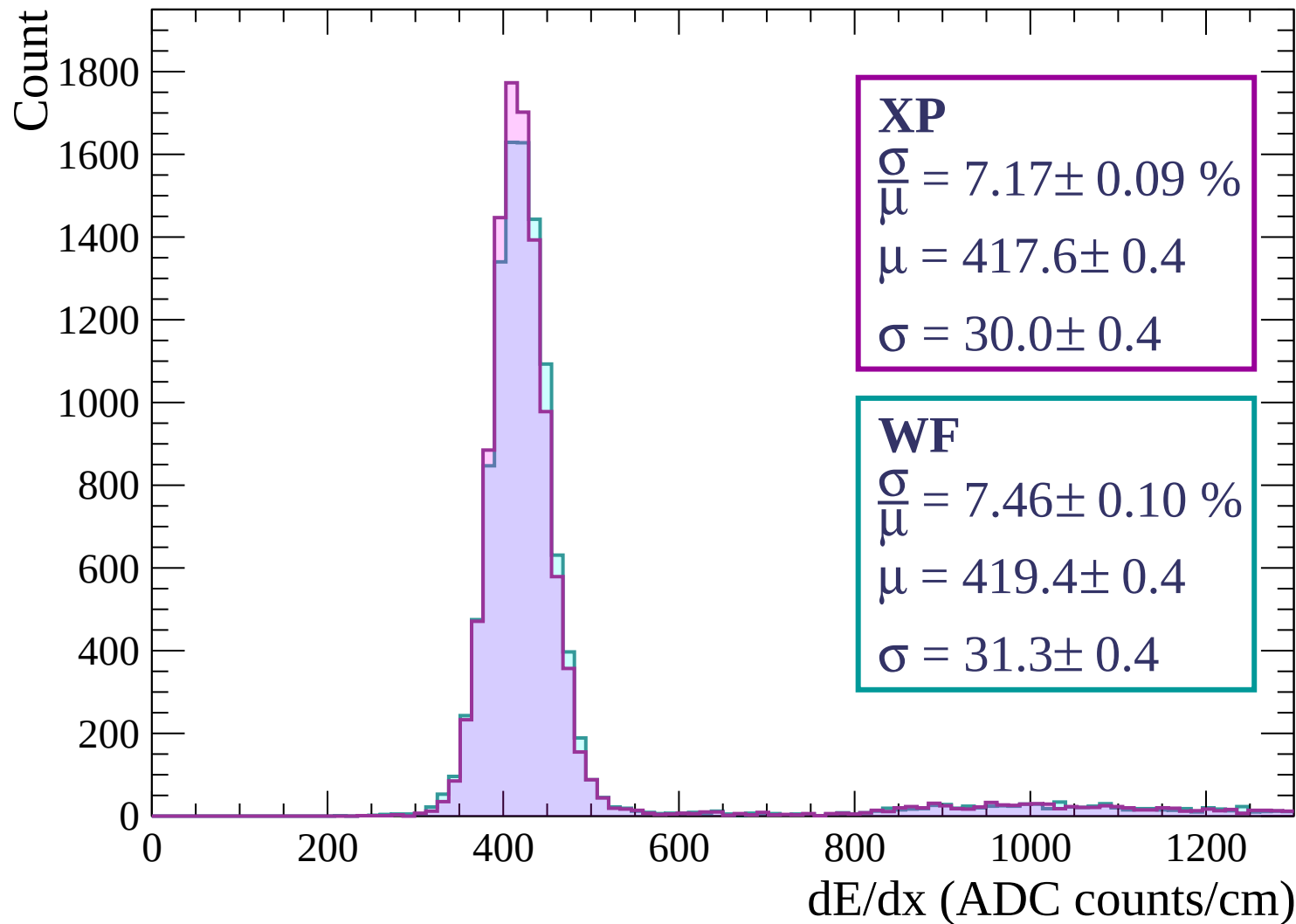
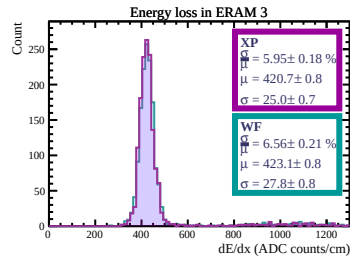
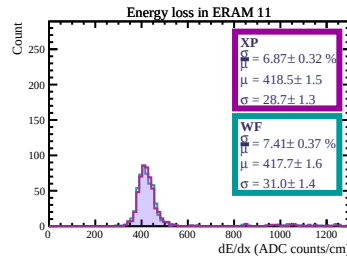
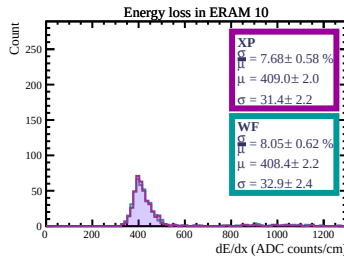
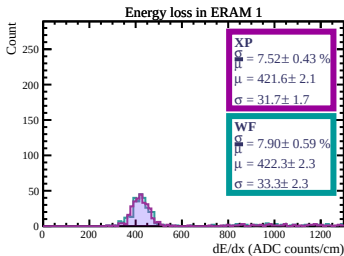
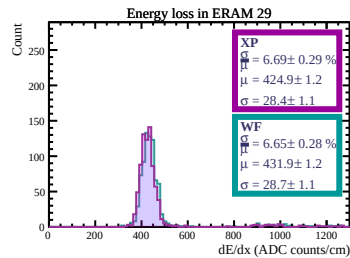
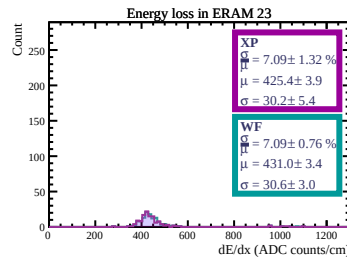
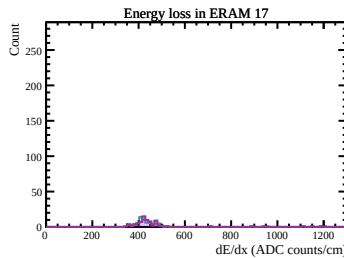
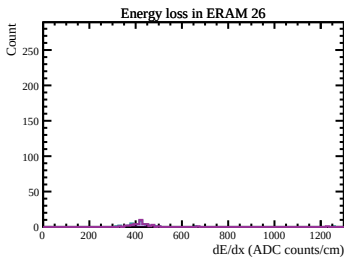
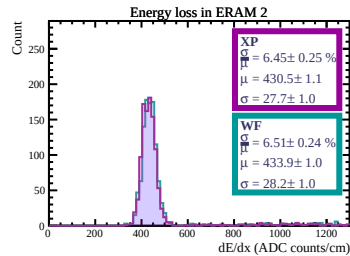
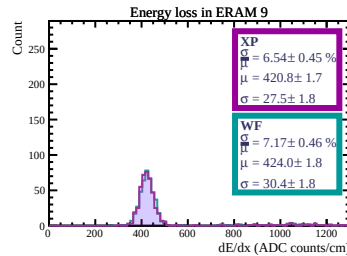
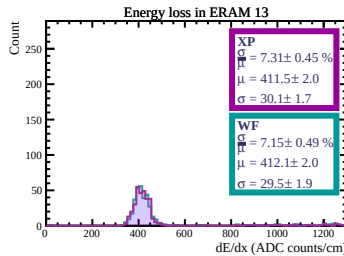
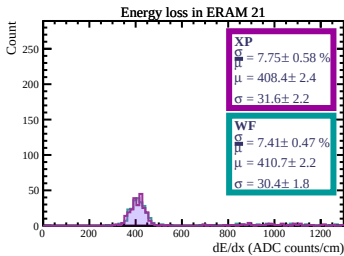
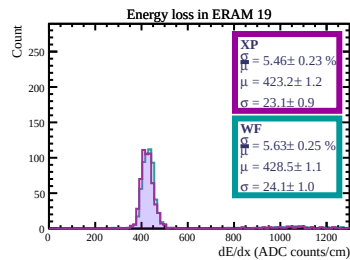
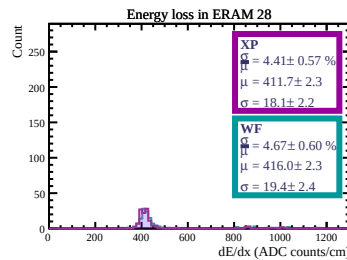
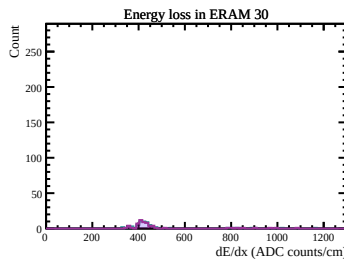
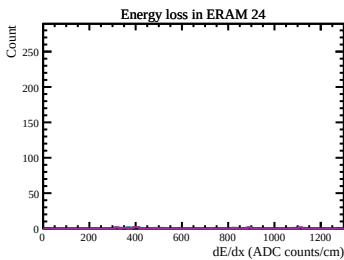
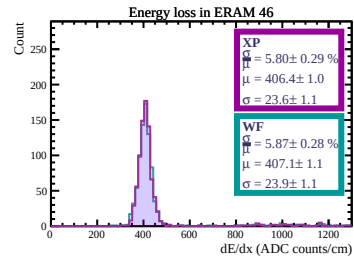
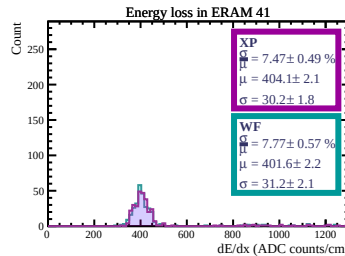
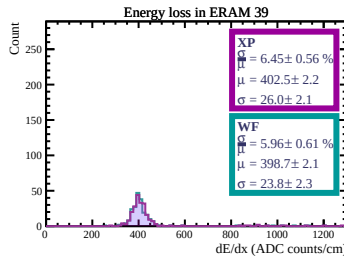
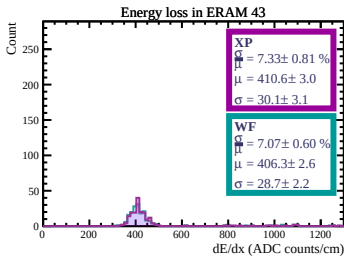
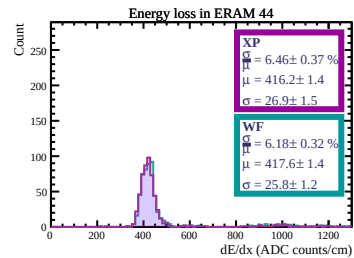
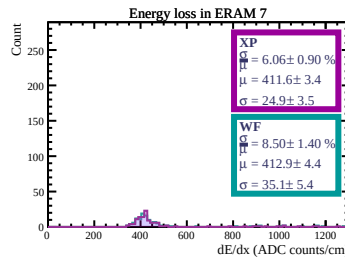
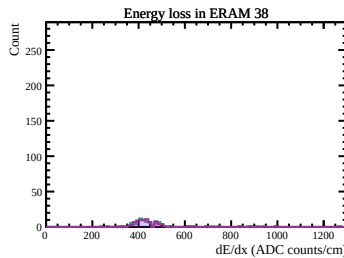
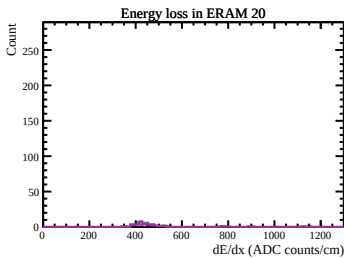
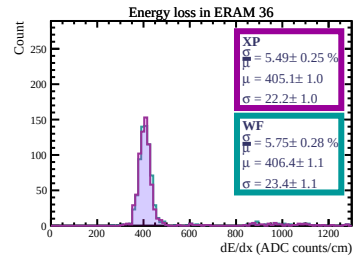
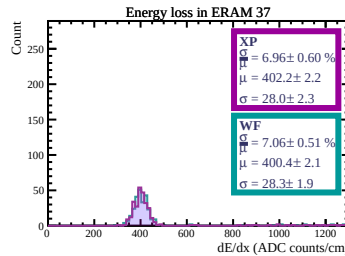
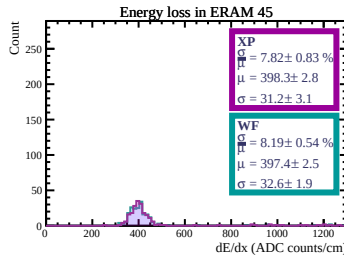
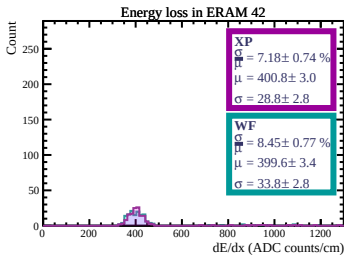
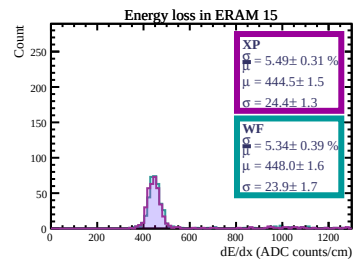
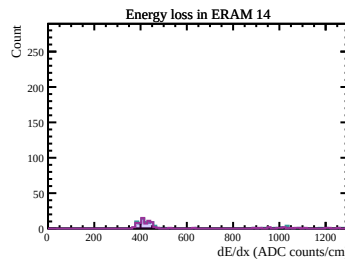
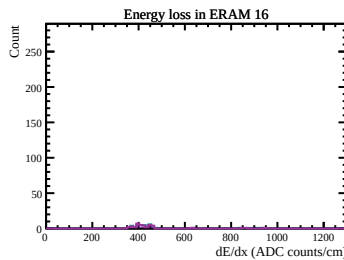
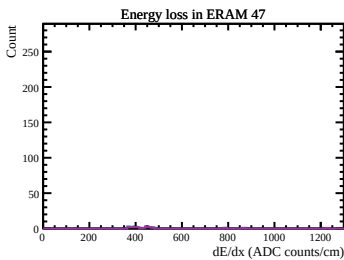
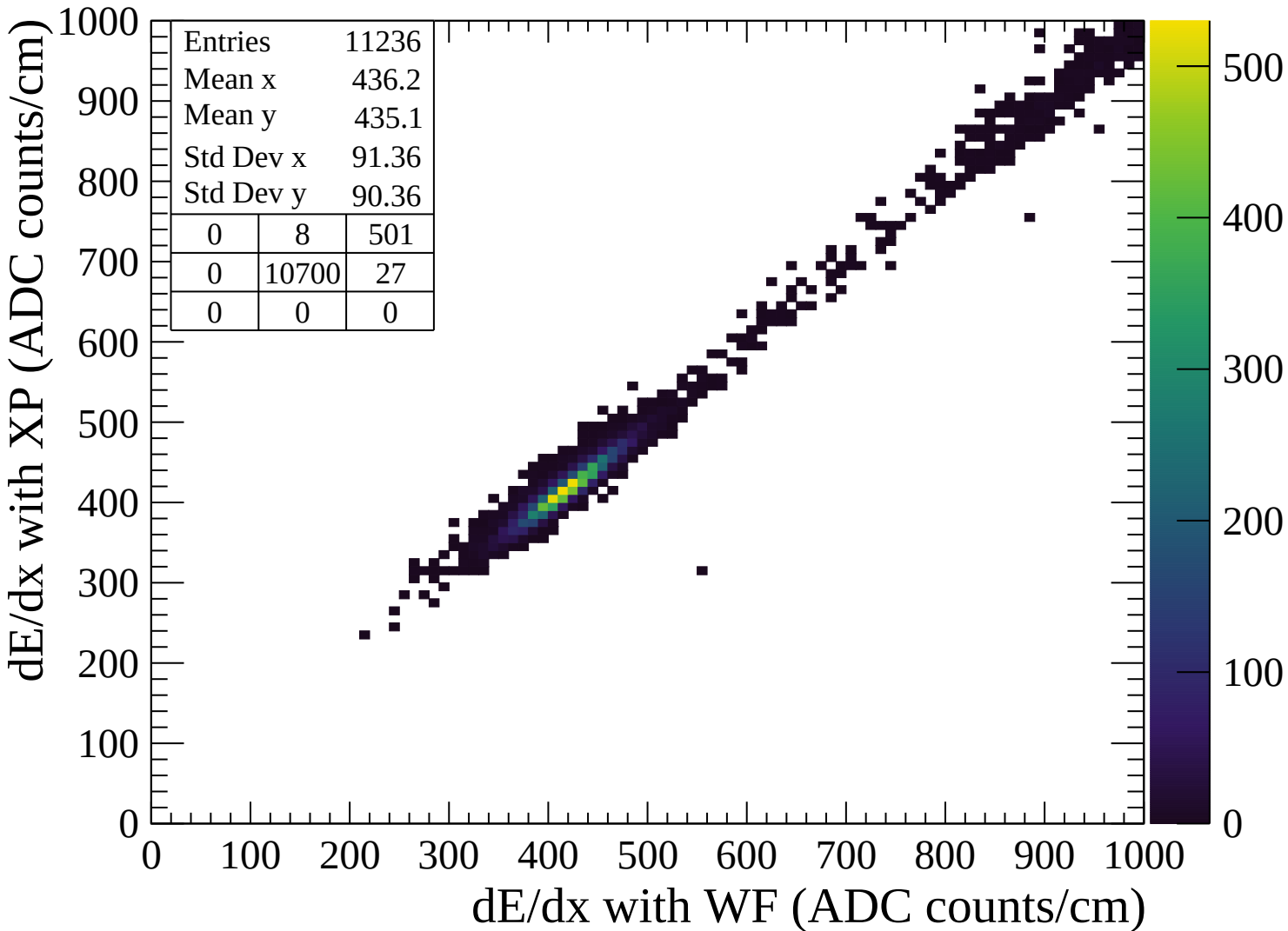
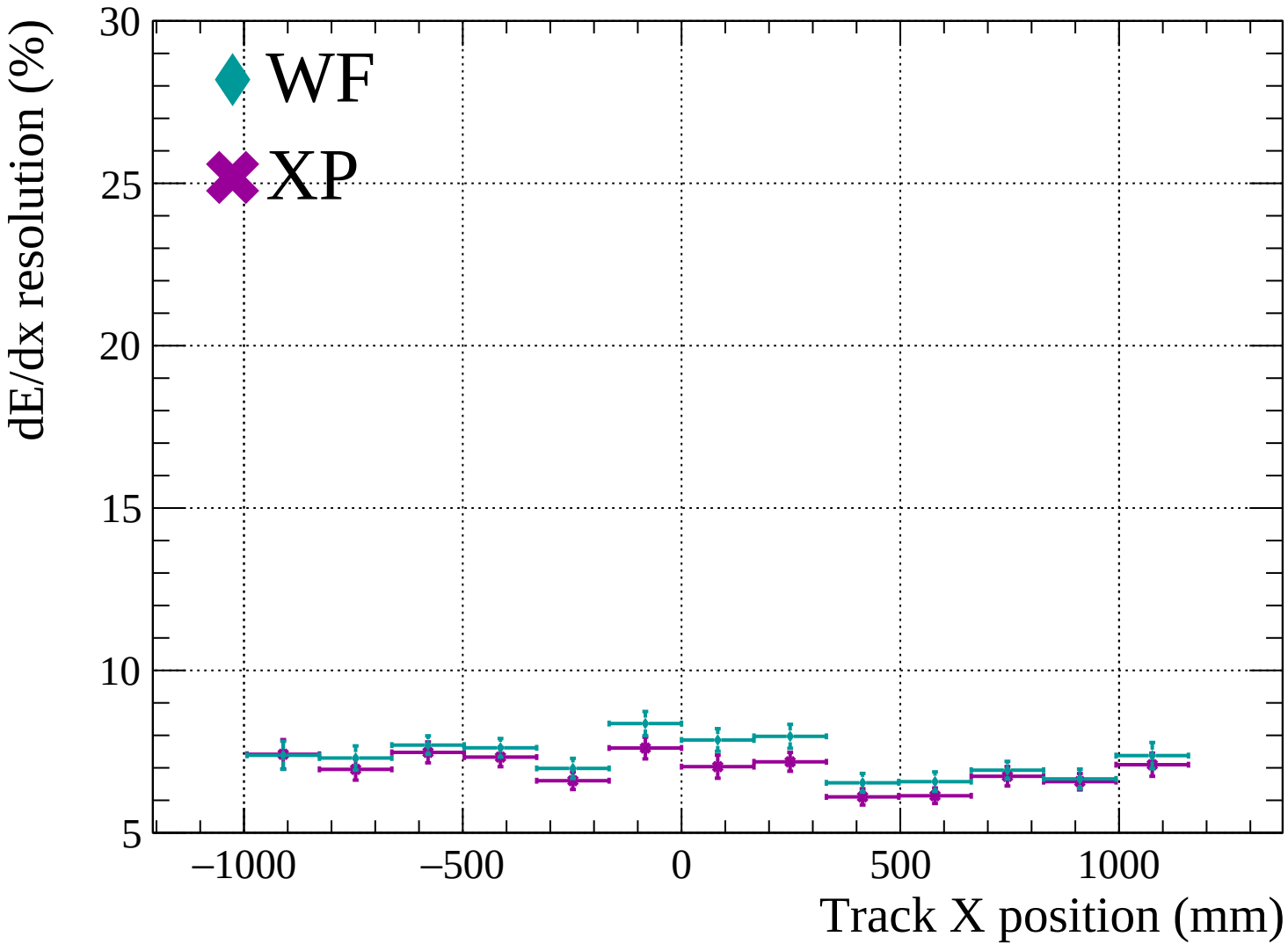


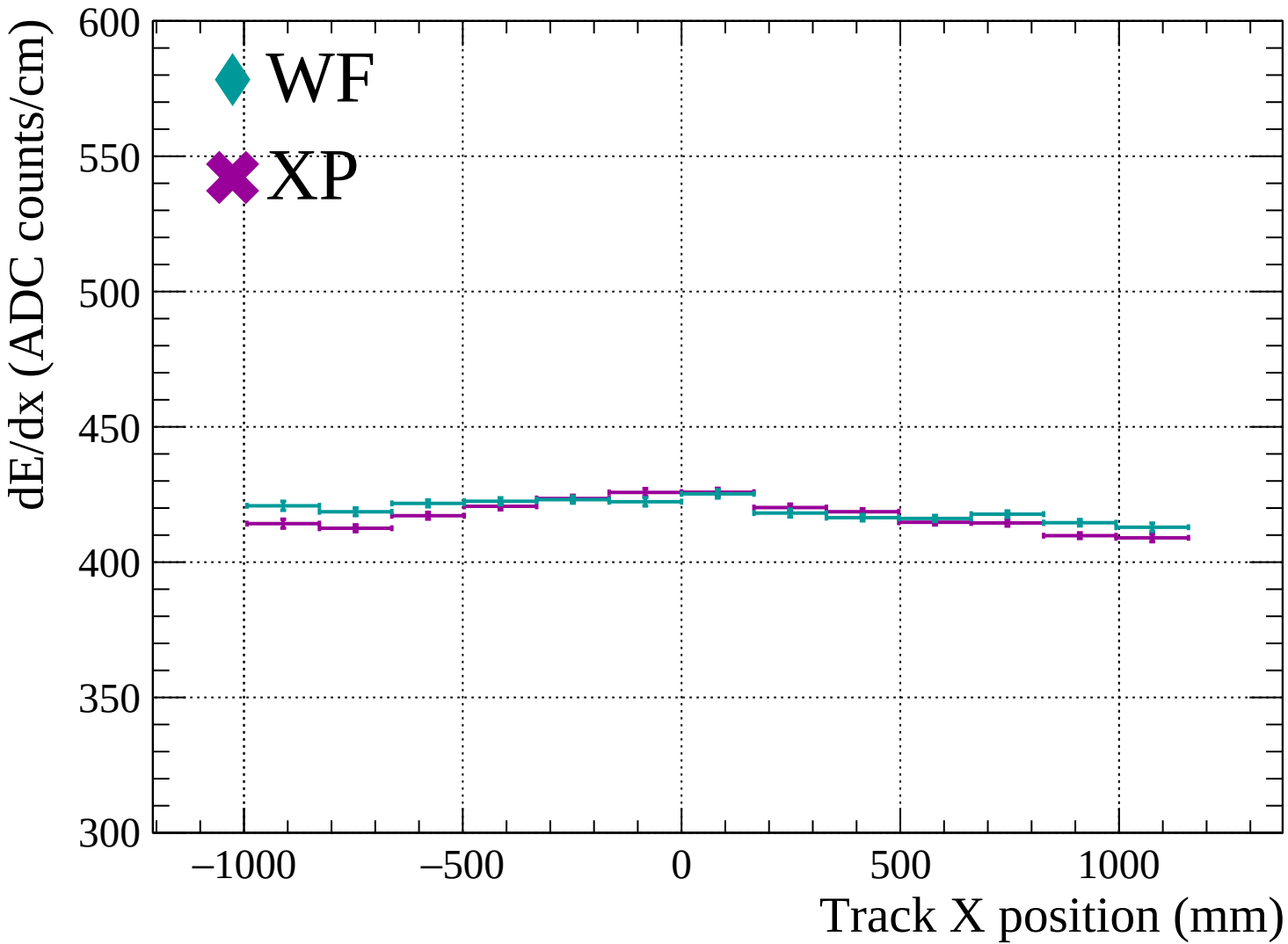


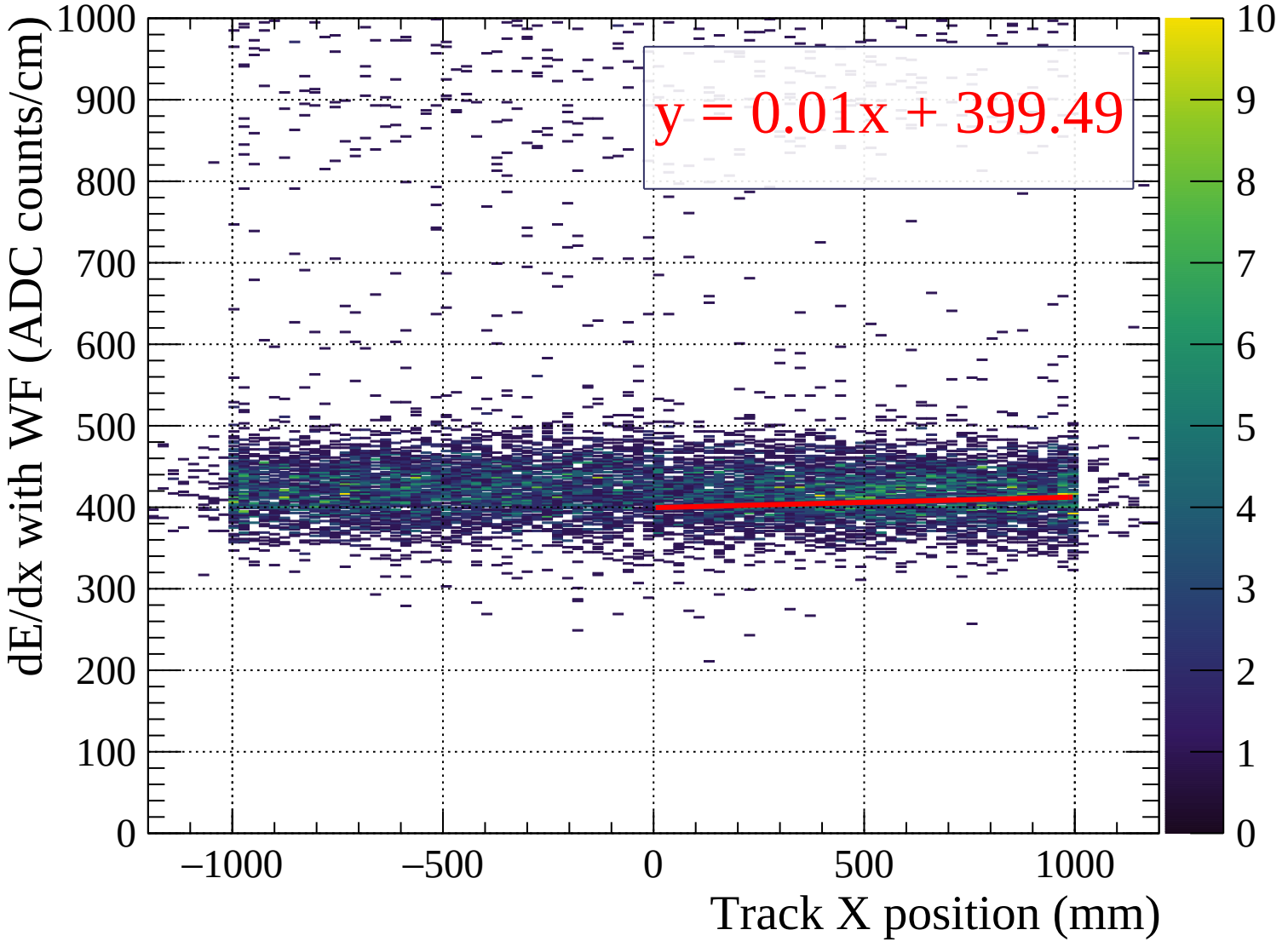


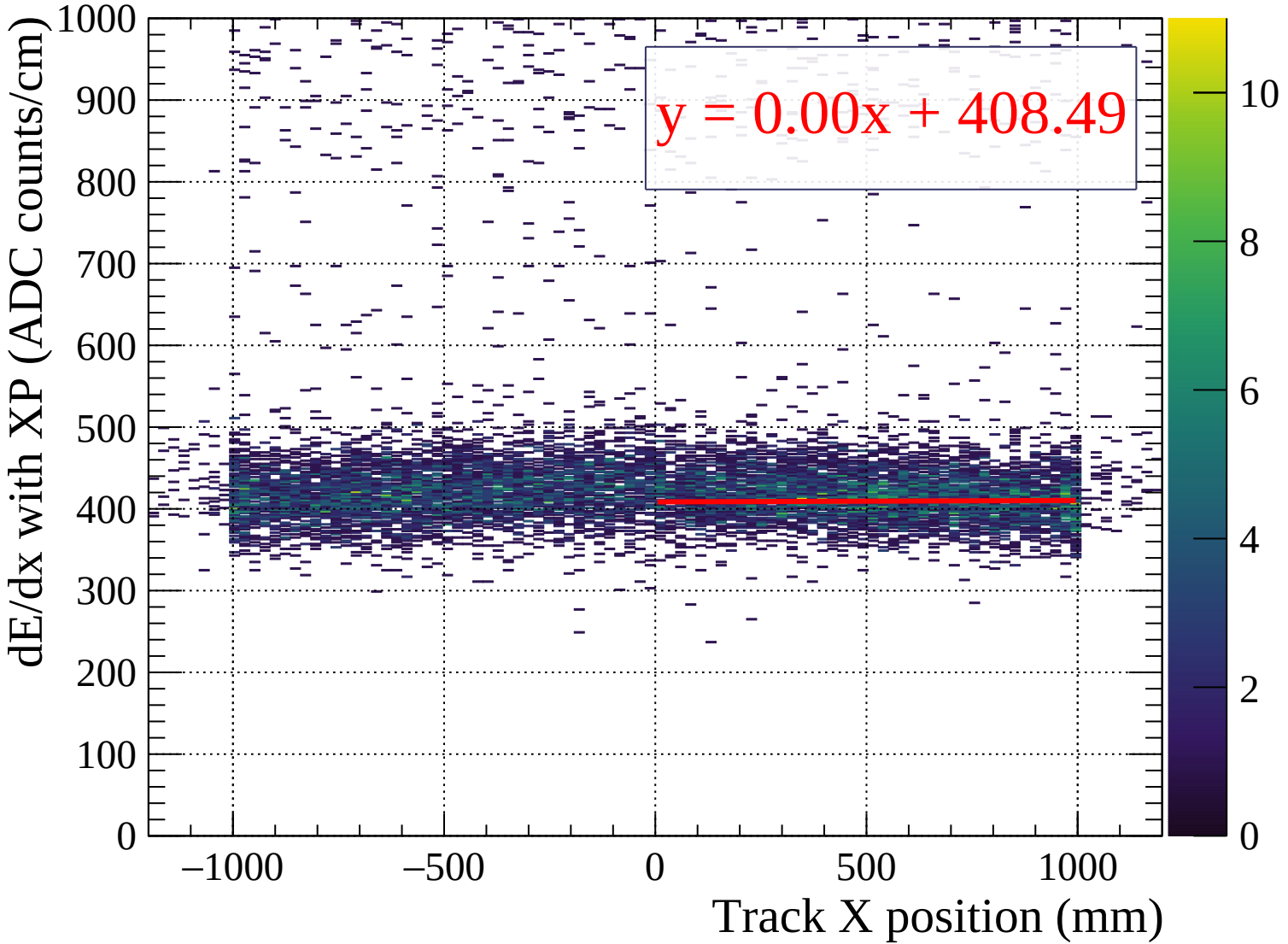


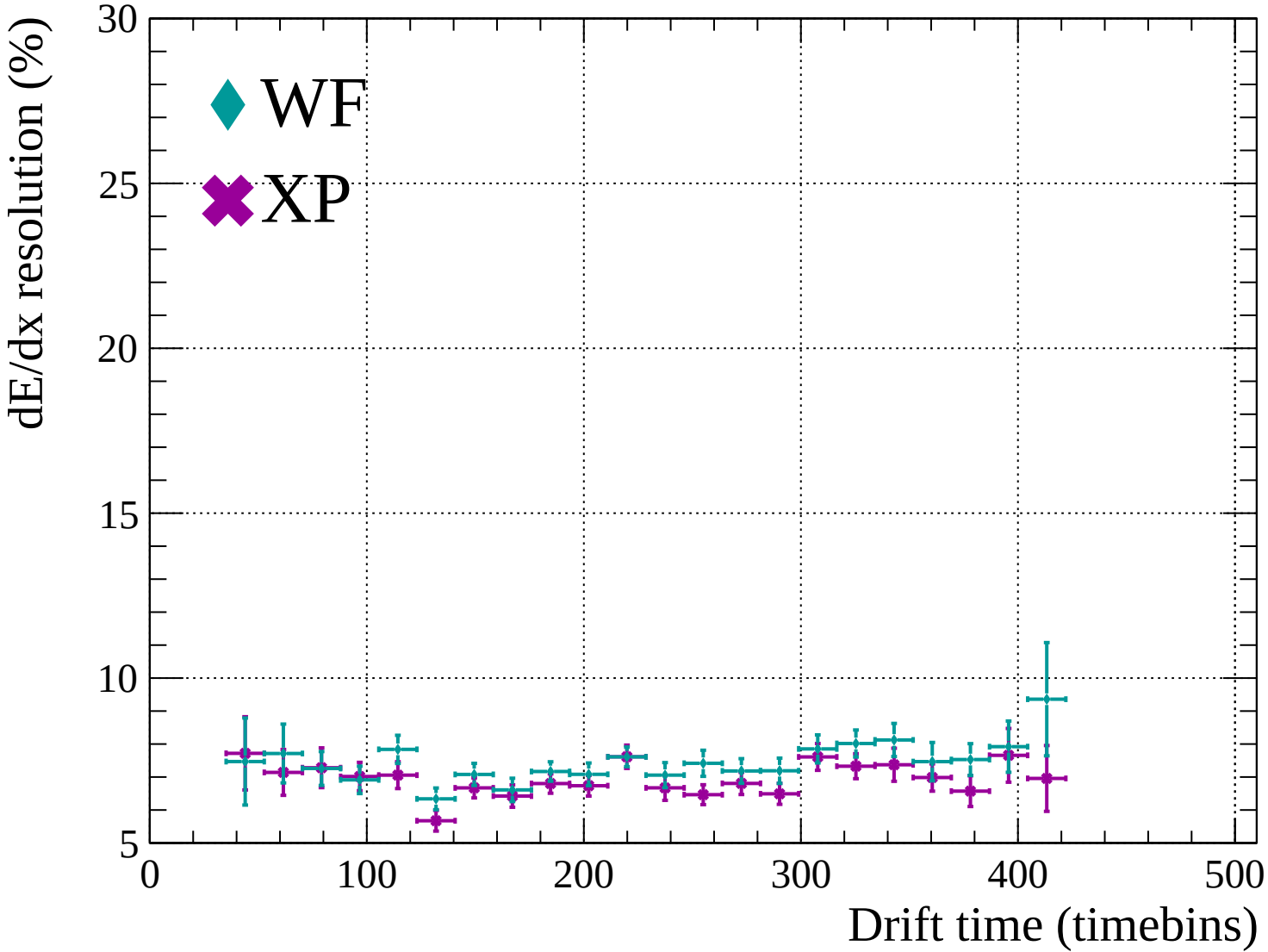


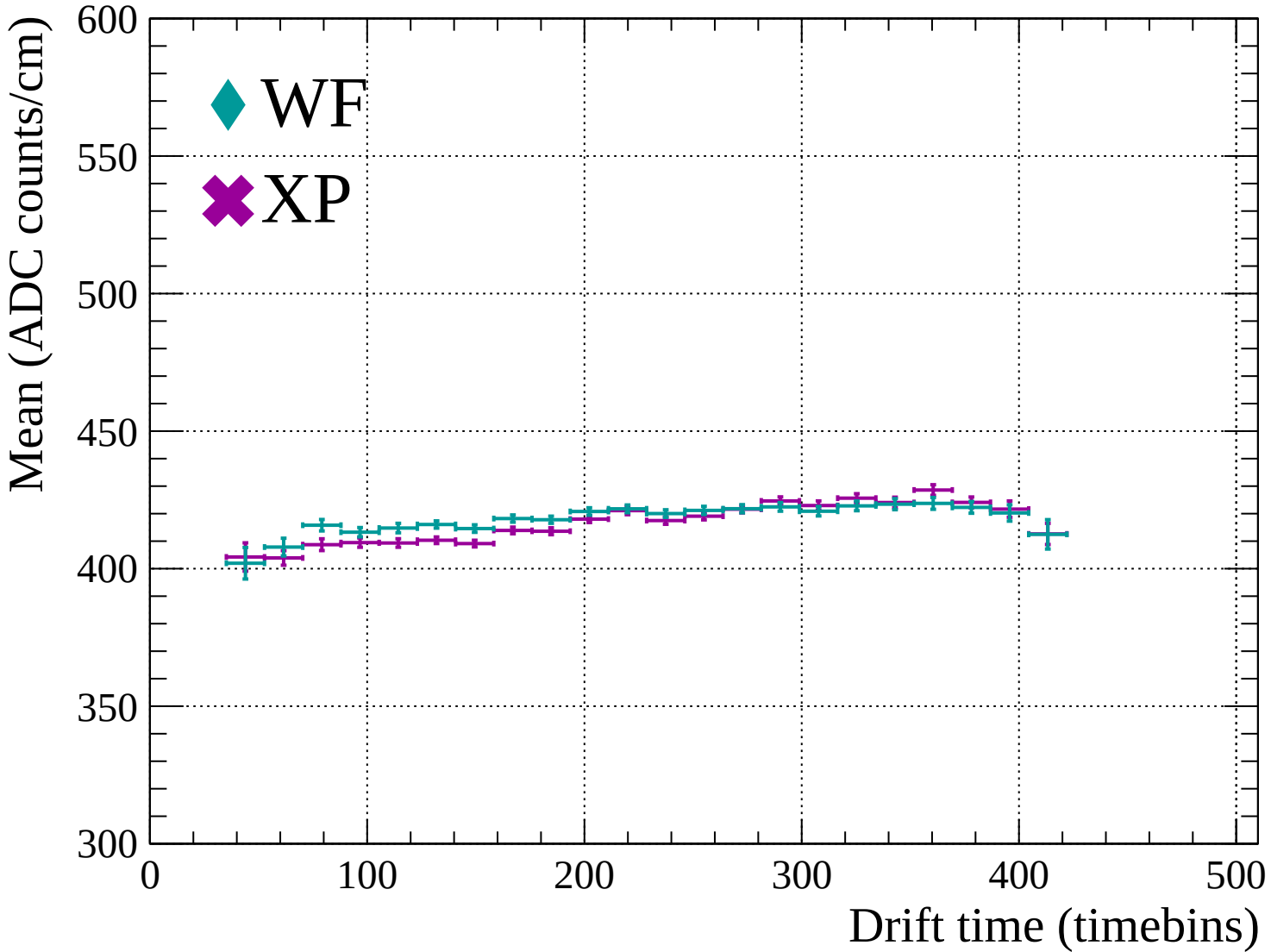


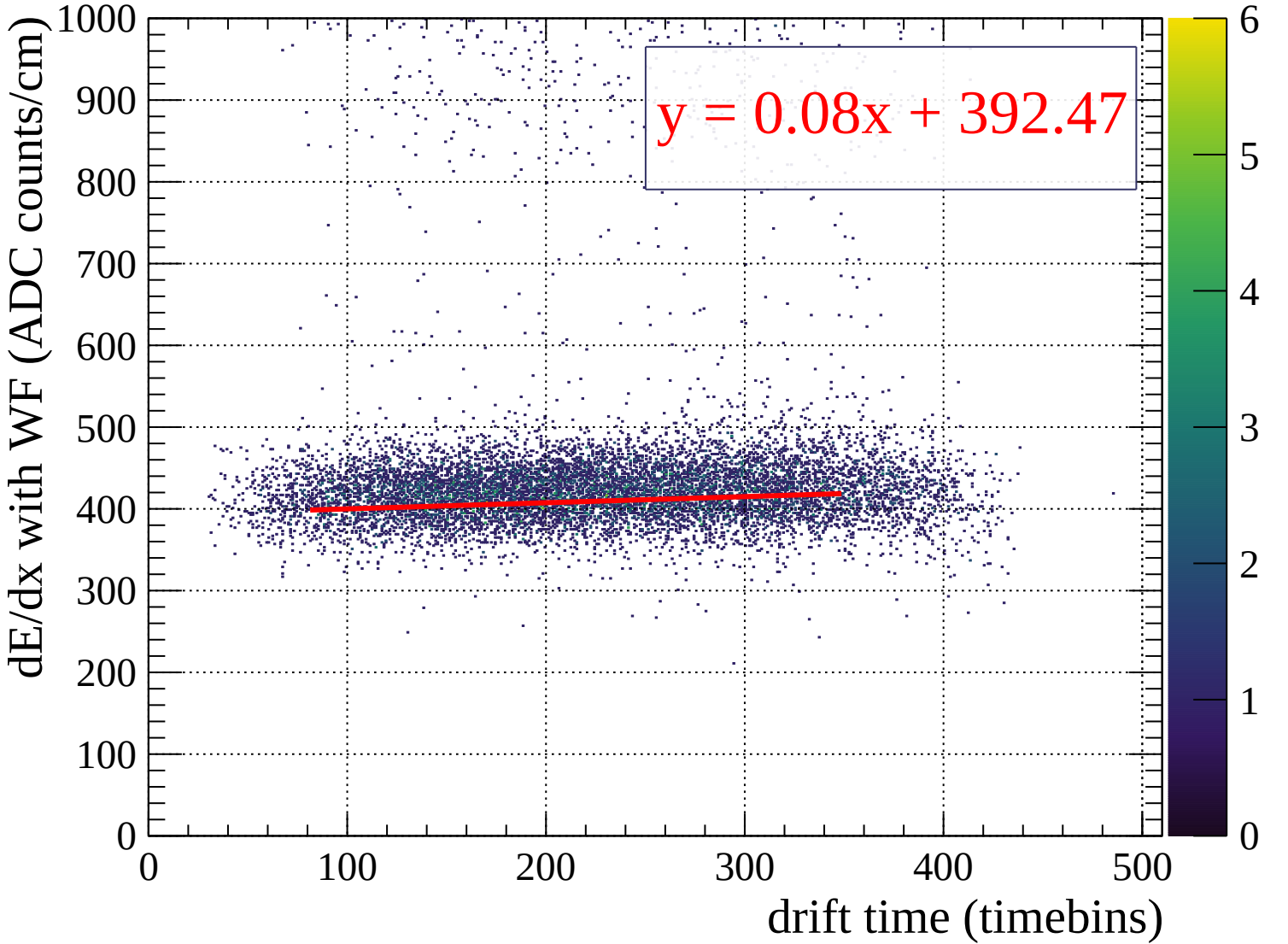


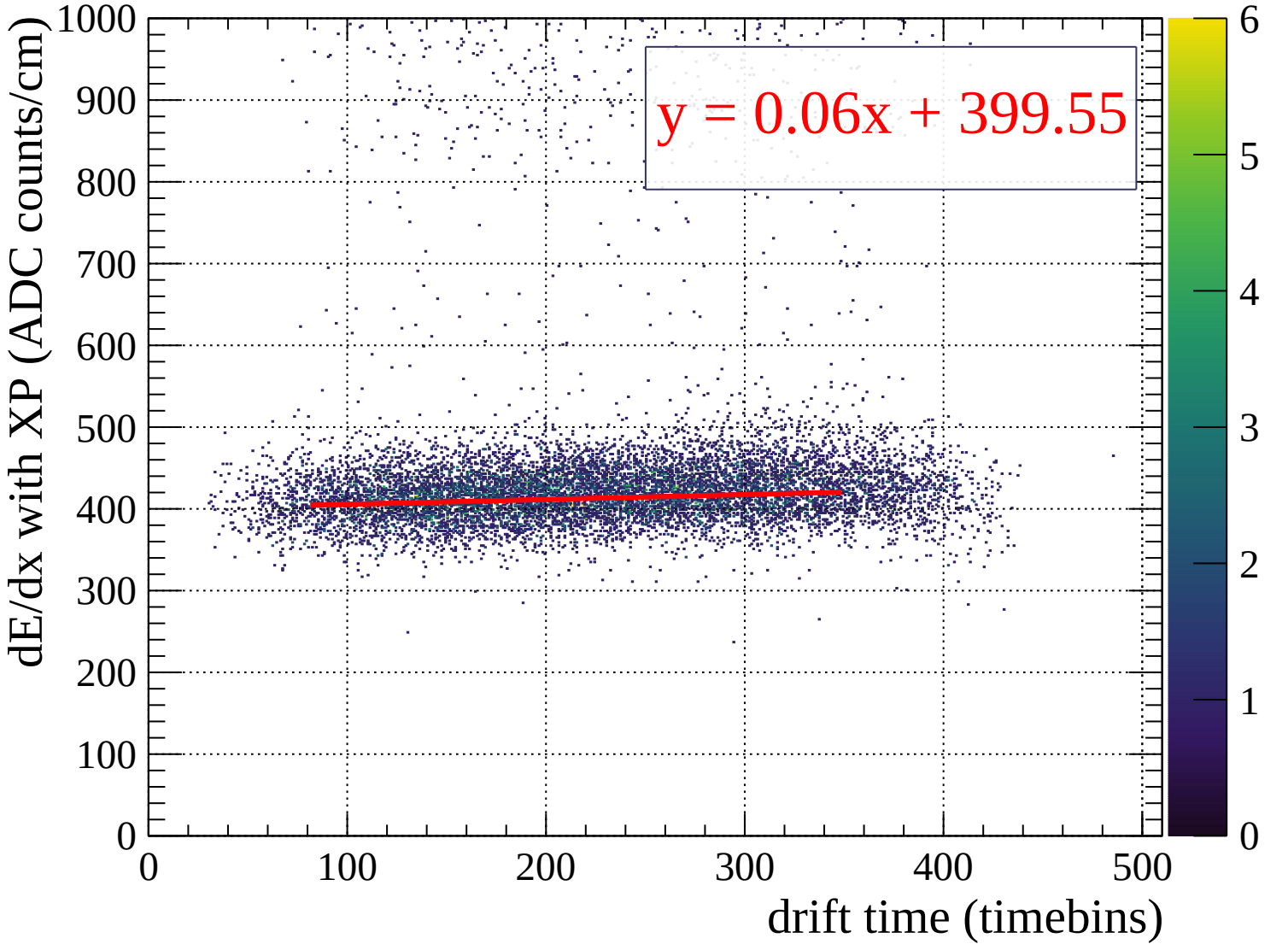


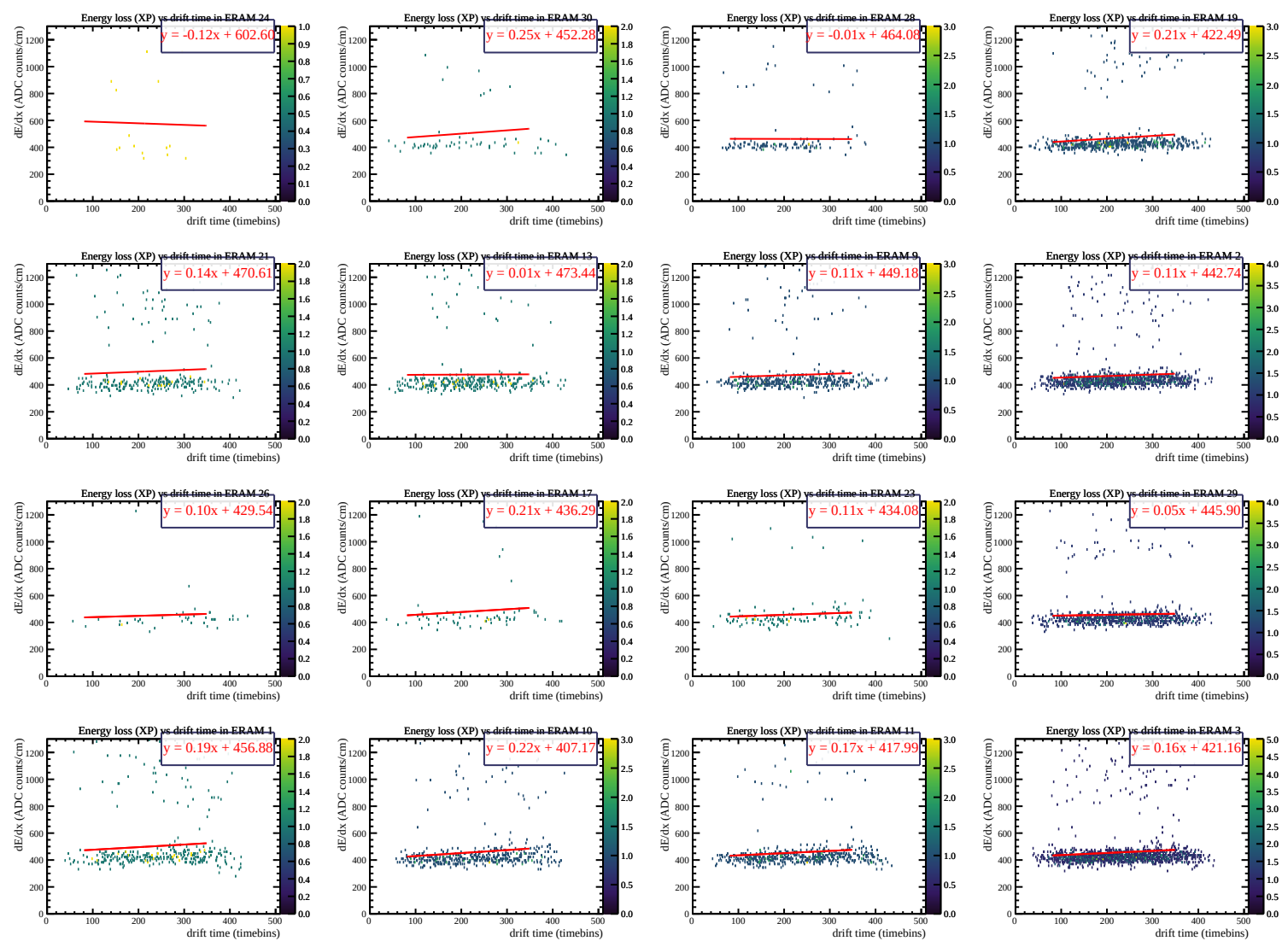


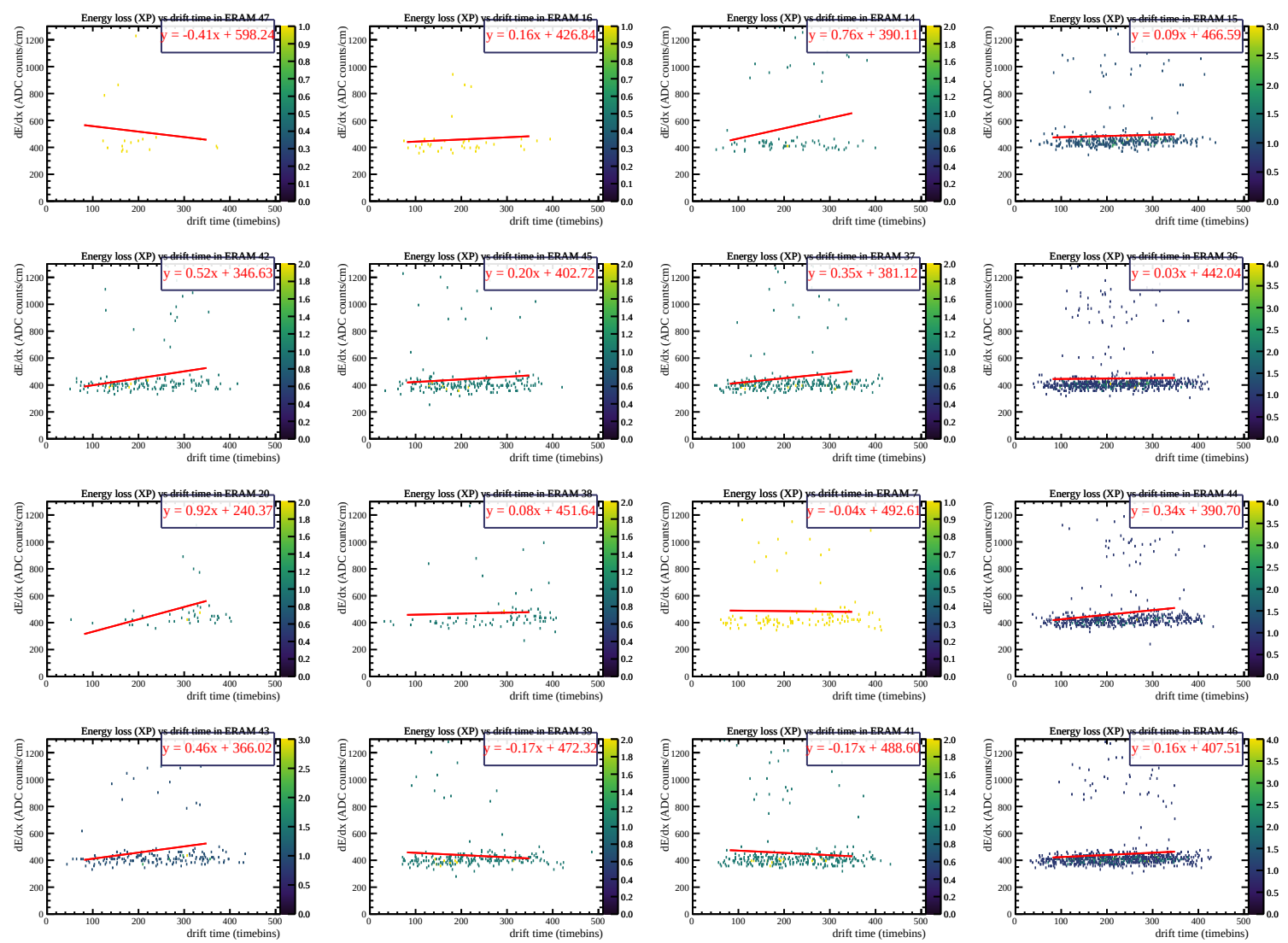


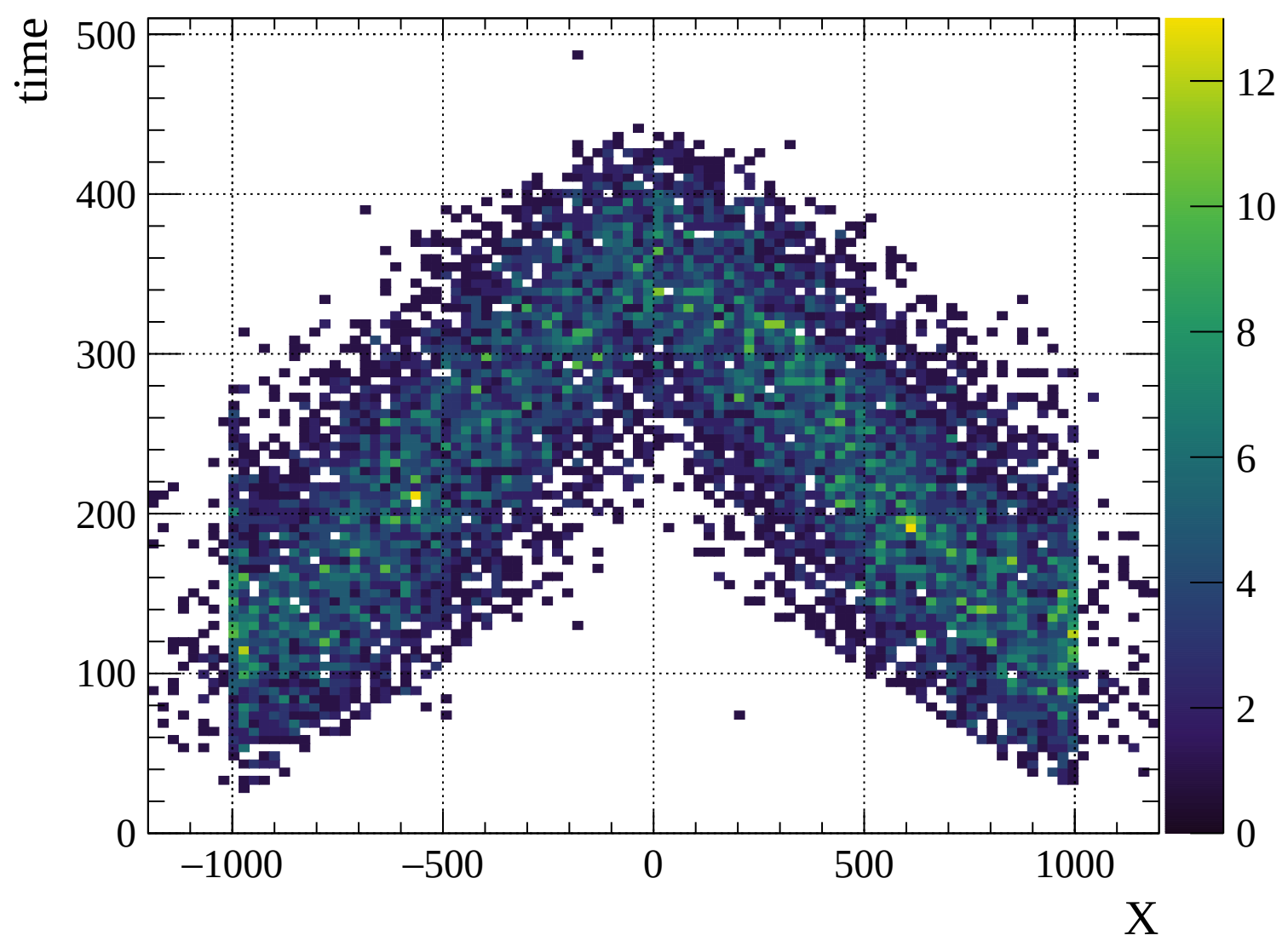


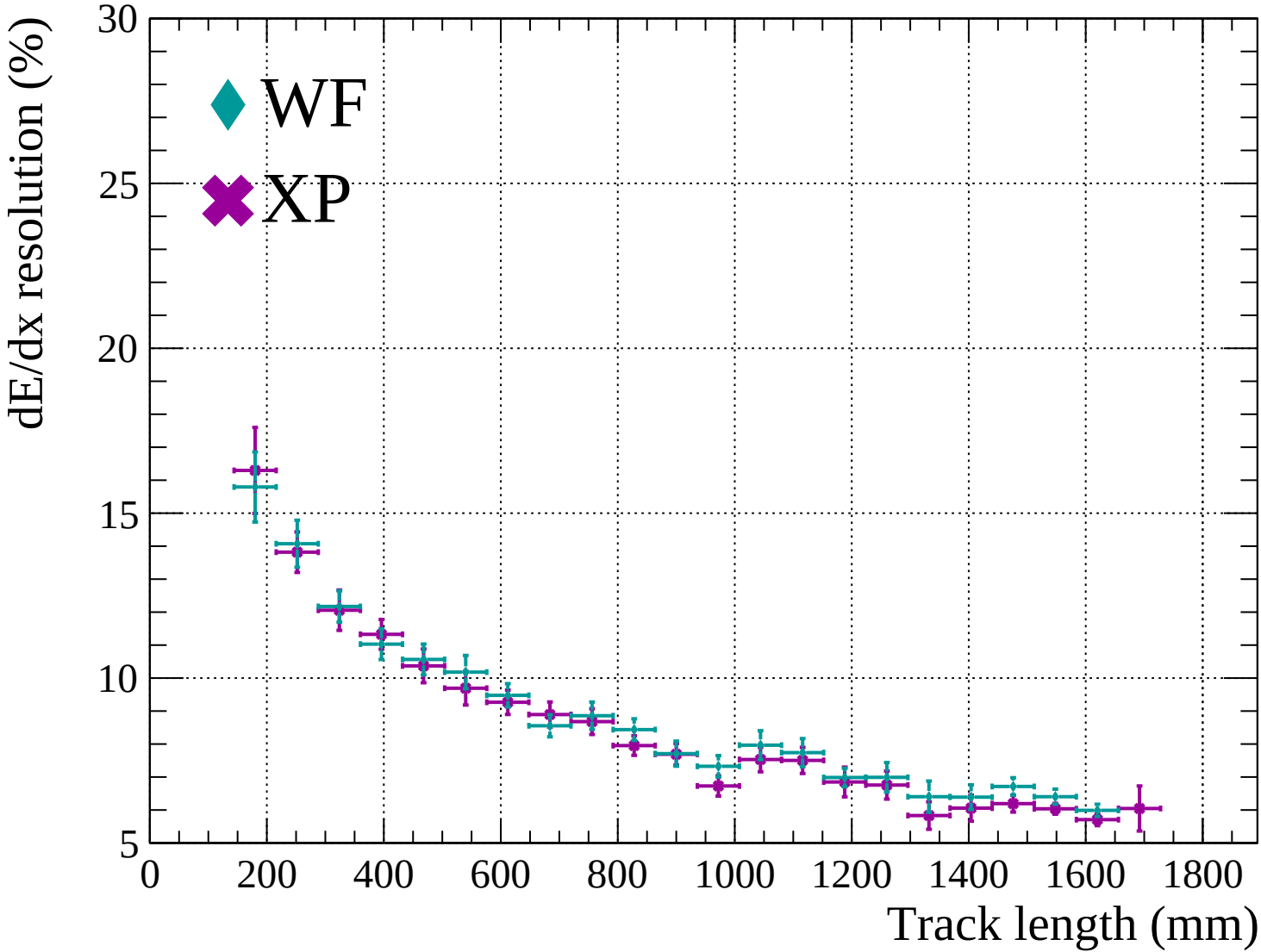


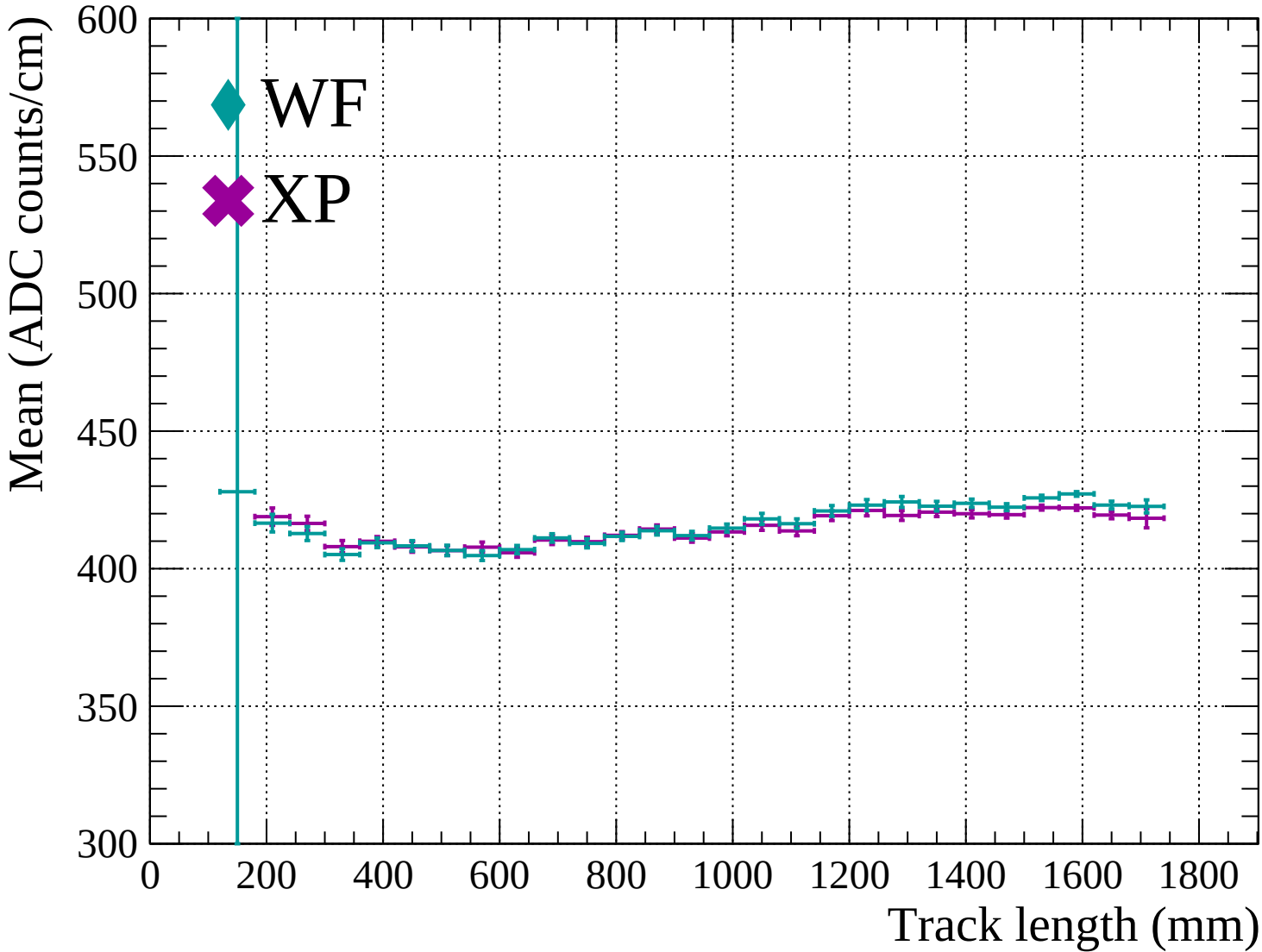


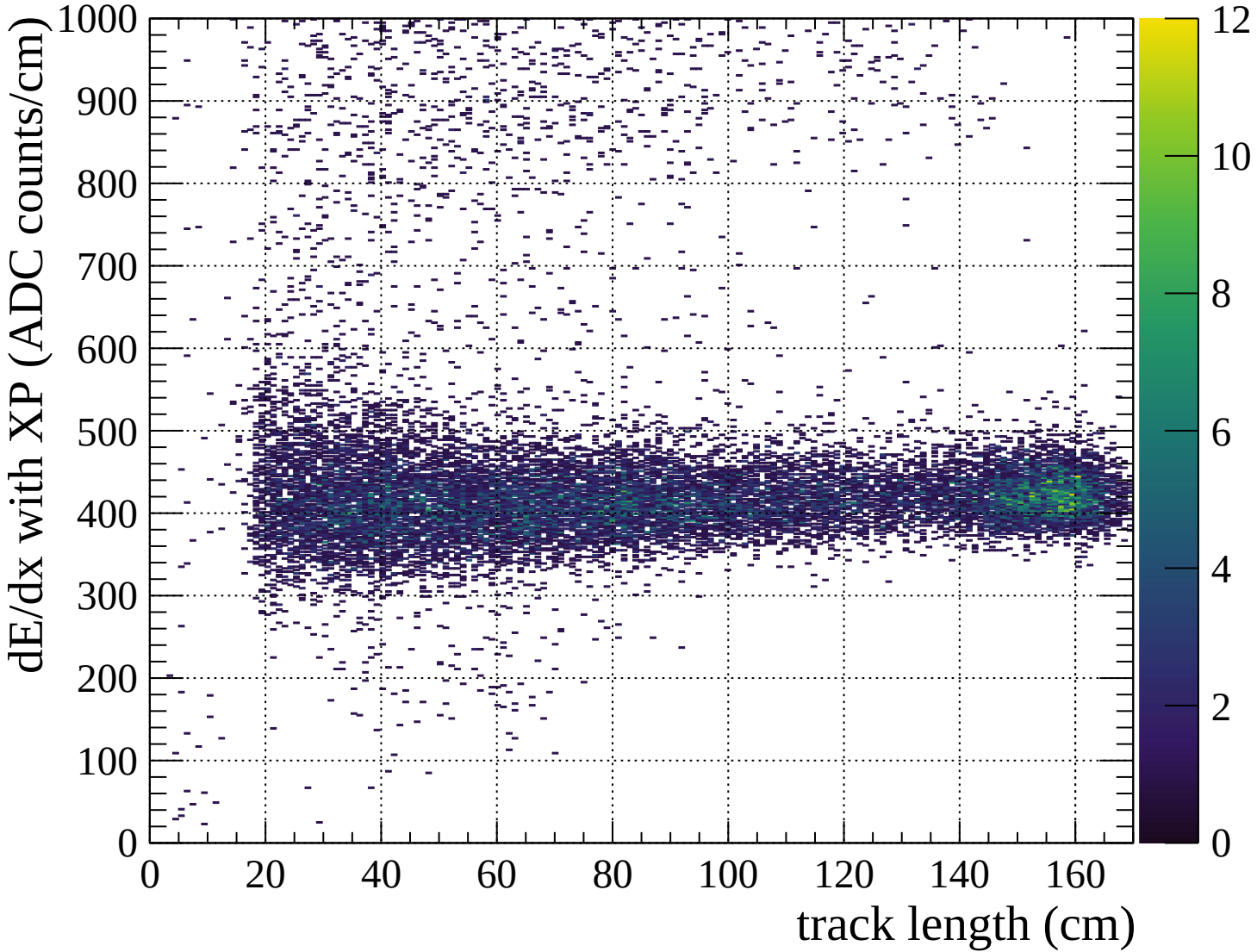


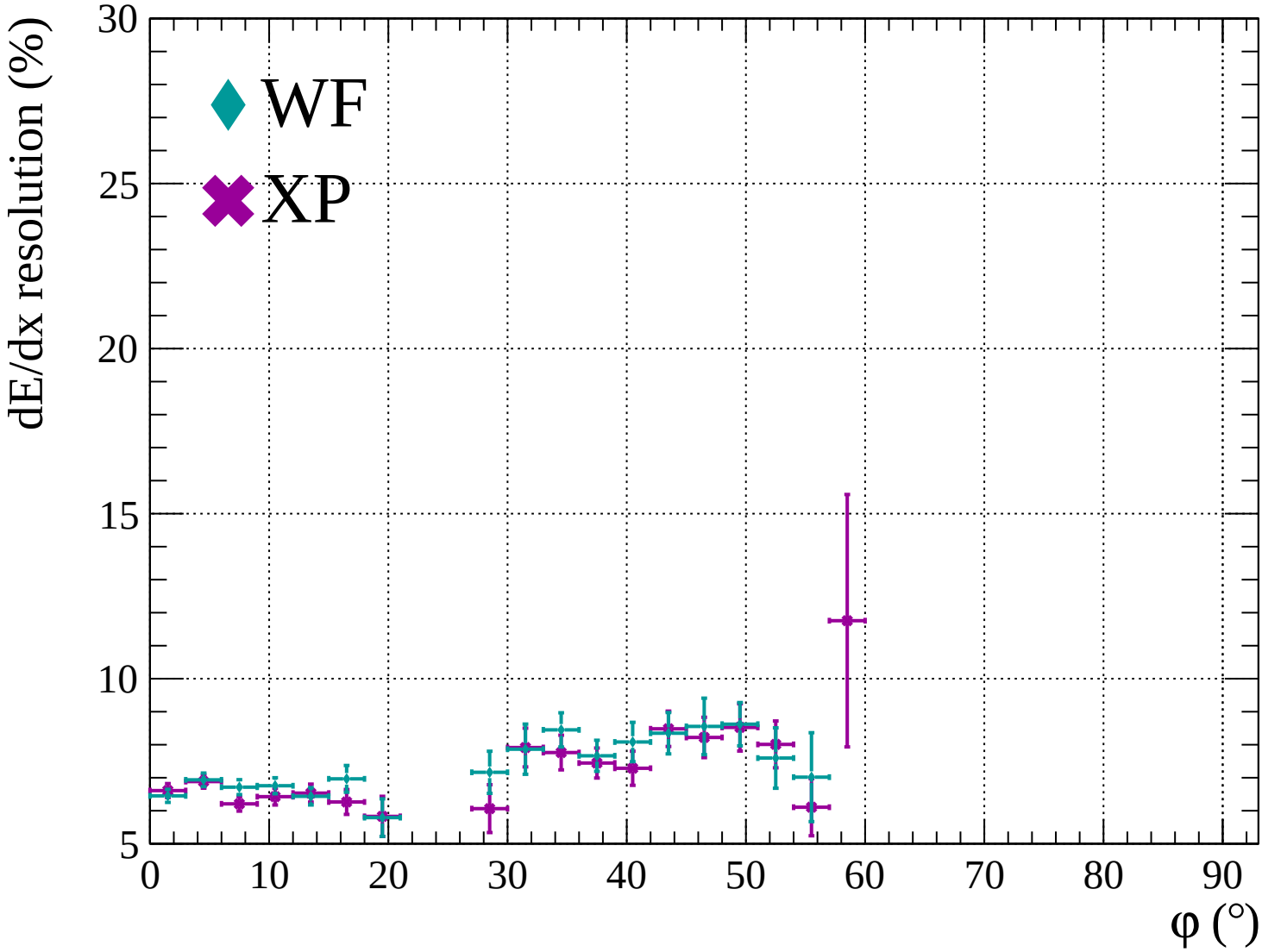


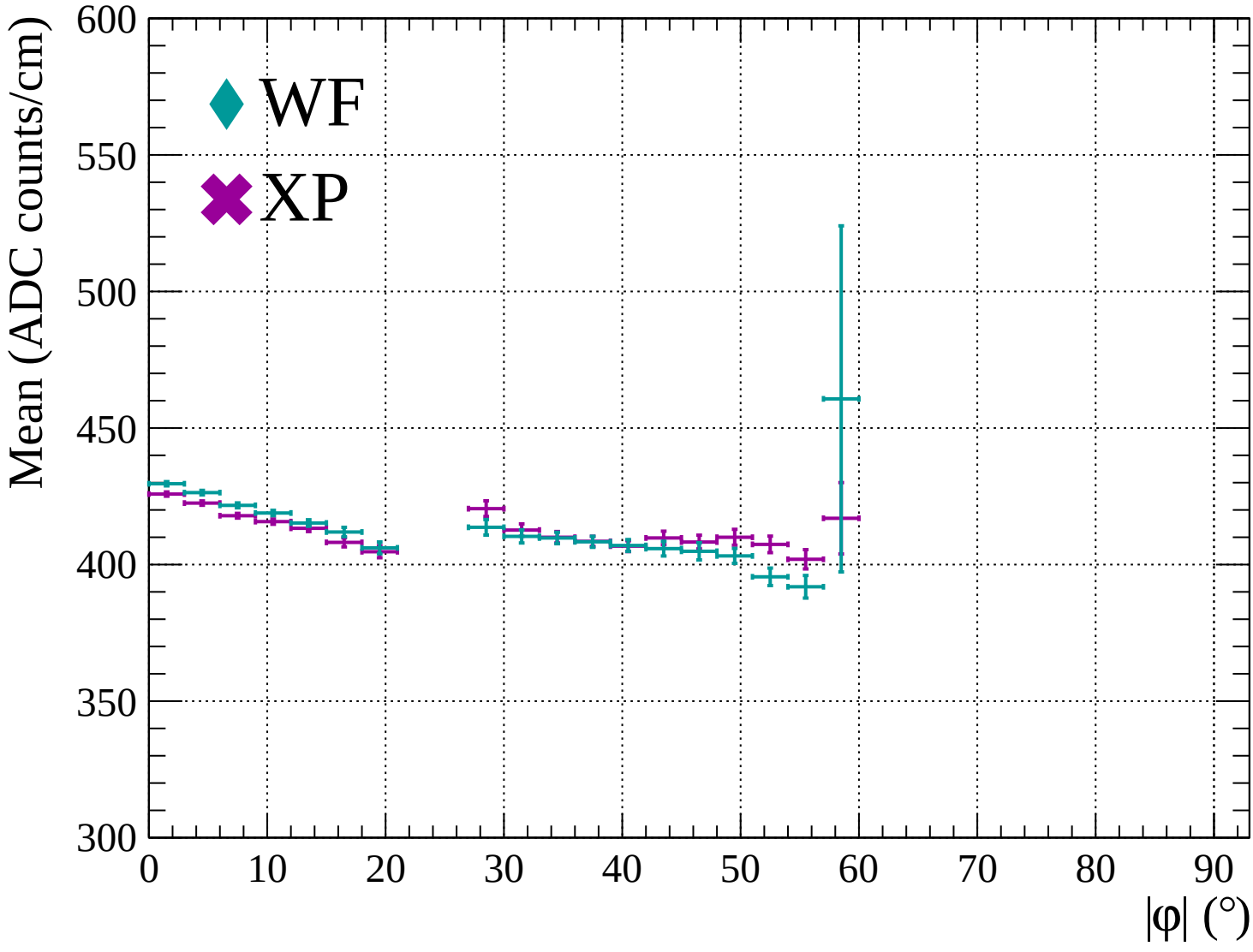


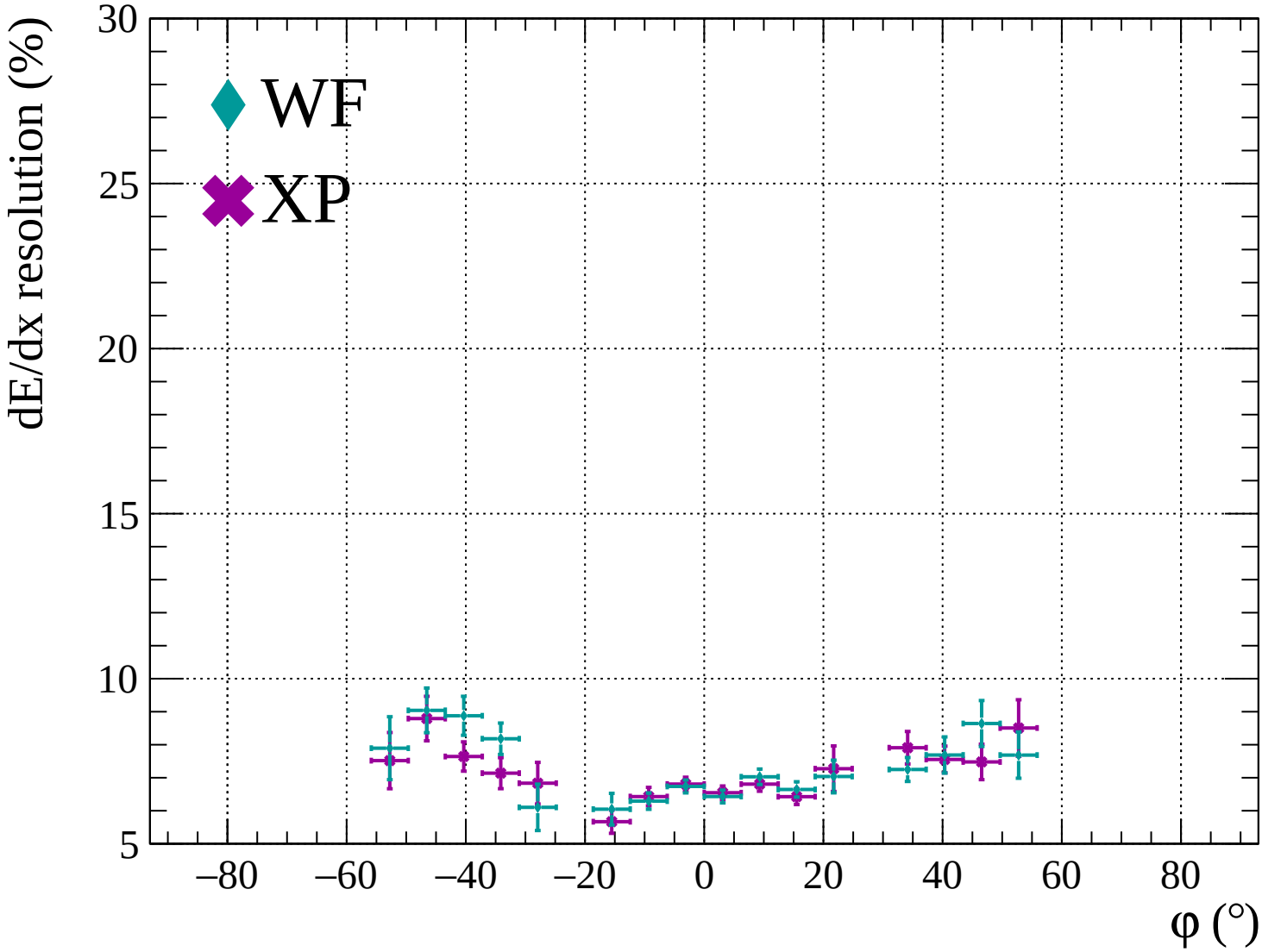


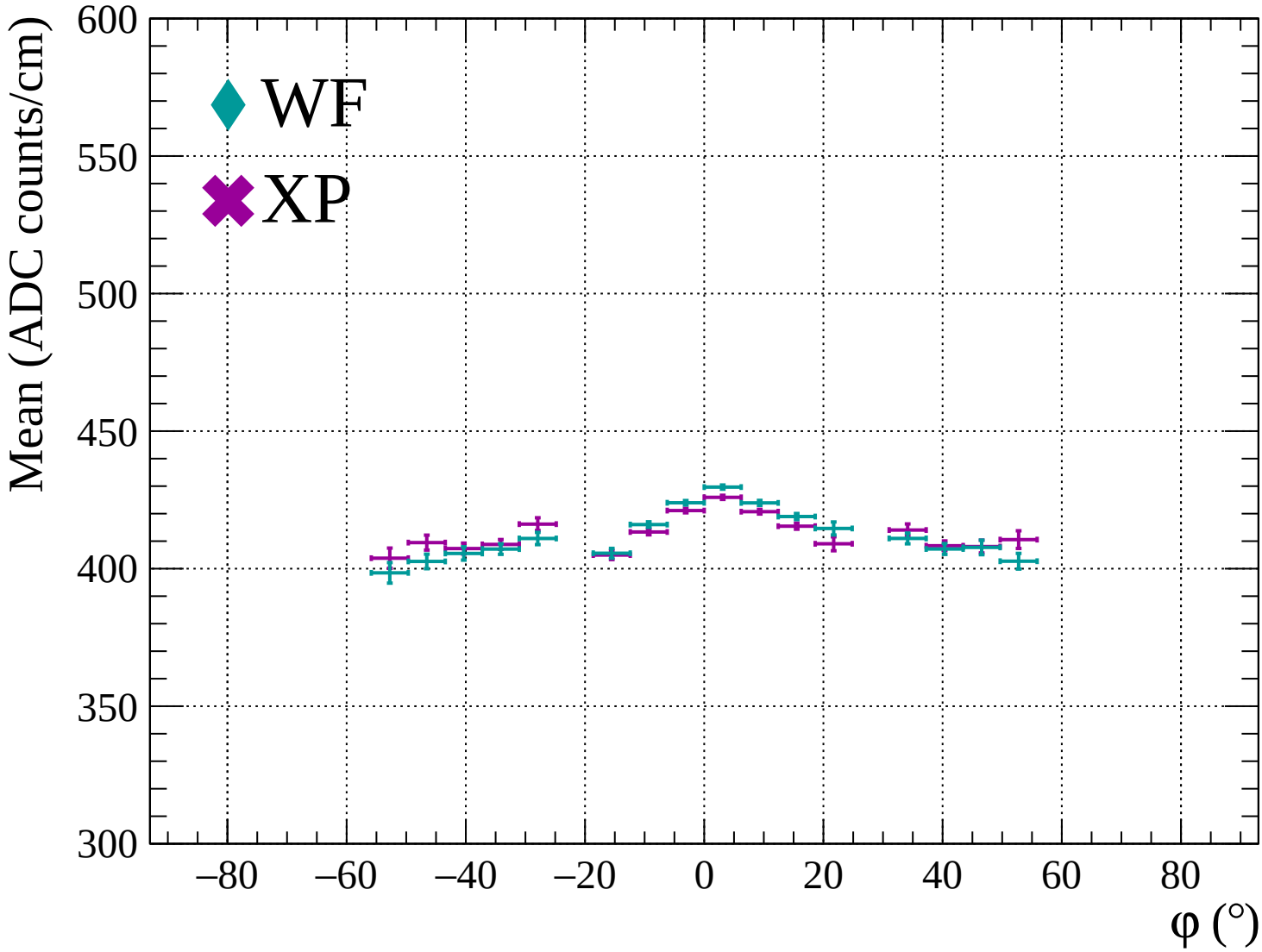


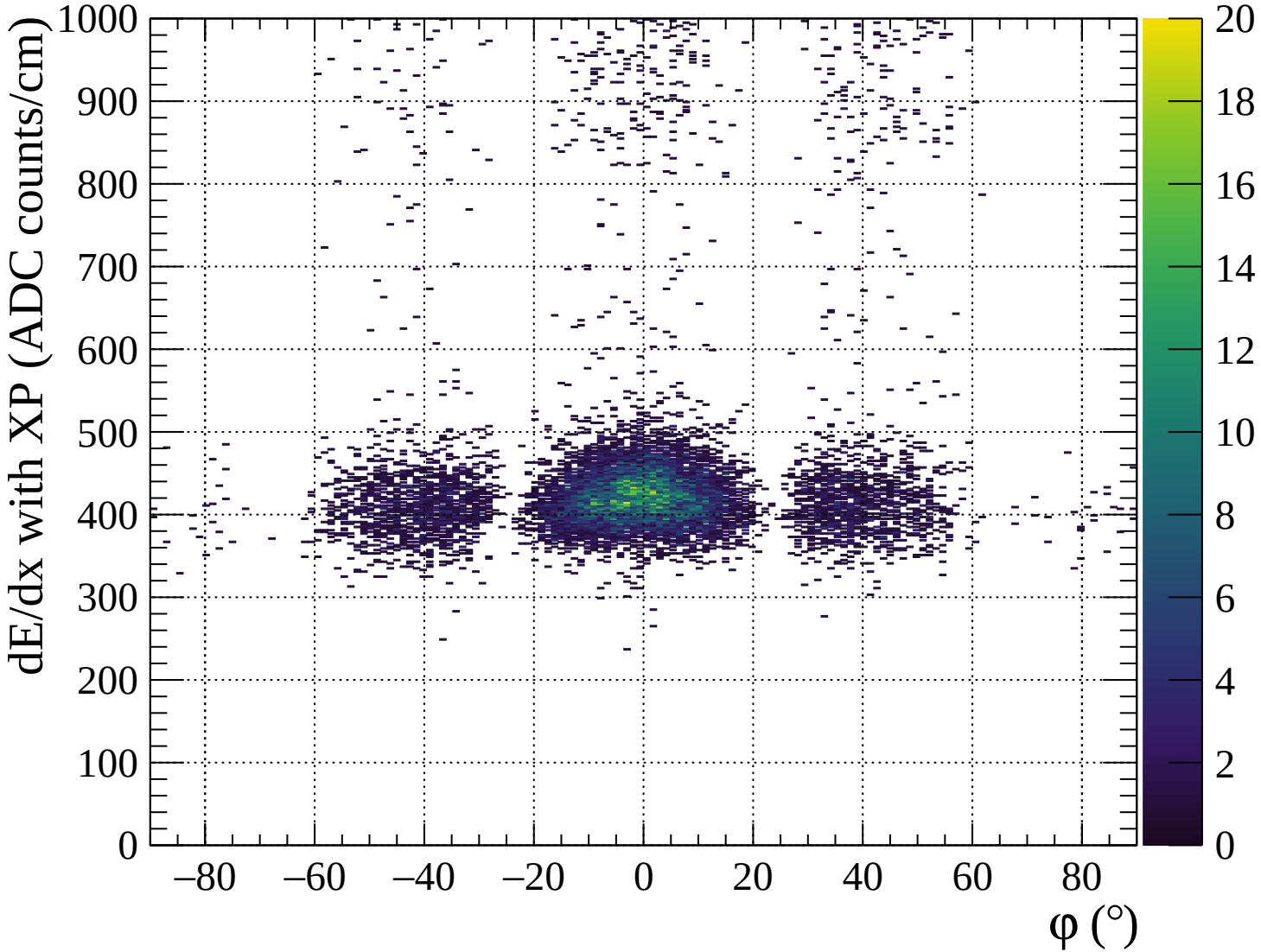


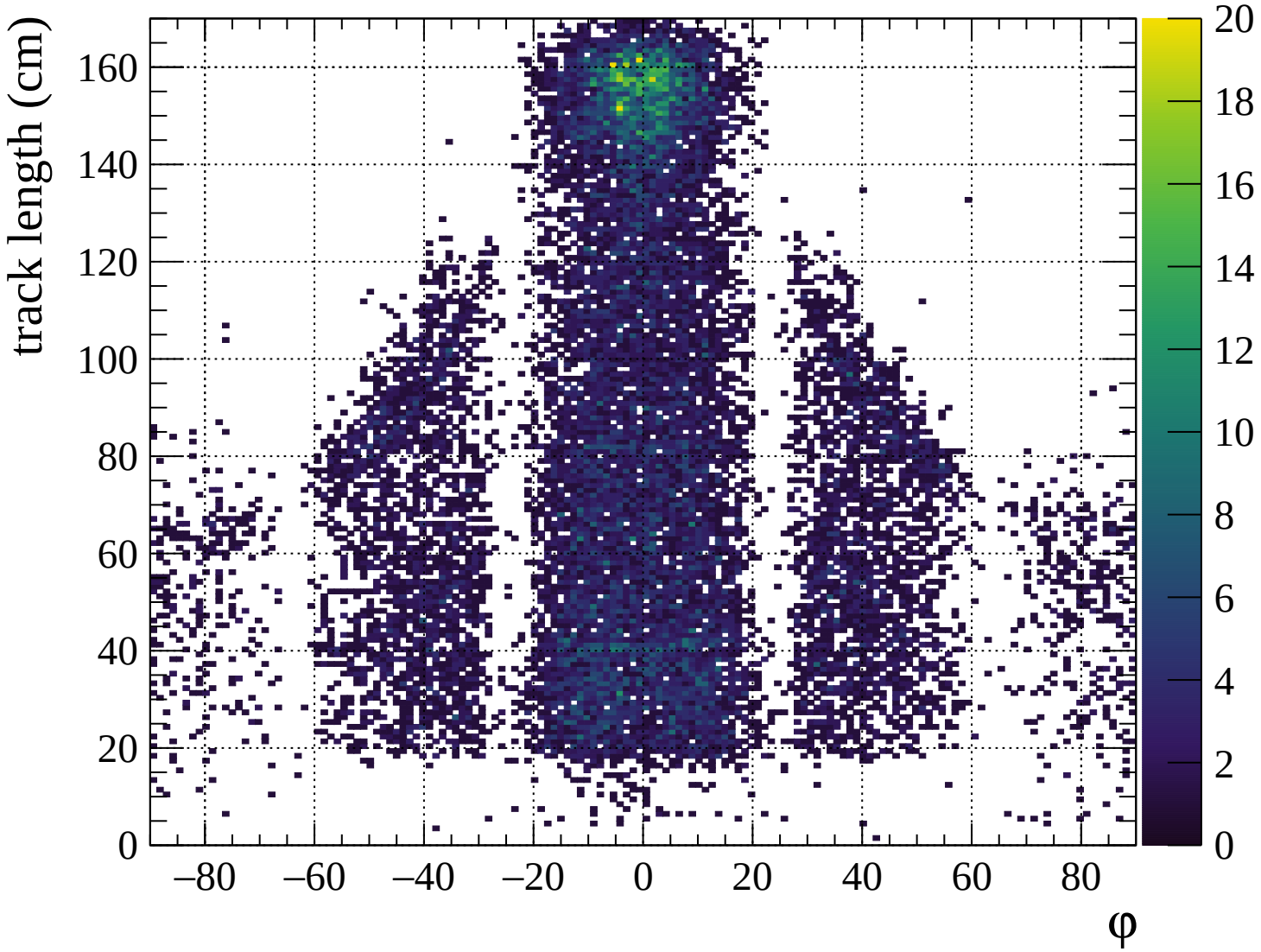


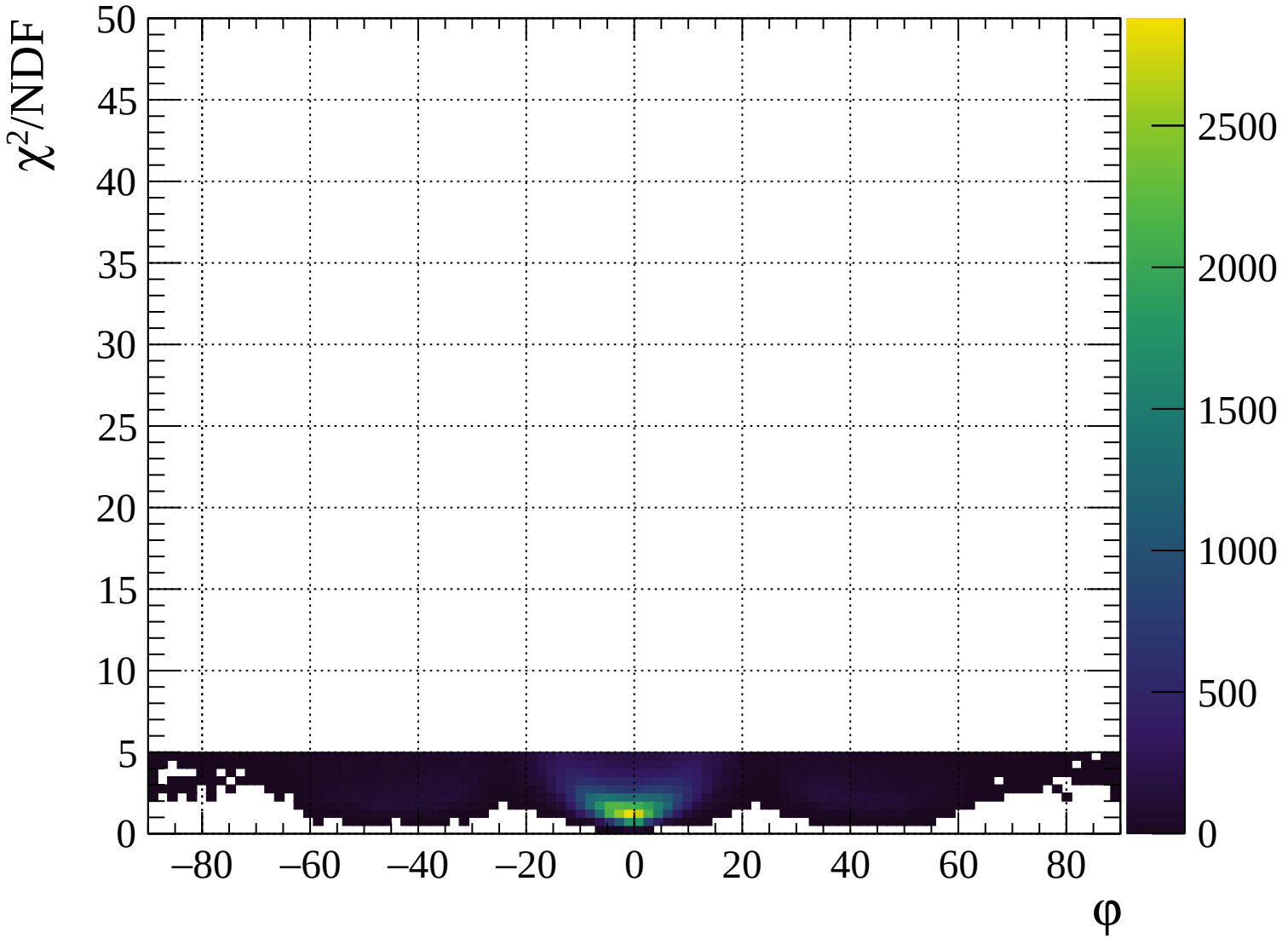


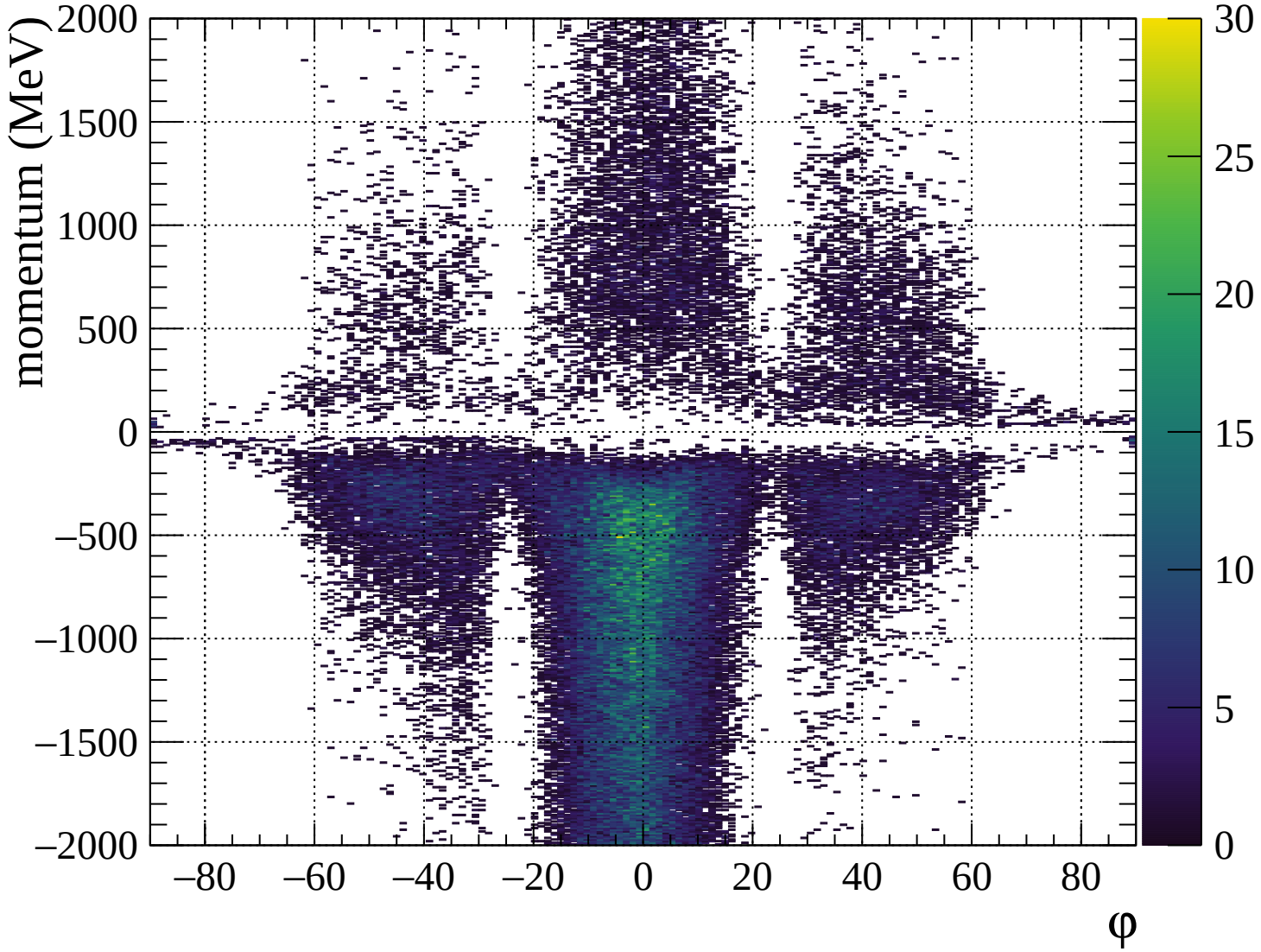


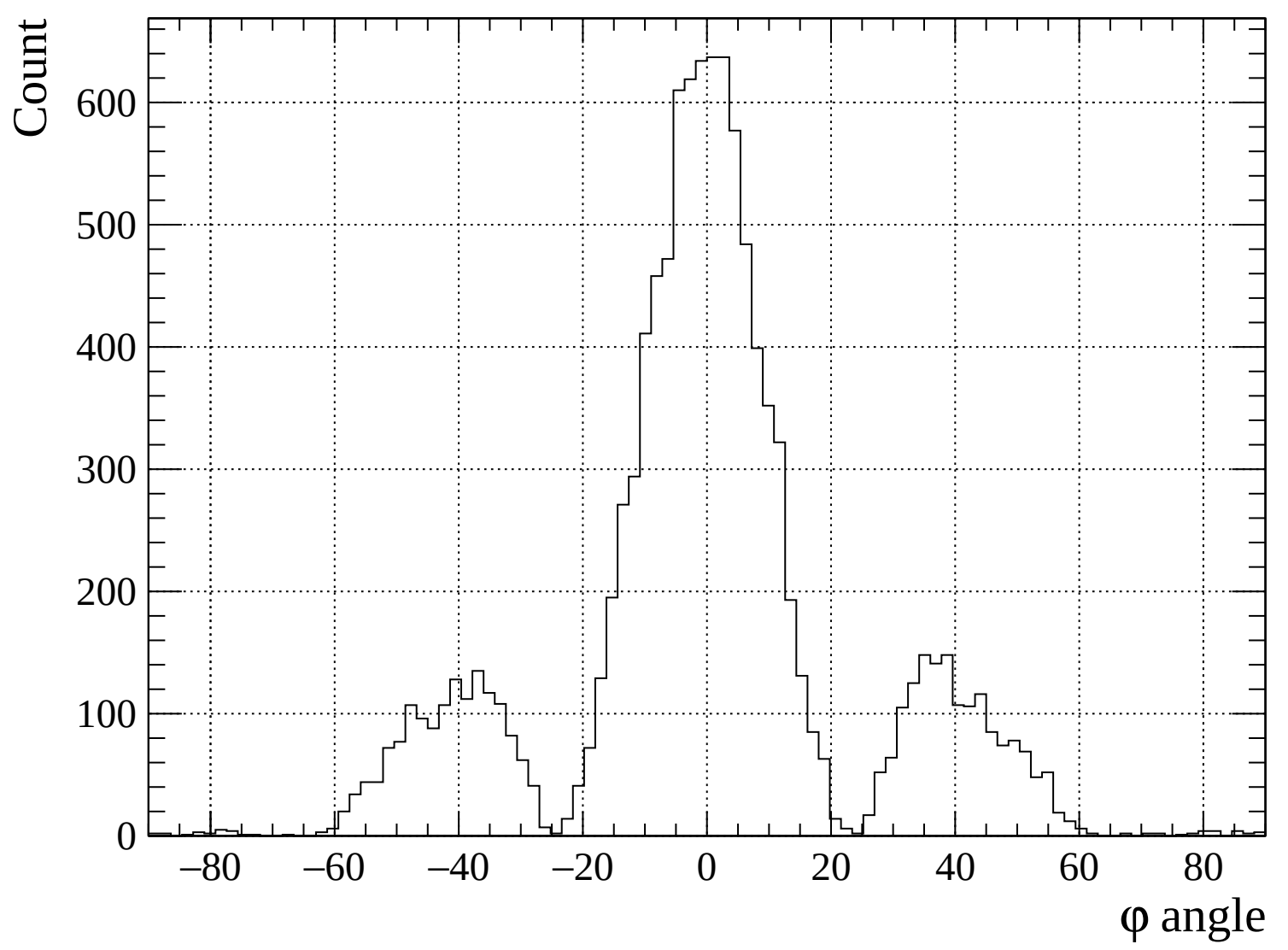


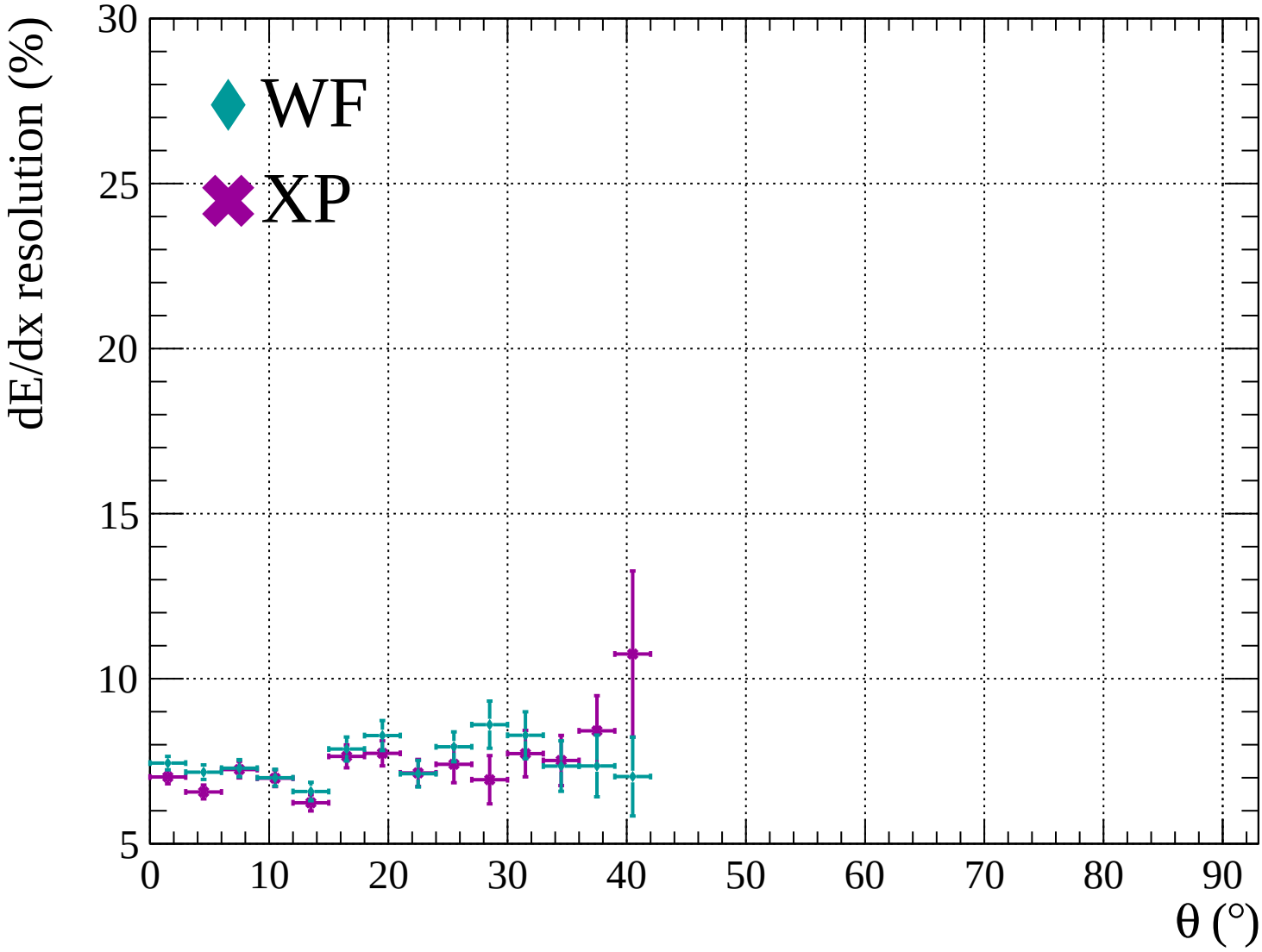


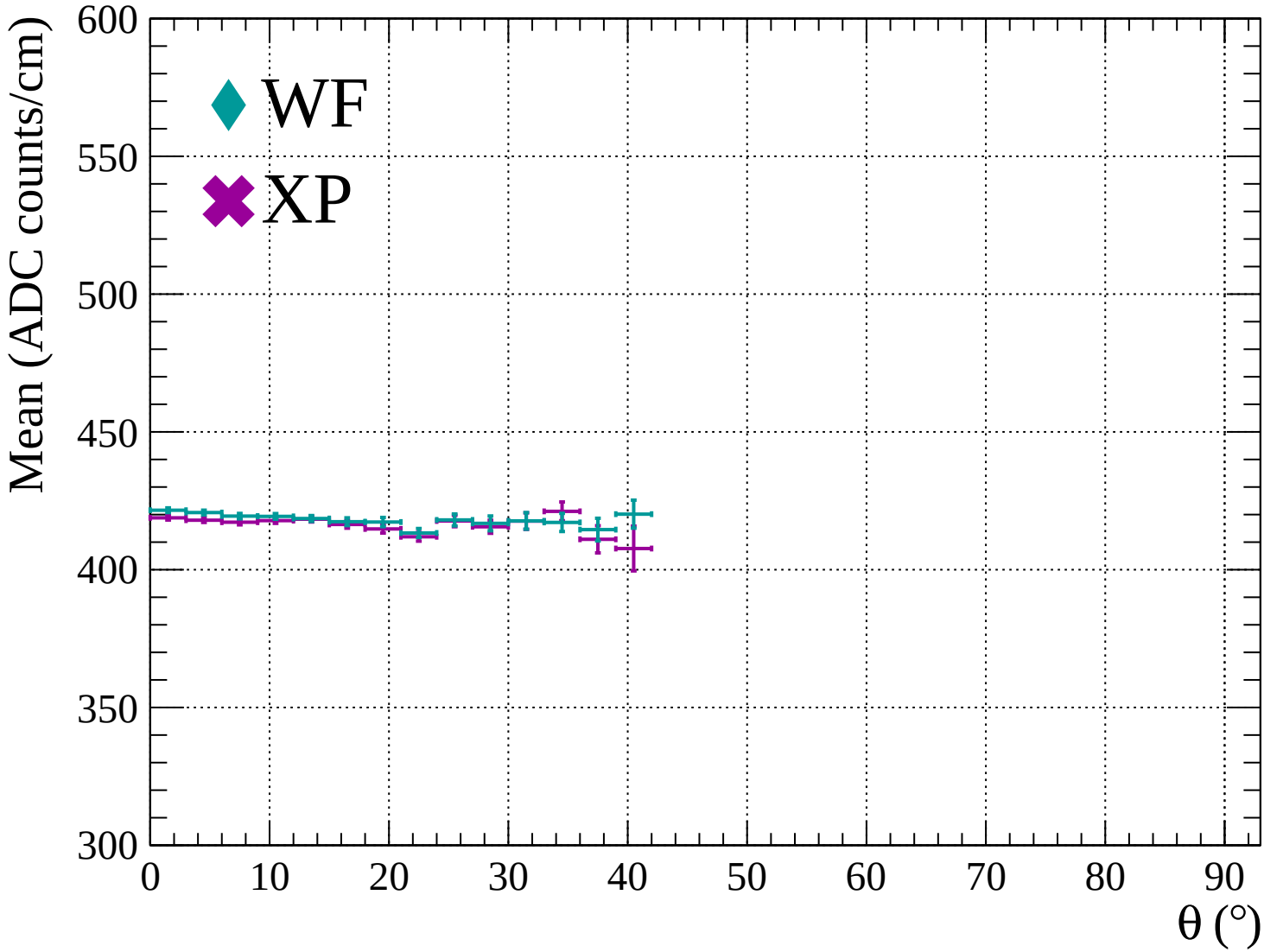


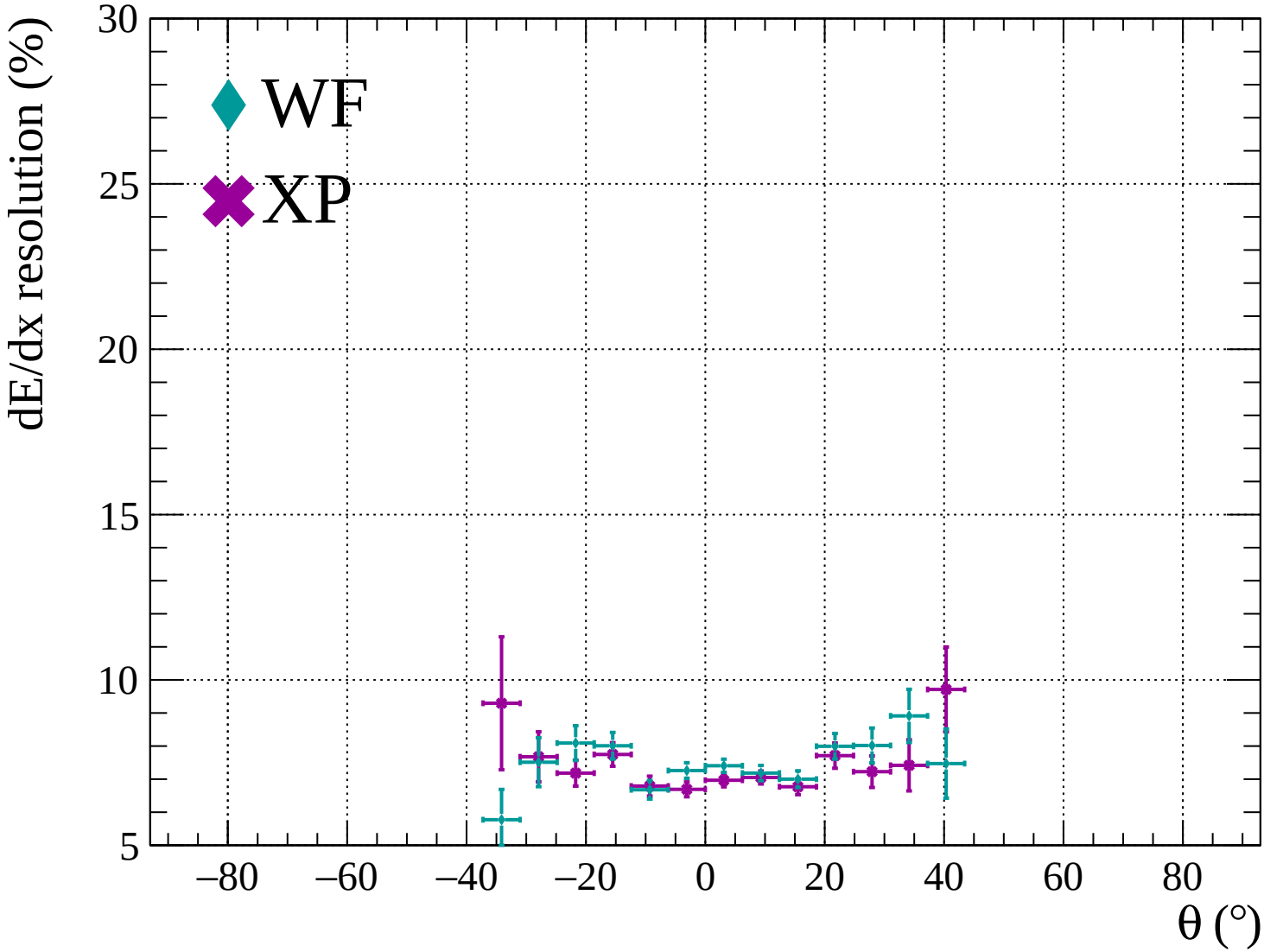


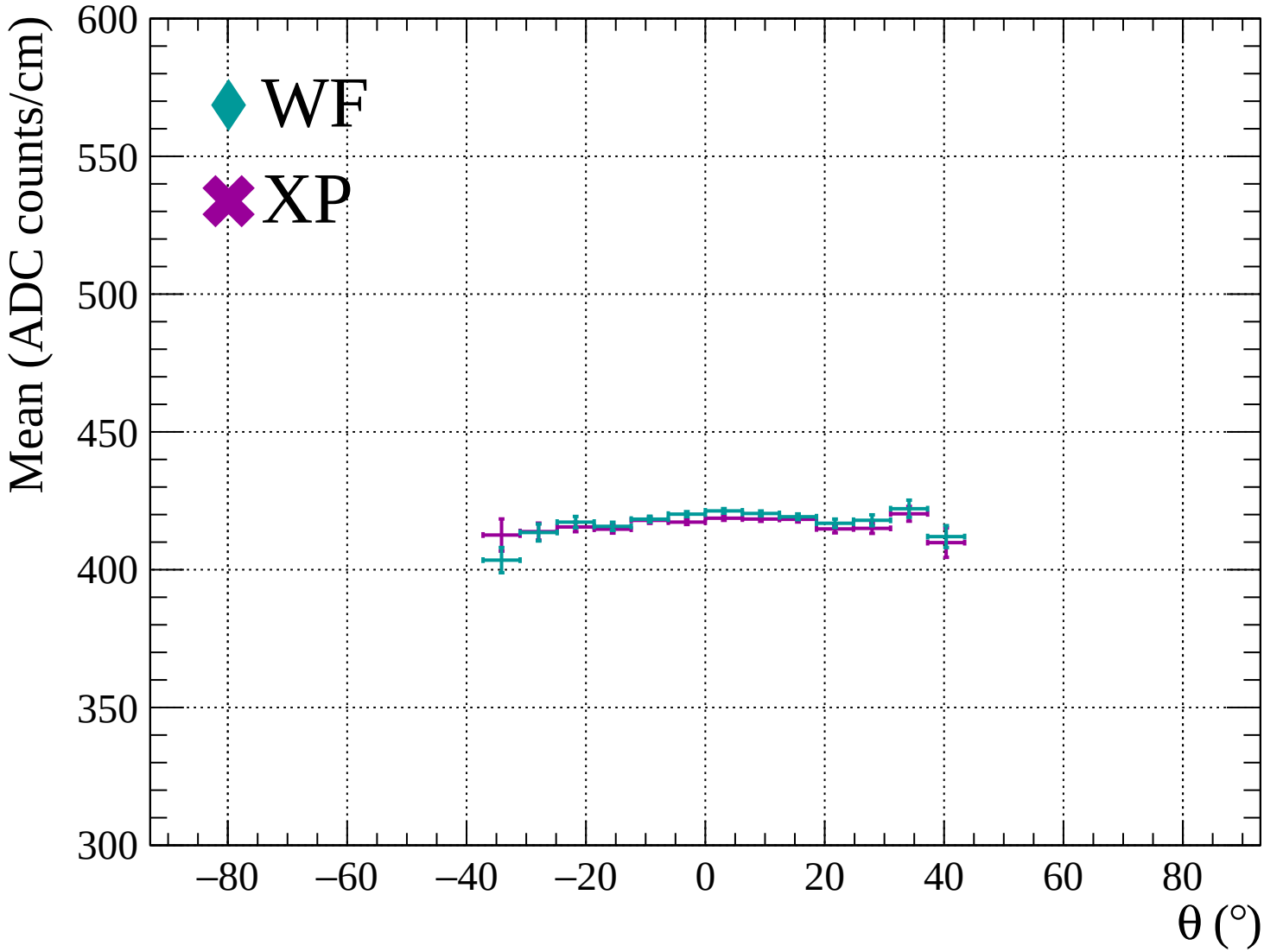


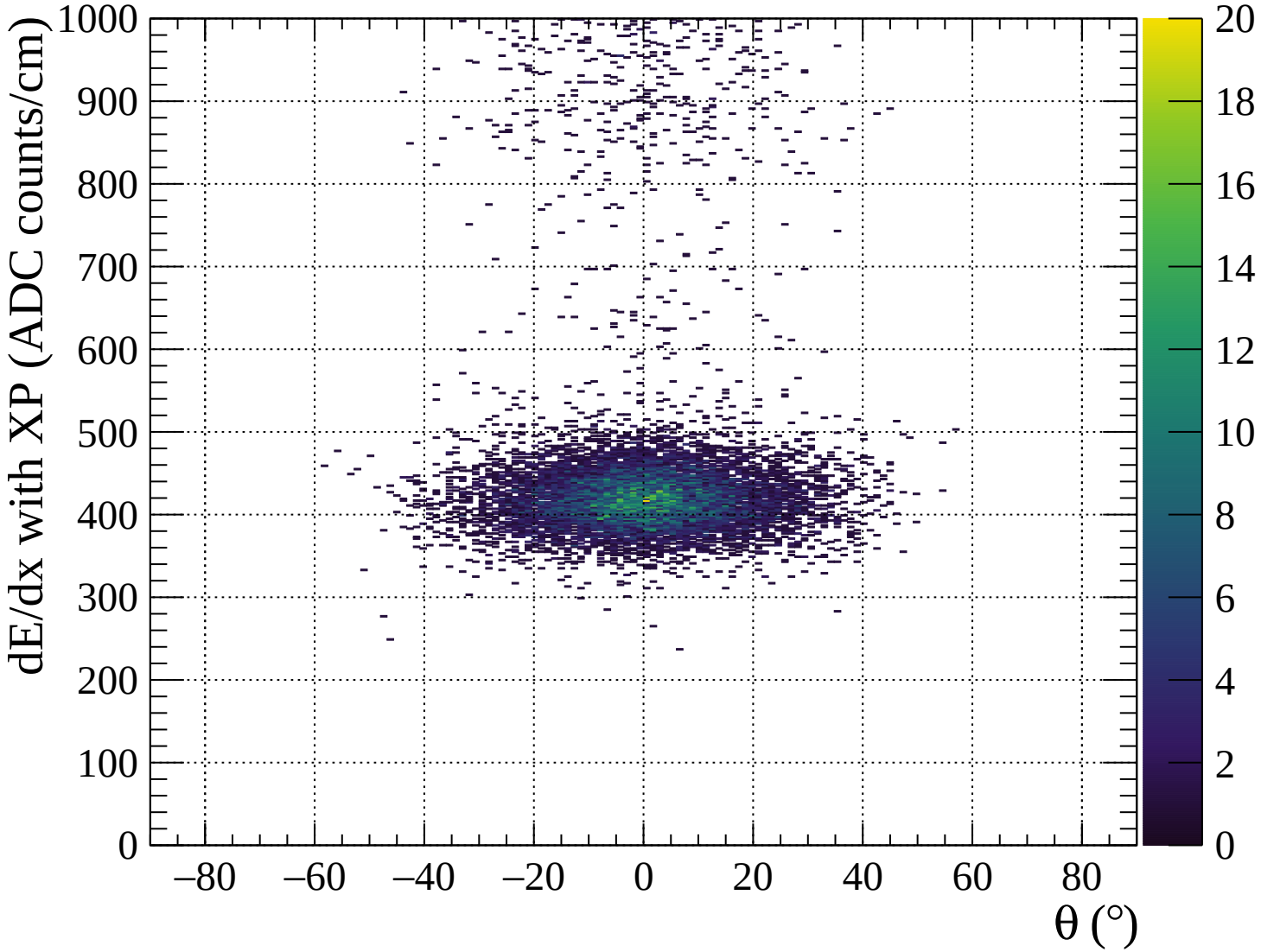


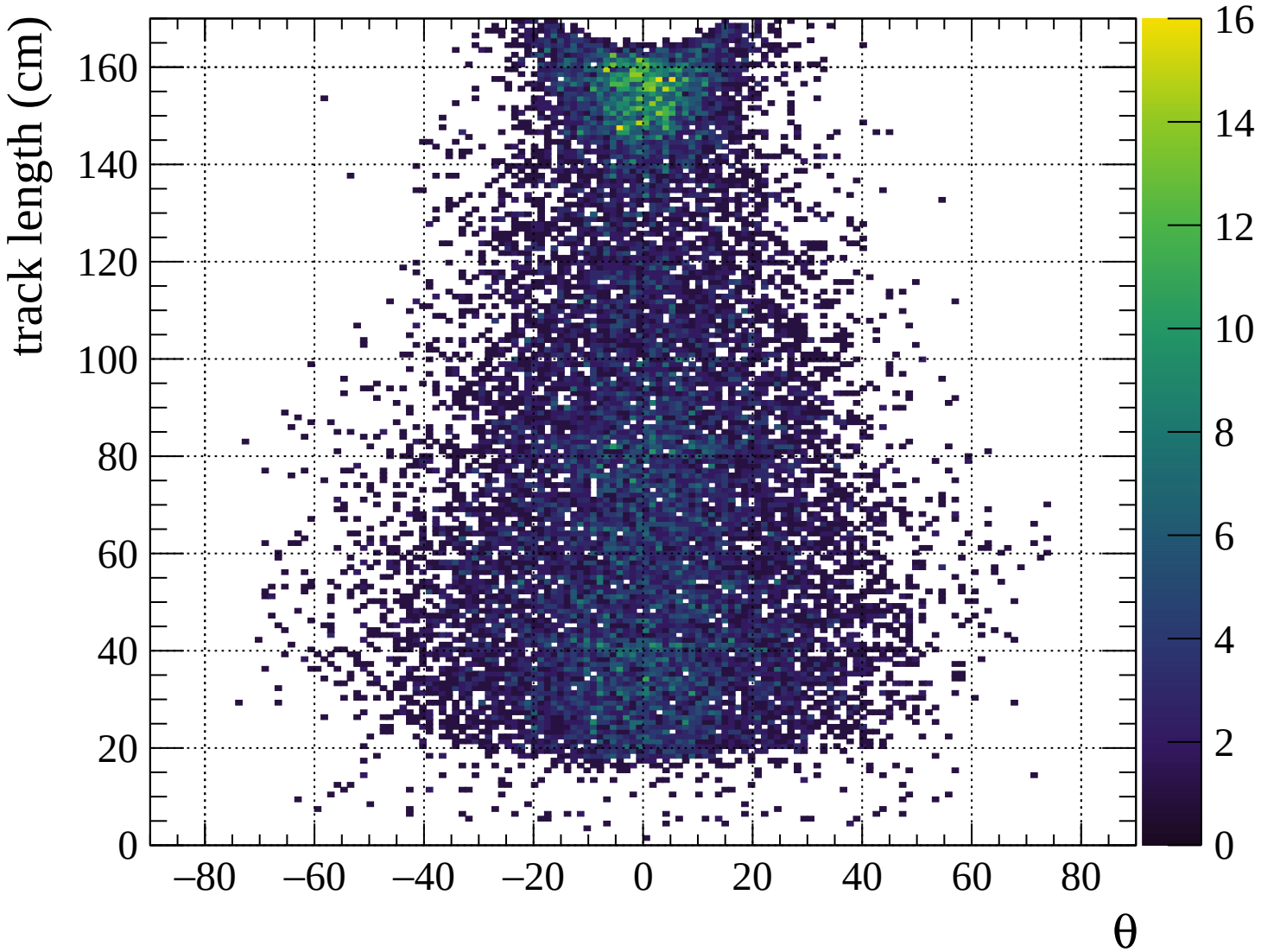


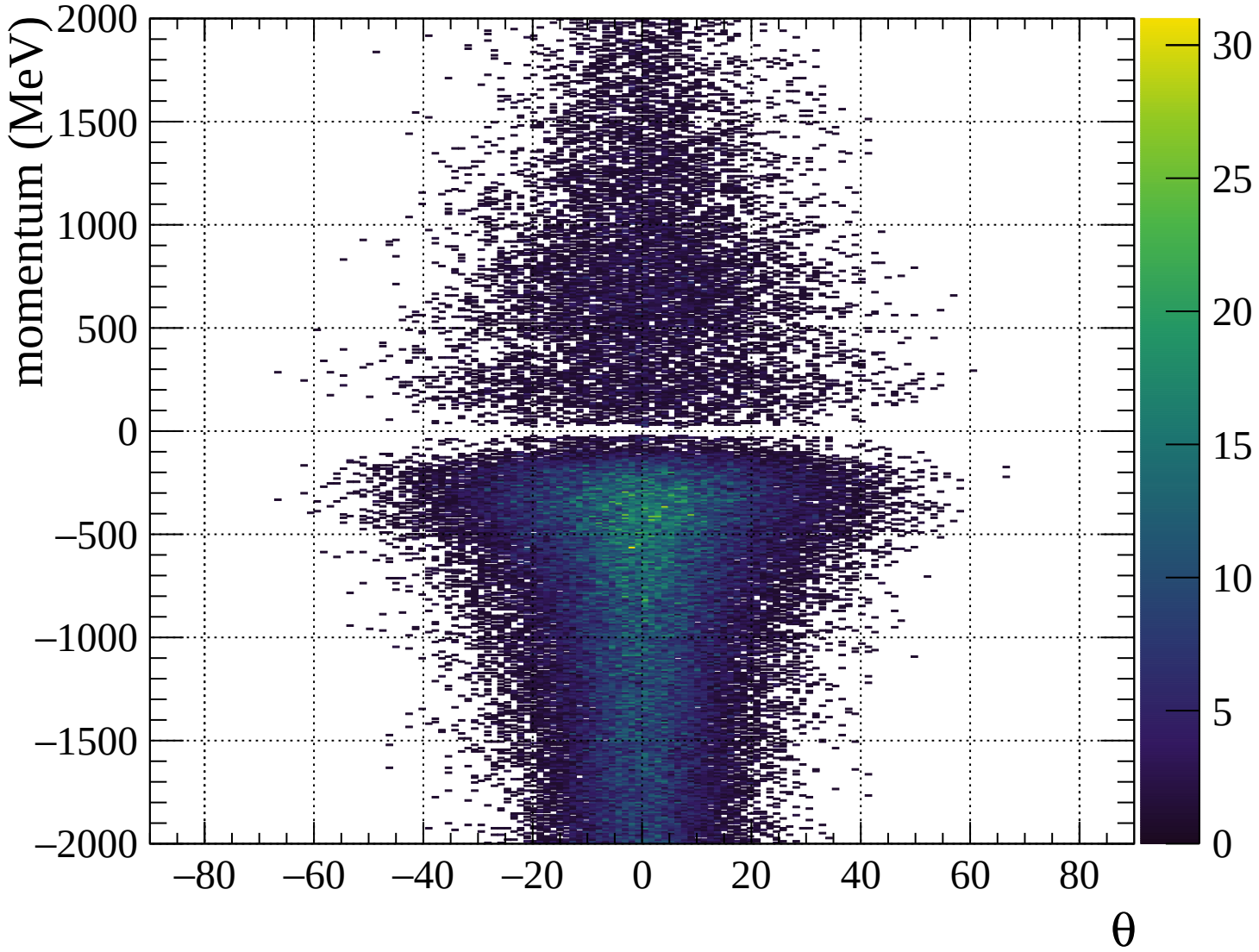


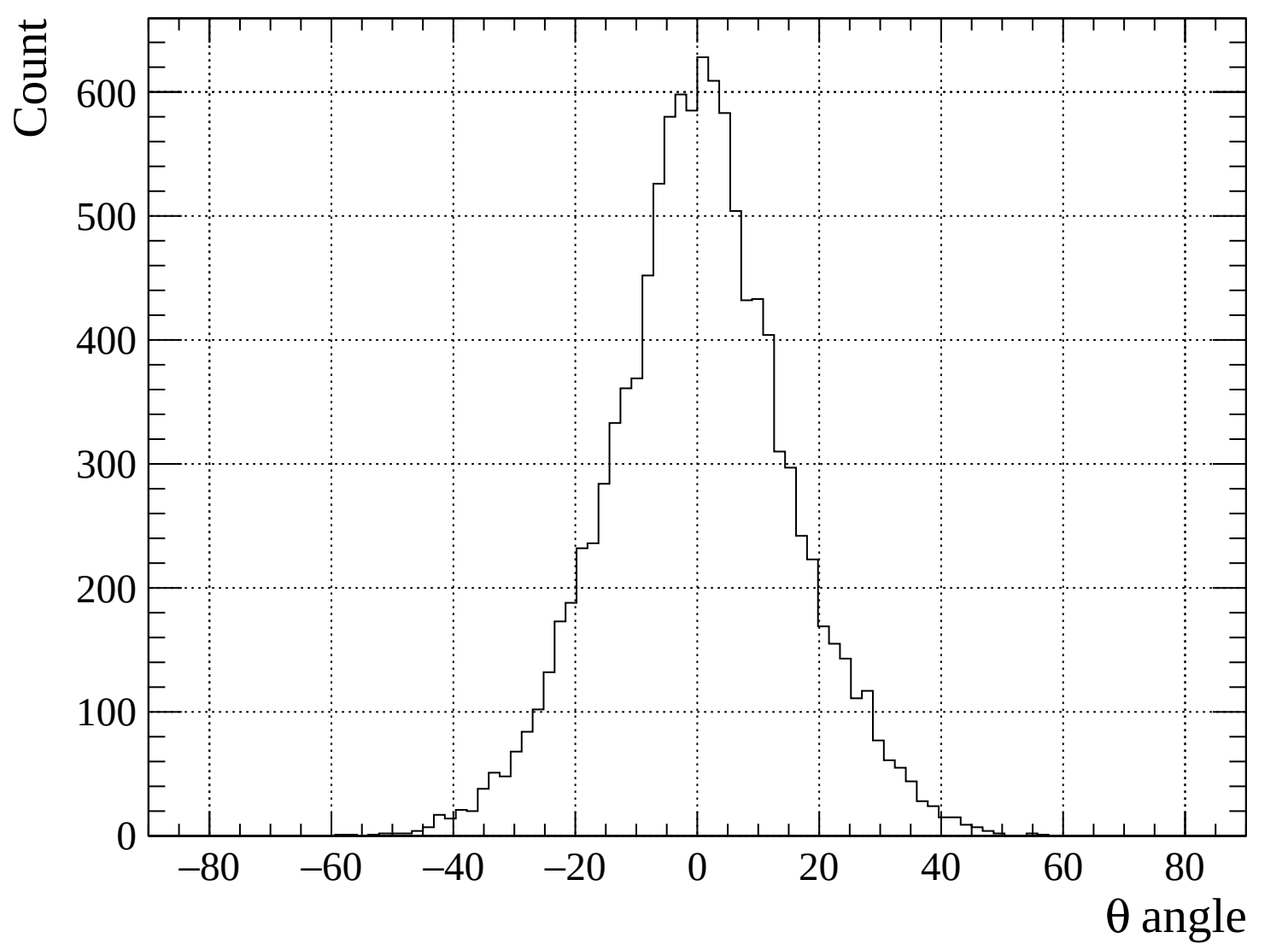


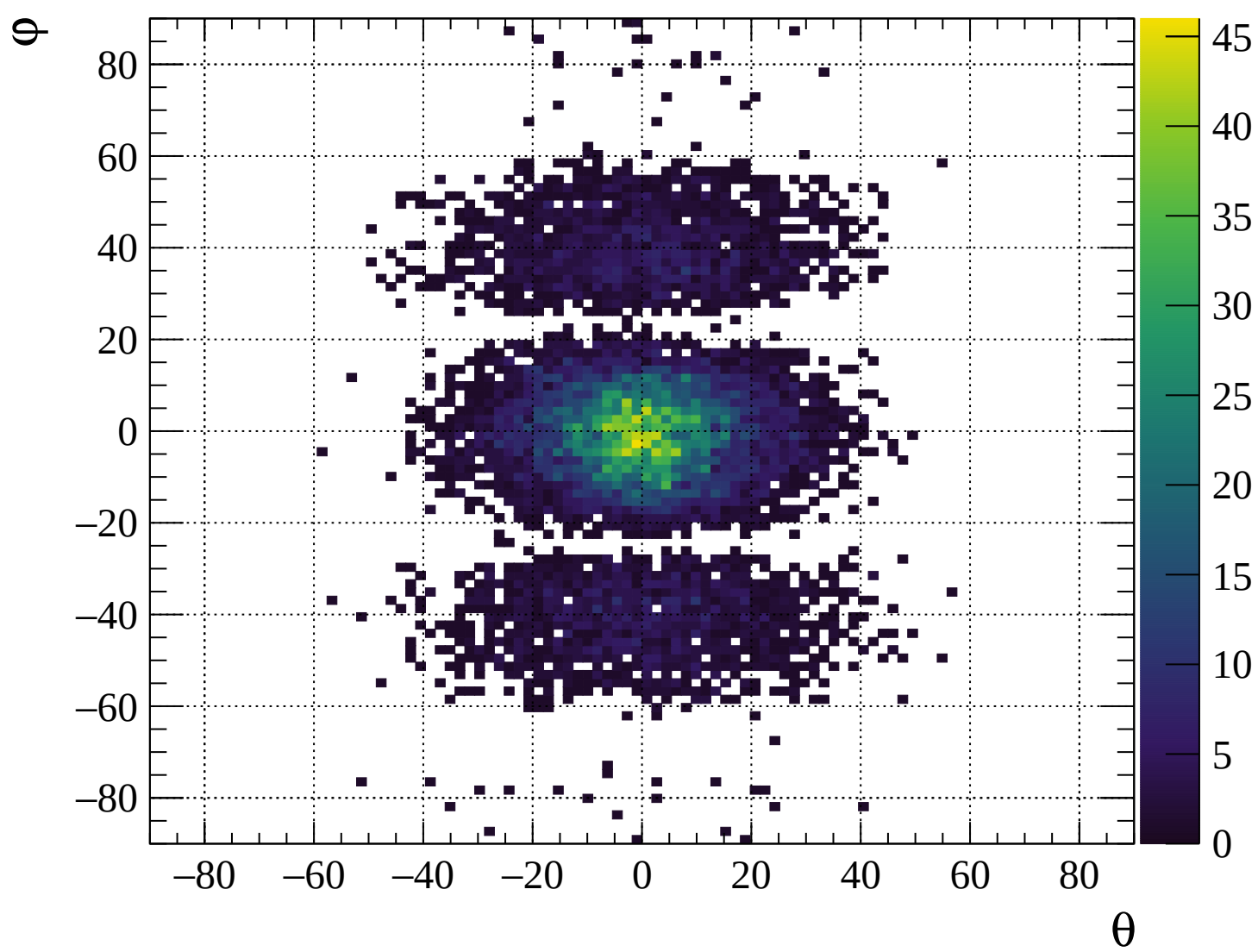


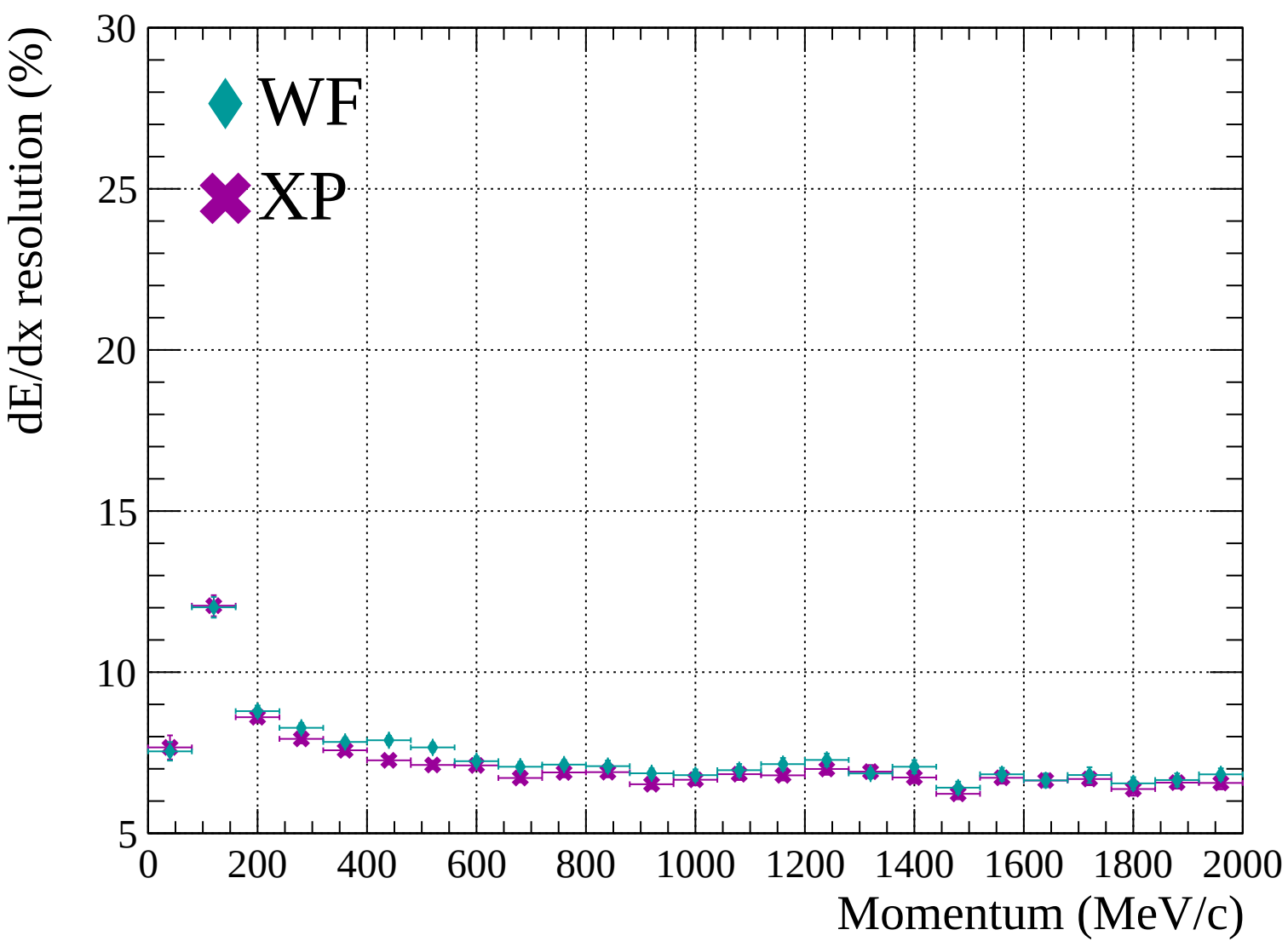


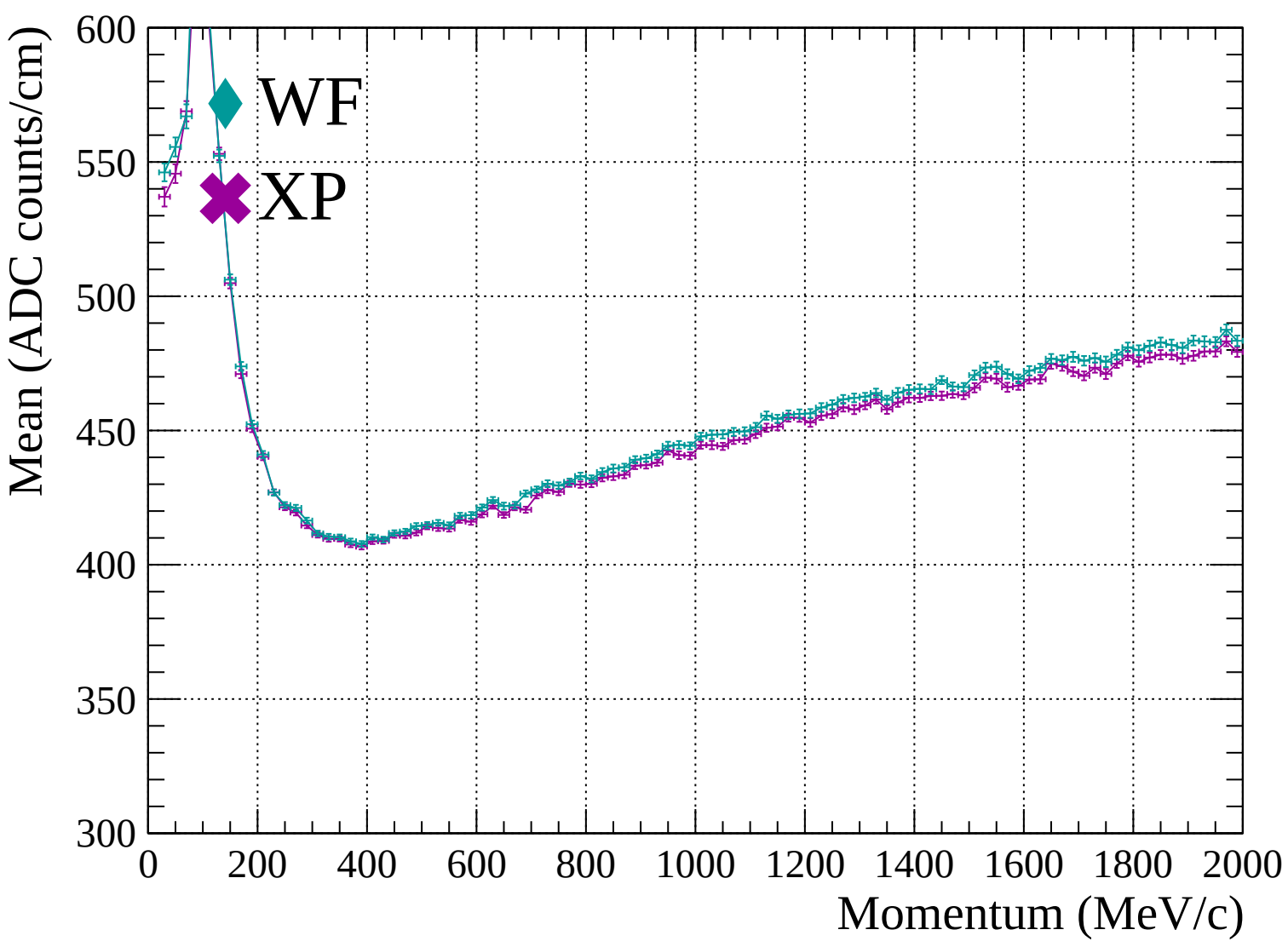


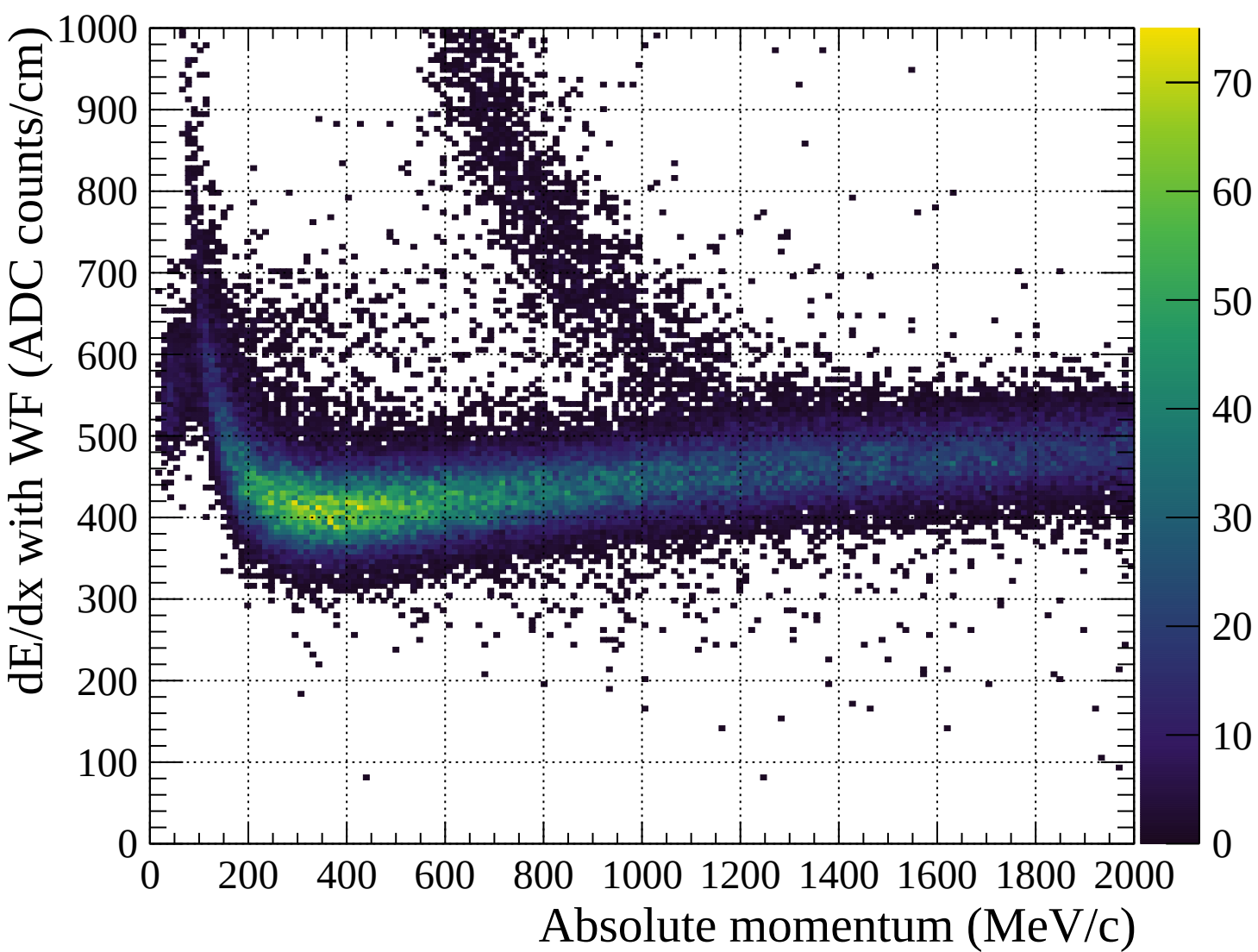


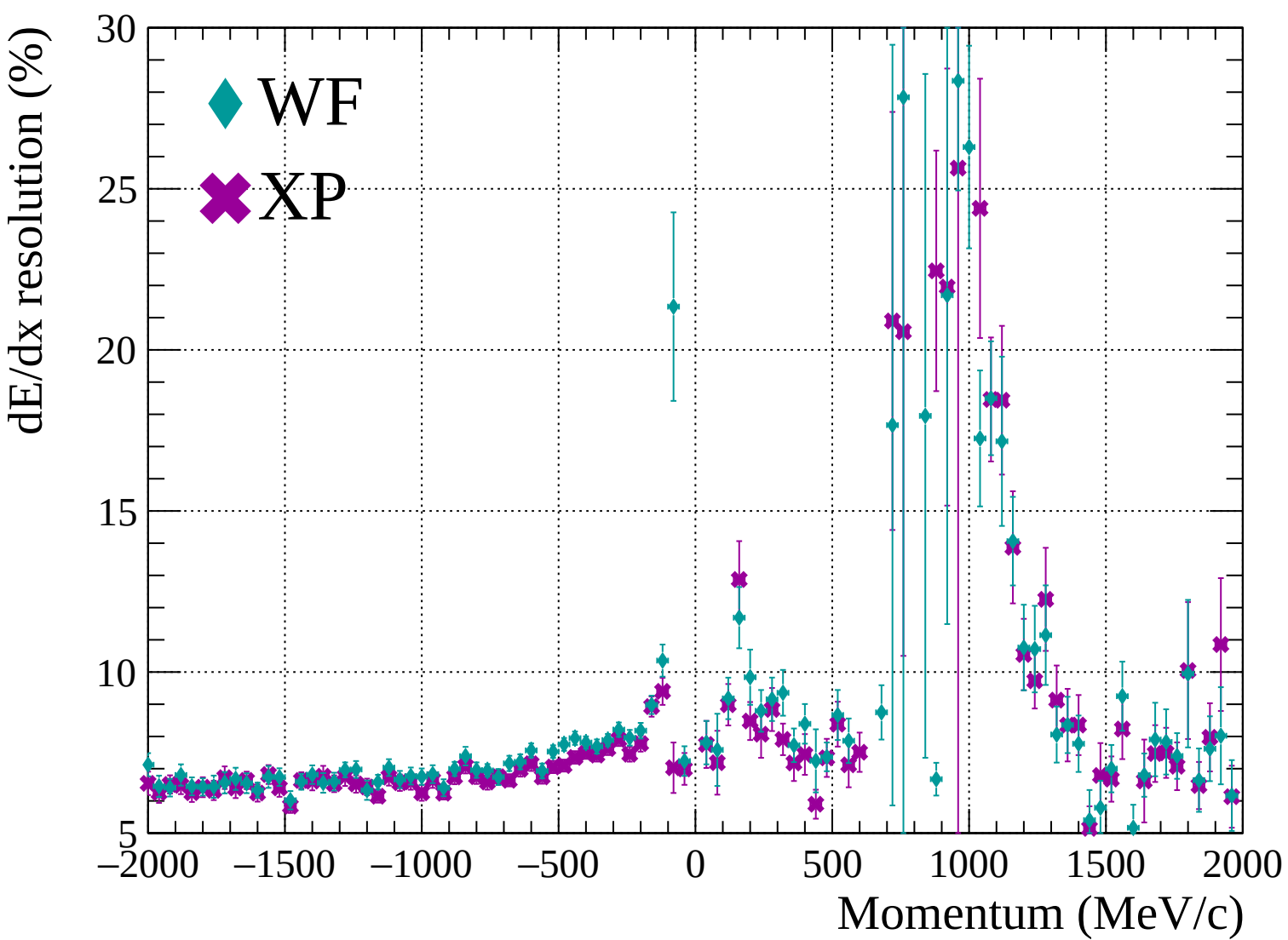


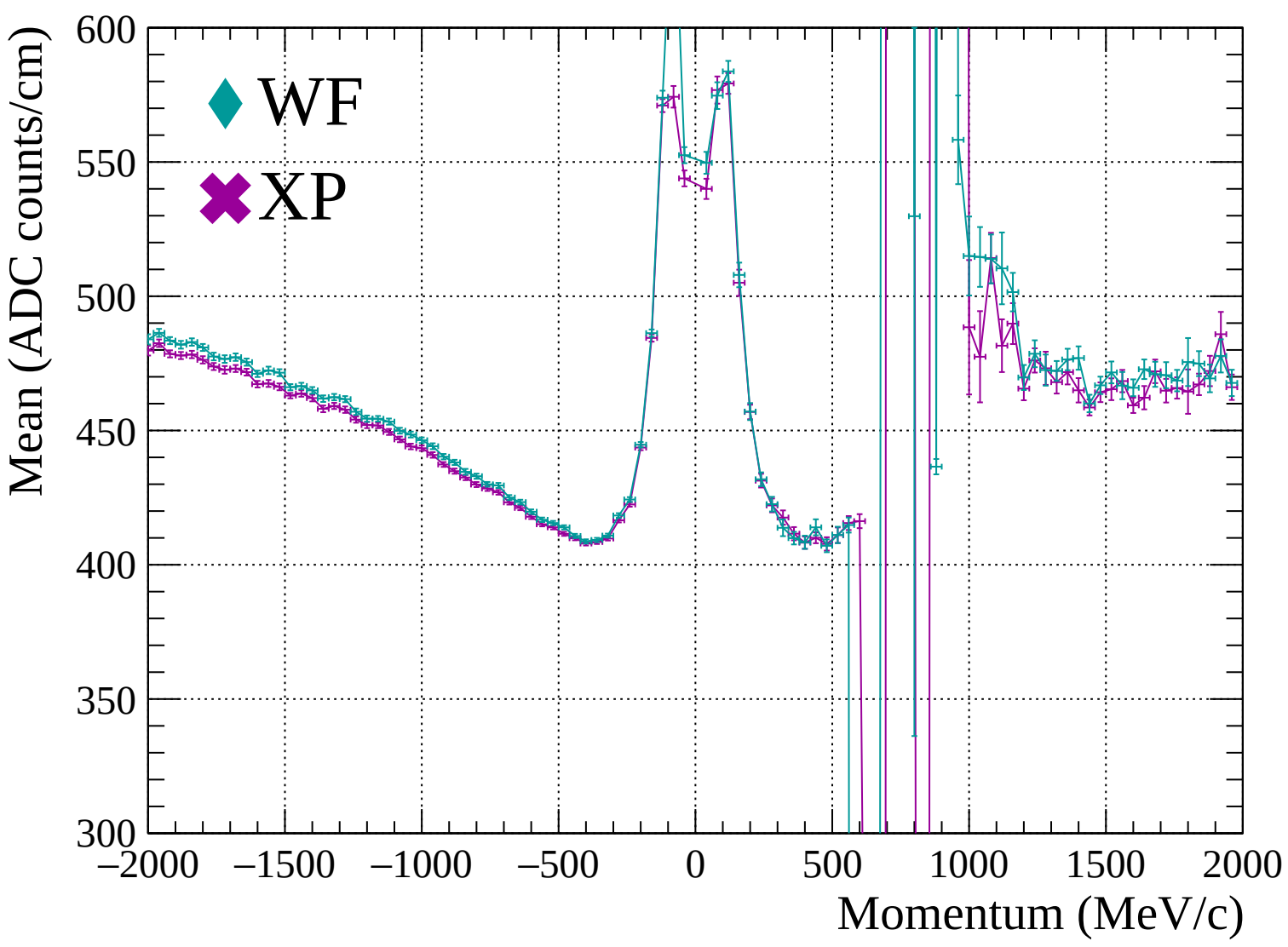


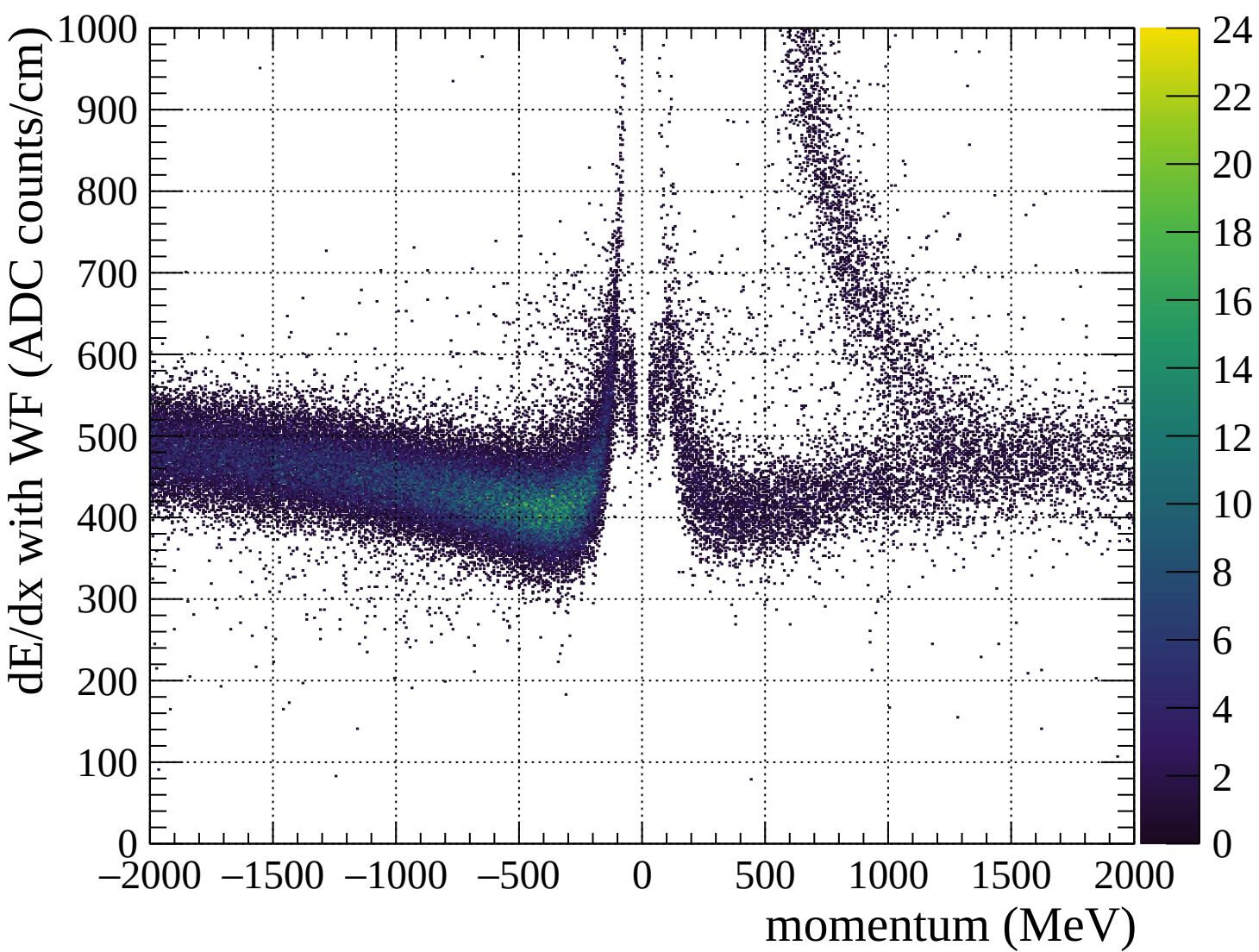


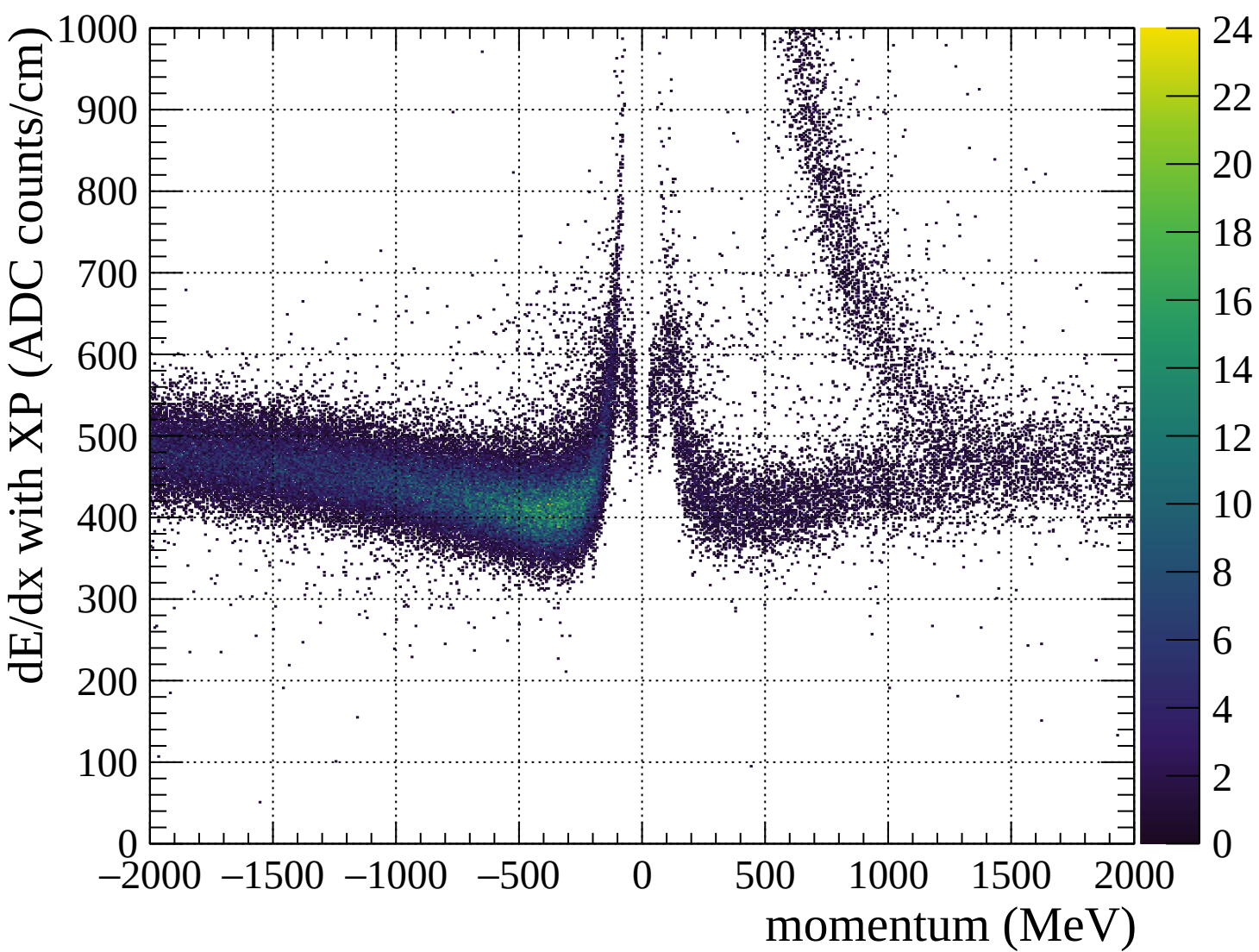


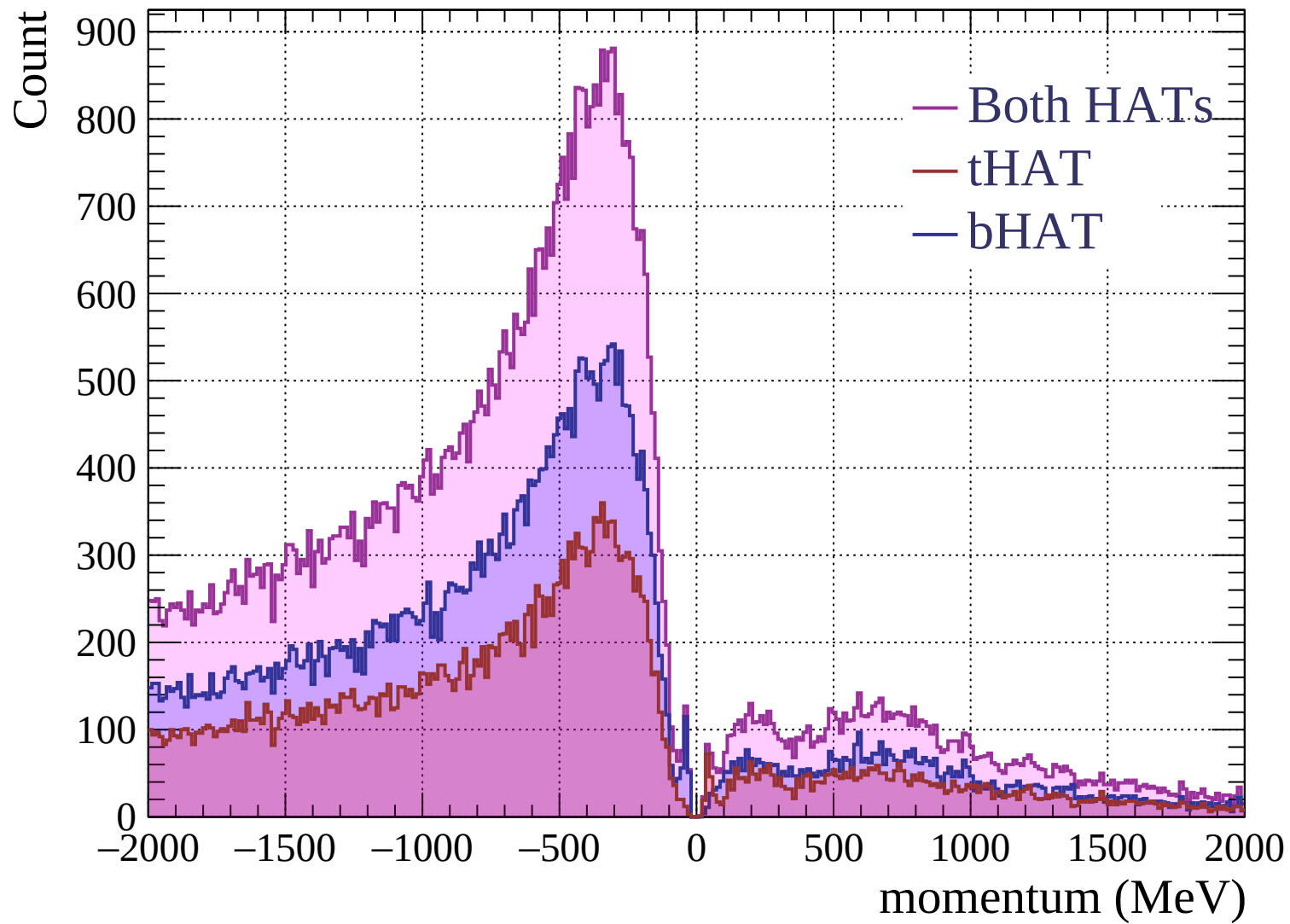


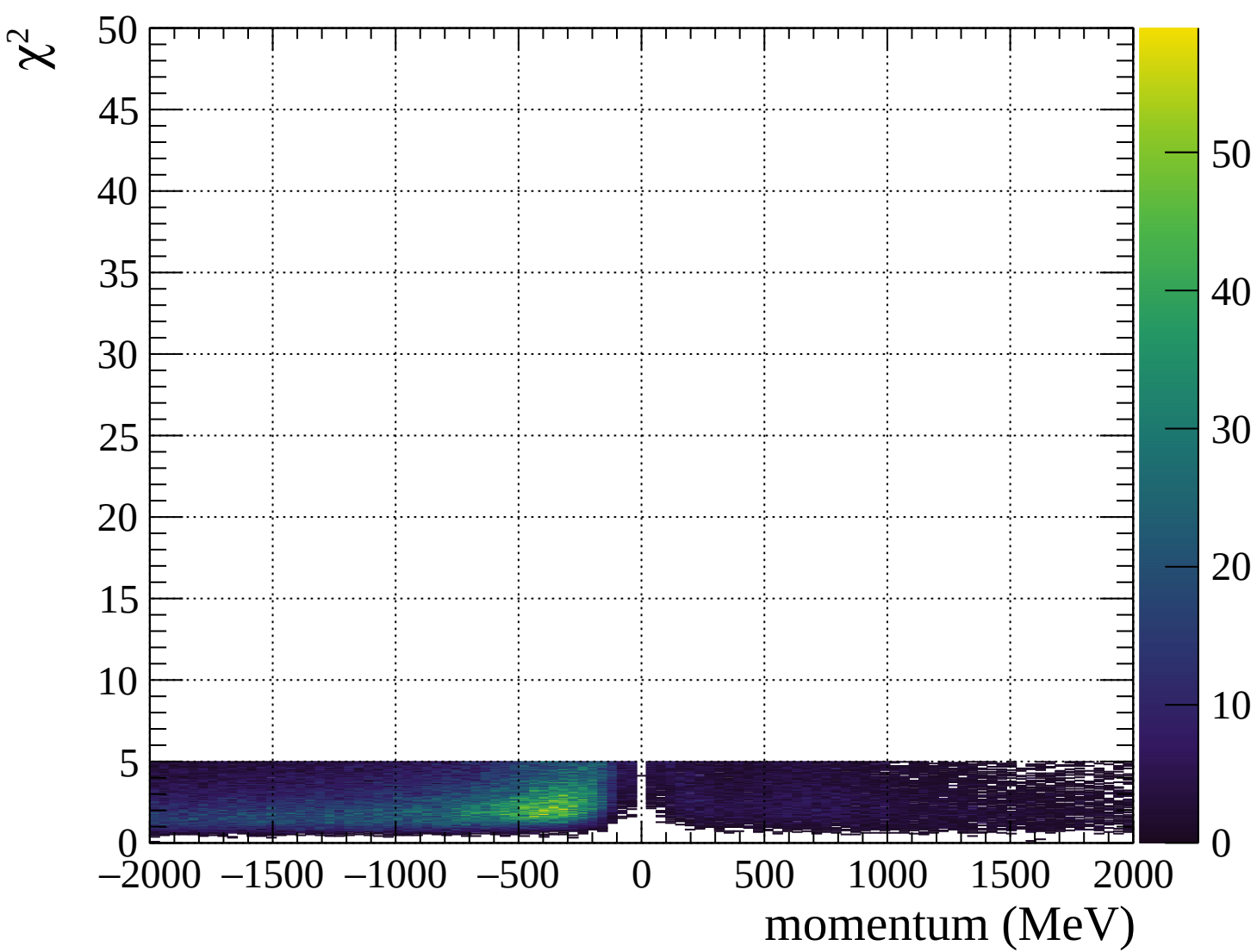




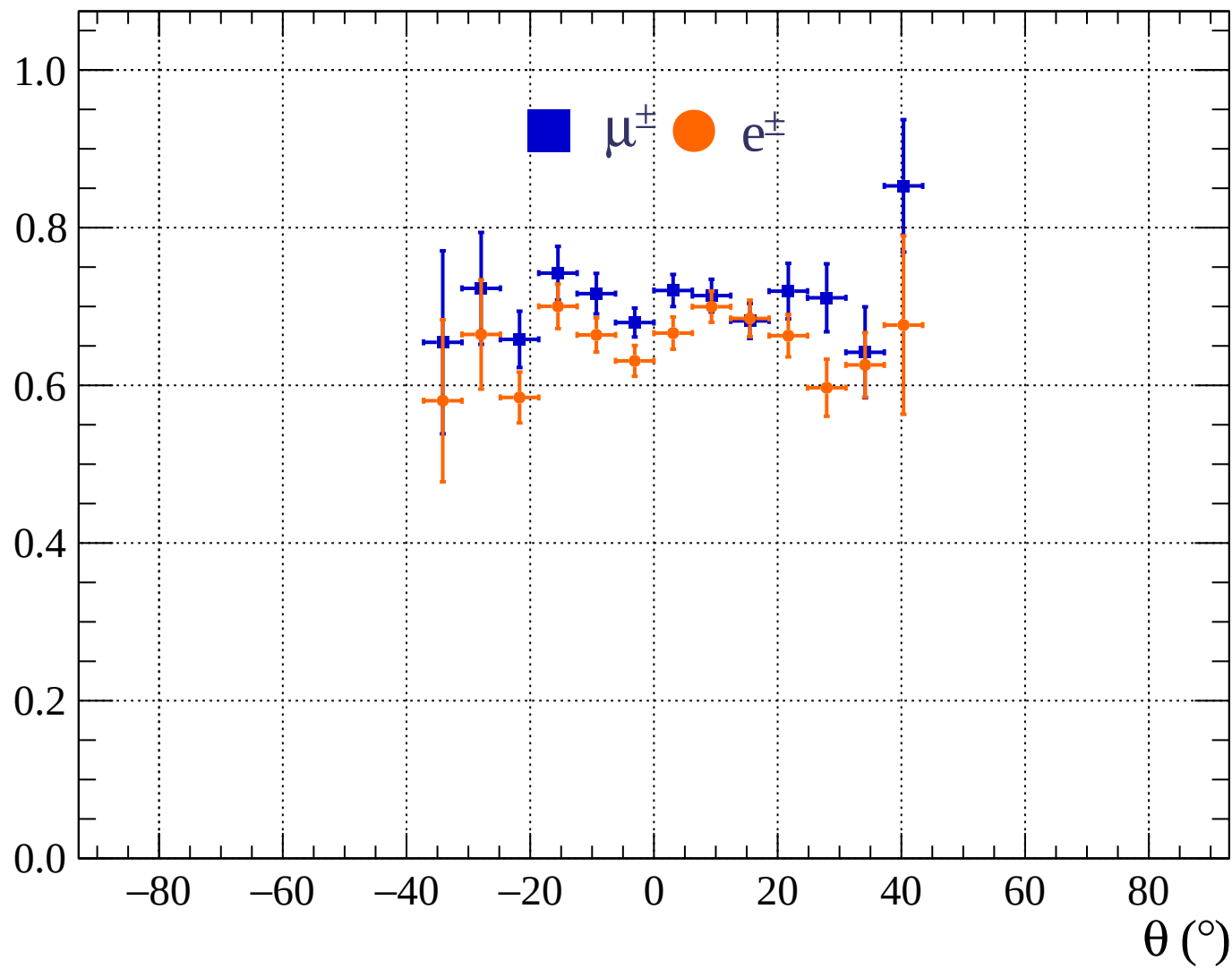


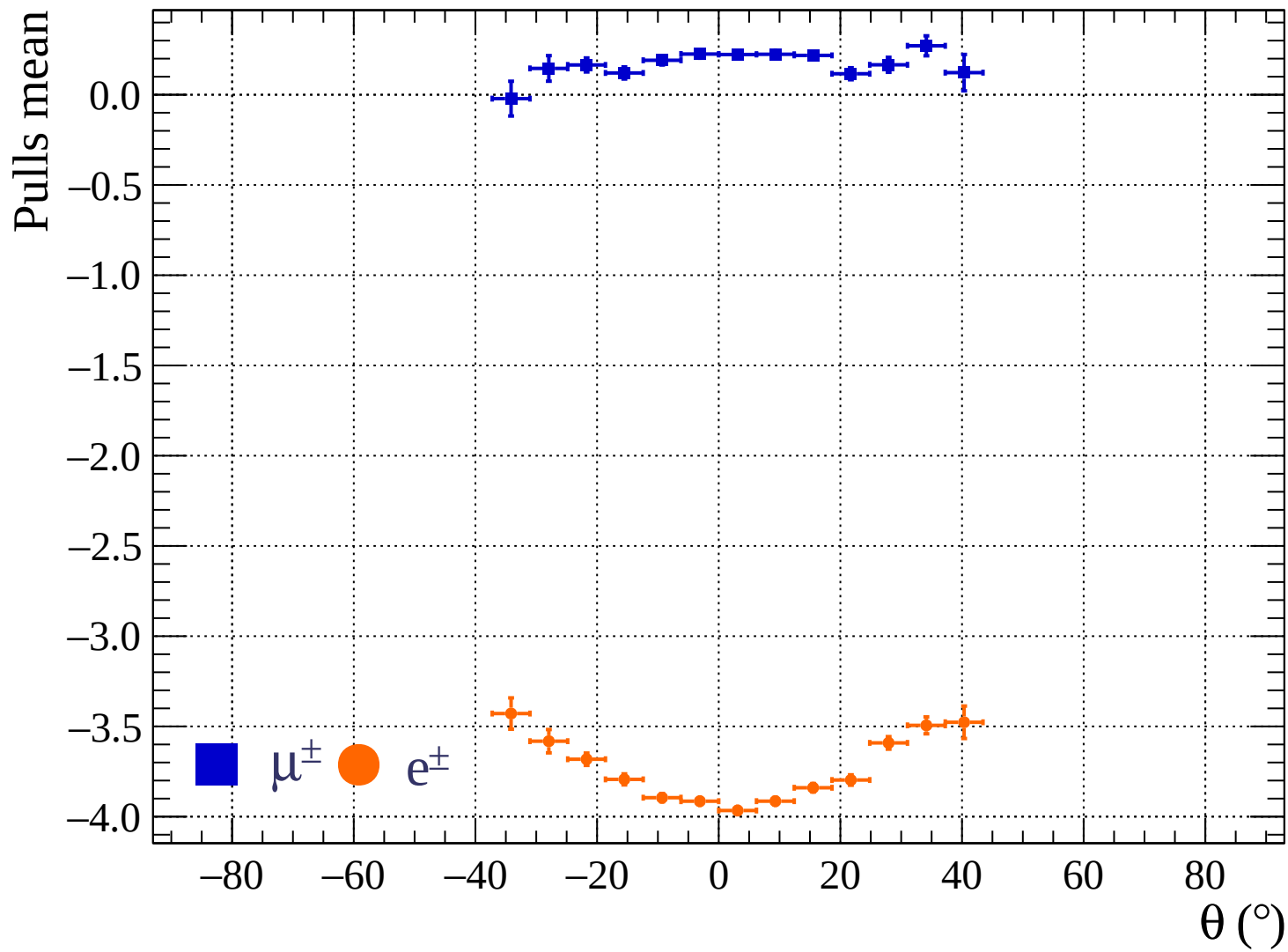




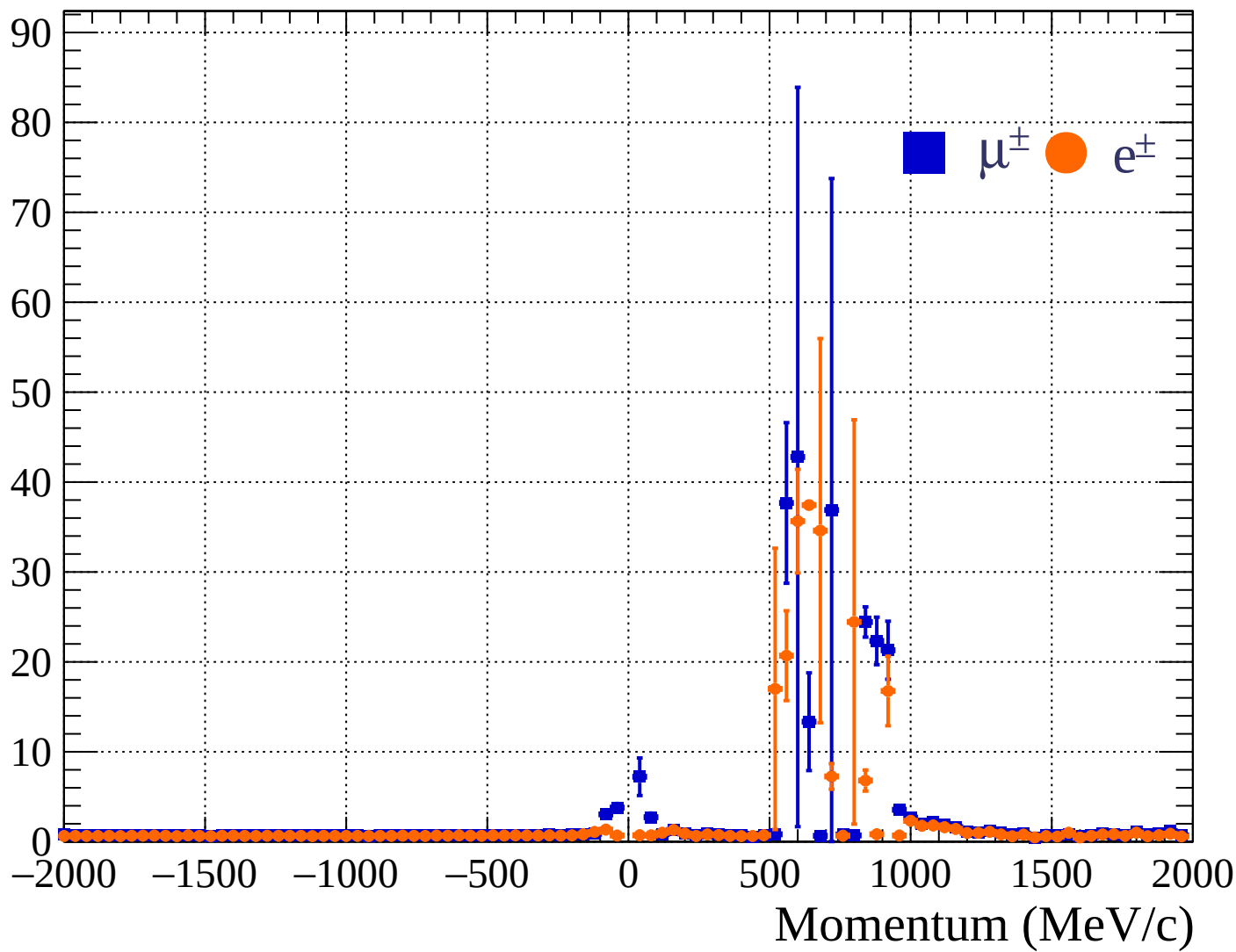


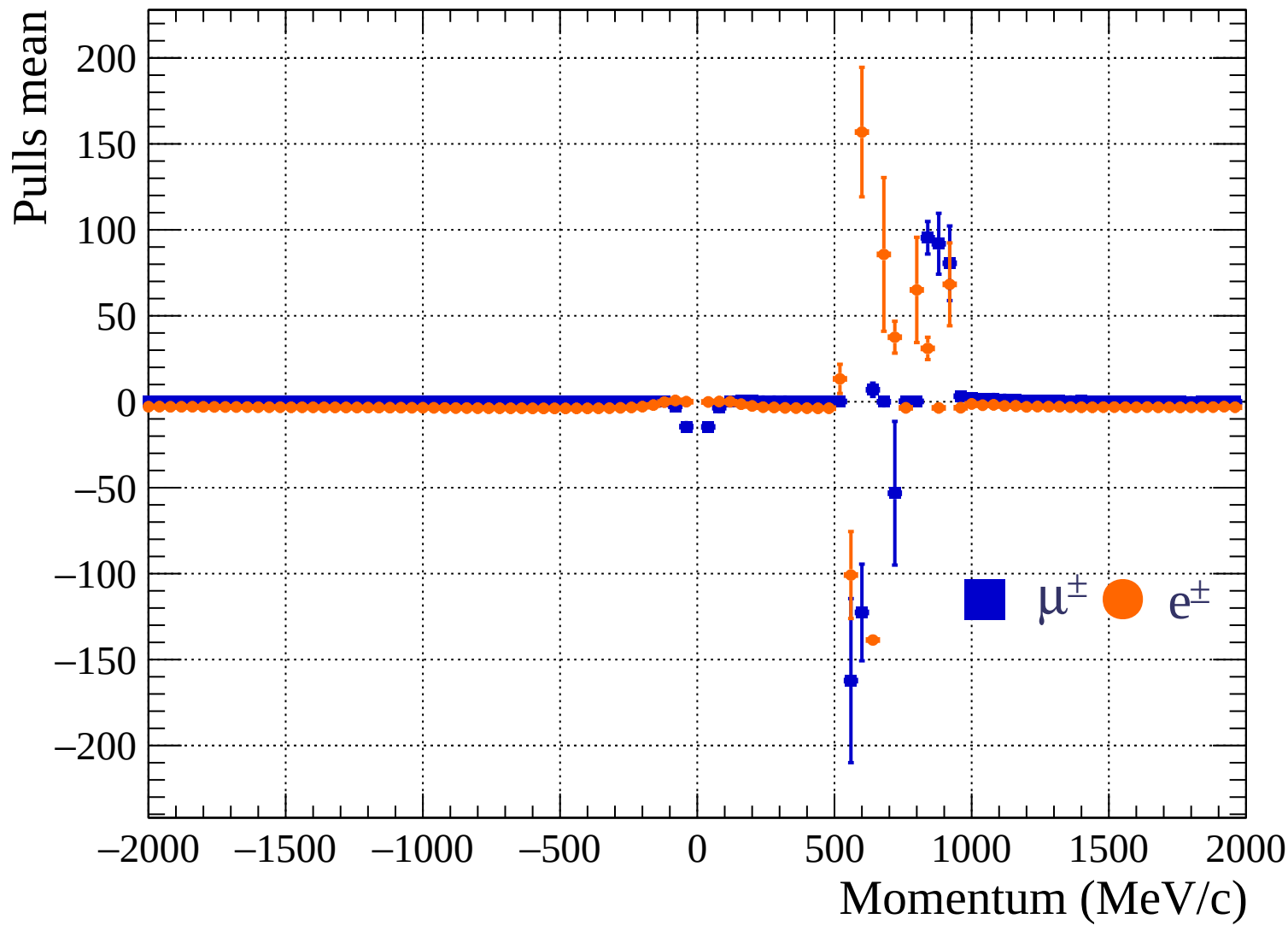
Pulls standard deviation

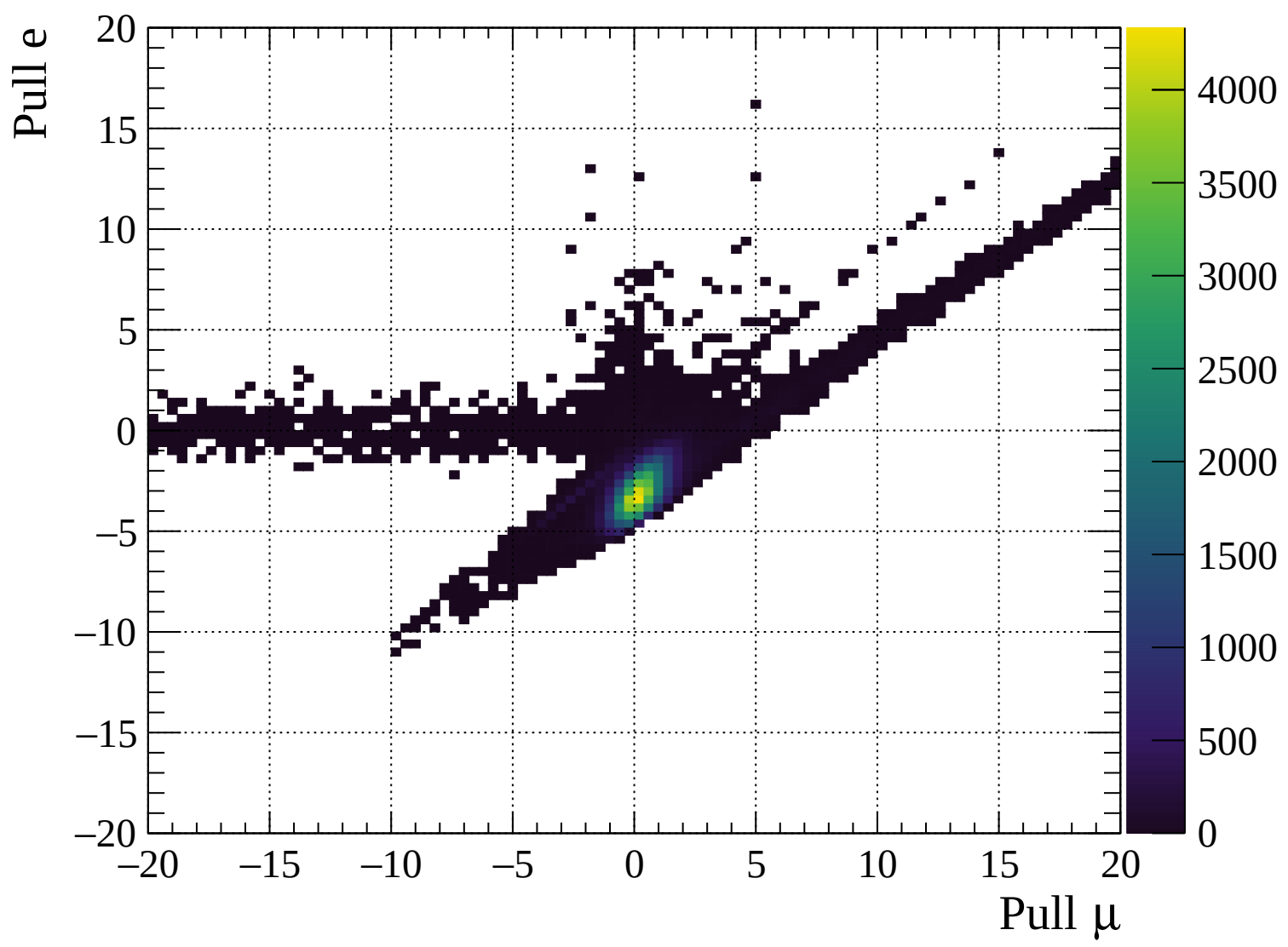


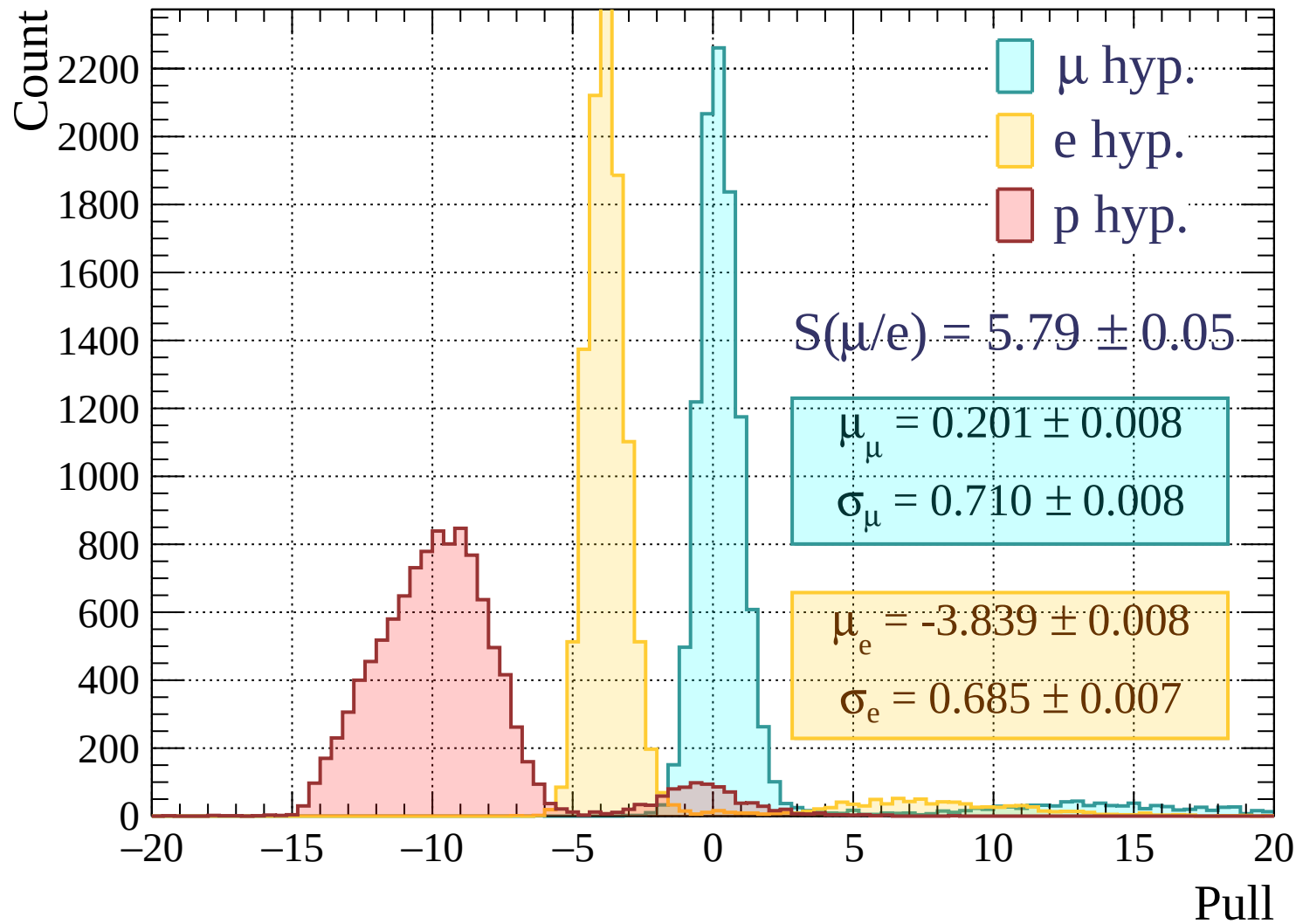


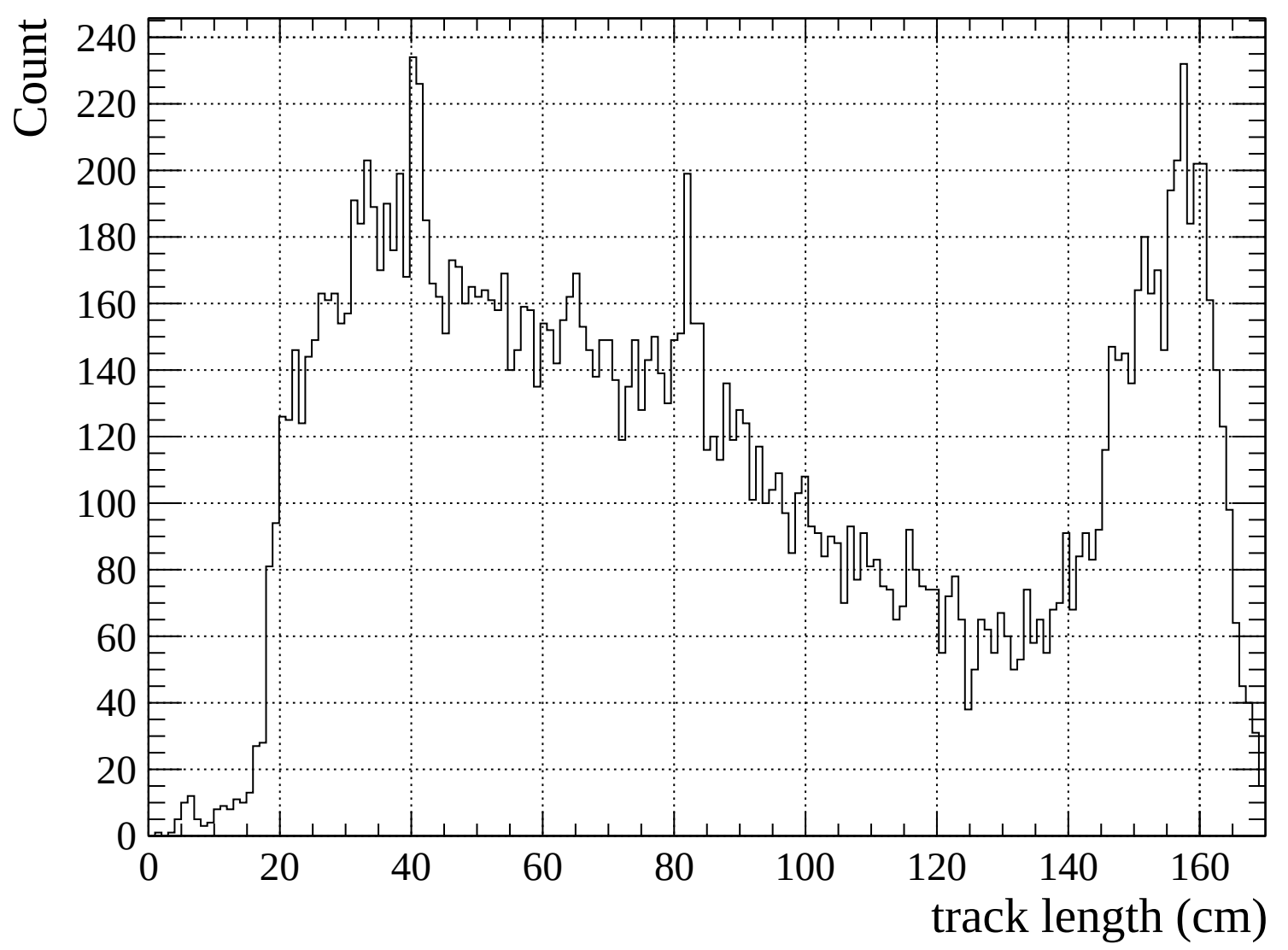
Pulls standard deviation

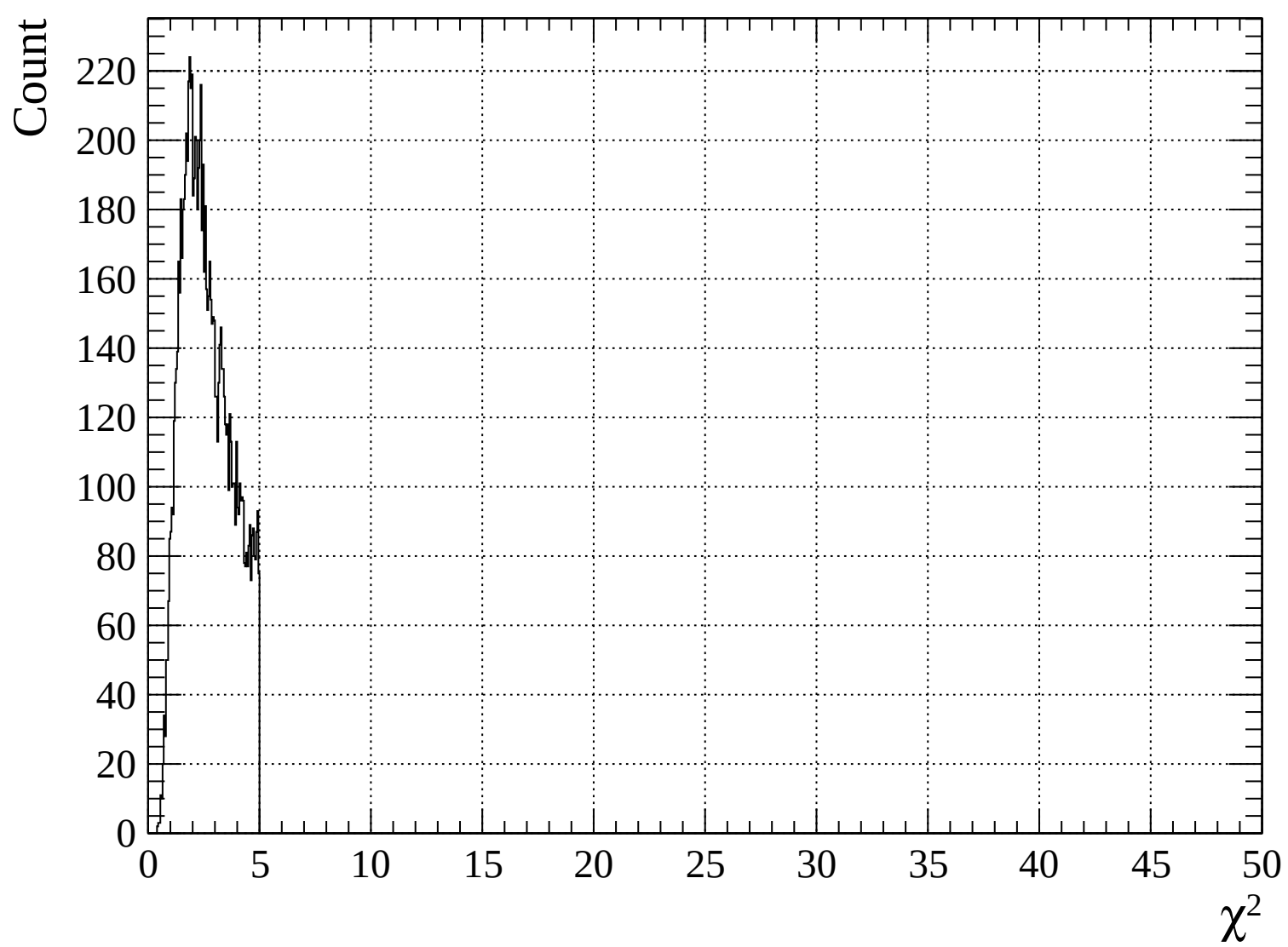


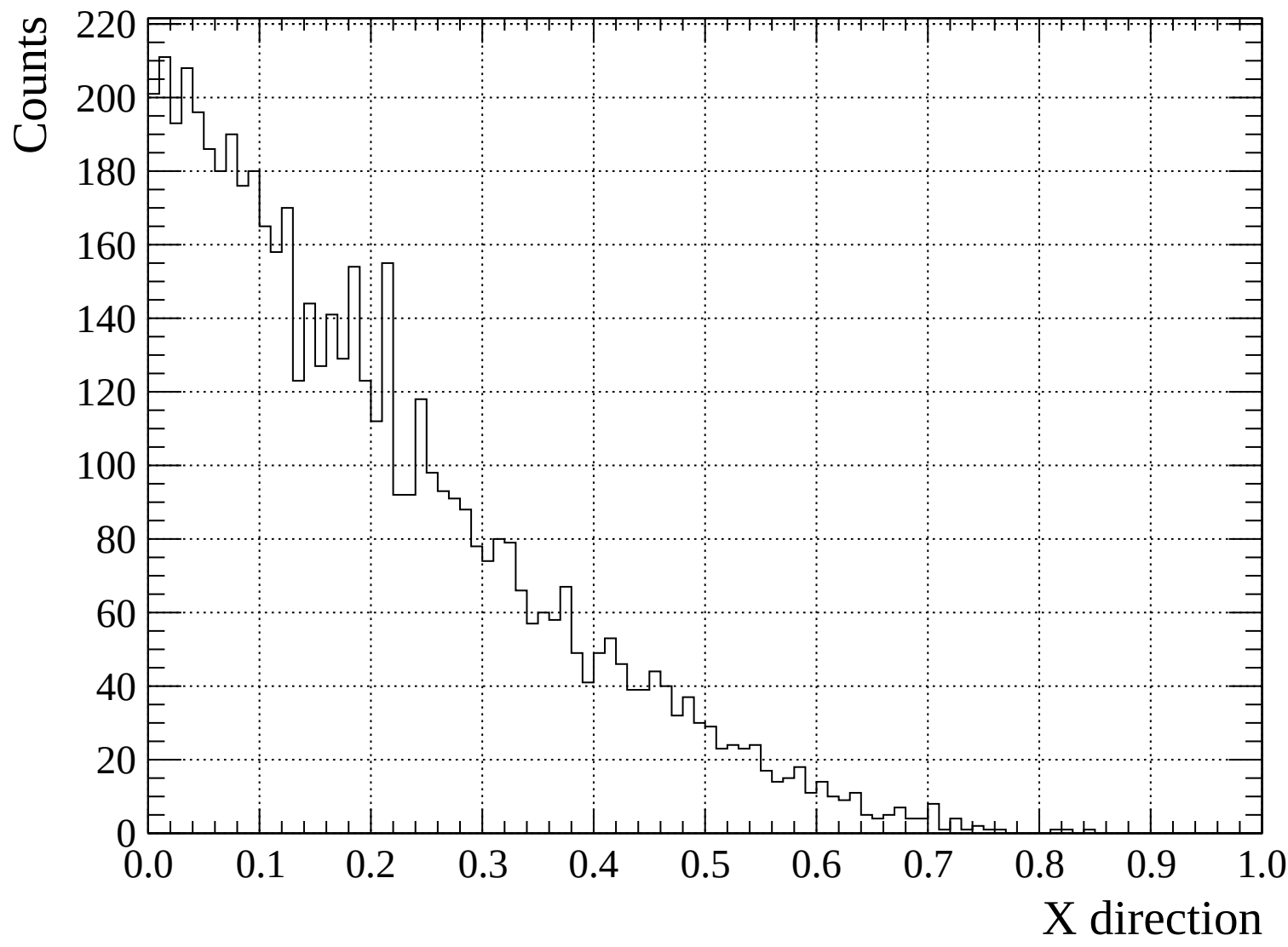


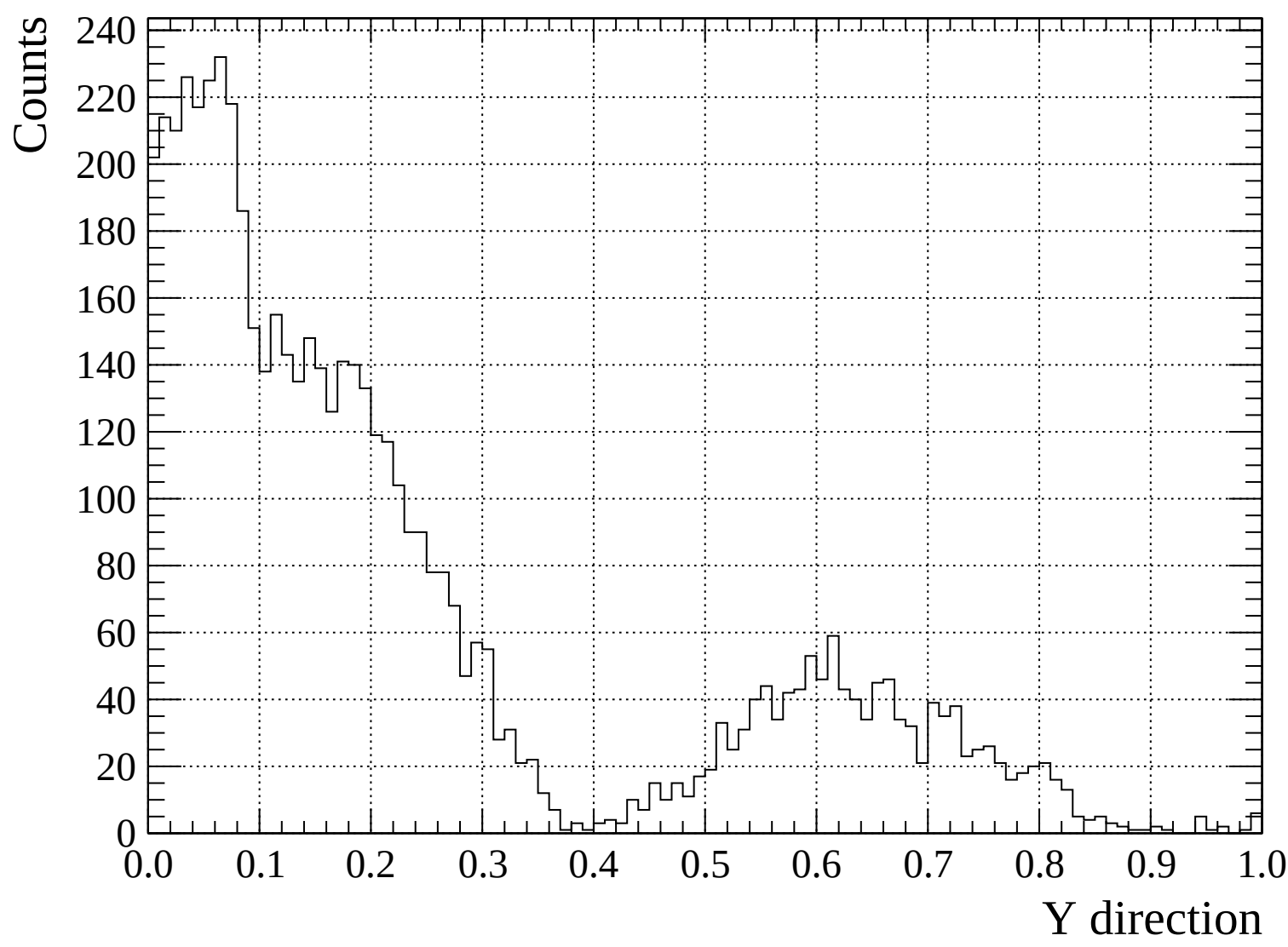


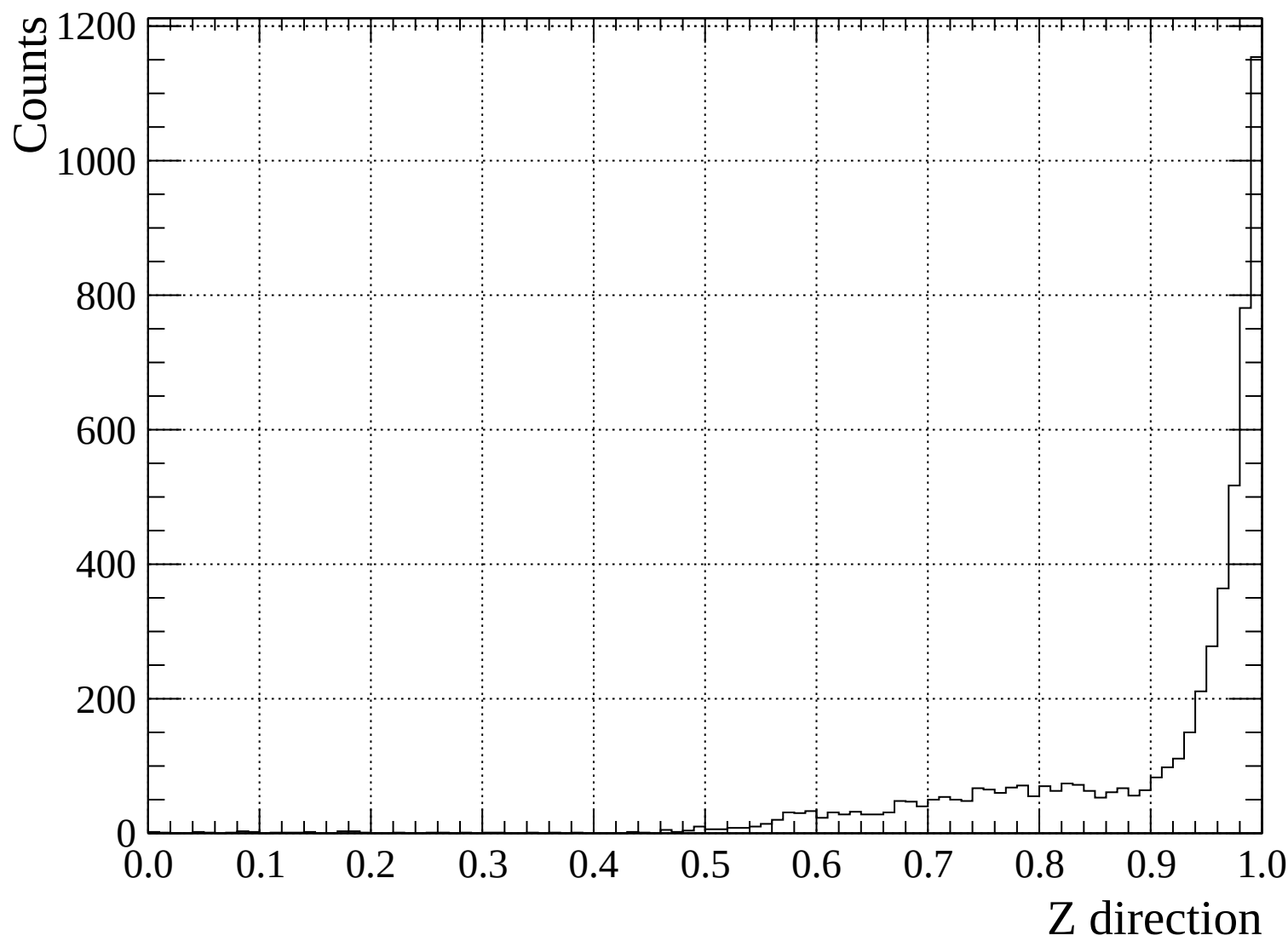


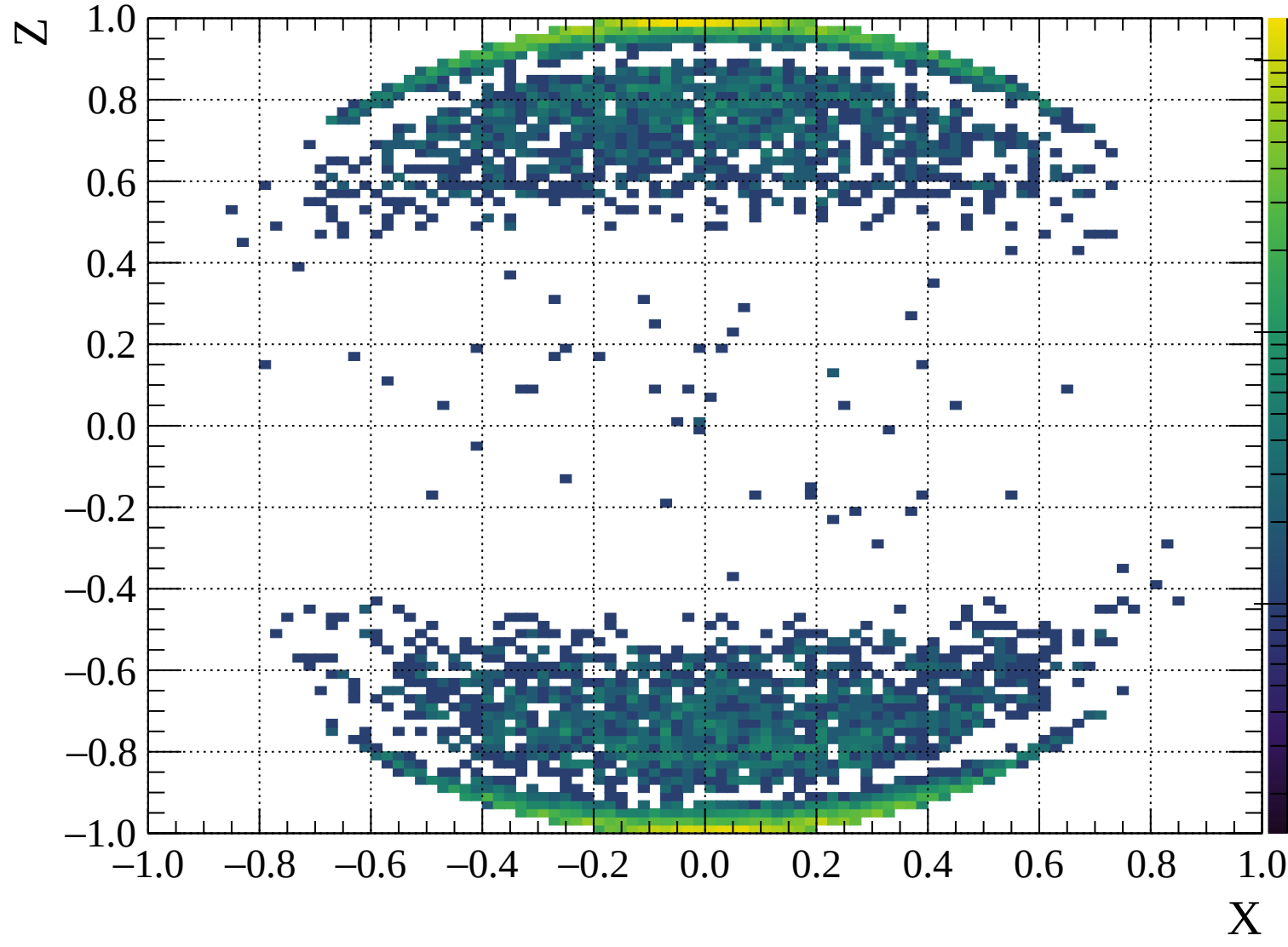


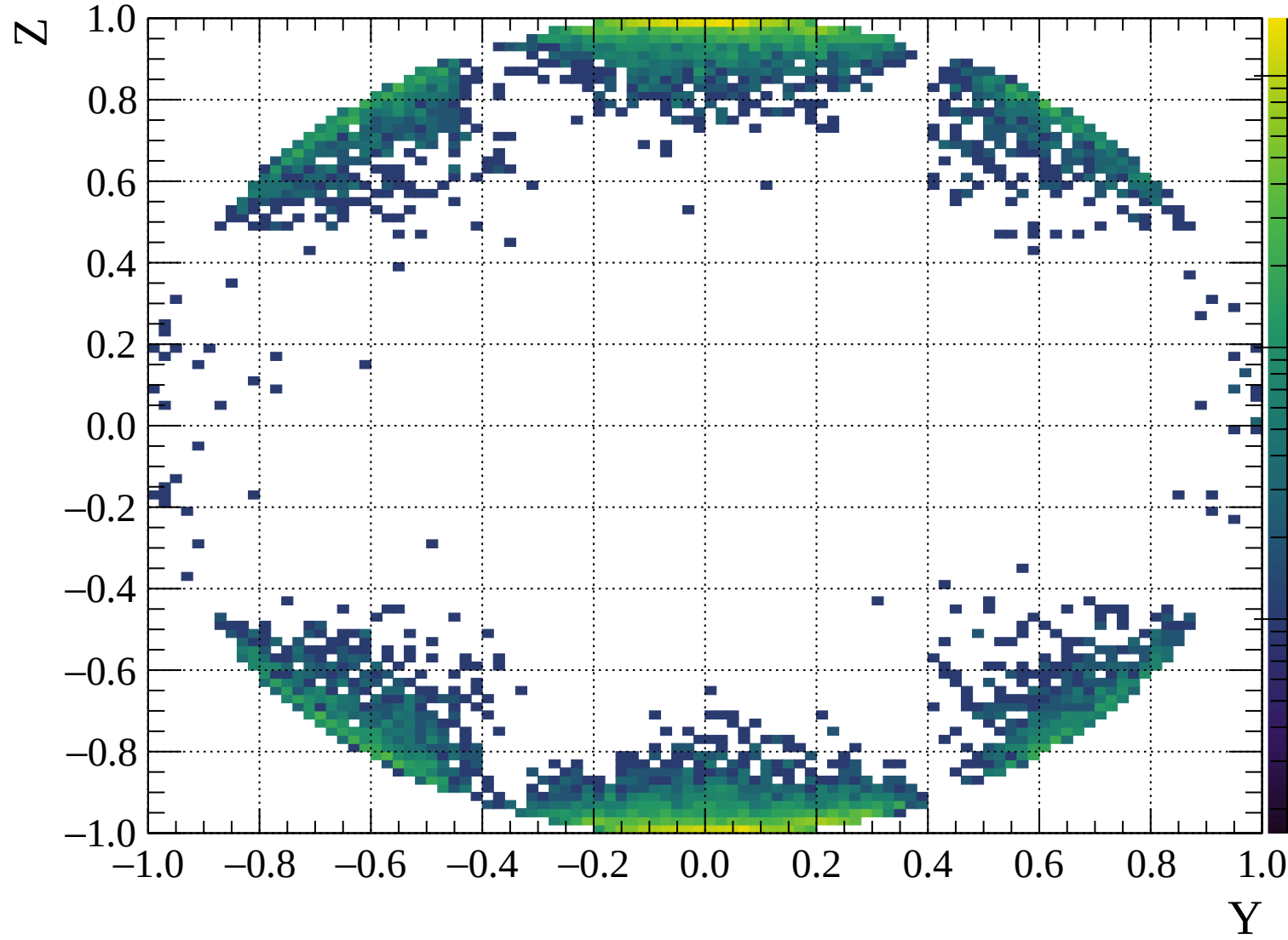


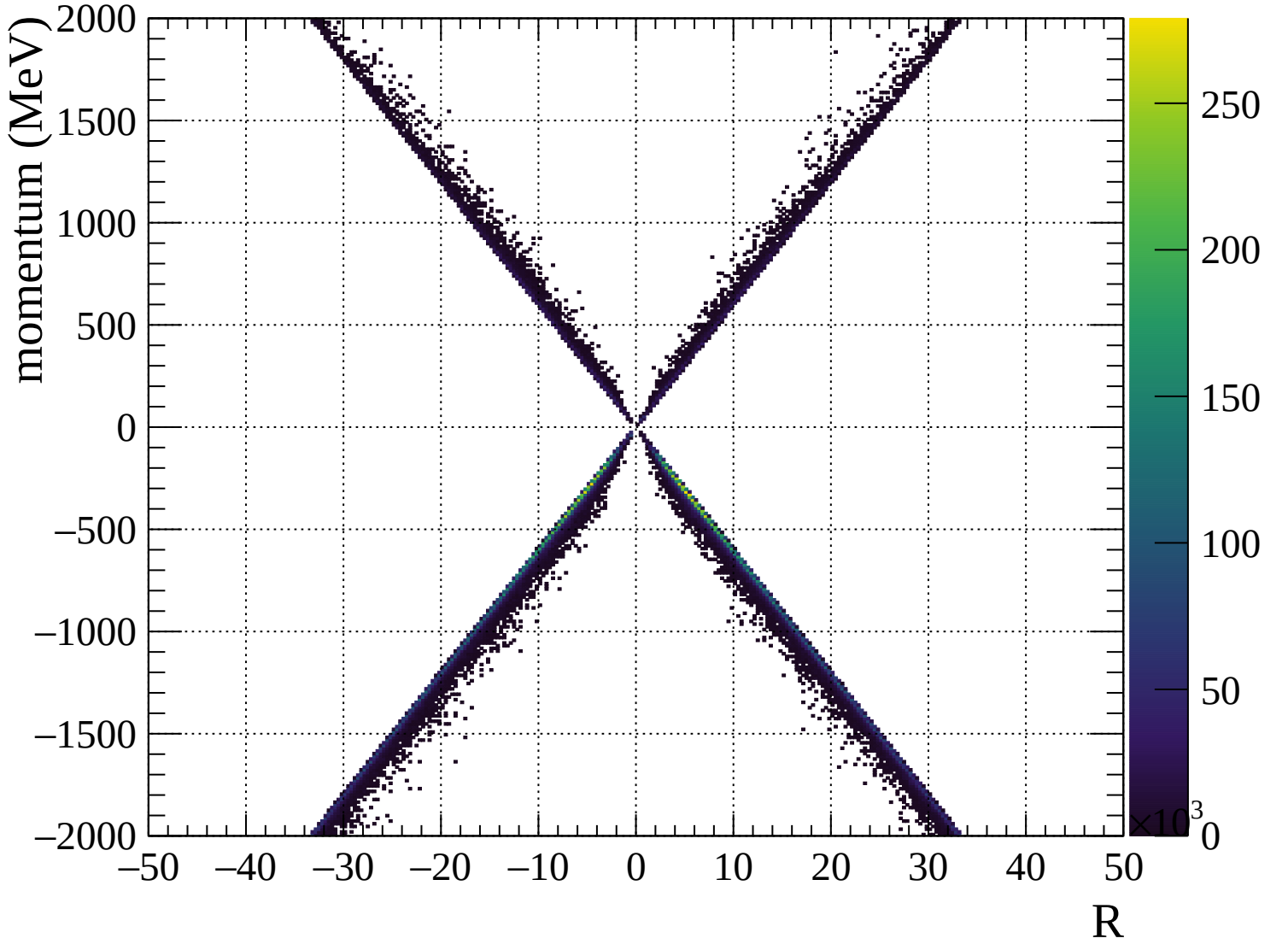


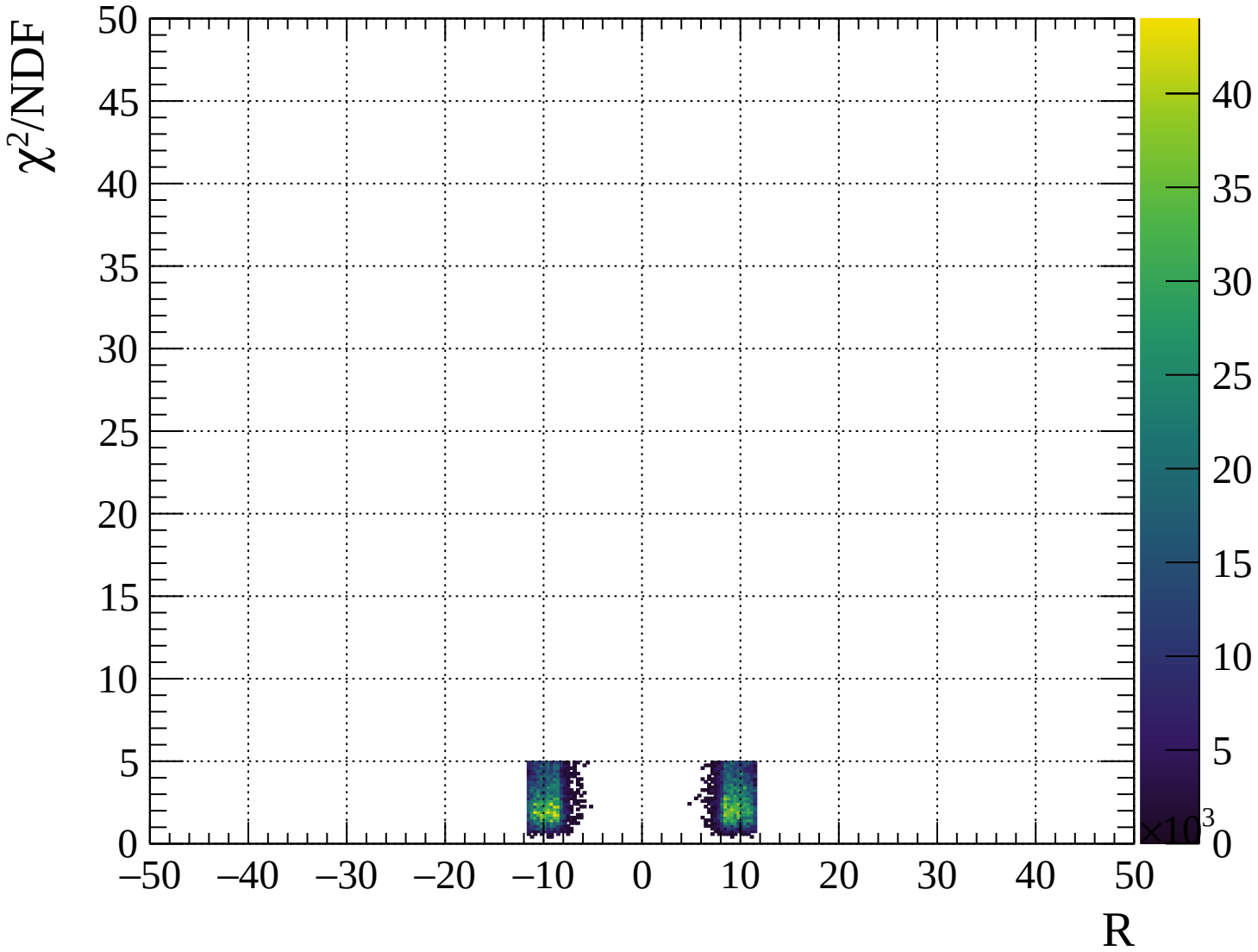












Energy loss | $0 < p < 80$

Count

XP

$$\frac{\sigma}{\mu} = 7.66 \pm 0.37 \%$$

$$\mu = 548.5 \pm 2.0$$

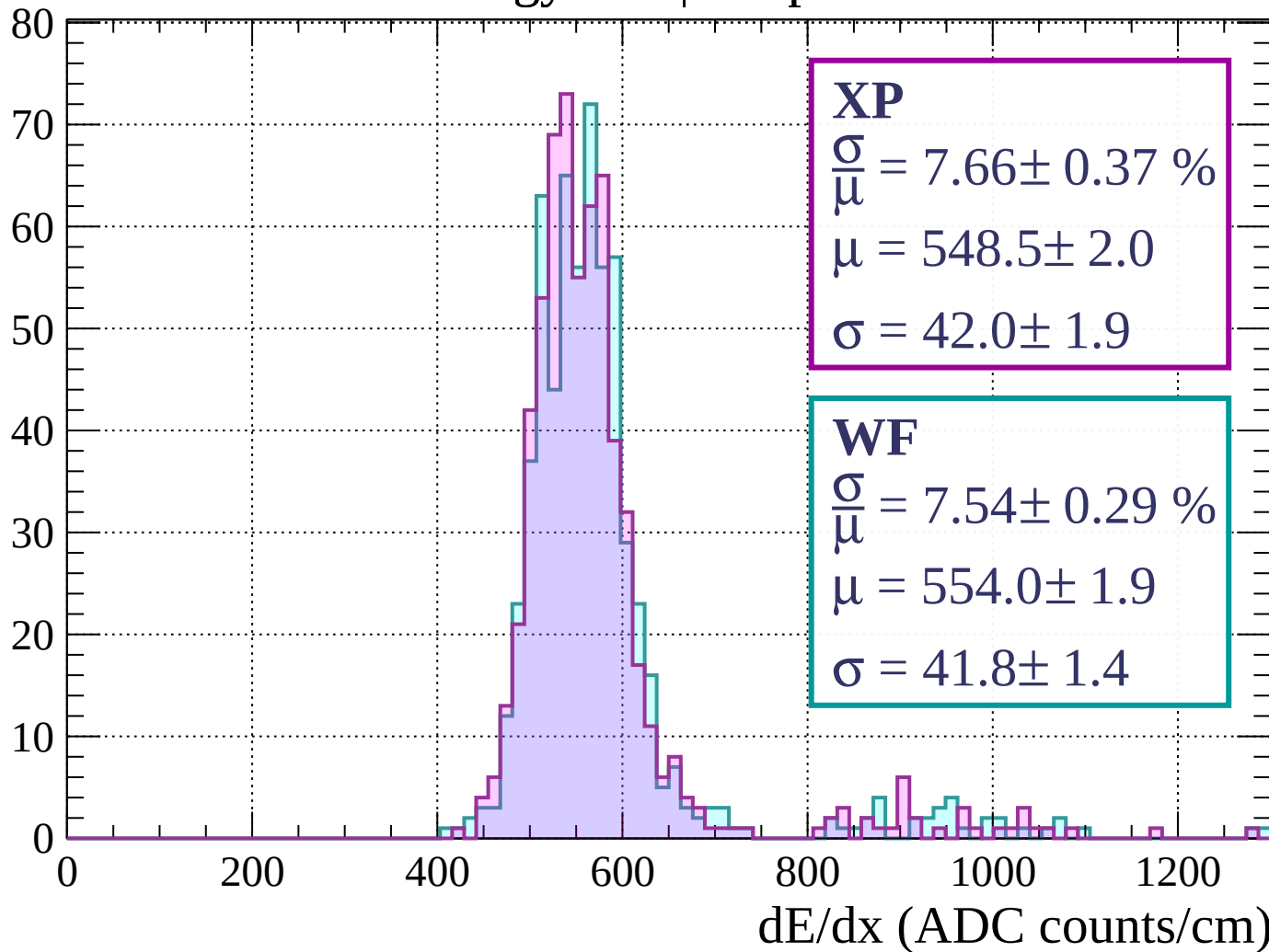
$$\sigma = 42.0 \pm 1.9$$

WF

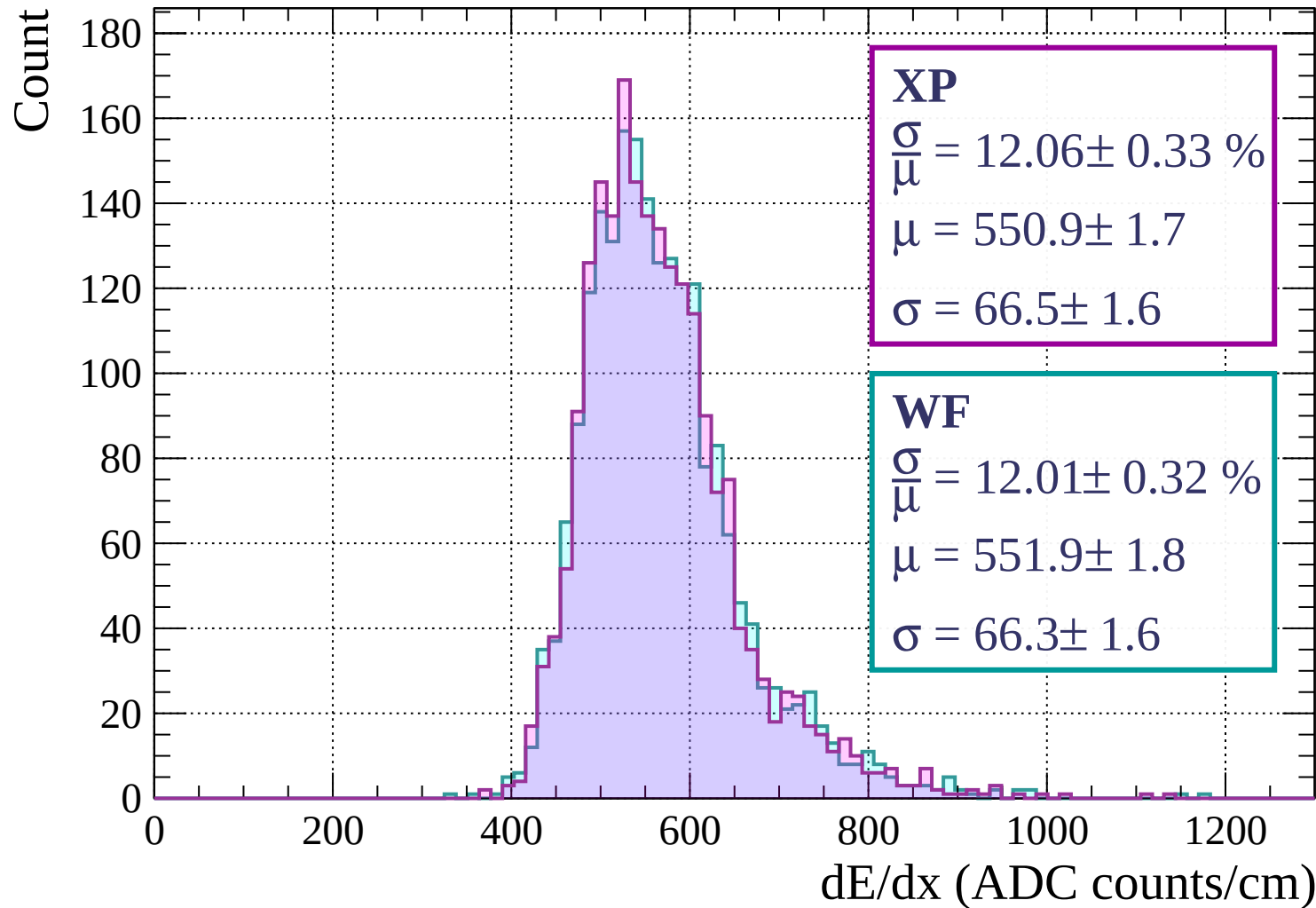
$$\frac{\sigma}{\mu} = 7.54 \pm 0.29 \%$$

$$\mu = 554.0 \pm 1.9$$

$$\sigma = 41.8 \pm 1.4$$



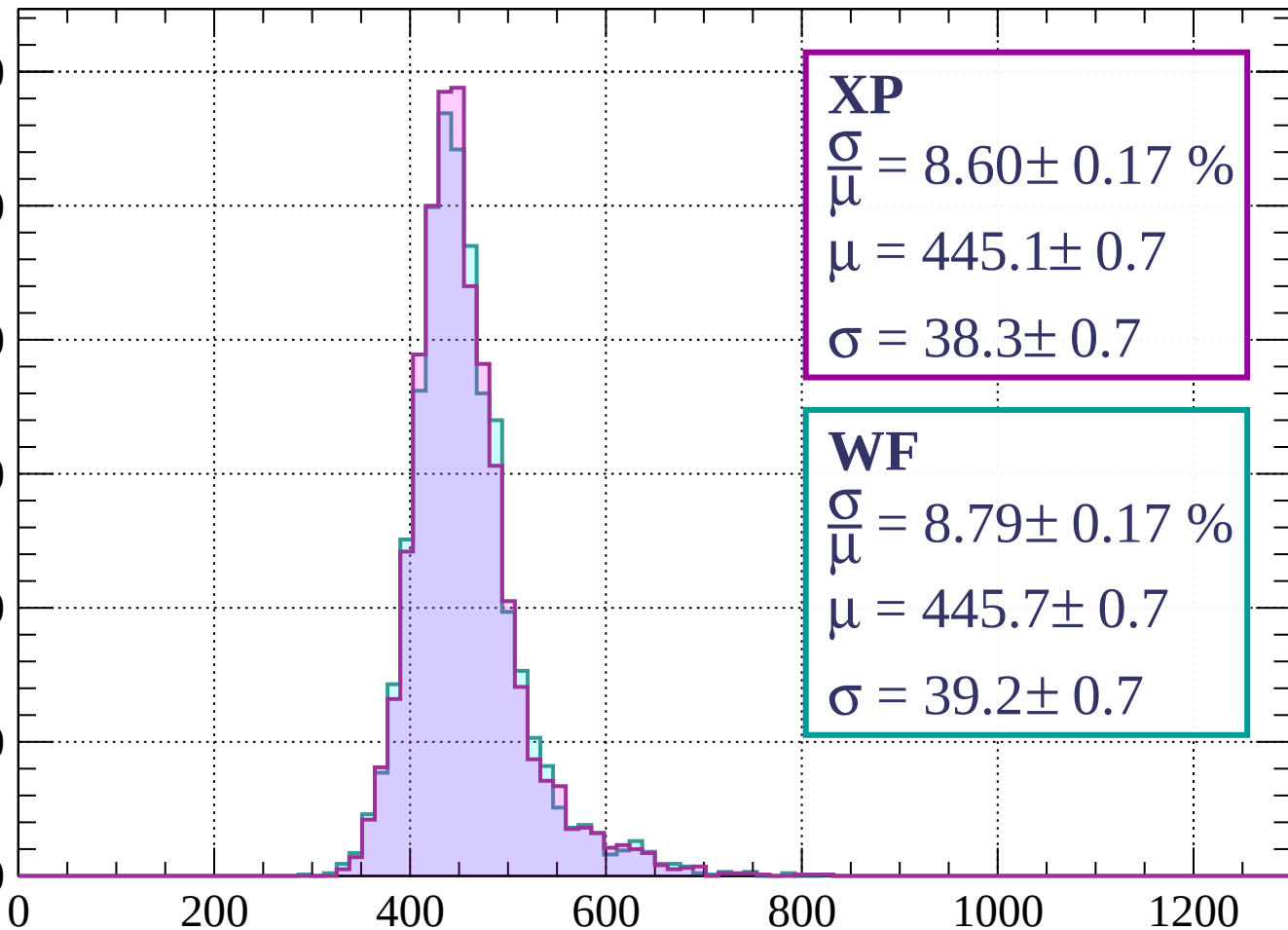
Energy loss | $80 < p < 160$



Energy loss | $160 < p < 240$

Count

600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 8.60 \pm 0.17 \%$$

$$\mu = 445.1 \pm 0.7$$

$$\sigma = 38.3 \pm 0.7$$

WF

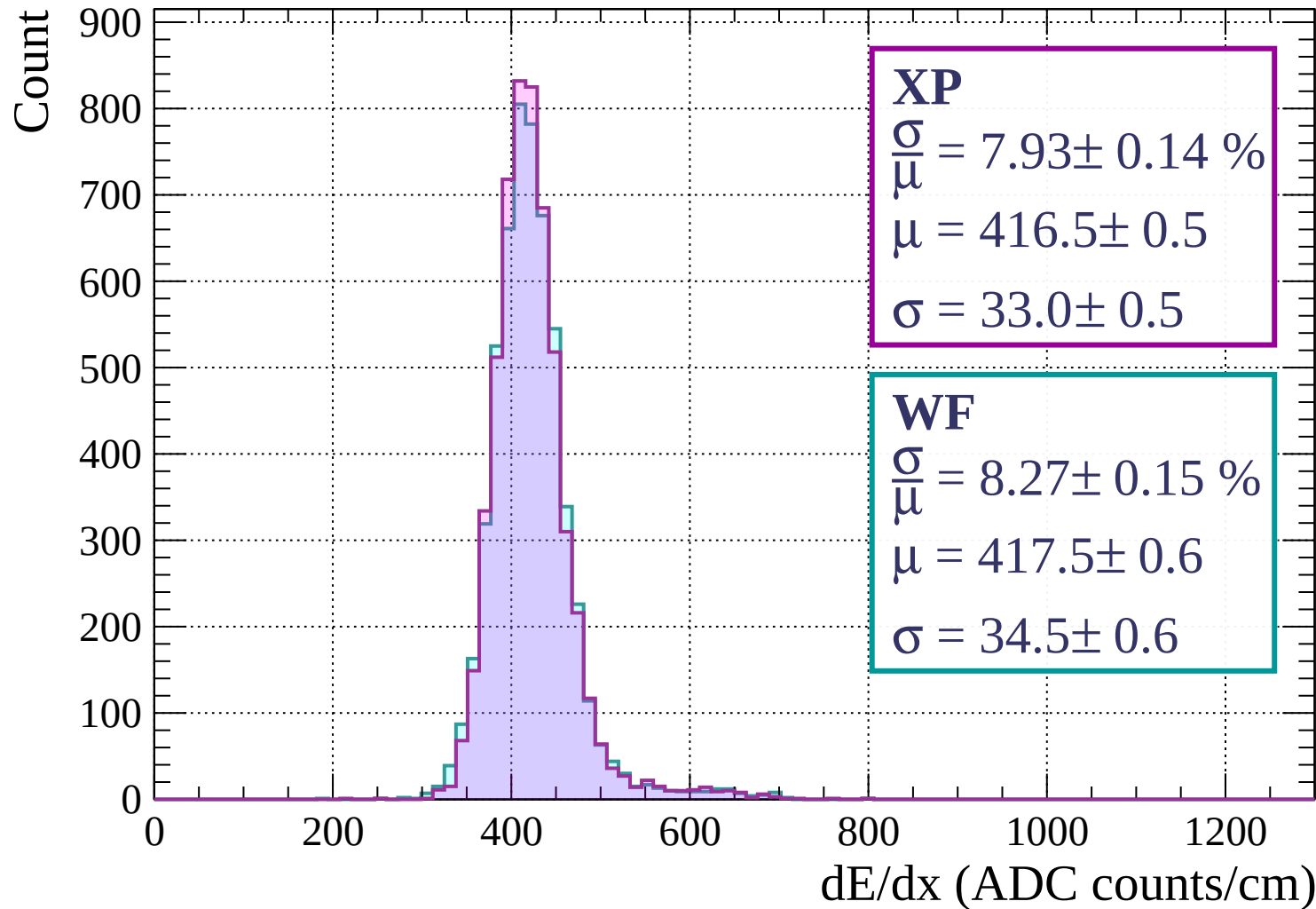
$$\frac{\sigma}{\mu} = 8.79 \pm 0.17 \%$$

$$\mu = 445.7 \pm 0.7$$

$$\sigma = 39.2 \pm 0.7$$

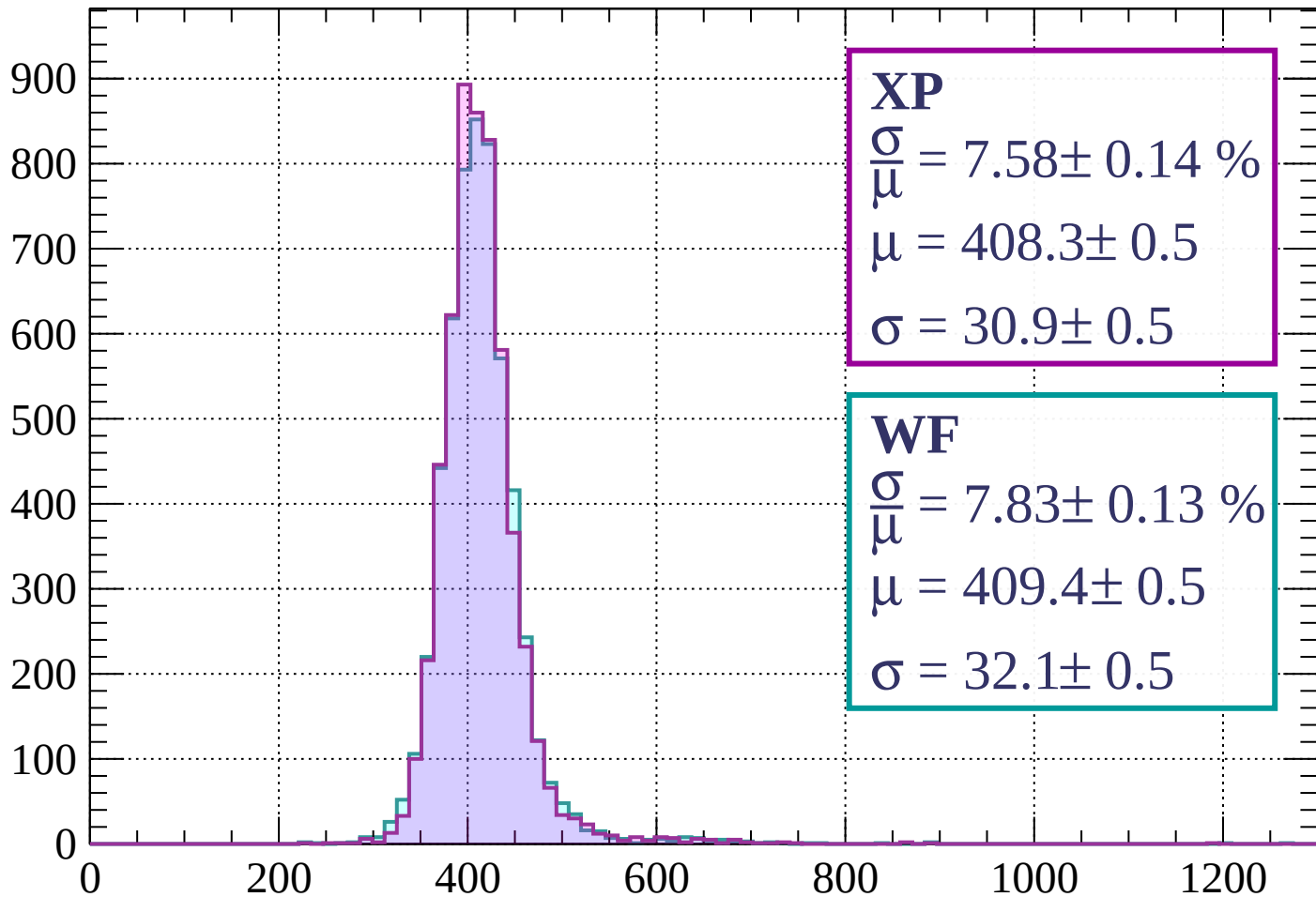
dE/dx (ADC counts/cm)

Energy loss | $240 < p < 320$



Energy loss | $320 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 7.58 \pm 0.14 \%$$

$$\mu = 408.3 \pm 0.5$$

$$\sigma = 30.9 \pm 0.5$$

WF

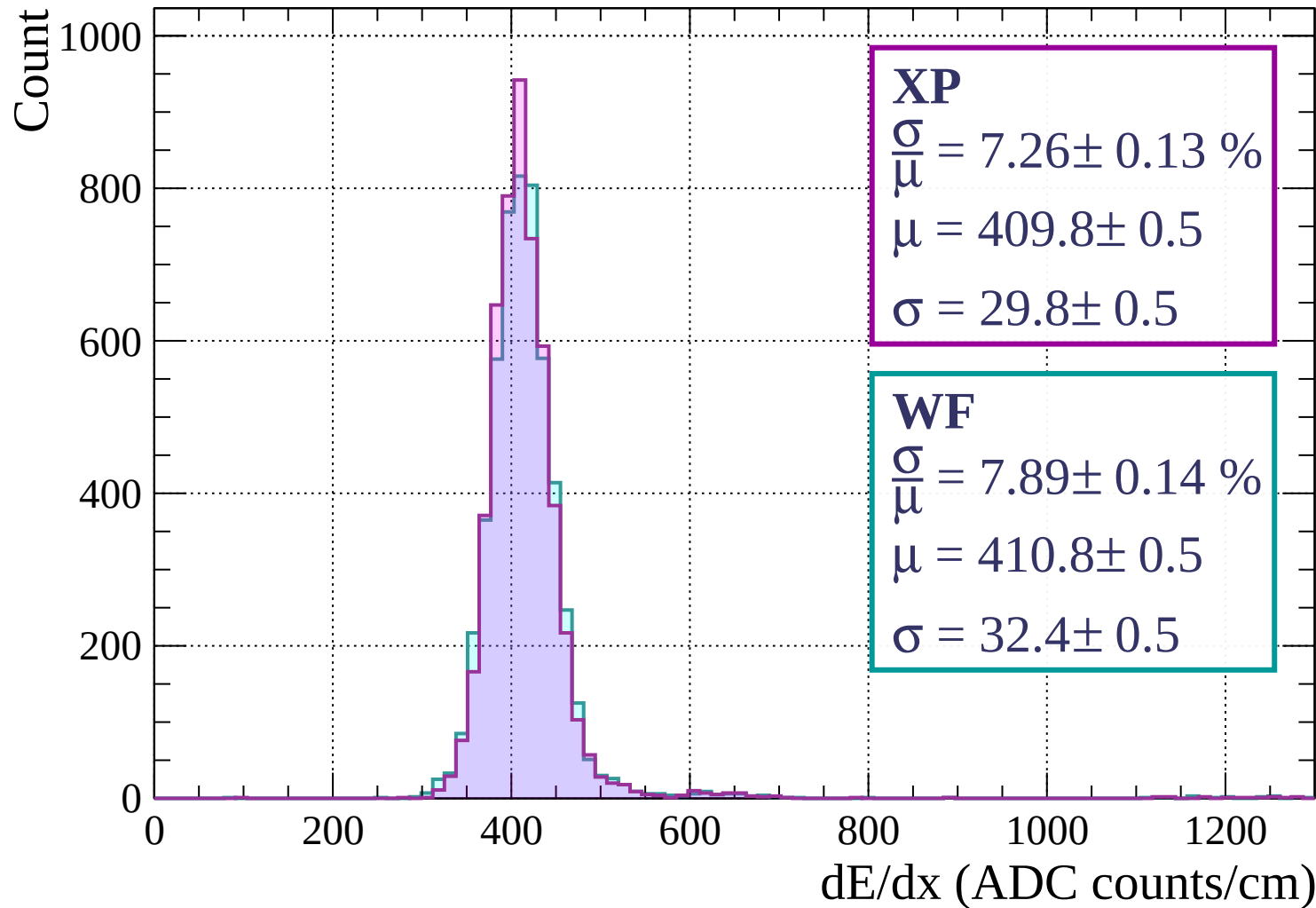
$$\frac{\sigma}{\mu} = 7.83 \pm 0.13 \%$$

$$\mu = 409.4 \pm 0.5$$

$$\sigma = 32.1 \pm 0.5$$

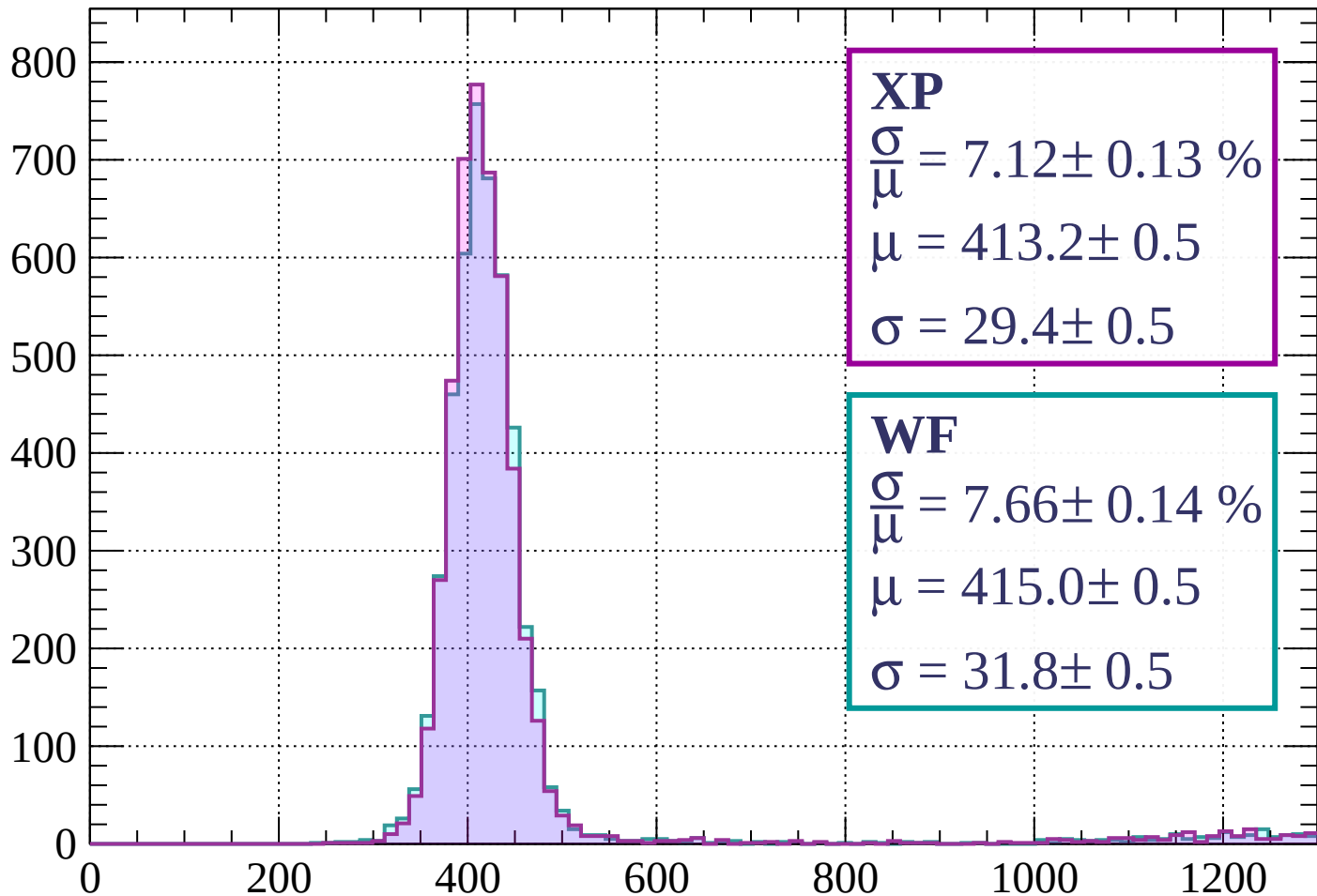
dE/dx (ADC counts/cm)

Energy loss | $400 < p < 480$



Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 7.12 \pm 0.13 \%$$

$$\mu = 413.2 \pm 0.5$$

$$\sigma = 29.4 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 7.66 \pm 0.14 \%$$

$$\mu = 415.0 \pm 0.5$$

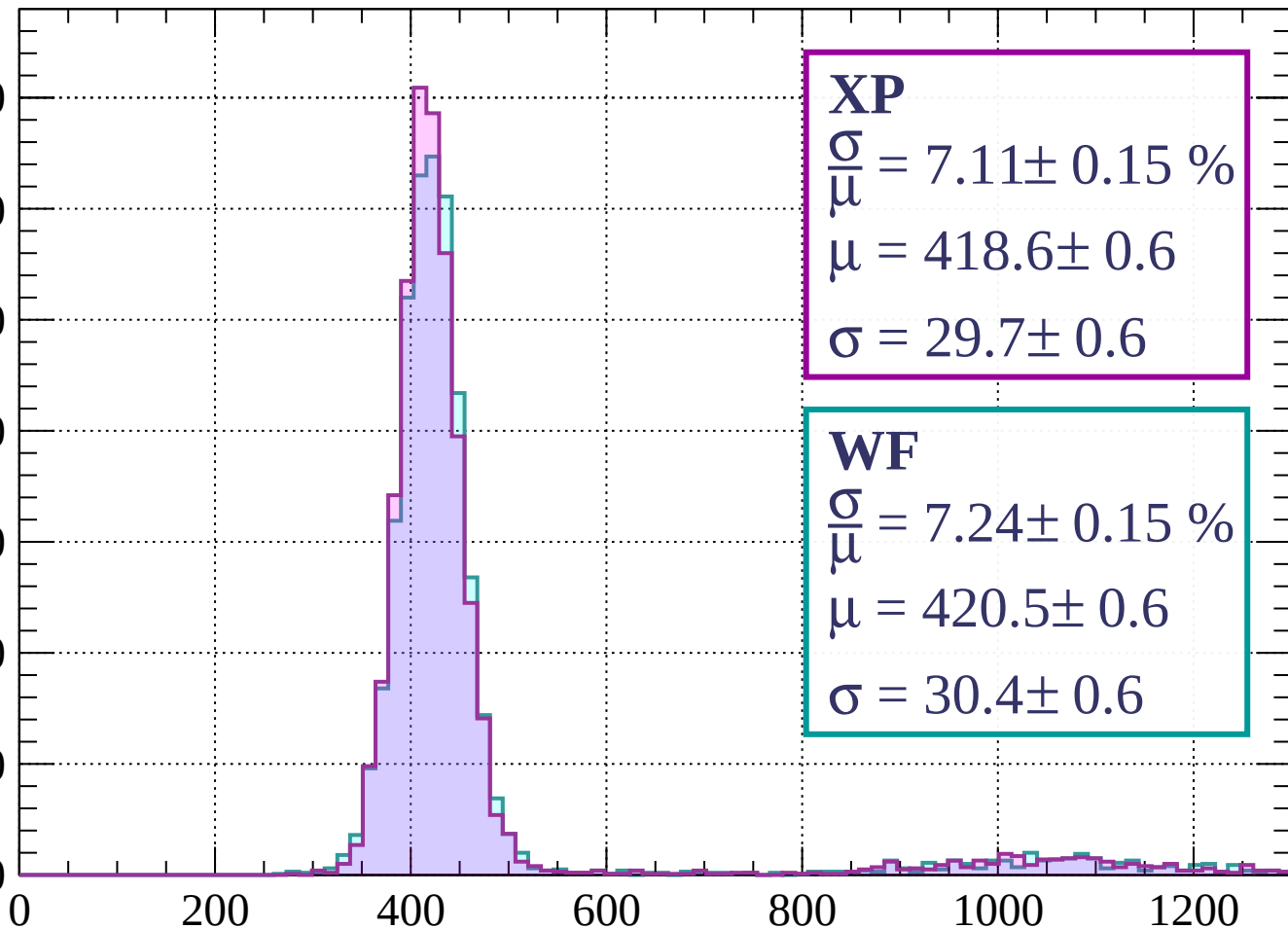
$$\sigma = 31.8 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 640$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\bar{\mu}} = 7.11 \pm 0.15 \%$$

$$\mu = 418.6 \pm 0.6$$

$$\sigma = 29.7 \pm 0.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 7.24 \pm 0.15 \%$$

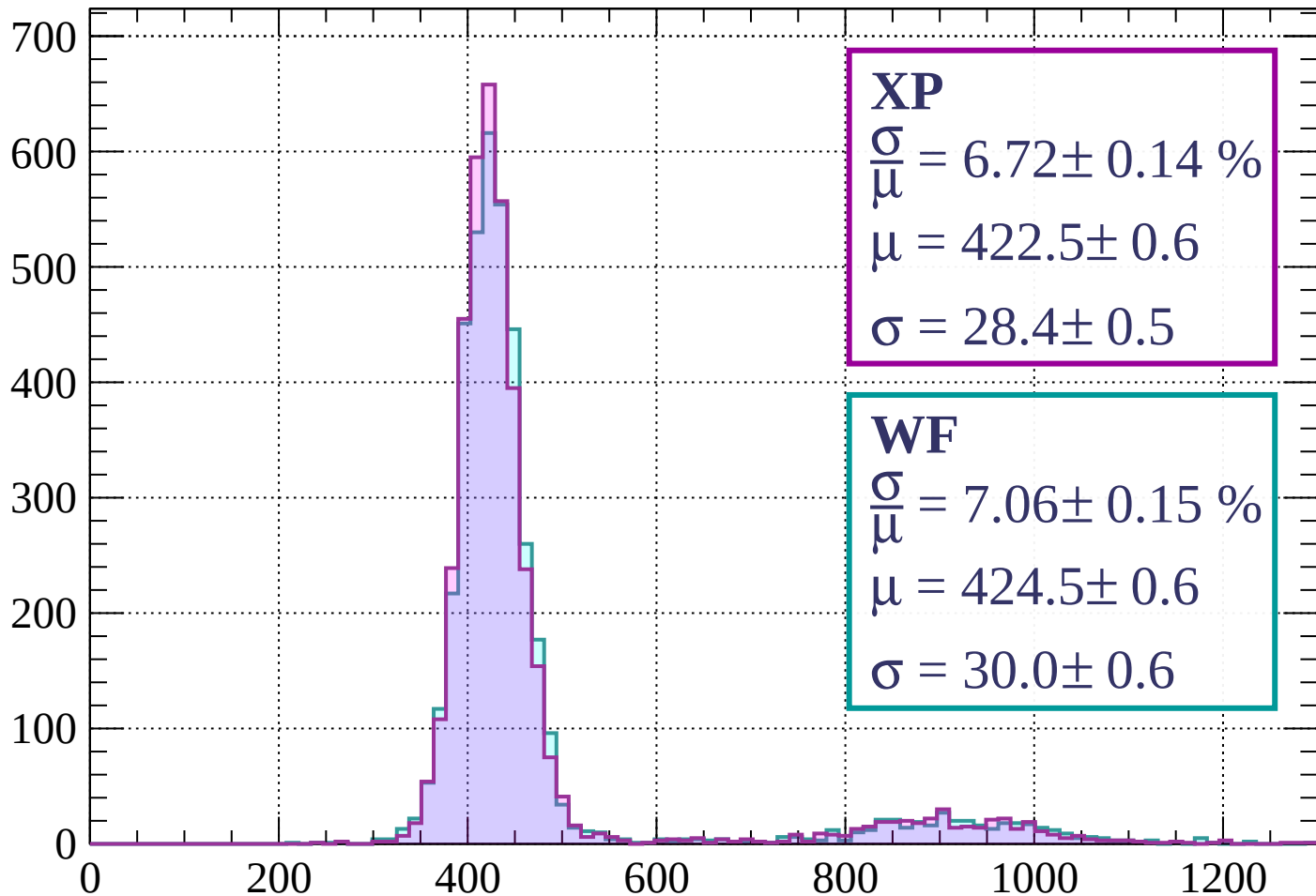
$$\mu = 420.5 \pm 0.6$$

$$\sigma = 30.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 6.72 \pm 0.14 \%$$

$$\mu = 422.5 \pm 0.6$$

$$\sigma = 28.4 \pm 0.5$$

WF

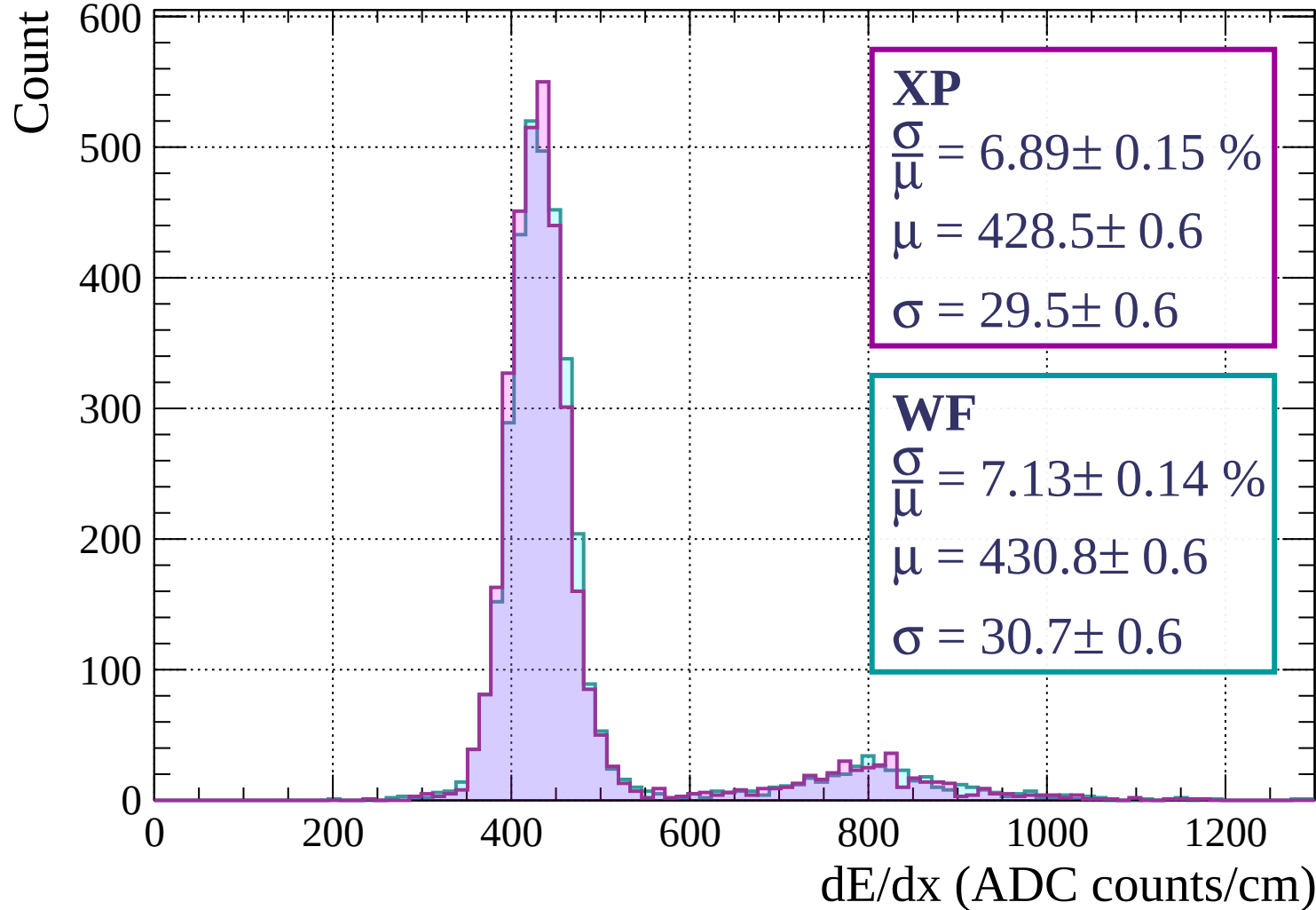
$$\frac{\sigma}{\mu} = 7.06 \pm 0.15 \%$$

$$\mu = 424.5 \pm 0.6$$

$$\sigma = 30.0 \pm 0.6$$

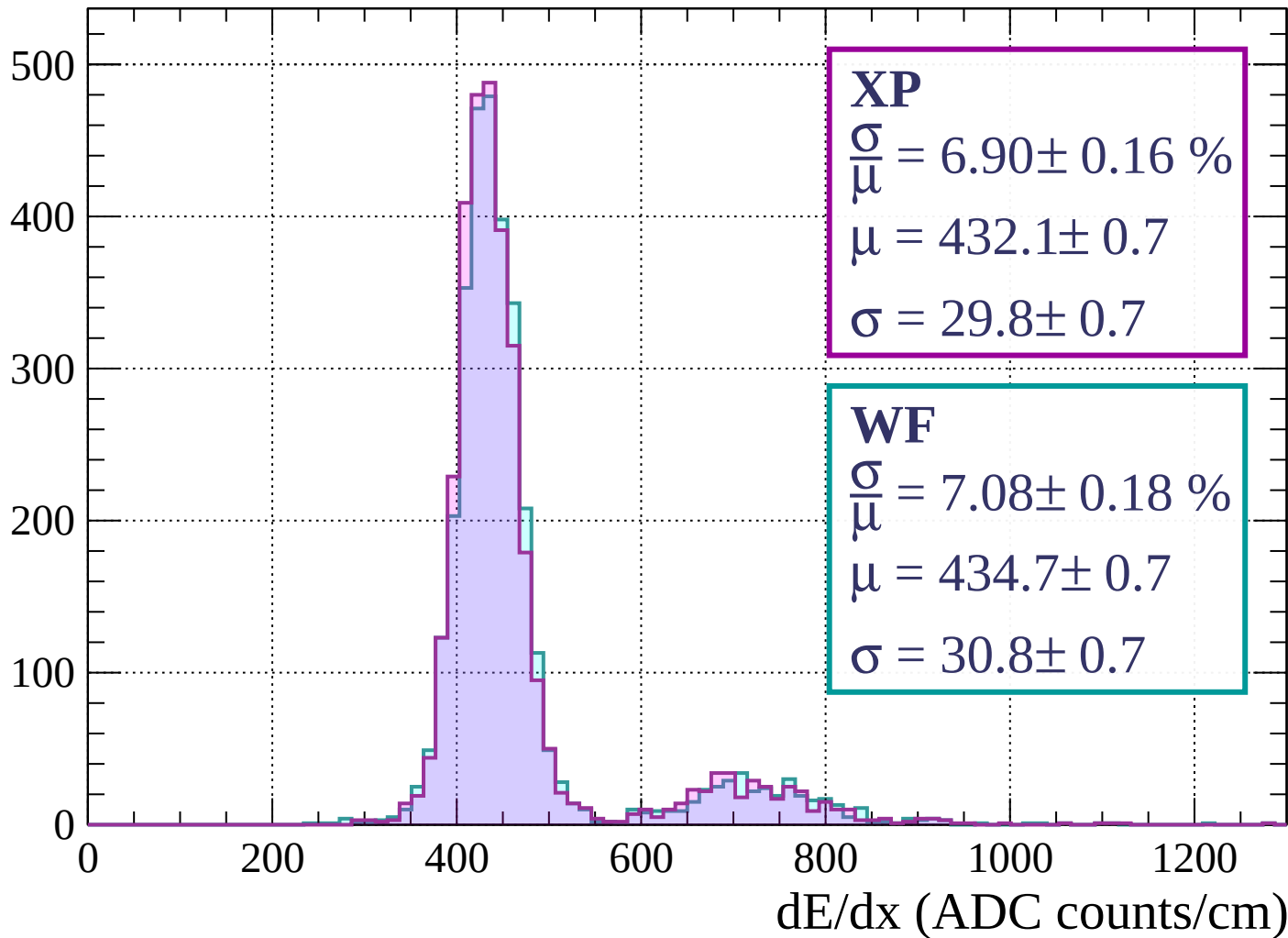
dE/dx (ADC counts/cm)

Energy loss | $720 < p < 800$



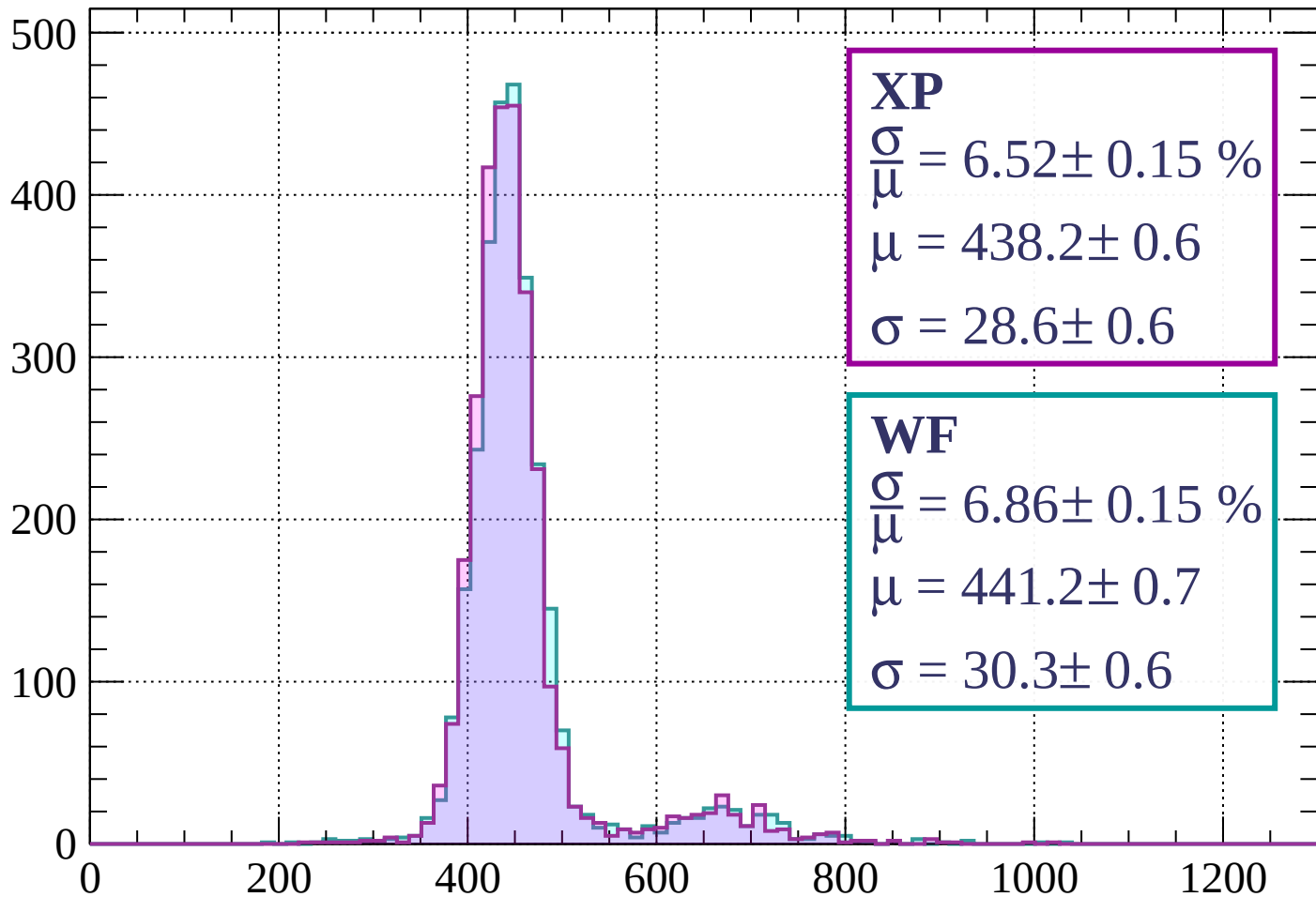
Energy loss | $800 < p < 880$

Count



Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 6.52 \pm 0.15 \%$$

$$\mu = 438.2 \pm 0.6$$

$$\sigma = 28.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.86 \pm 0.15 \%$$

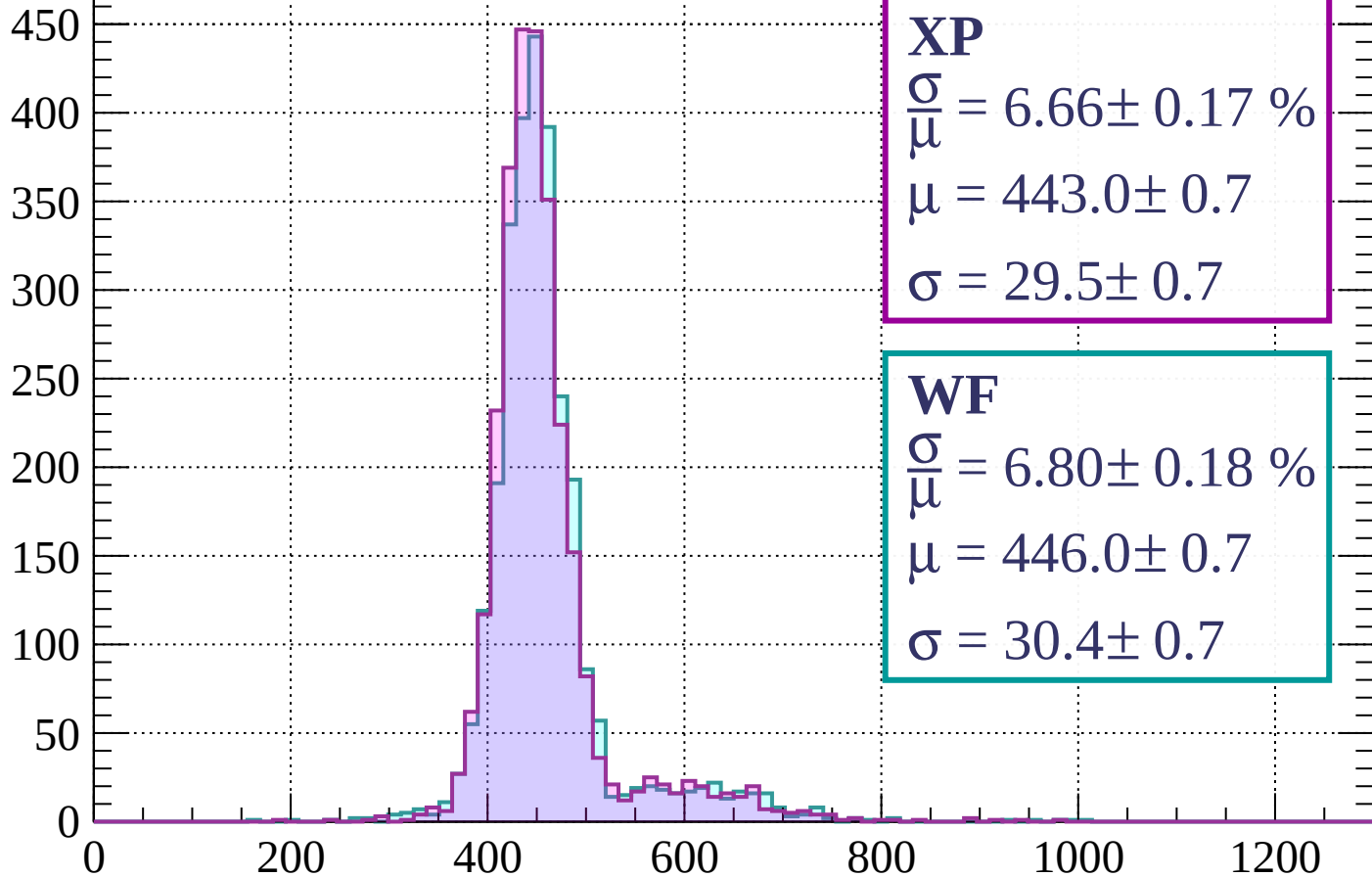
$$\mu = 441.2 \pm 0.7$$

$$\sigma = 30.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 6.66 \pm 0.17 \%$$

$$\mu = 443.0 \pm 0.7$$

$$\sigma = 29.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.80 \pm 0.18 \%$$

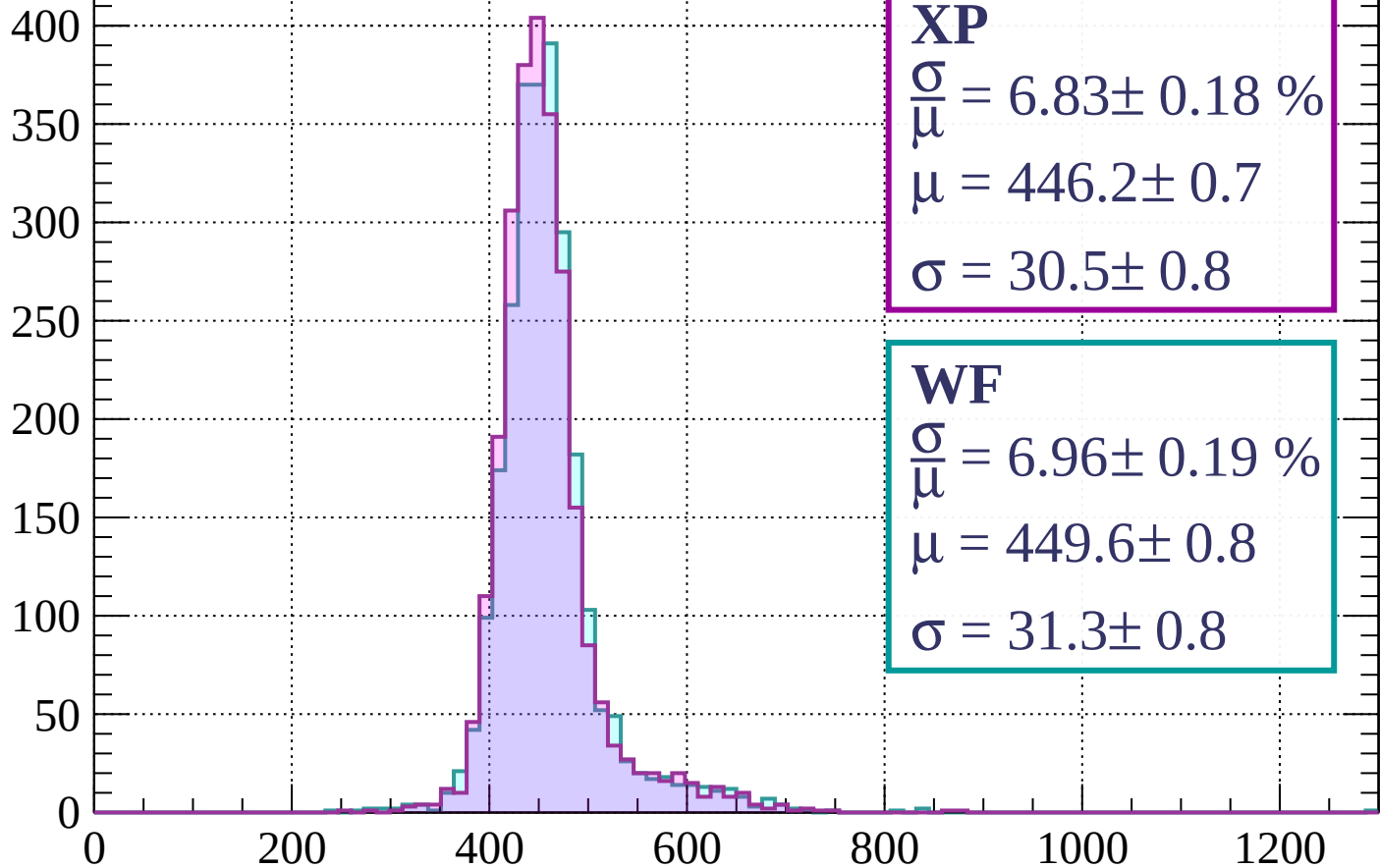
$$\mu = 446.0 \pm 0.7$$

$$\sigma = 30.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 6.83 \pm 0.18 \%$$

$$\mu = 446.2 \pm 0.7$$

$$\sigma = 30.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.96 \pm 0.19 \%$$

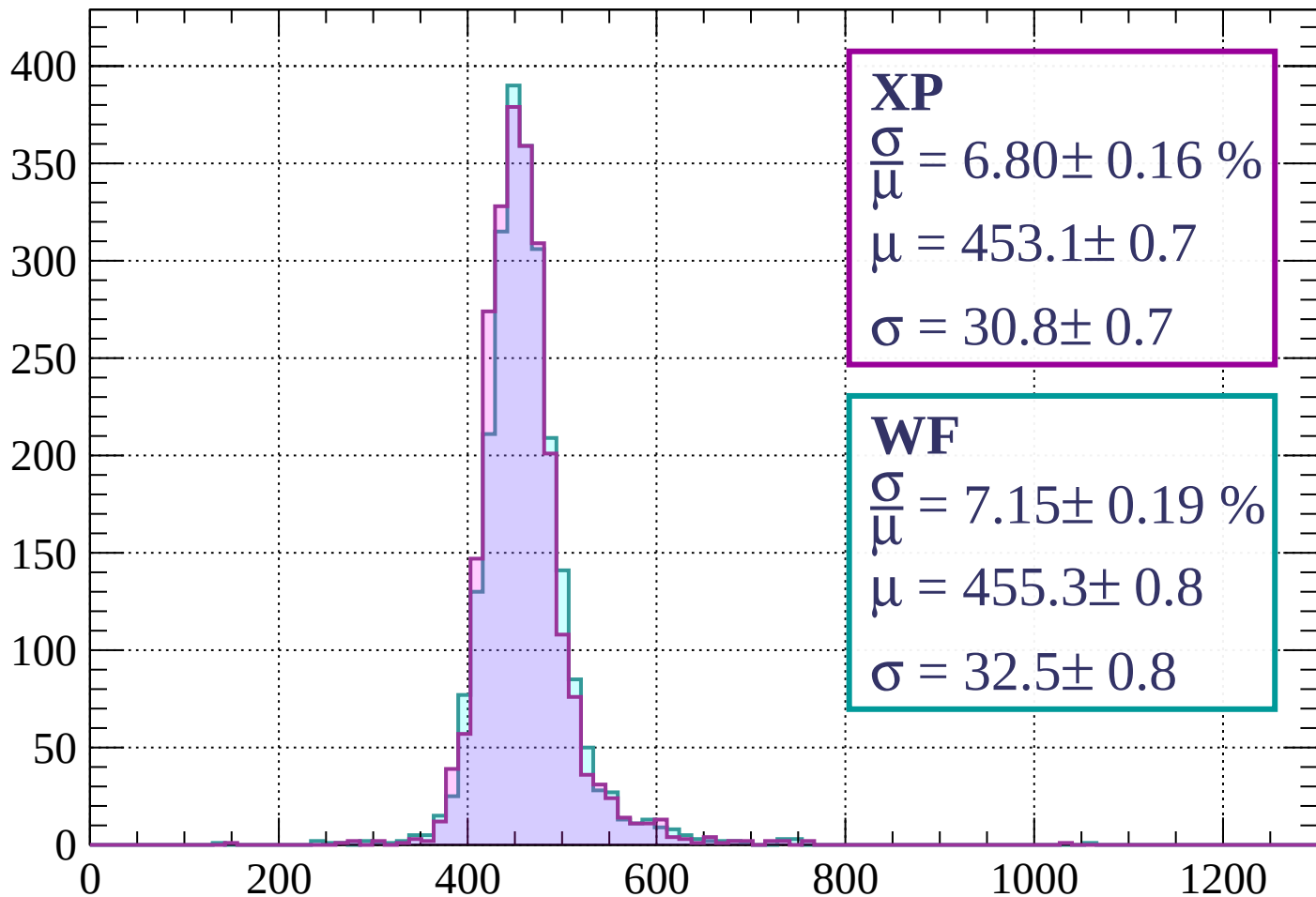
$$\mu = 449.6 \pm 0.8$$

$$\sigma = 31.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$

Count

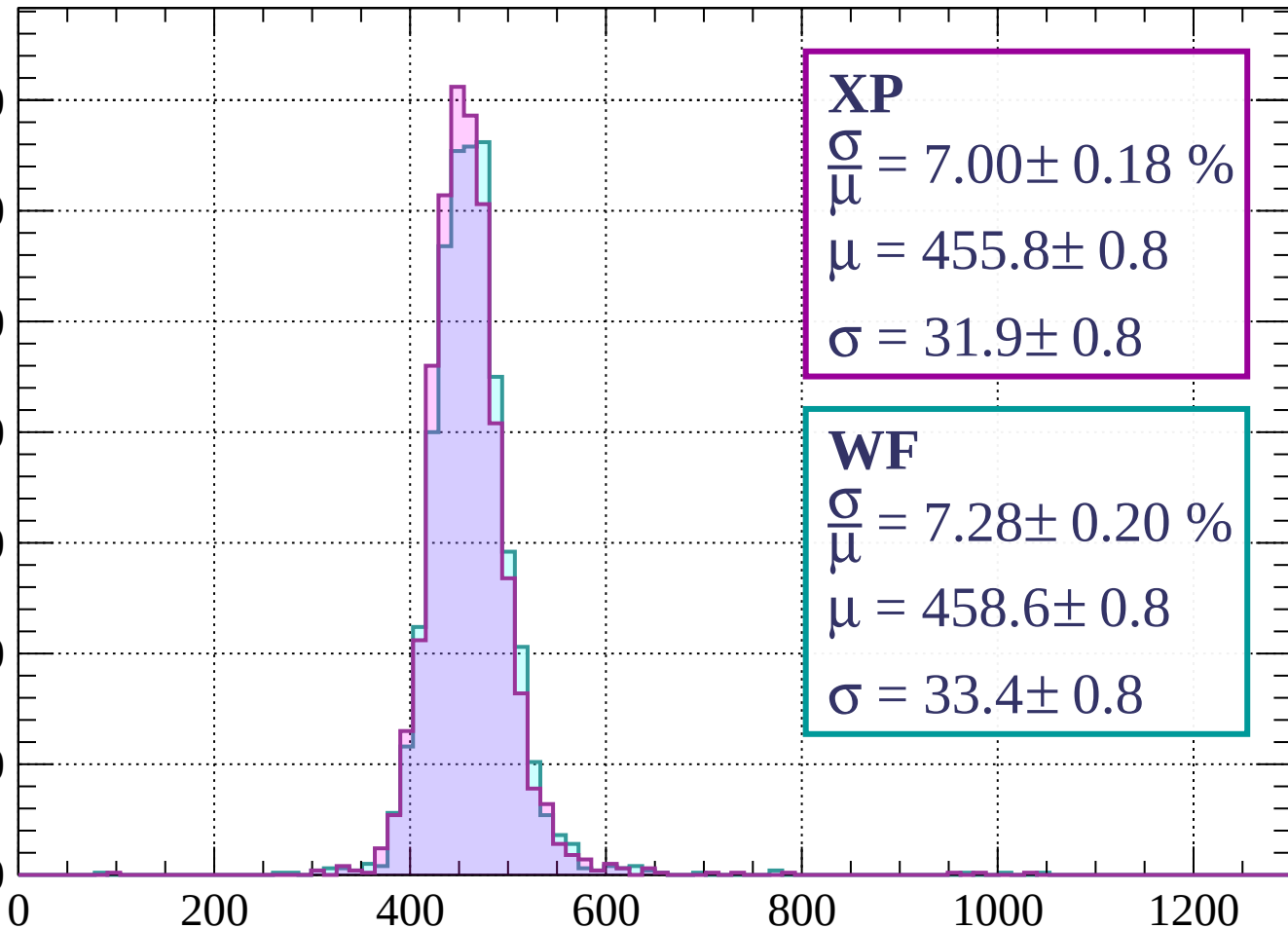


dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 7.00 \pm 0.18 \%$$

$$\mu = 455.8 \pm 0.8$$

$$\sigma = 31.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.28 \pm 0.20 \%$$

$$\mu = 458.6 \pm 0.8$$

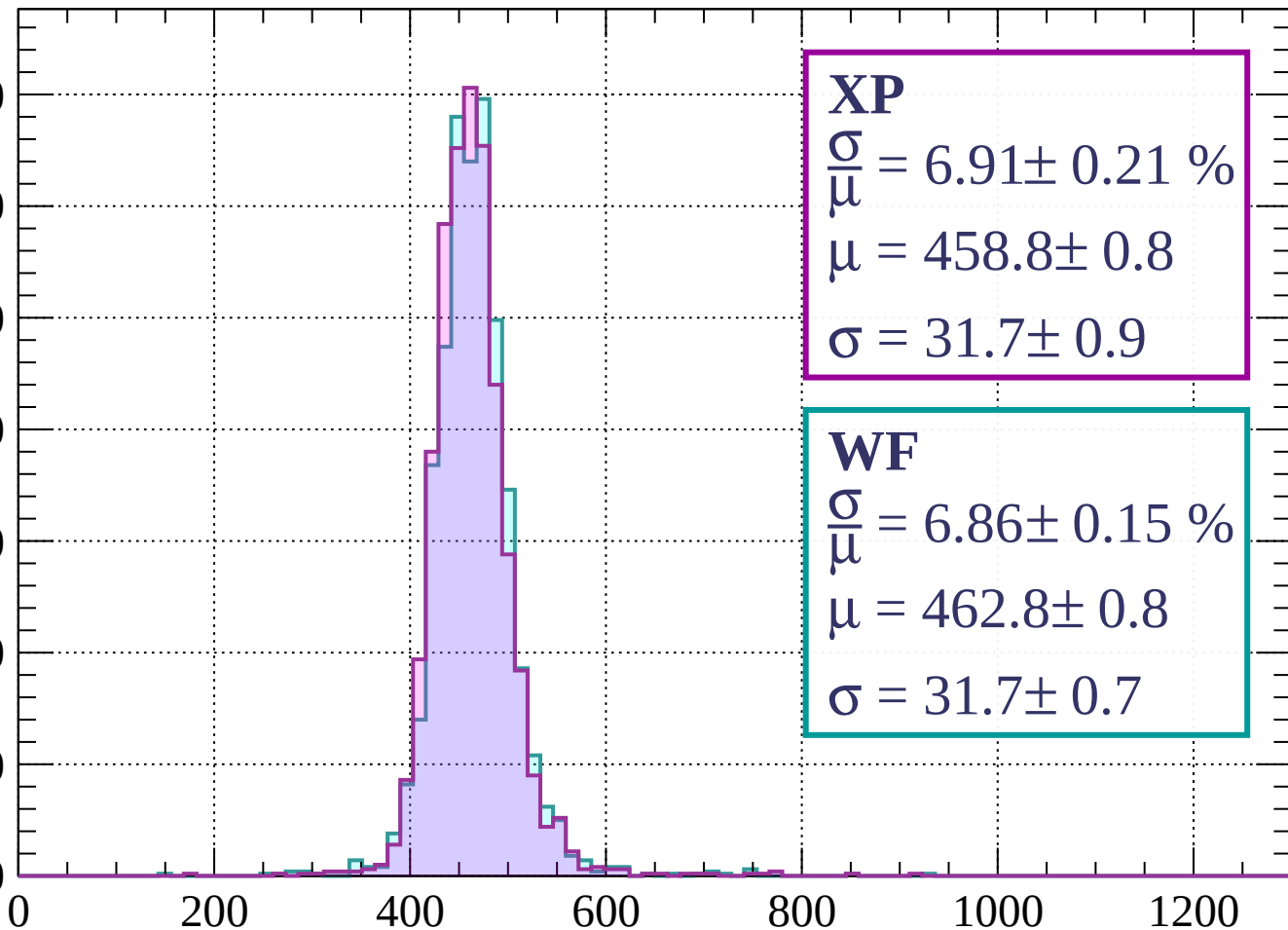
$$\sigma = 33.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1360$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 6.91 \pm 0.21 \%$$

$$\mu = 458.8 \pm 0.8$$

$$\sigma = 31.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.86 \pm 0.15 \%$$

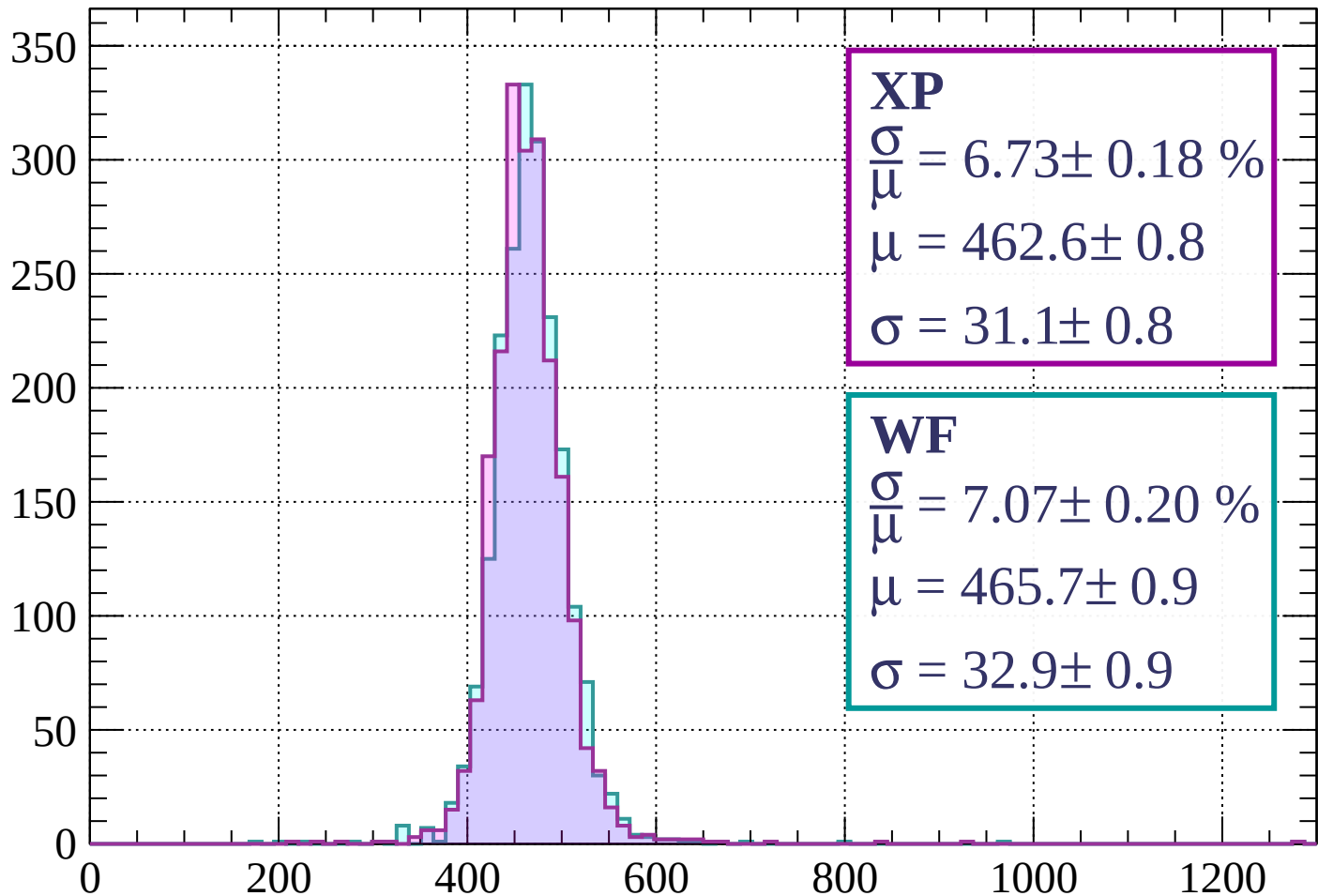
$$\mu = 462.8 \pm 0.8$$

$$\sigma = 31.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 6.73 \pm 0.18 \%$$

$$\mu = 462.6 \pm 0.8$$

$$\sigma = 31.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.07 \pm 0.20 \%$$

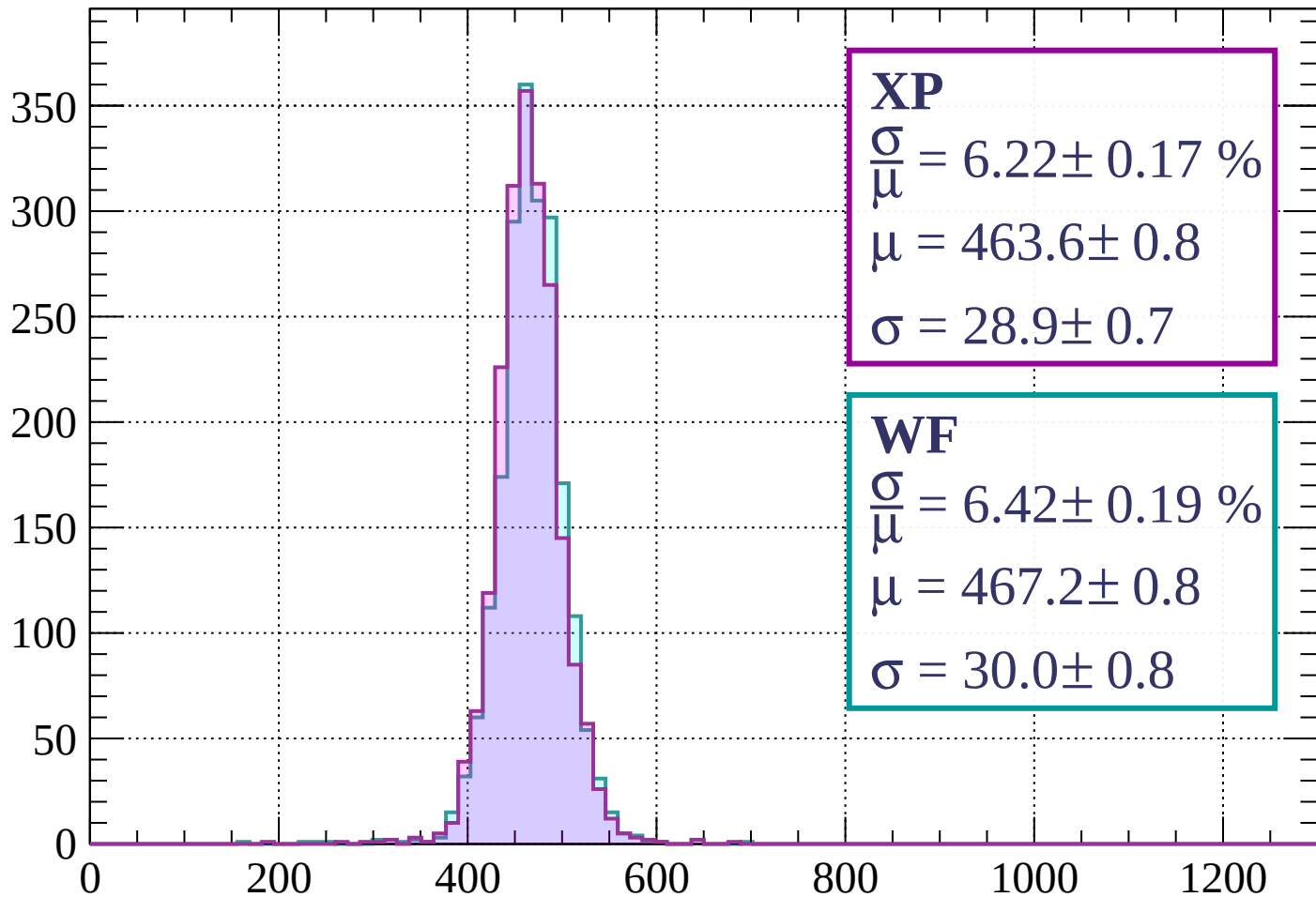
$$\mu = 465.7 \pm 0.9$$

$$\sigma = 32.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 6.22 \pm 0.17 \%$$

$$\mu = 463.6 \pm 0.8$$

$$\sigma = 28.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.42 \pm 0.19 \%$$

$$\mu = 467.2 \pm 0.8$$

$$\sigma = 30.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$

Count

300

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 6.72 \pm 0.18 \%$$

$$\mu = 467.8 \pm 0.8$$

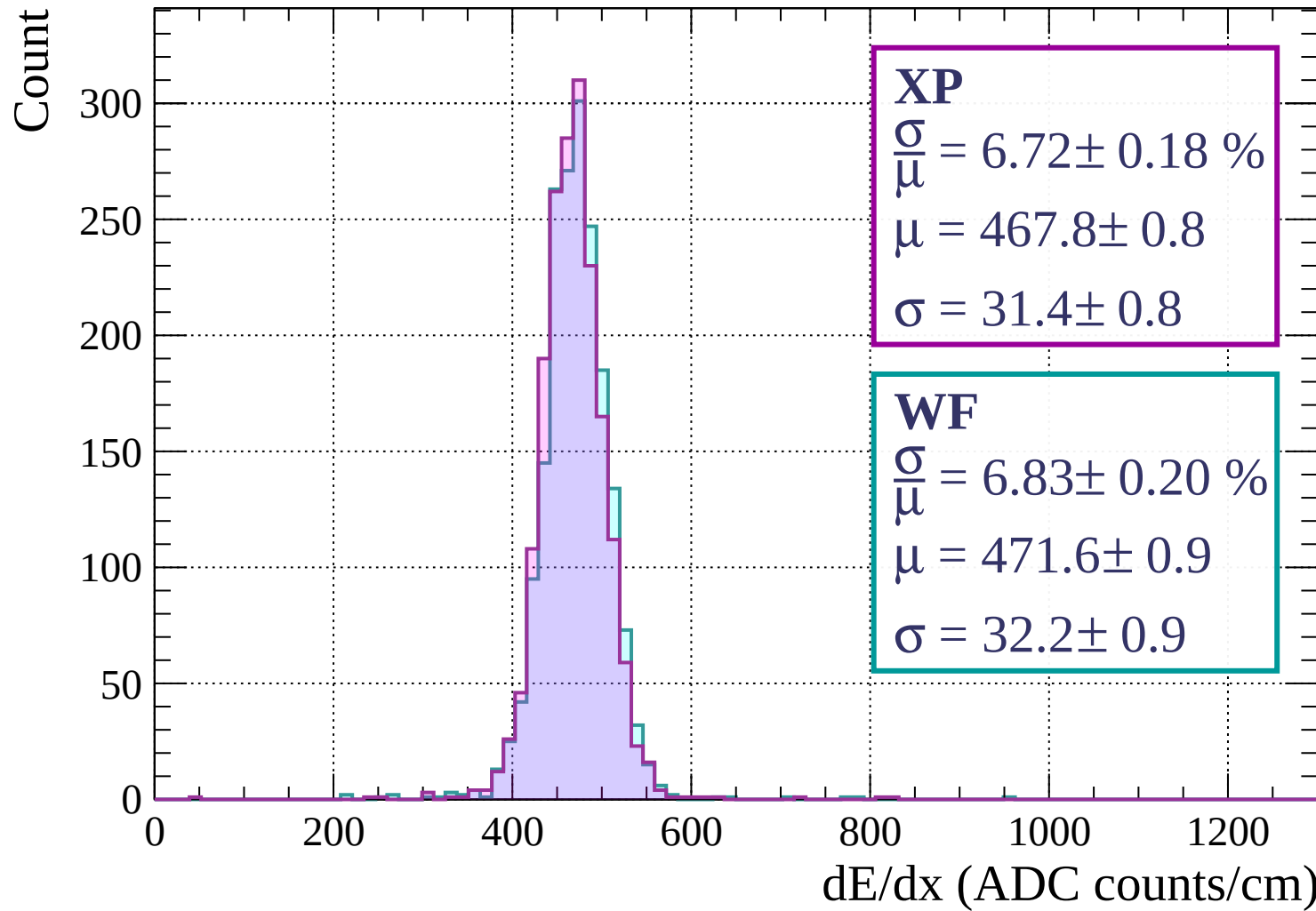
$$\sigma = 31.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.83 \pm 0.20 \%$$

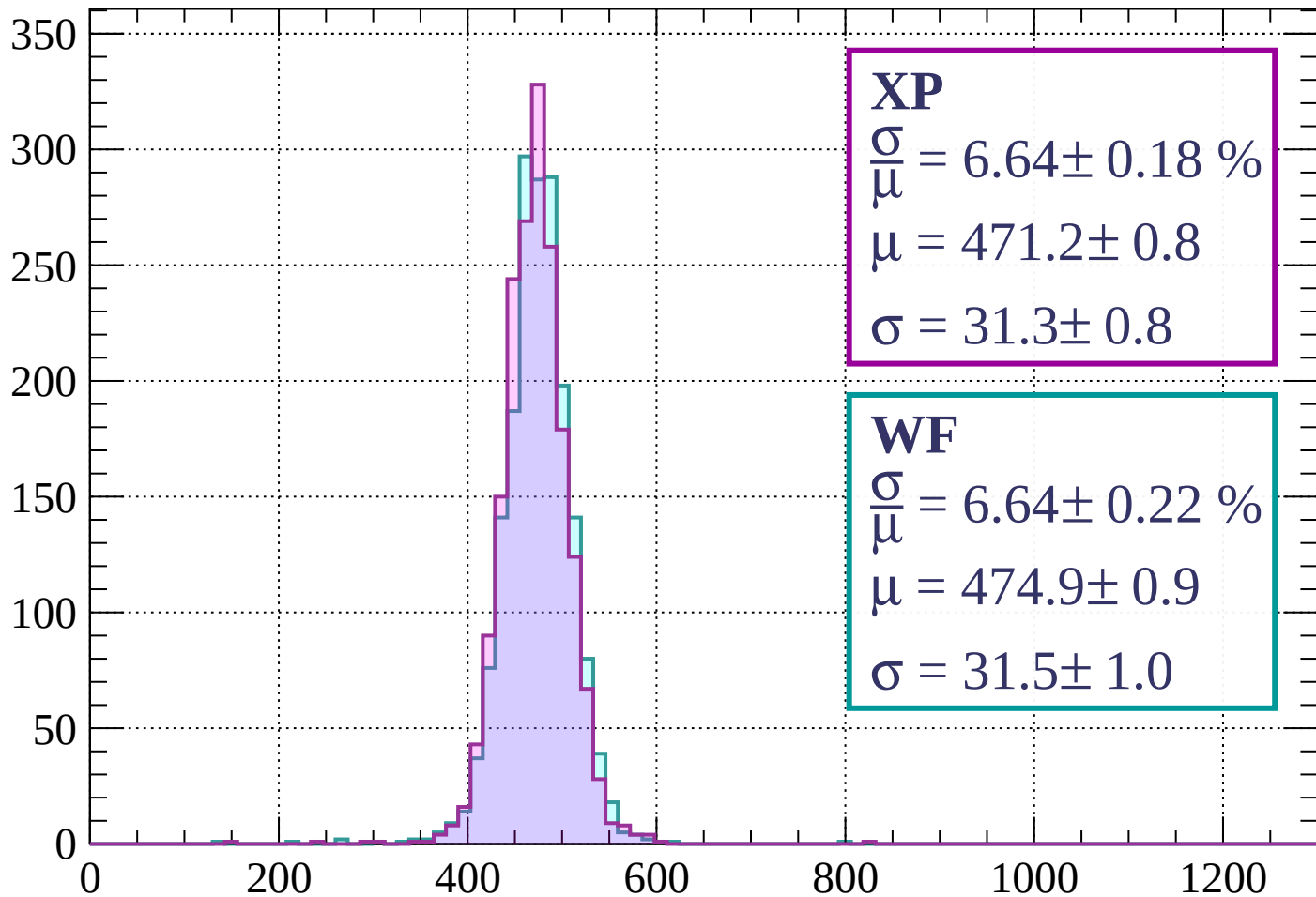
$$\mu = 471.6 \pm 0.9$$

$$\sigma = 32.2 \pm 0.9$$



Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 6.64 \pm 0.18 \%$$

$$\mu = 471.2 \pm 0.8$$

$$\sigma = 31.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.64 \pm 0.22 \%$$

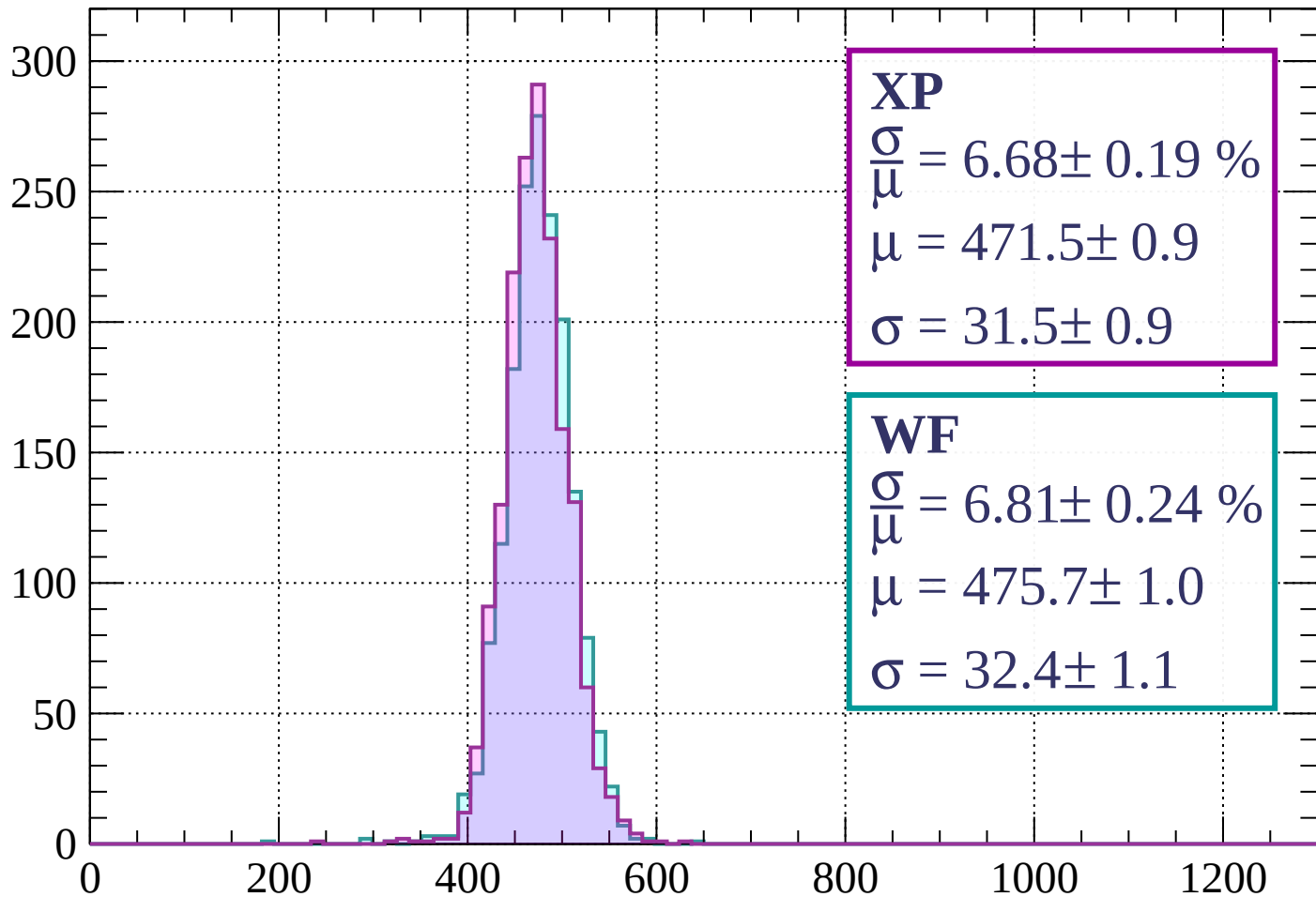
$$\mu = 474.9 \pm 0.9$$

$$\sigma = 31.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 6.68 \pm 0.19 \%$$

$$\mu = 471.5 \pm 0.9$$

$$\sigma = 31.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.81 \pm 0.24 \%$$

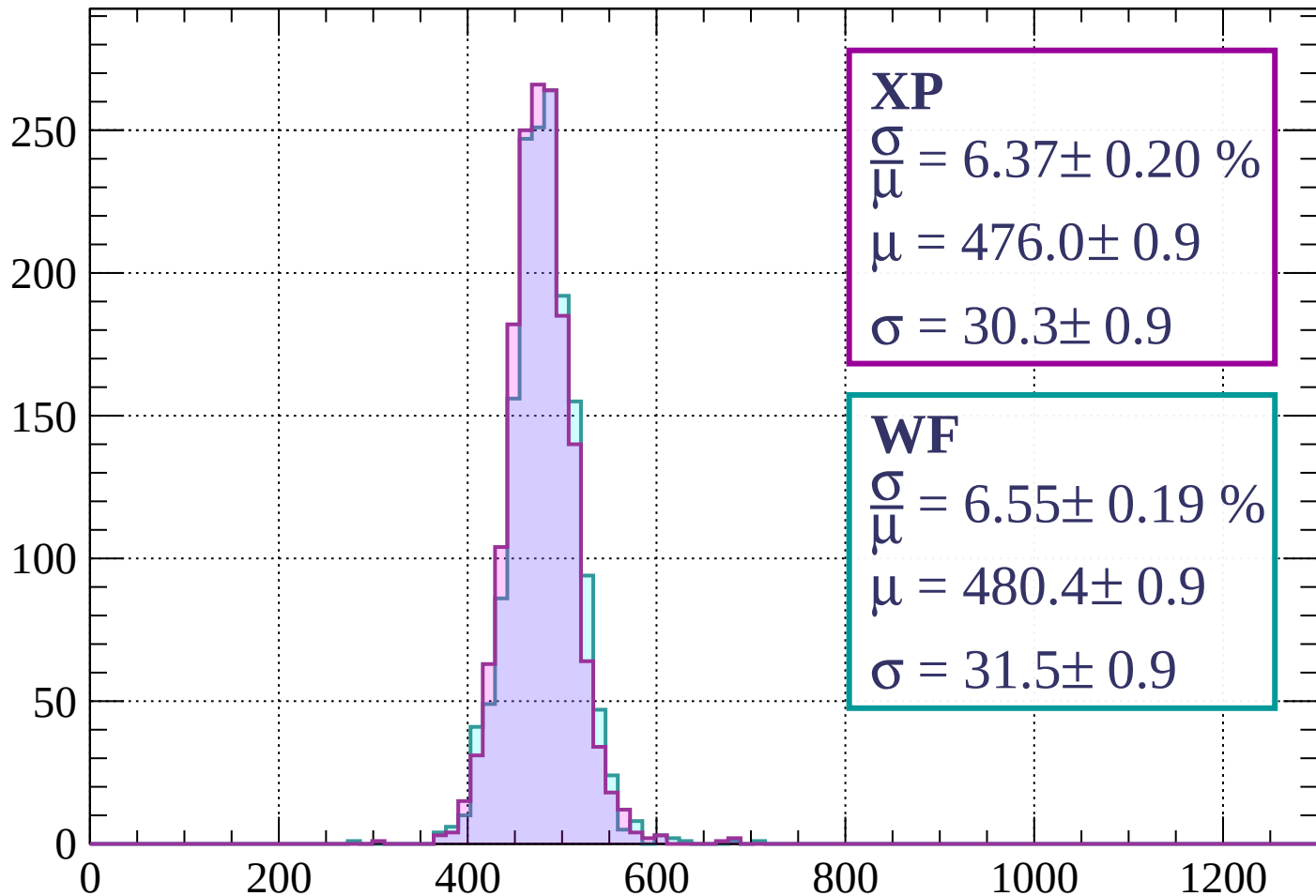
$$\mu = 475.7 \pm 1.0$$

$$\sigma = 32.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 6.37 \pm 0.20 \%$$

$$\mu = 476.0 \pm 0.9$$

$$\sigma = 30.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.55 \pm 0.19 \%$$

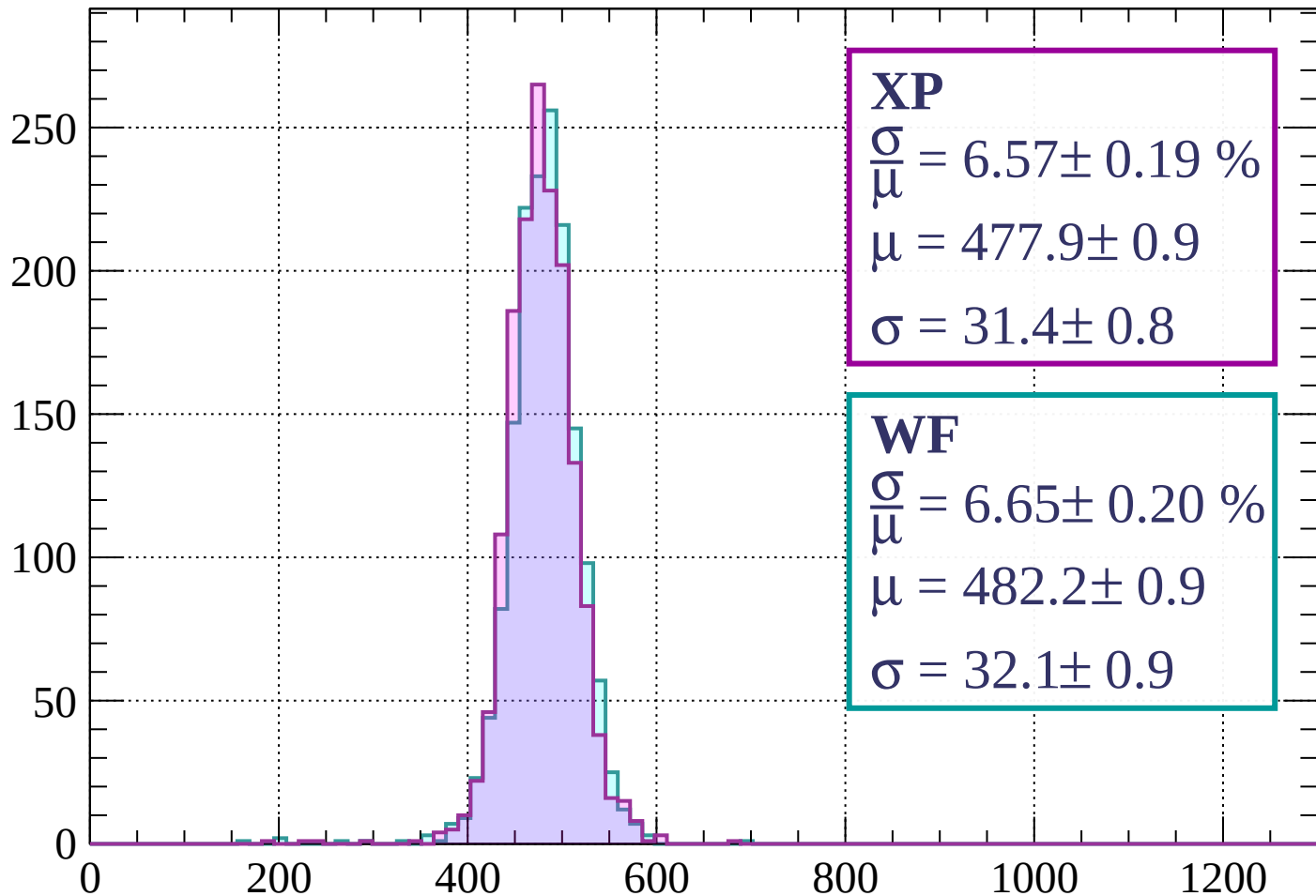
$$\mu = 480.4 \pm 0.9$$

$$\sigma = 31.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 6.57 \pm 0.19 \%$$

$$\mu = 477.9 \pm 0.9$$

$$\sigma = 31.4 \pm 0.8$$

WF

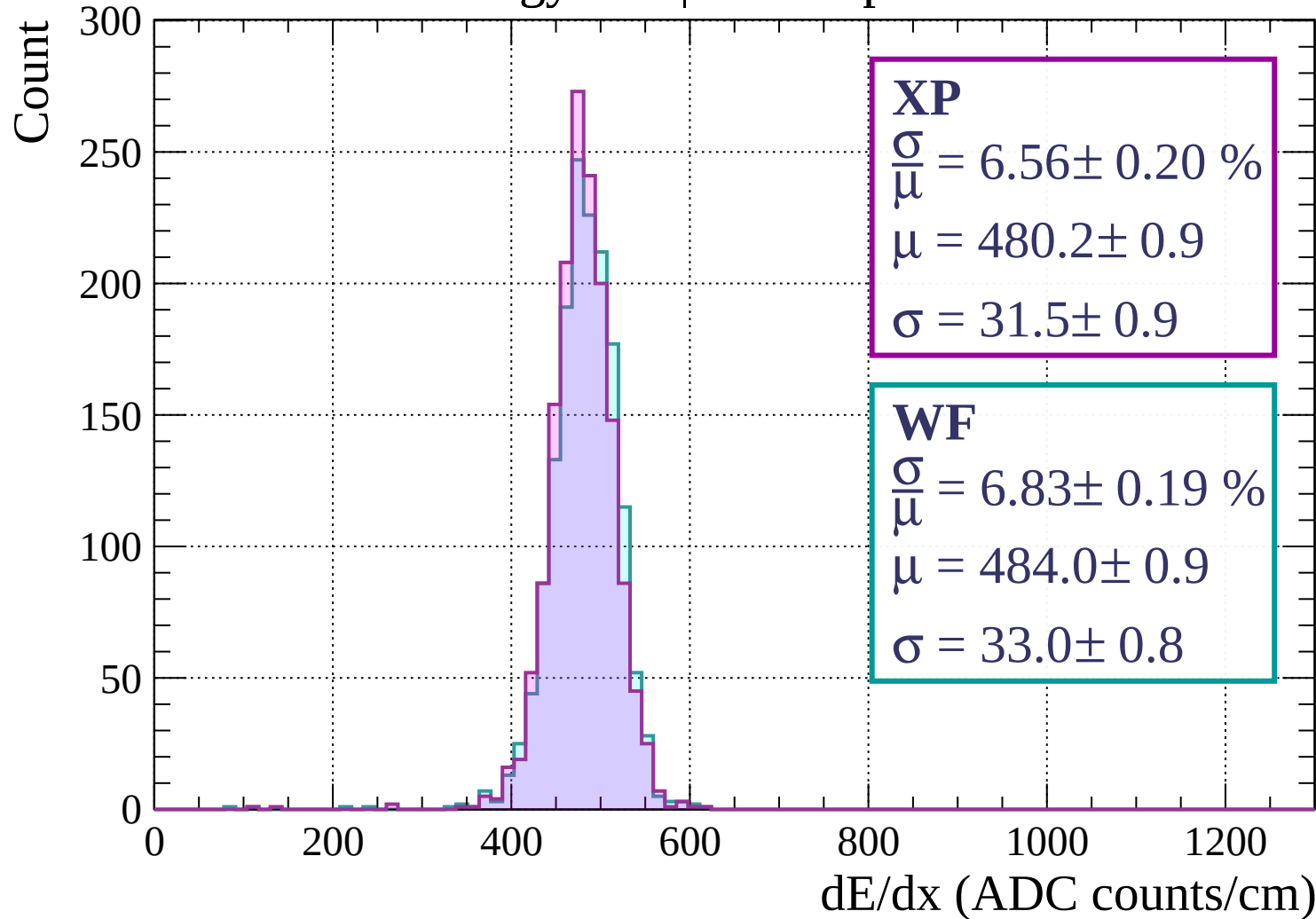
$$\frac{\sigma}{\mu} = 6.65 \pm 0.20 \%$$

$$\mu = 482.2 \pm 0.9$$

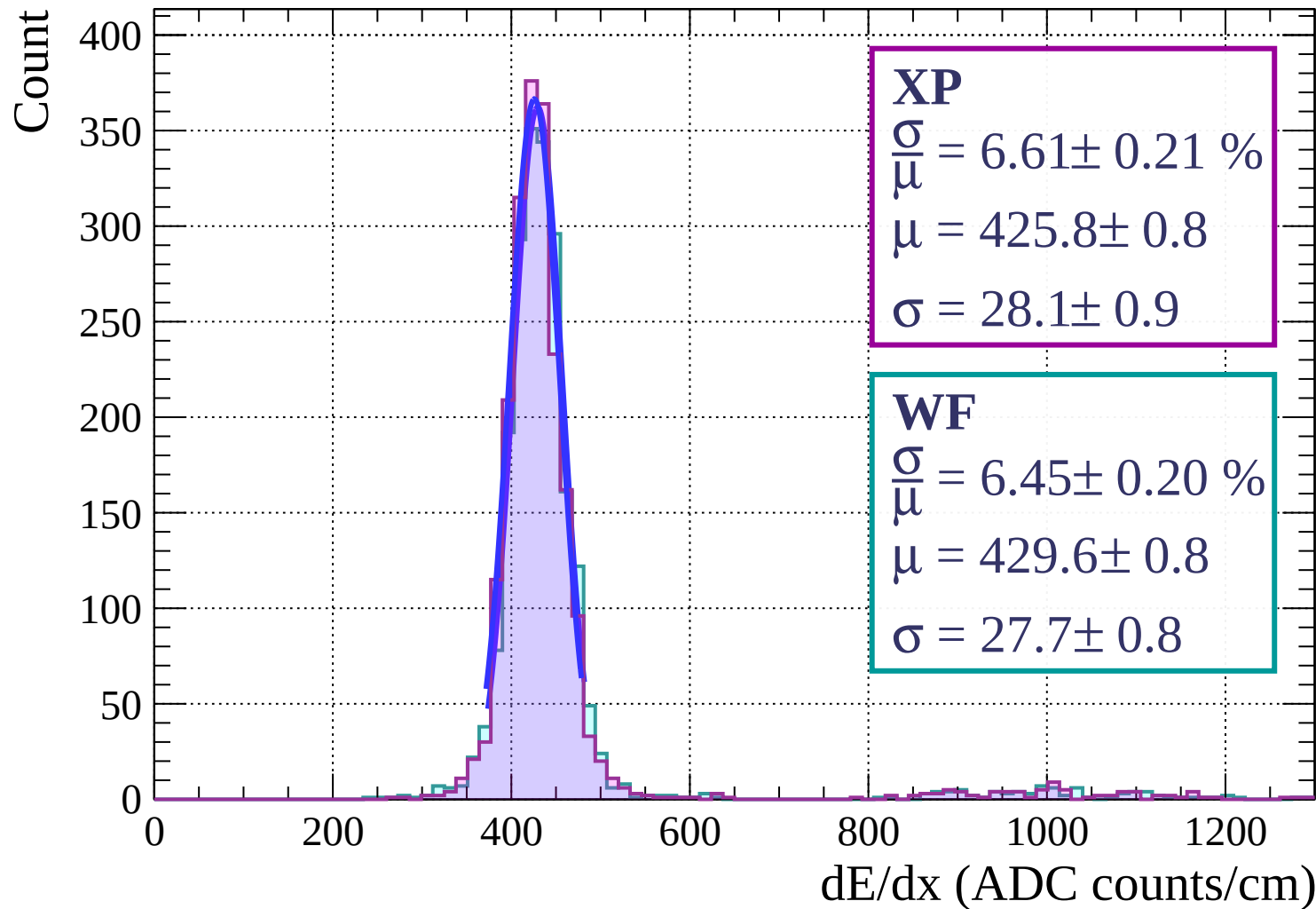
$$\sigma = 32.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 2000$

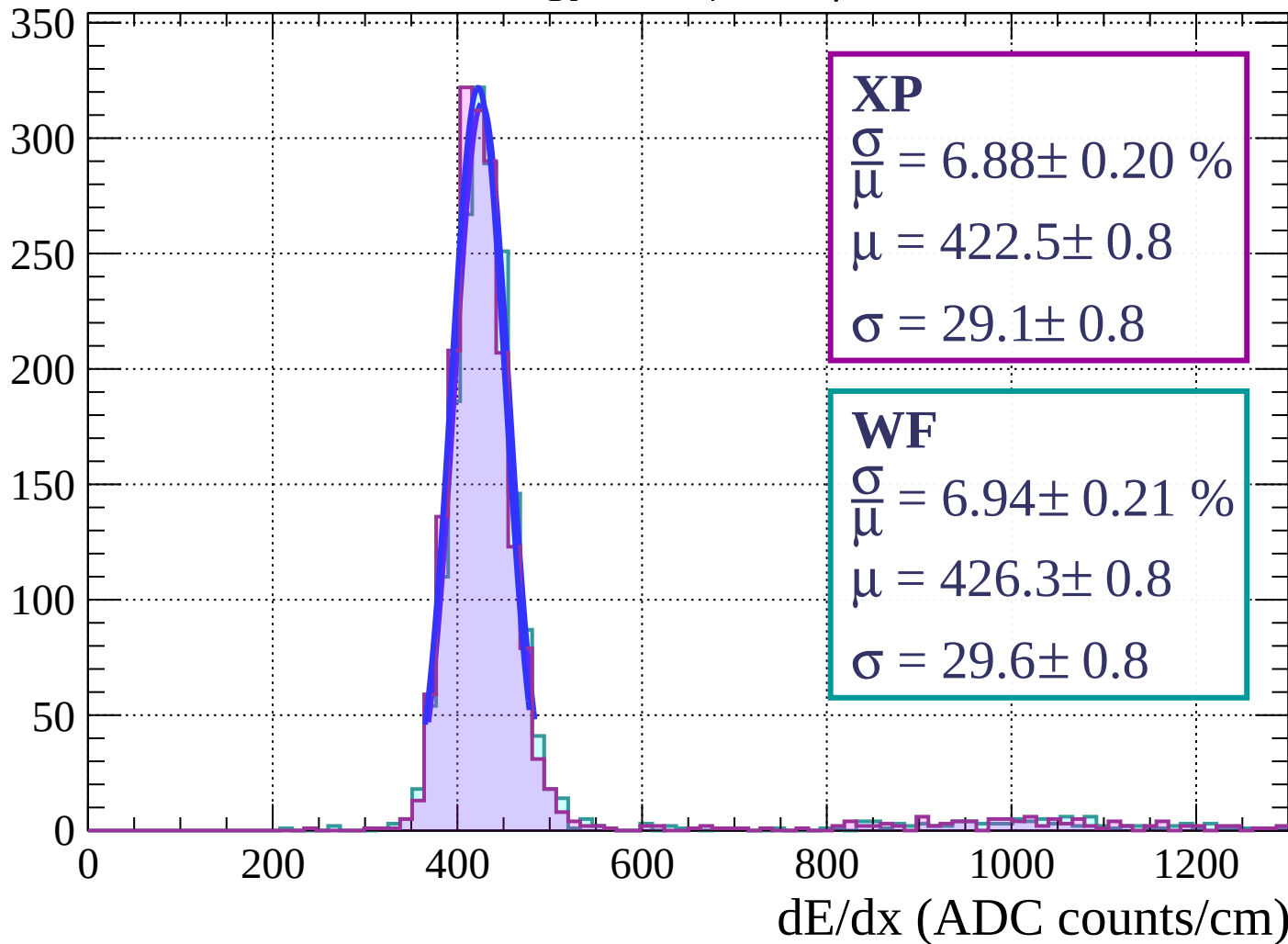


Energy loss | $0 < \phi < 3$

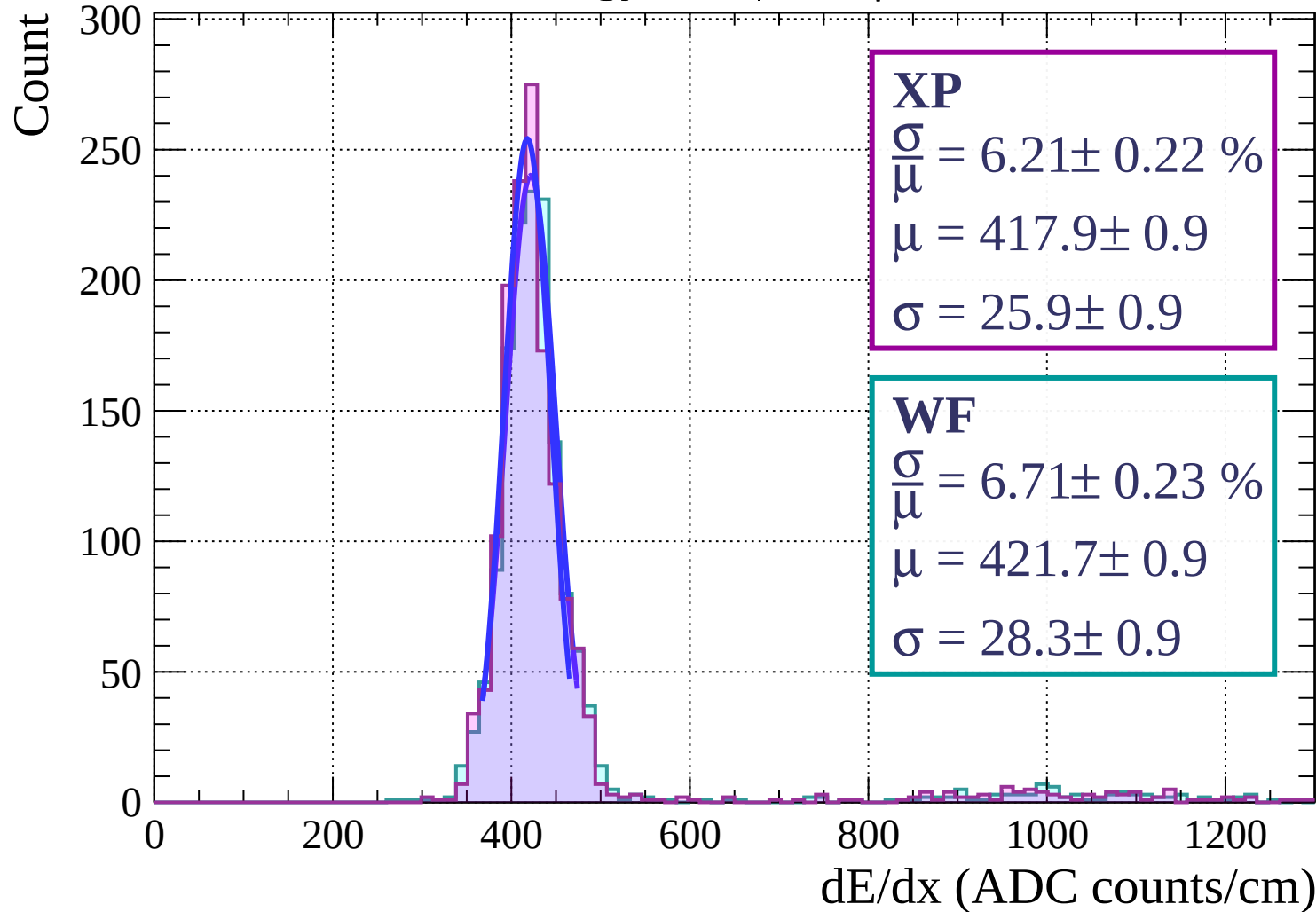


Energy loss | $3 < \phi < 6$

Count

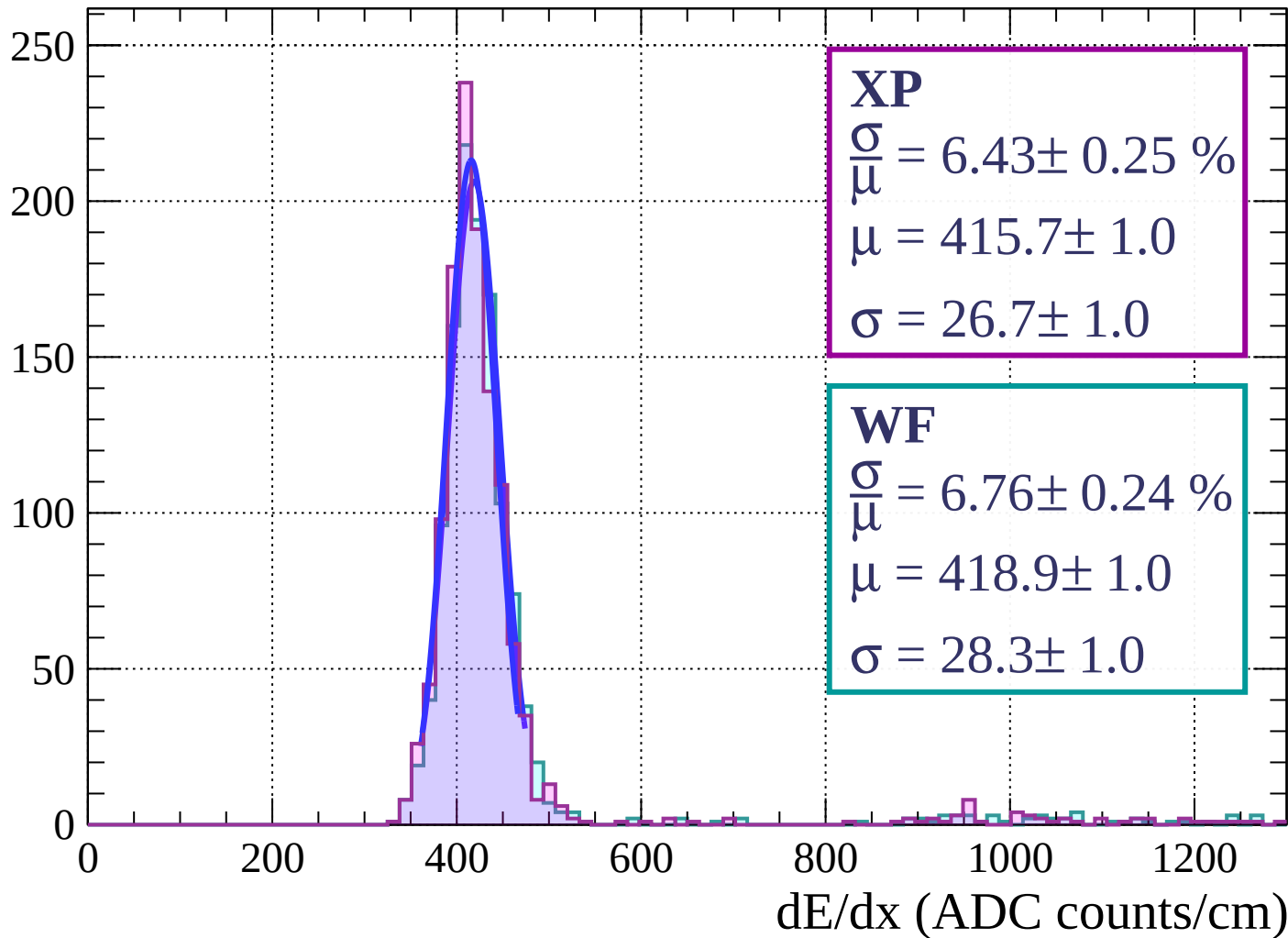


Energy loss | $6 < \phi < 9$



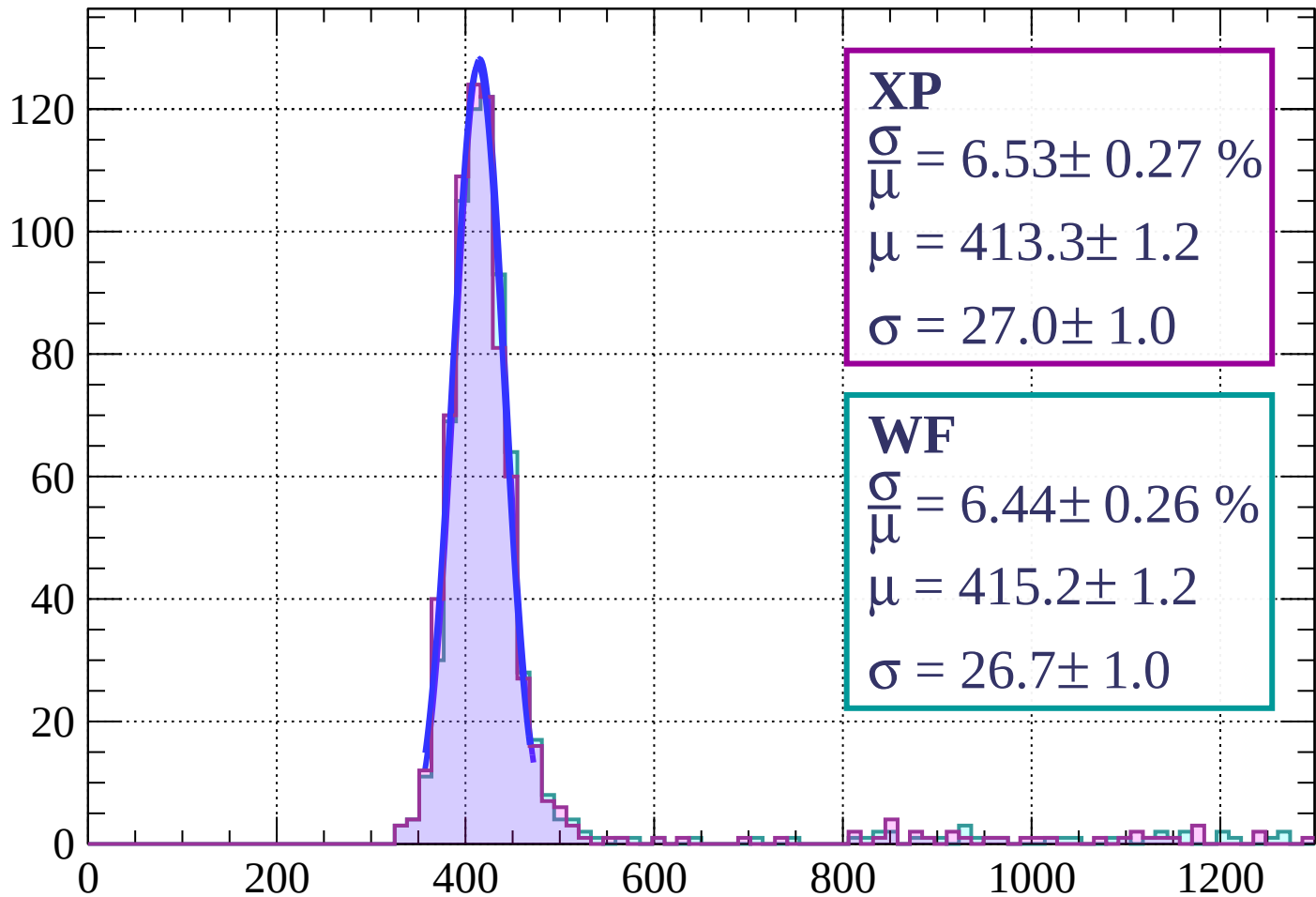
Energy loss | $9 < \phi < 12$

Count



Energy loss | $12 < \phi < 15$

Count



XP

$$\frac{\sigma}{\mu} = 6.53 \pm 0.27 \%$$

$$\mu = 413.3 \pm 1.2$$

$$\sigma = 27.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.44 \pm 0.26 \%$$

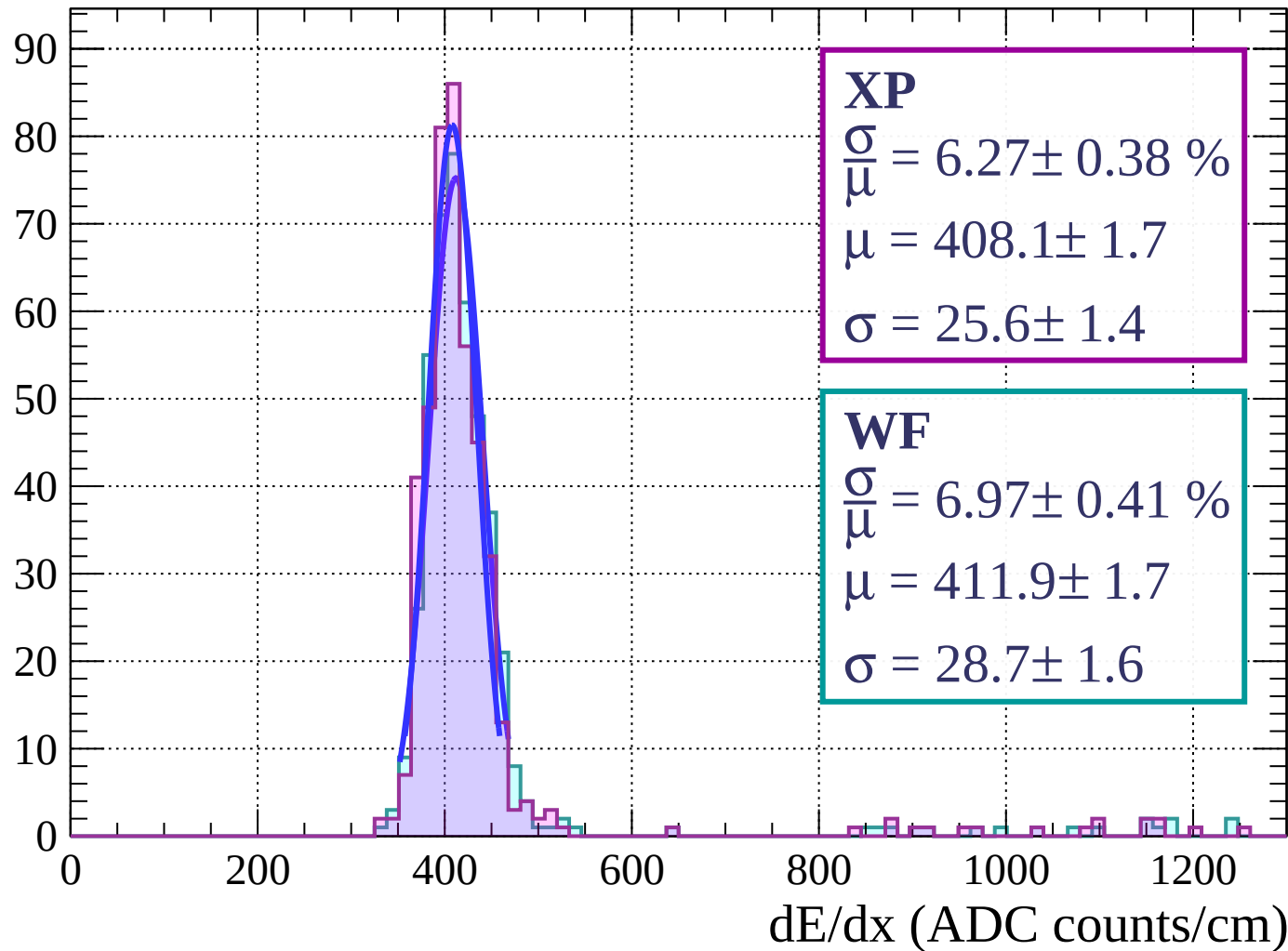
$$\mu = 415.2 \pm 1.2$$

$$\sigma = 26.7 \pm 1.0$$

dE/dx (ADC counts/cm)

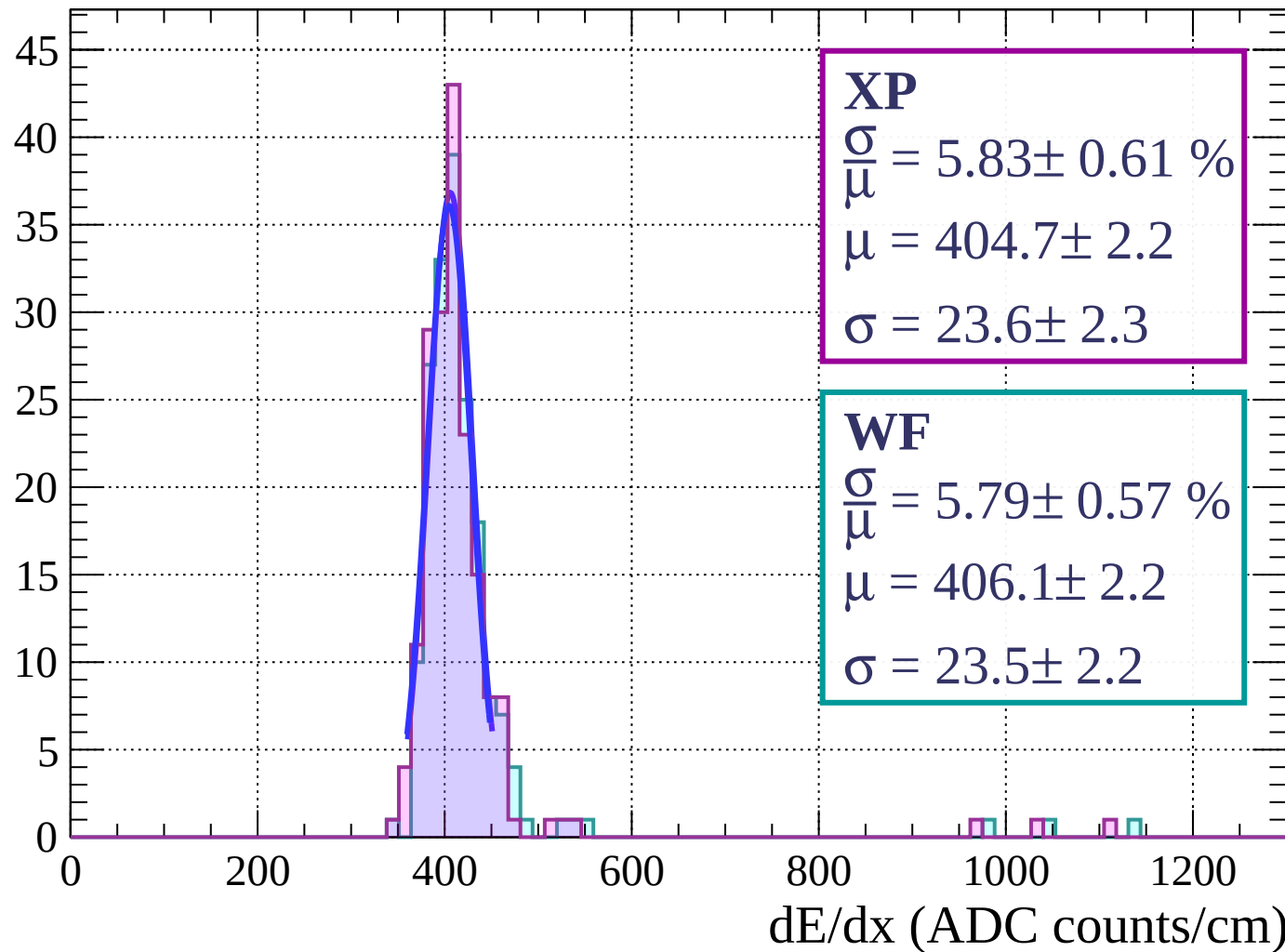
Energy loss | $15 < \phi < 18$

Count



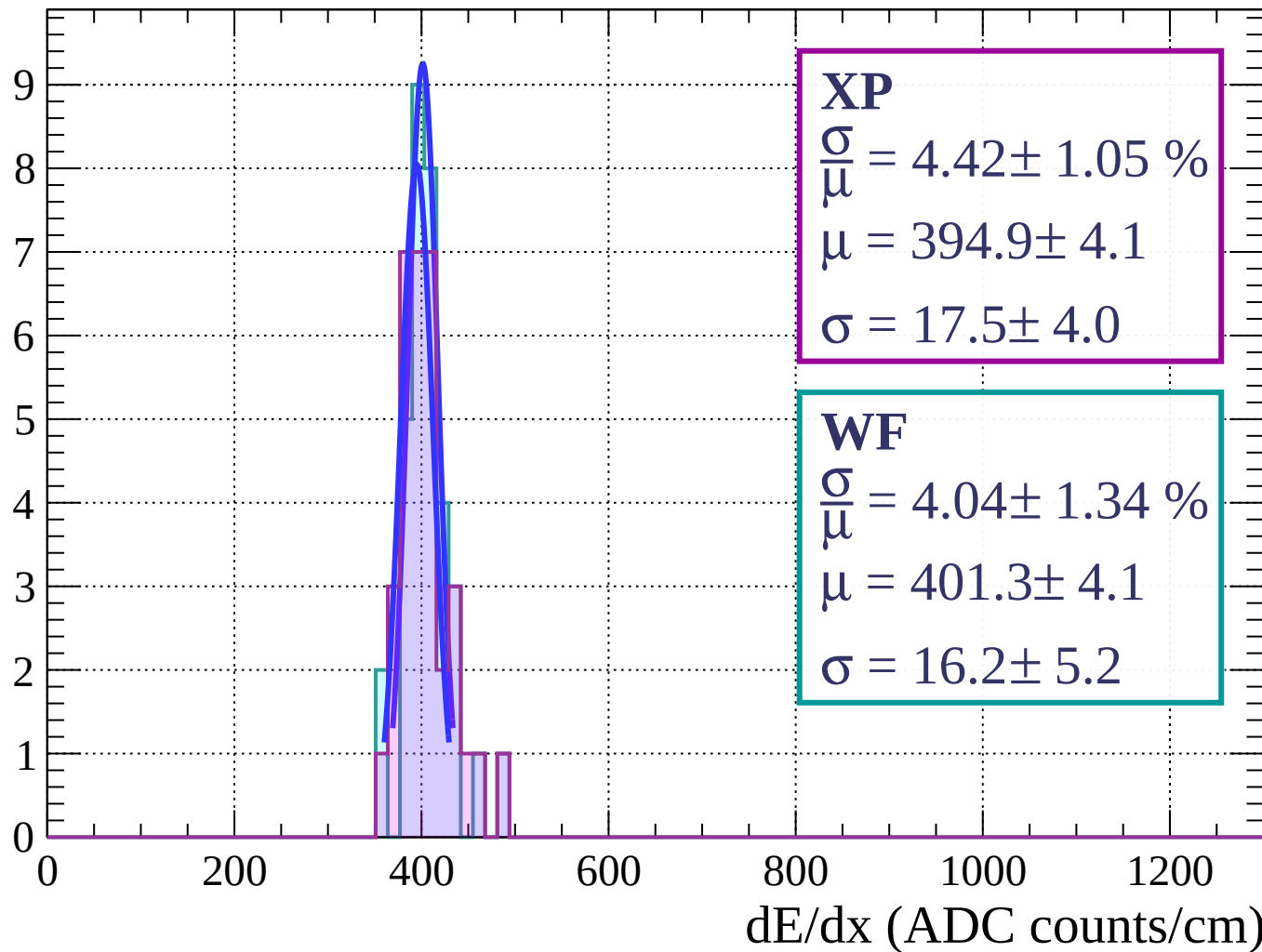
Energy loss | $18 < \phi < 21$

Count



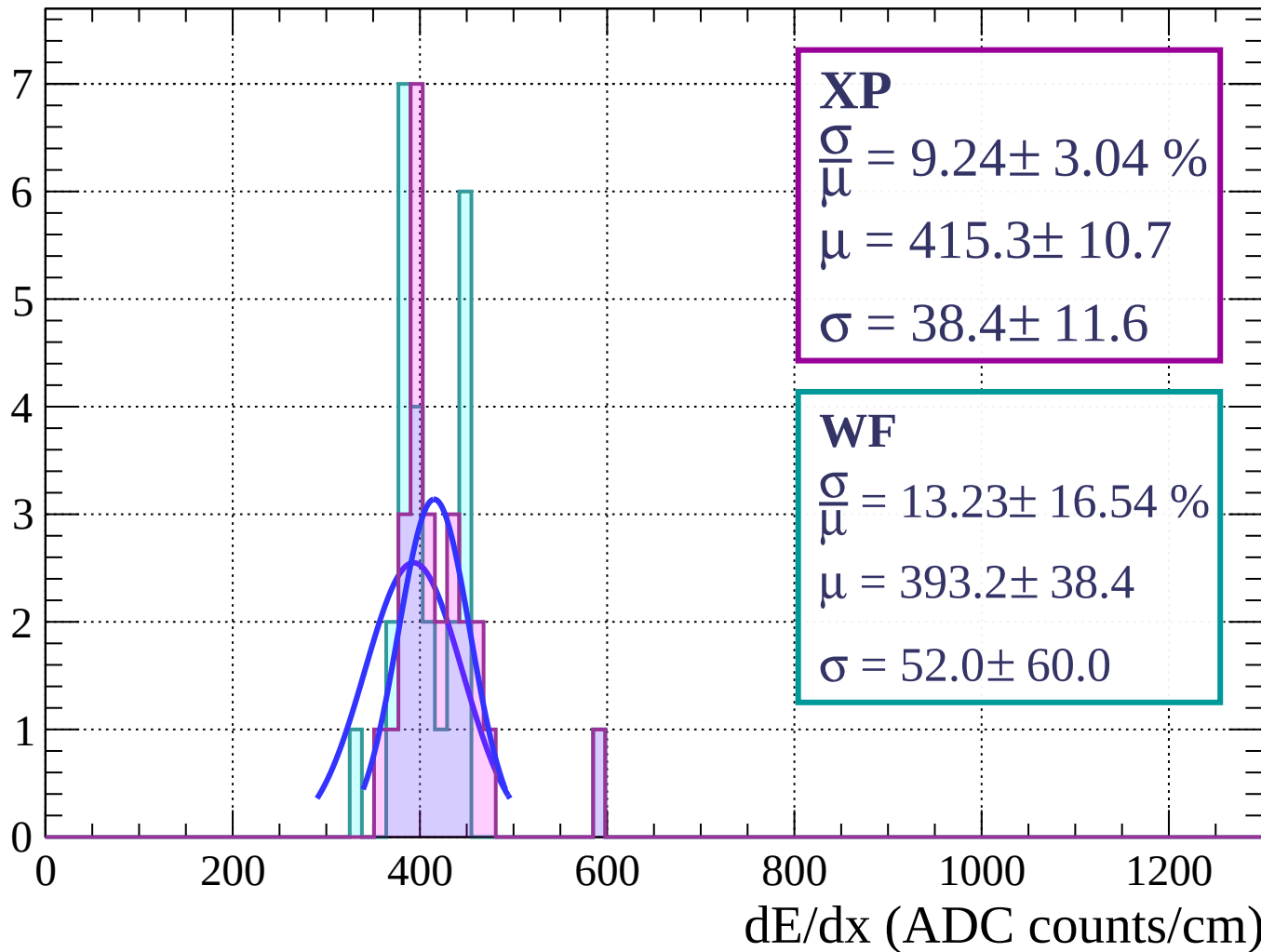
Energy loss | $21 < \phi < 24$

Count



Energy loss | $24 < \phi < 27$

Count



Energy loss | $27 < \phi < 30$

Count

XP

$$\frac{\sigma}{\mu} = 6.06 \pm 0.72 \%$$

$$\mu = 420.5 \pm 2.9$$

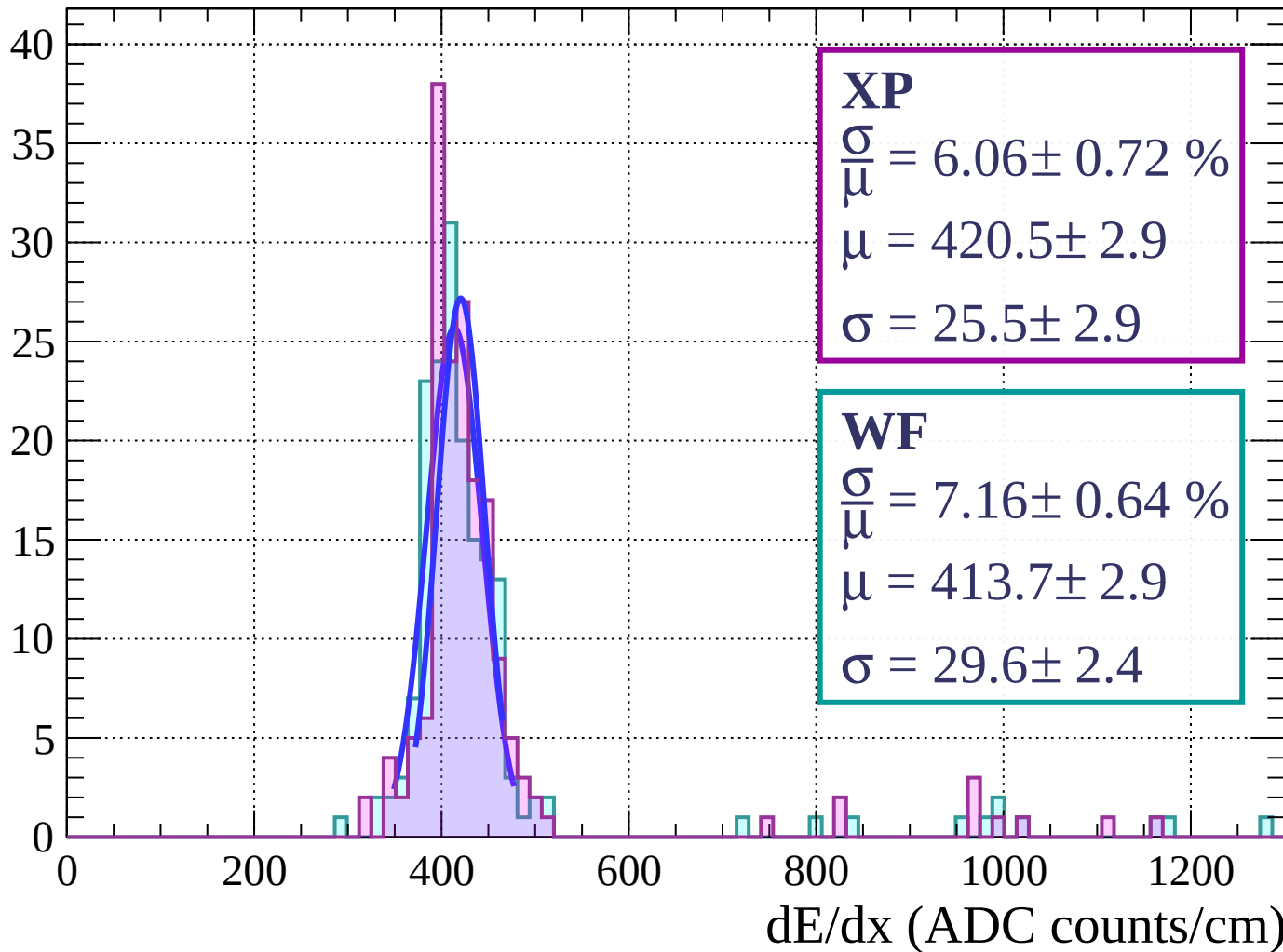
$$\sigma = 25.5 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 7.16 \pm 0.64 \%$$

$$\mu = 413.7 \pm 2.9$$

$$\sigma = 29.6 \pm 2.4$$



Energy loss | $30 < \phi < 33$

Count

XP

$$\frac{\sigma}{\mu} = 7.91 \pm 0.59 \%$$

$$\mu = 412.6 \pm 2.3$$

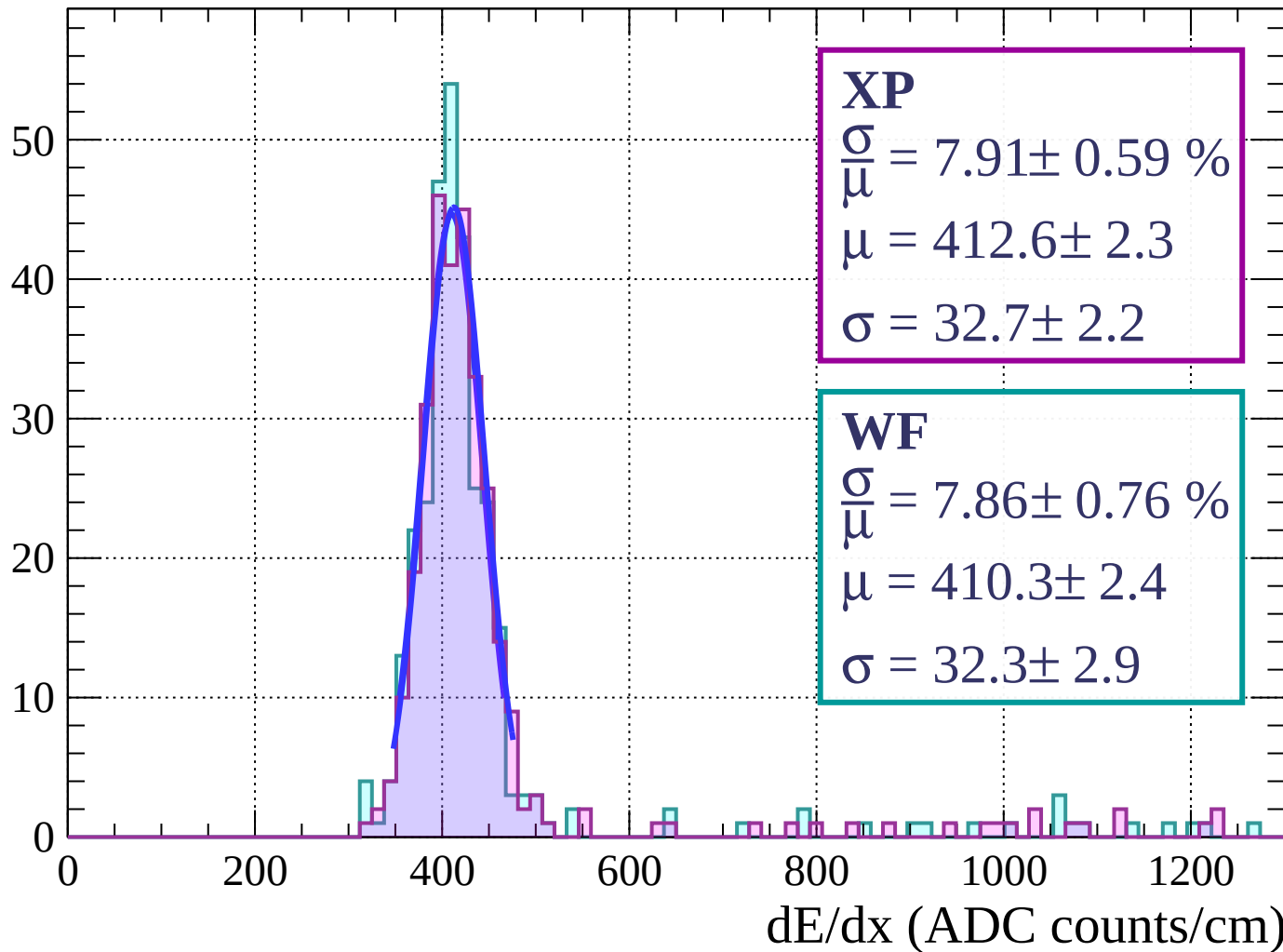
$$\sigma = 32.7 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 7.86 \pm 0.76 \%$$

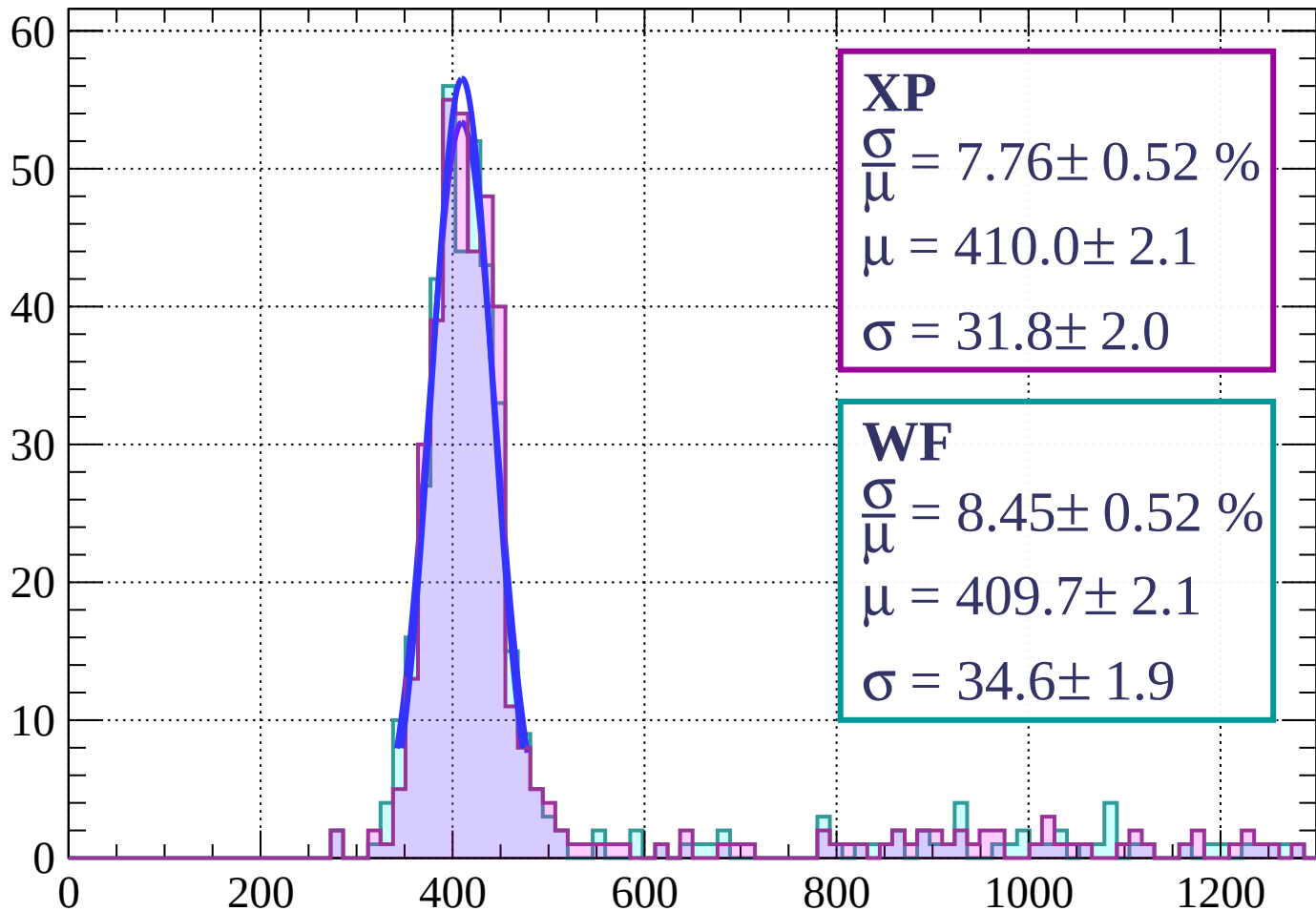
$$\mu = 410.3 \pm 2.4$$

$$\sigma = 32.3 \pm 2.9$$



Energy loss | $33 < \phi < 36$

Count



XP

$$\frac{\sigma}{\mu} = 7.76 \pm 0.52 \%$$

$$\mu = 410.0 \pm 2.1$$

$$\sigma = 31.8 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 8.45 \pm 0.52 \%$$

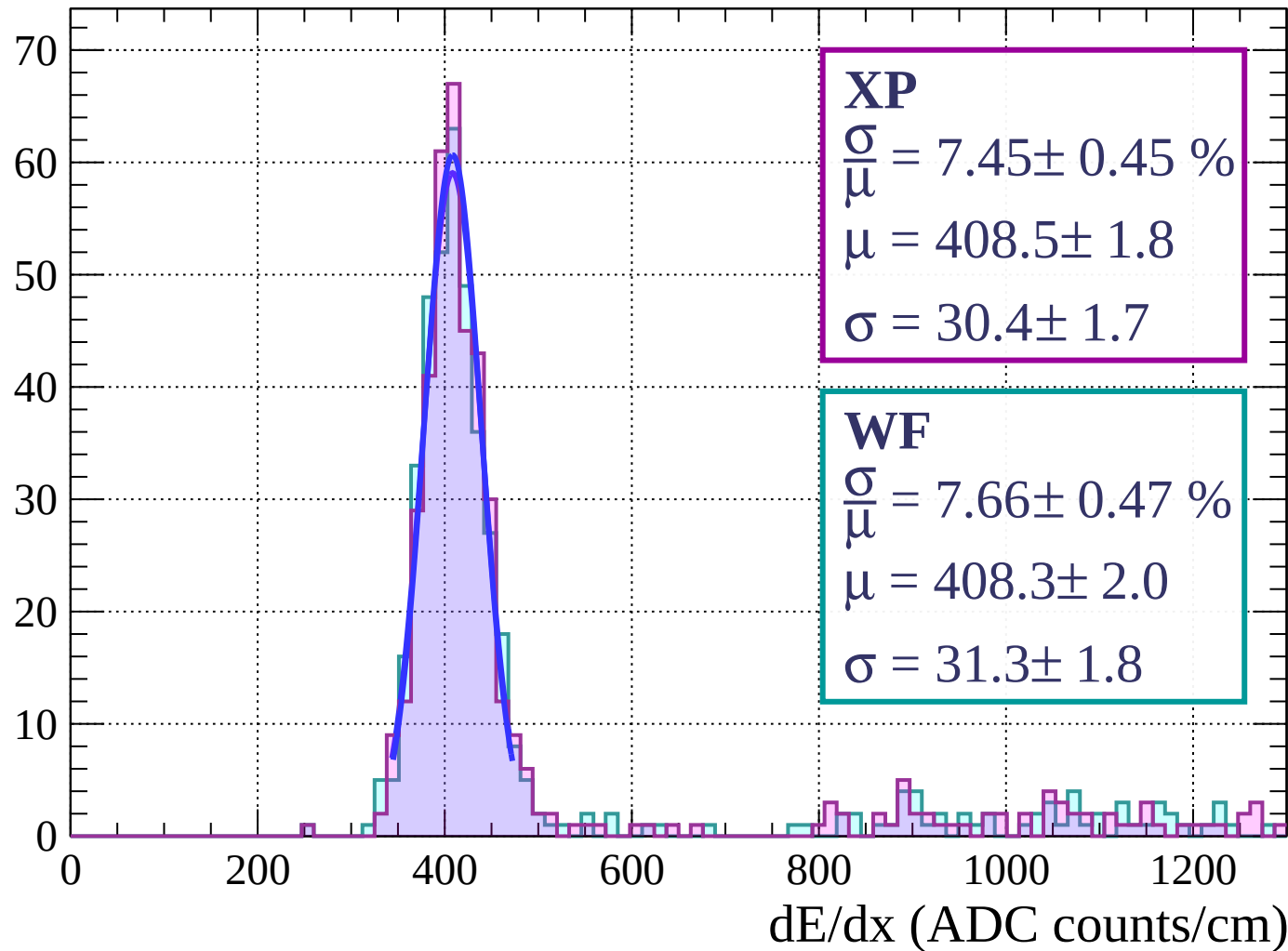
$$\mu = 409.7 \pm 2.1$$

$$\sigma = 34.6 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $36 < \phi < 39$

Count



Energy loss | $39 < \phi < 42$

Count

XP

$$\frac{\sigma}{\mu} = 7.29 \pm 0.52 \%$$

$$\mu = 406.8 \pm 2.0$$

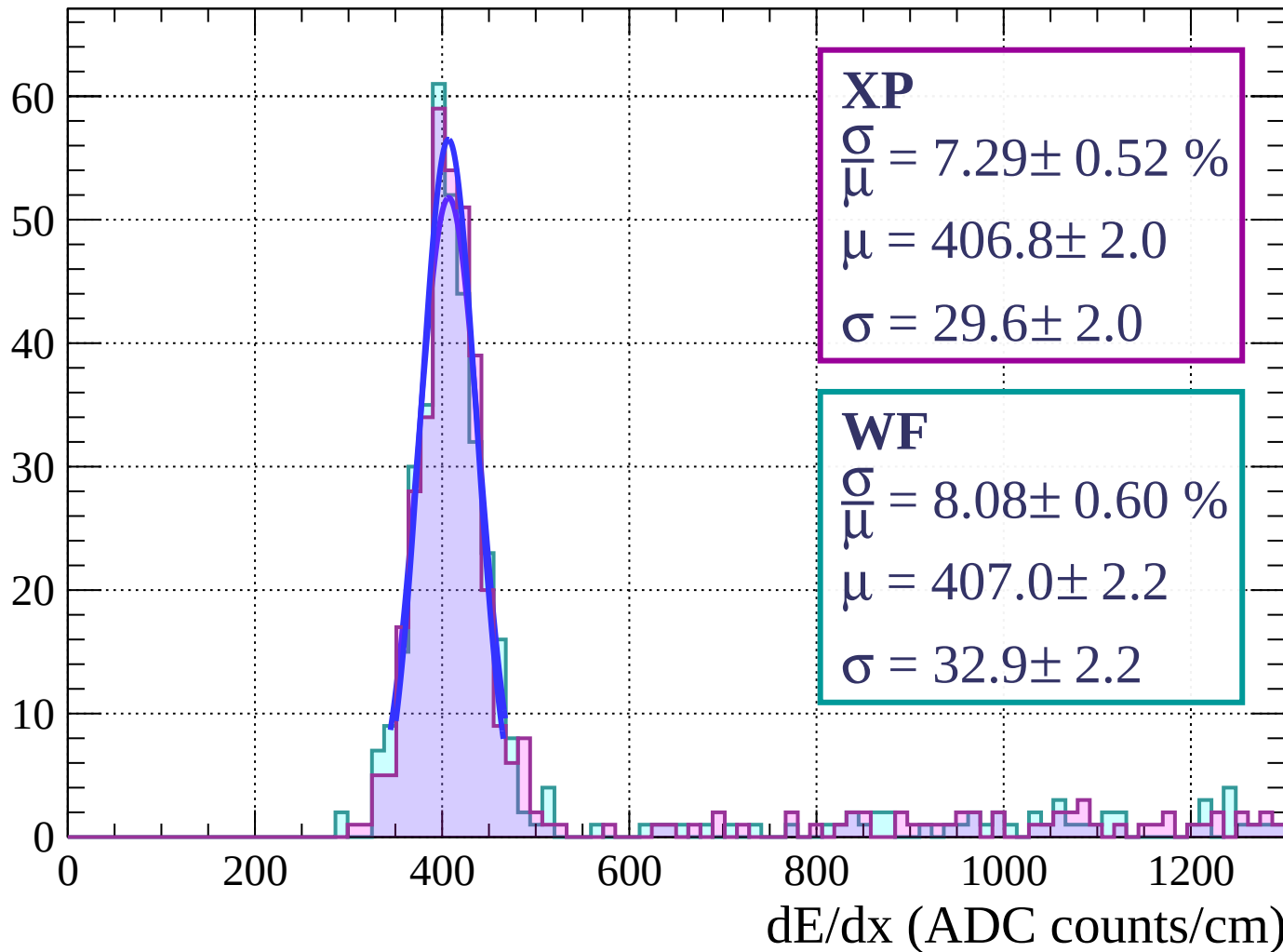
$$\sigma = 29.6 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 8.08 \pm 0.60 \%$$

$$\mu = 407.0 \pm 2.2$$

$$\sigma = 32.9 \pm 2.2$$



Energy loss | $42 < \phi < 45$

Count

XP

$$\frac{\sigma}{\mu} = 8.48 \pm 0.54 \%$$

$$\mu = 409.7 \pm 2.5$$

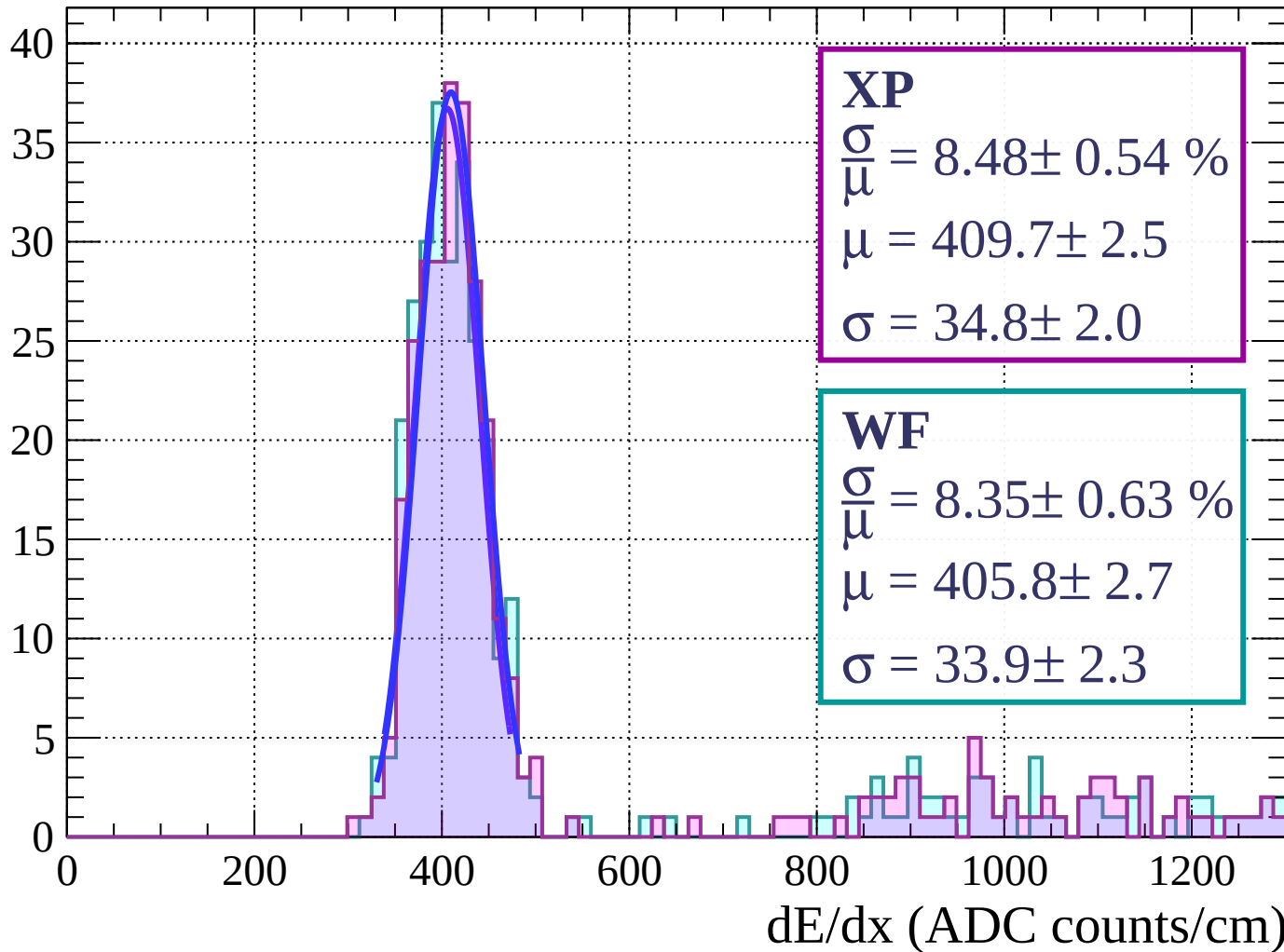
$$\sigma = 34.8 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 8.35 \pm 0.63 \%$$

$$\mu = 405.8 \pm 2.7$$

$$\sigma = 33.9 \pm 2.3$$



Energy loss | $45 < \phi < 48$

Count

XP

$$\frac{\sigma}{\mu} = 8.22 \pm 0.61 \%$$

$$\mu = 408.3 \pm 2.5$$

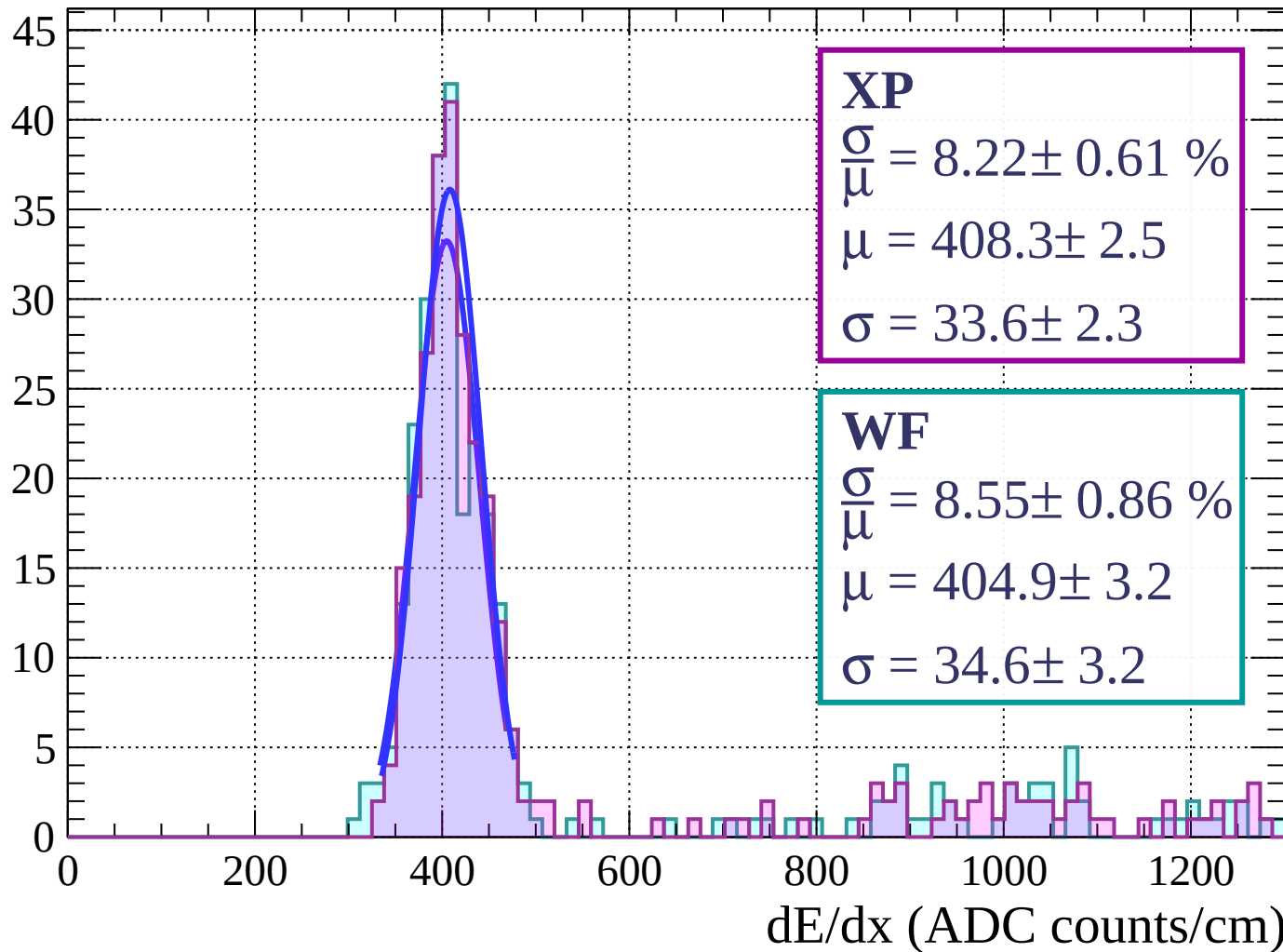
$$\sigma = 33.6 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 8.55 \pm 0.86 \%$$

$$\mu = 404.9 \pm 3.2$$

$$\sigma = 34.6 \pm 3.2$$



Energy loss | $48 < \phi < 51$

Count

XP

$$\frac{\sigma}{\mu} = 8.52 \pm 0.72 \%$$

$$\mu = 410.0 \pm 2.9$$

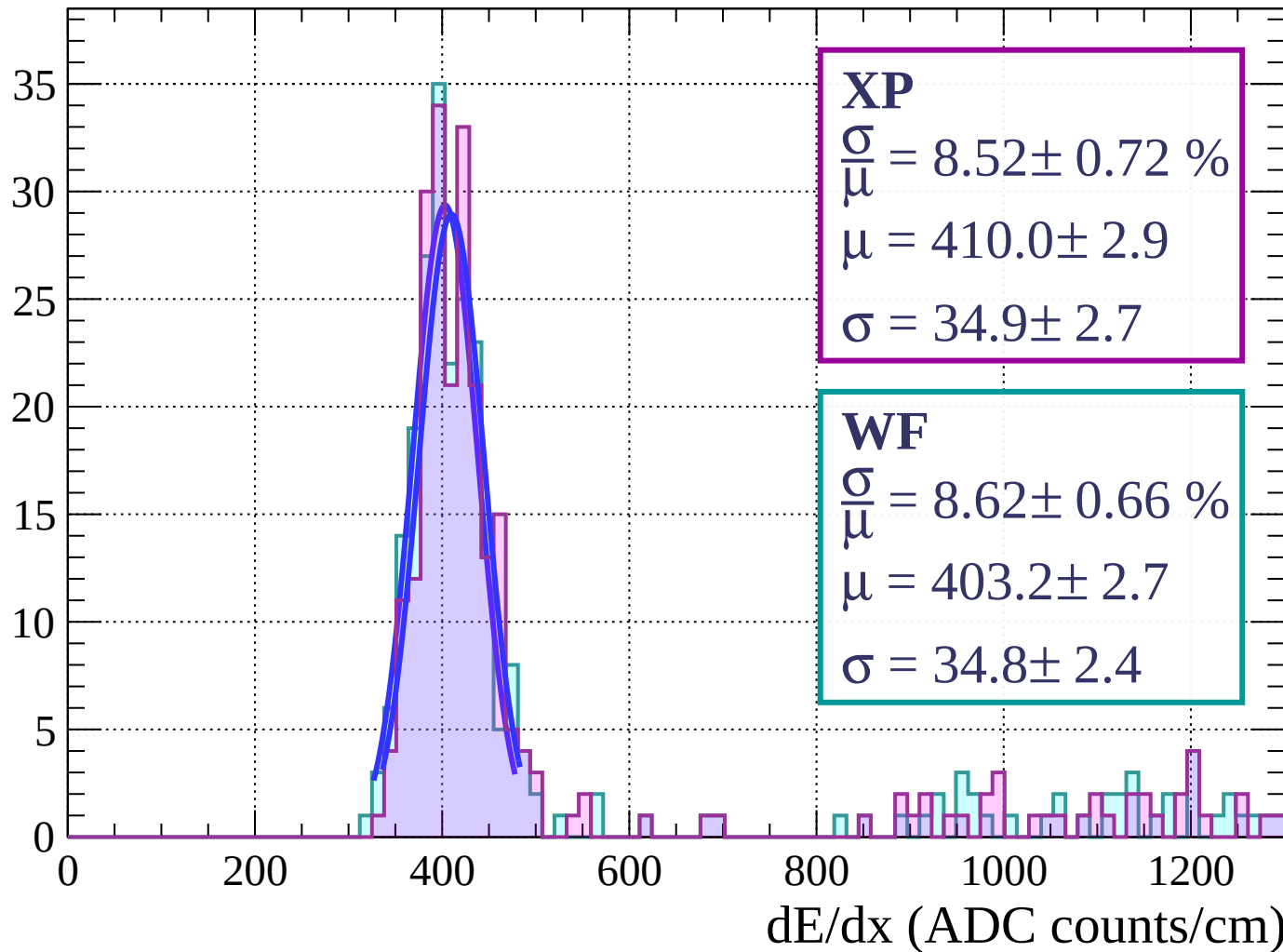
$$\sigma = 34.9 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 8.62 \pm 0.66 \%$$

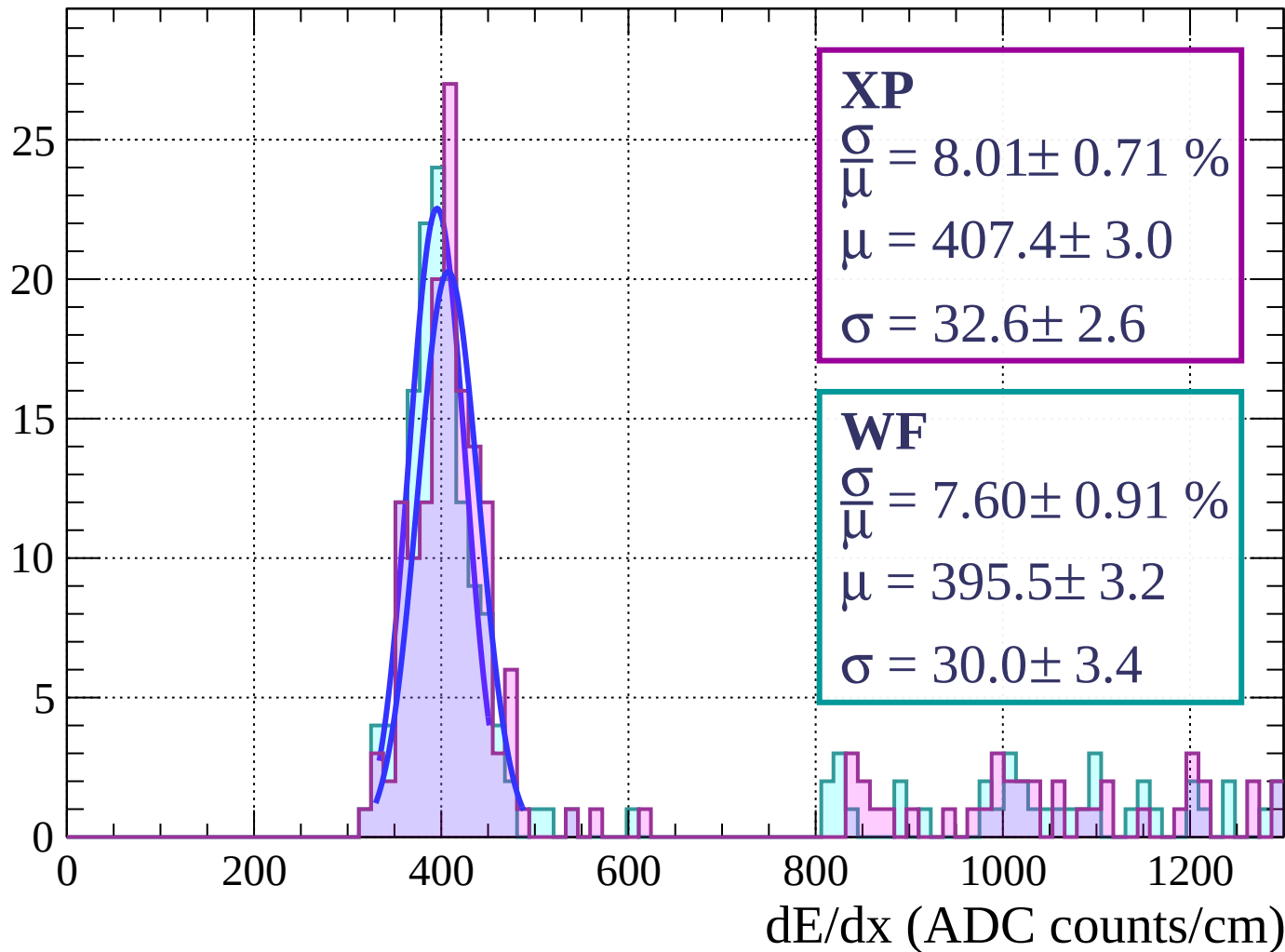
$$\mu = 403.2 \pm 2.7$$

$$\sigma = 34.8 \pm 2.4$$



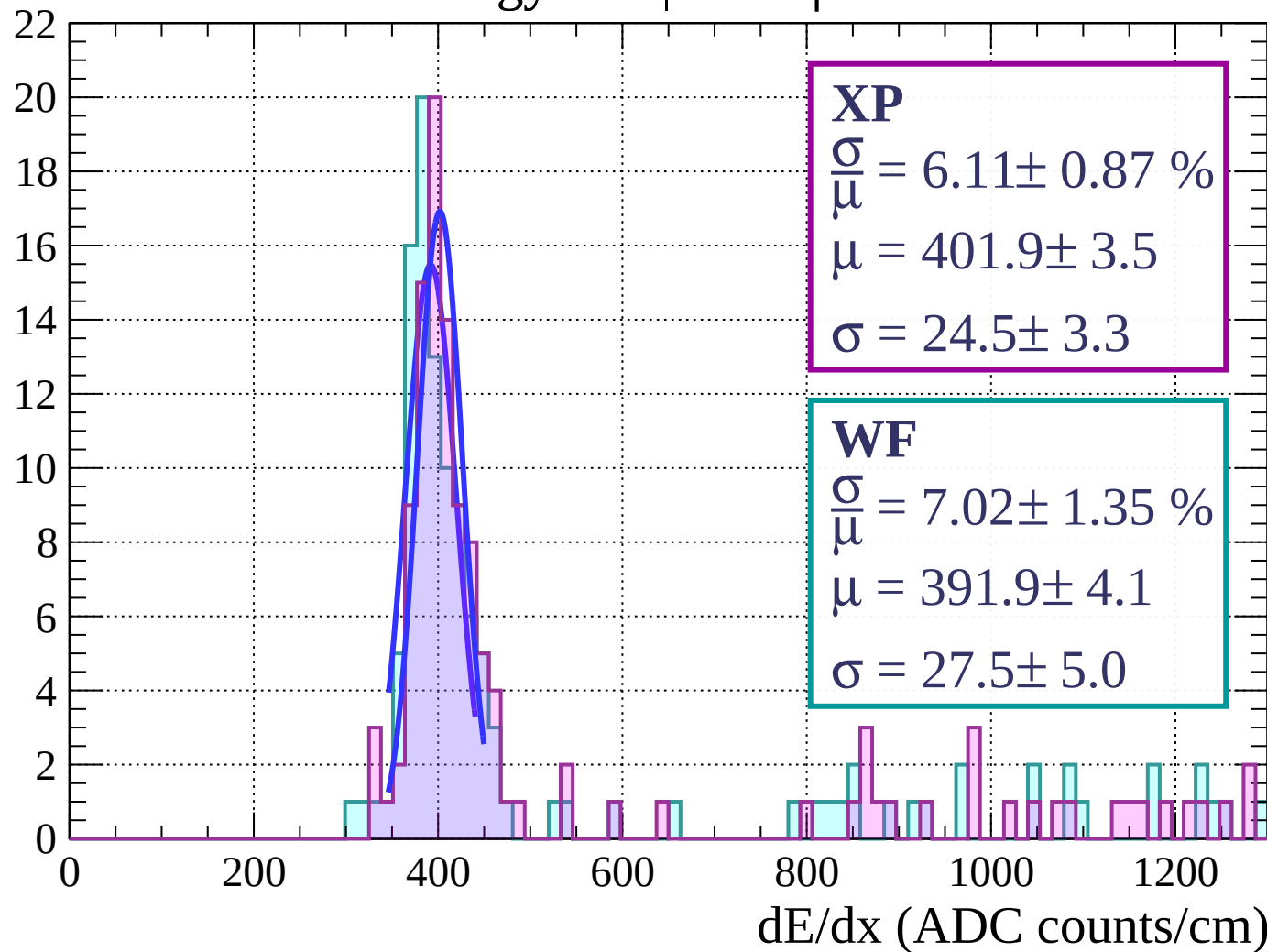
Energy loss | $51 < \phi < 54$

Count



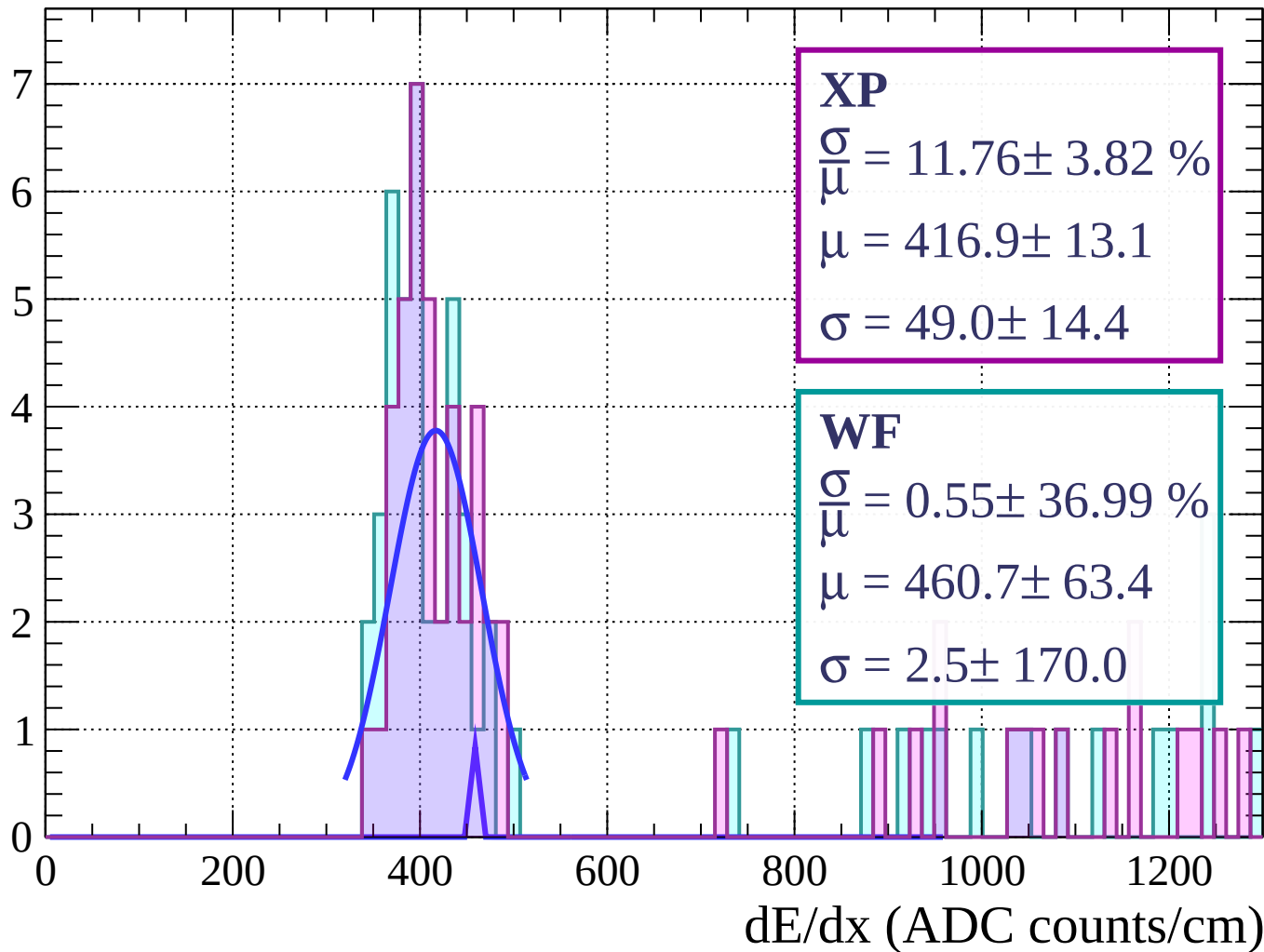
Energy loss | $54 < \phi < 57$

Count



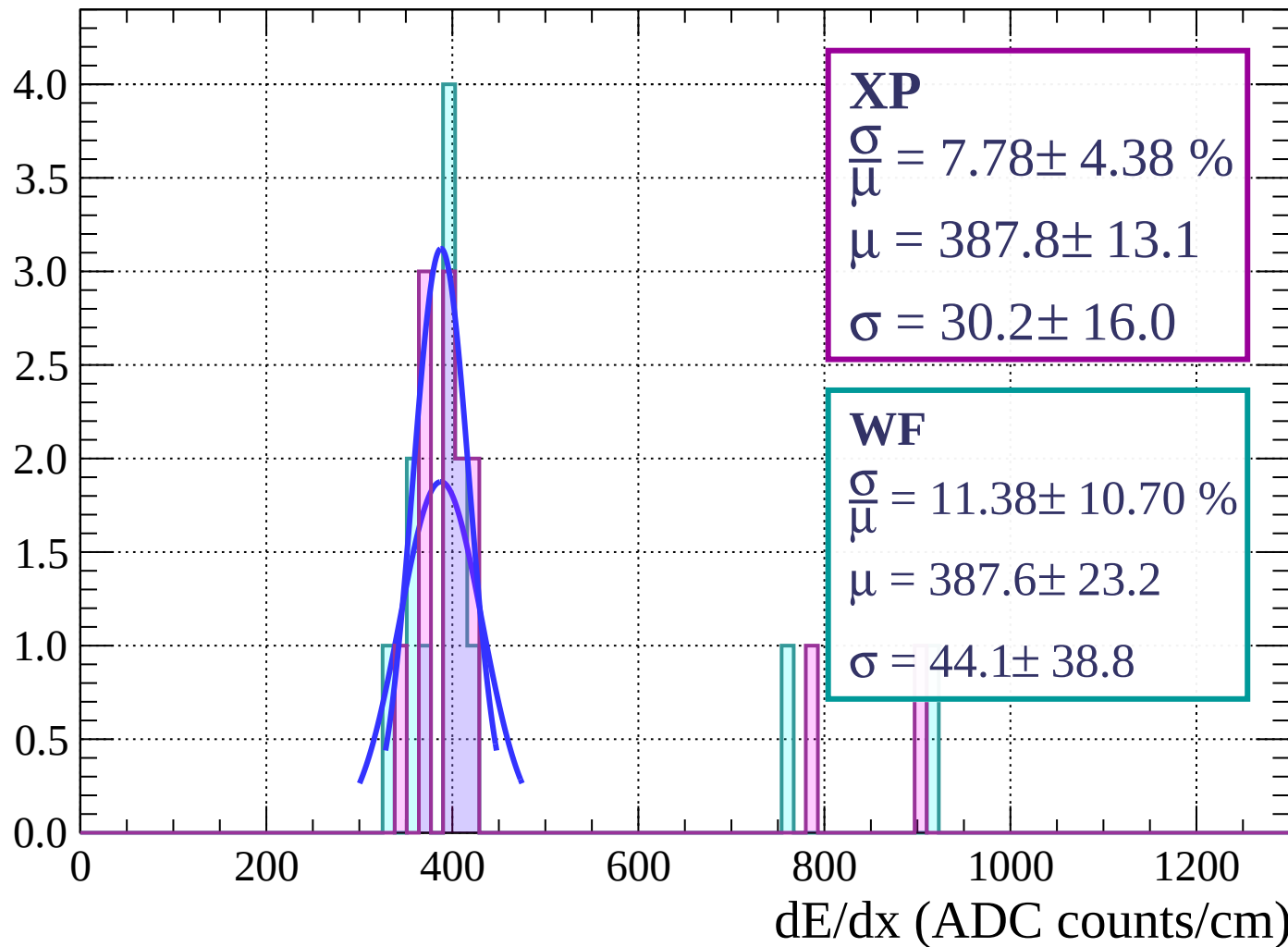
Energy loss | $57 < \phi < 60$

Count



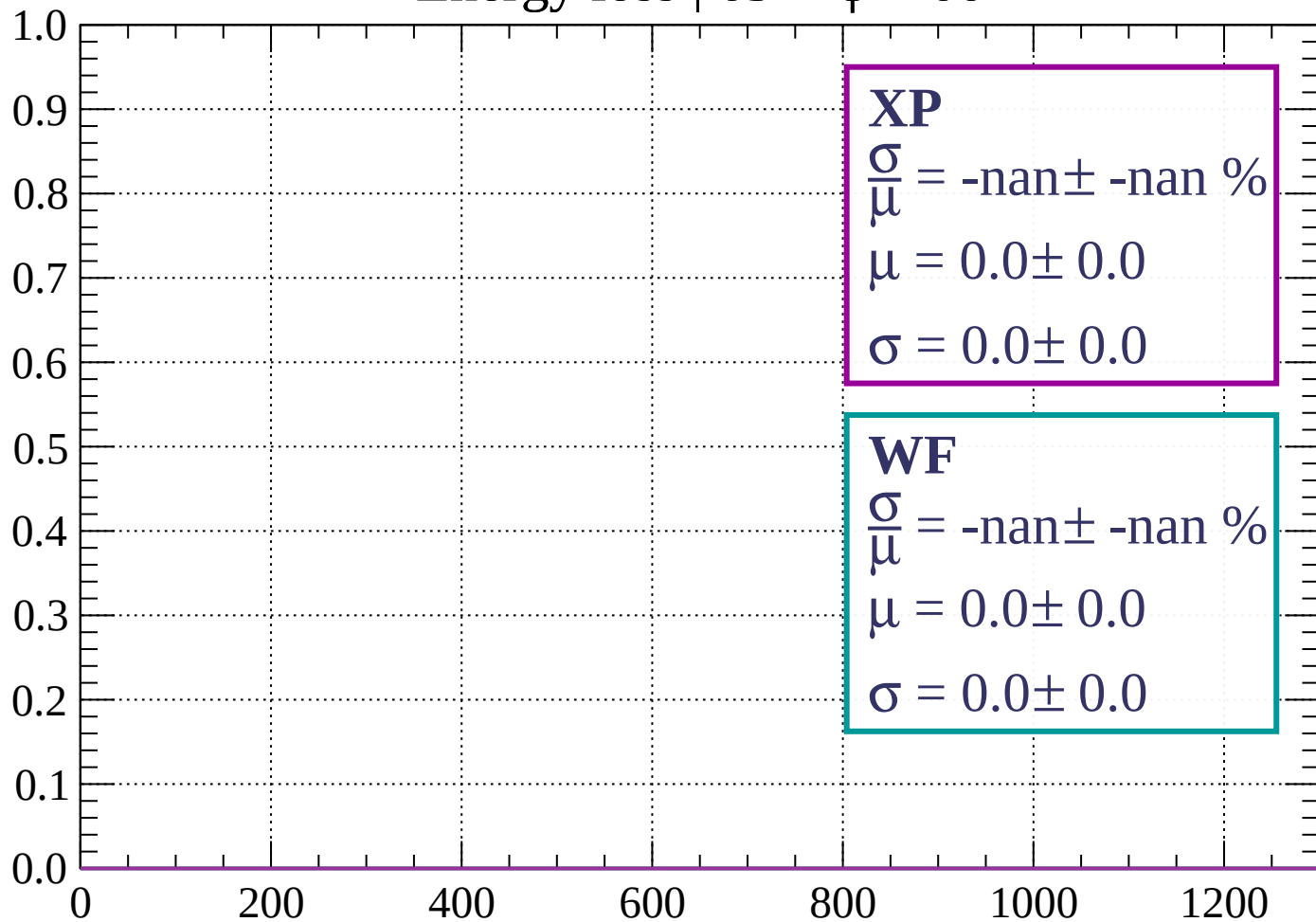
Energy loss | $60 < \phi < 63$

Count



Energy loss | $63 < \phi < 66$

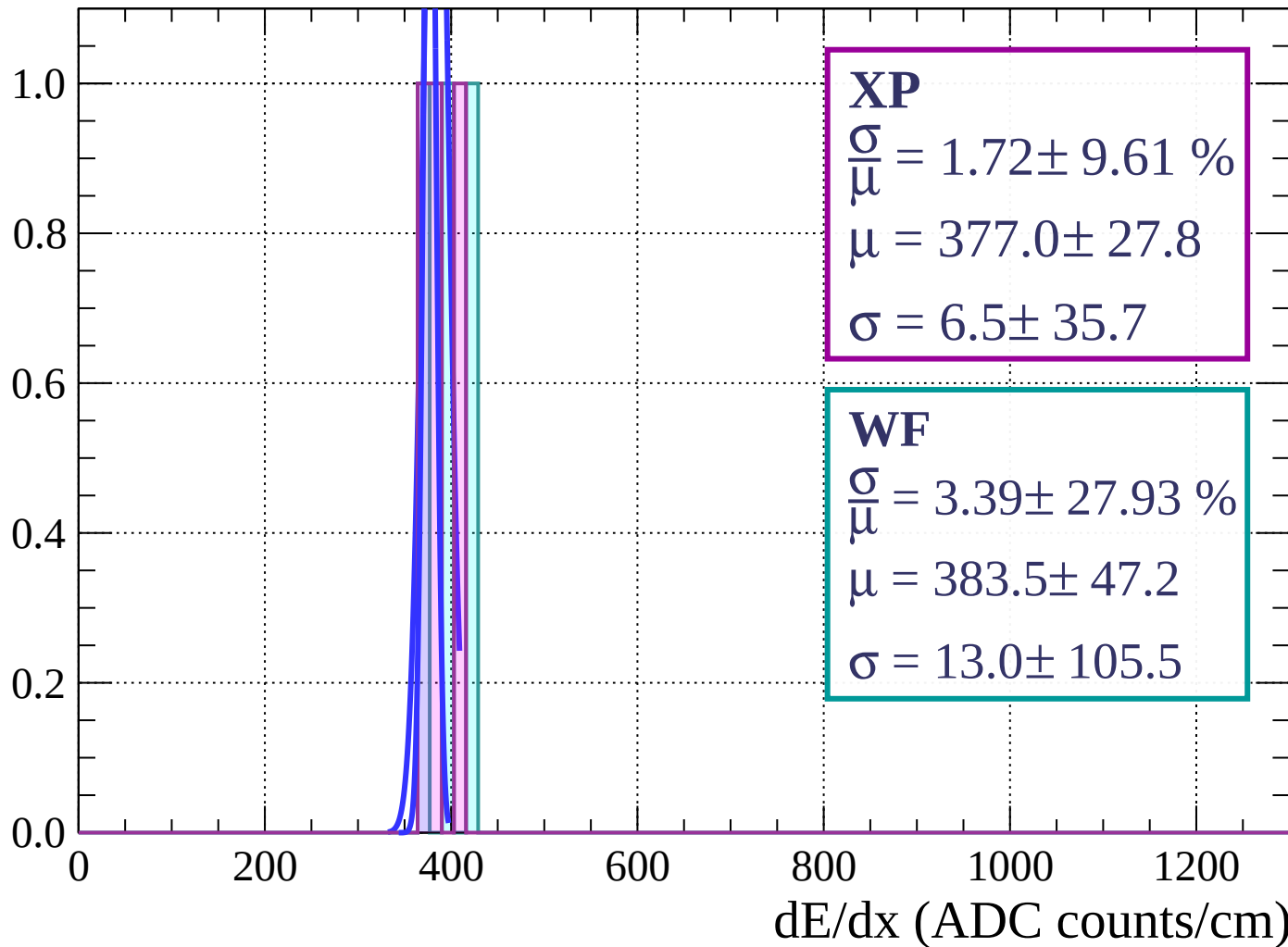
Count



dE/dx (ADC counts/cm)

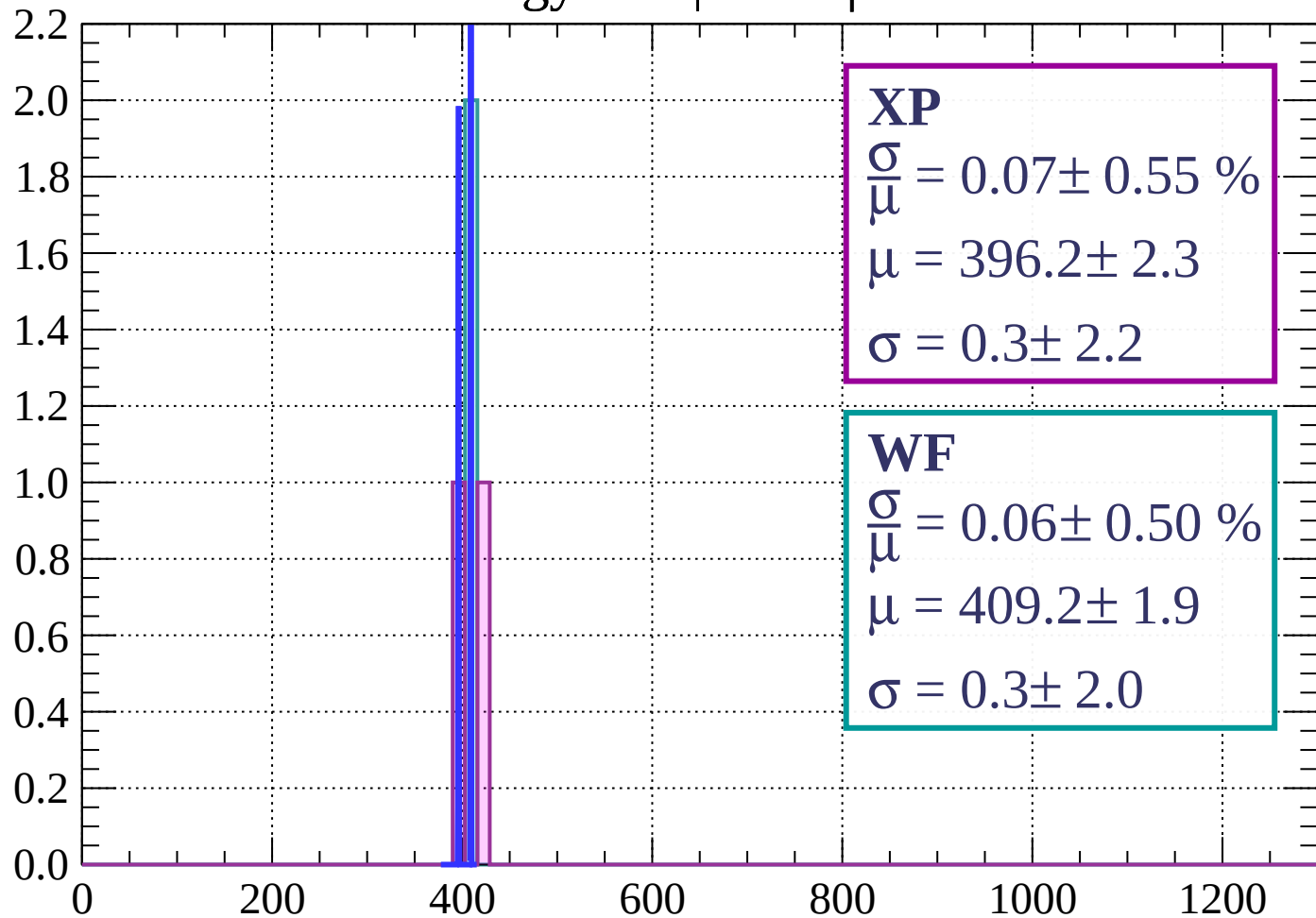
Energy loss | $66 < \phi < 69$

Count



Energy loss | $69 < \phi < 72$

Count



XP

$$\frac{\sigma}{\mu} = 0.07 \pm 0.55 \%$$

$$\mu = 396.2 \pm 2.3$$

$$\sigma = 0.3 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 0.06 \pm 0.50 \%$$

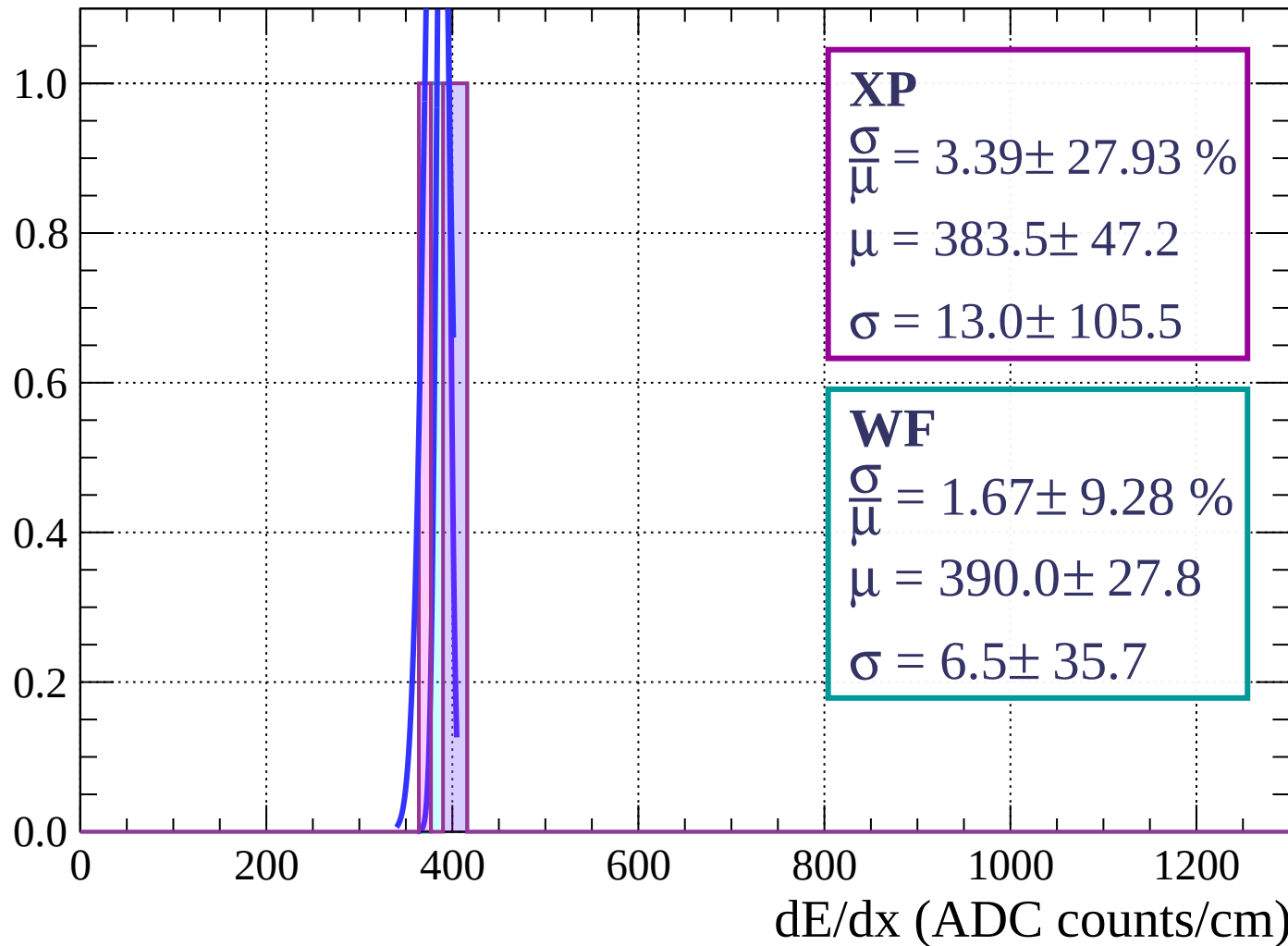
$$\mu = 409.2 \pm 1.9$$

$$\sigma = 0.3 \pm 2.0$$

dE/dx (ADC counts/cm)

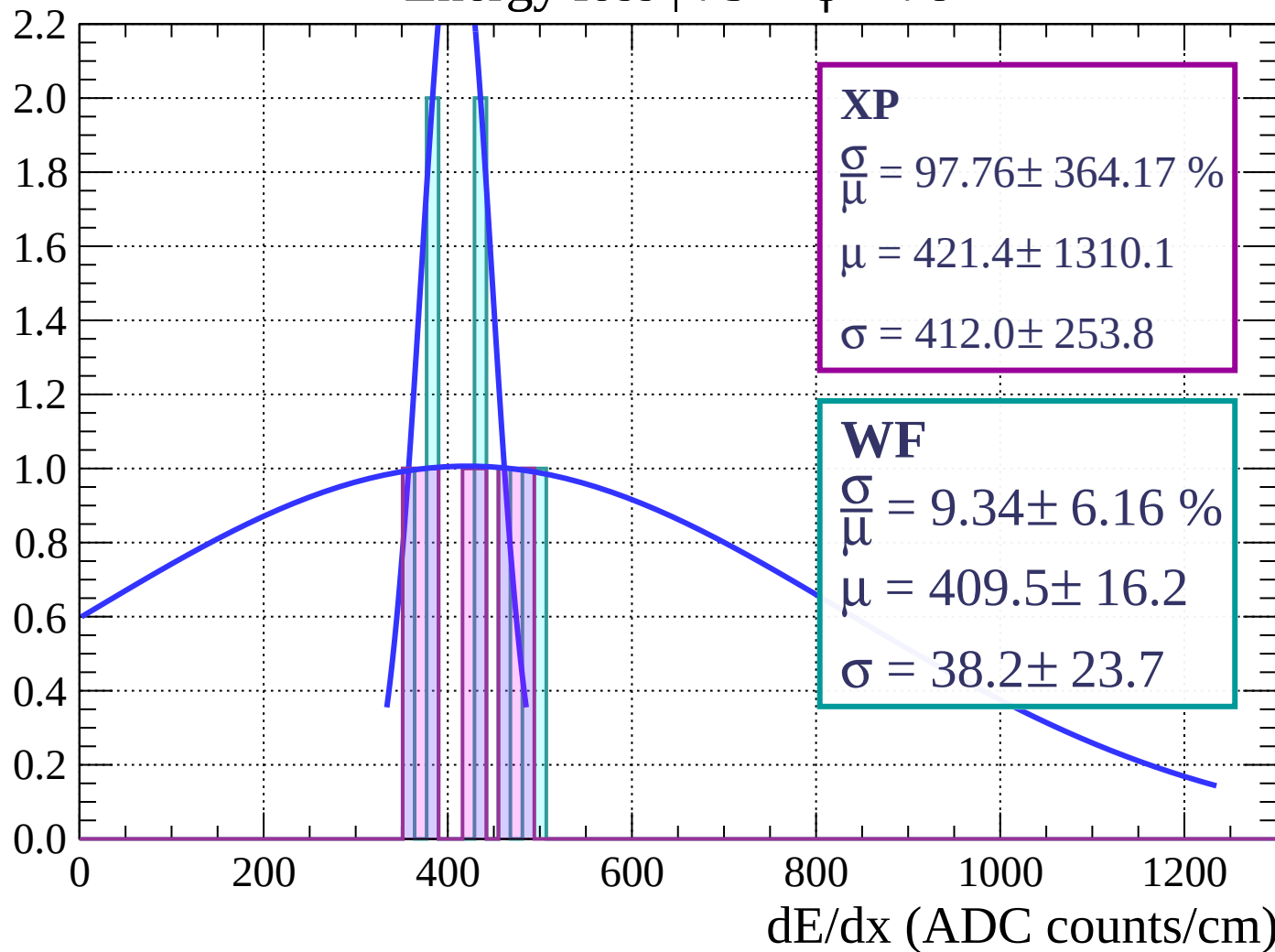
Energy loss | $72 < \phi < 75$

Count



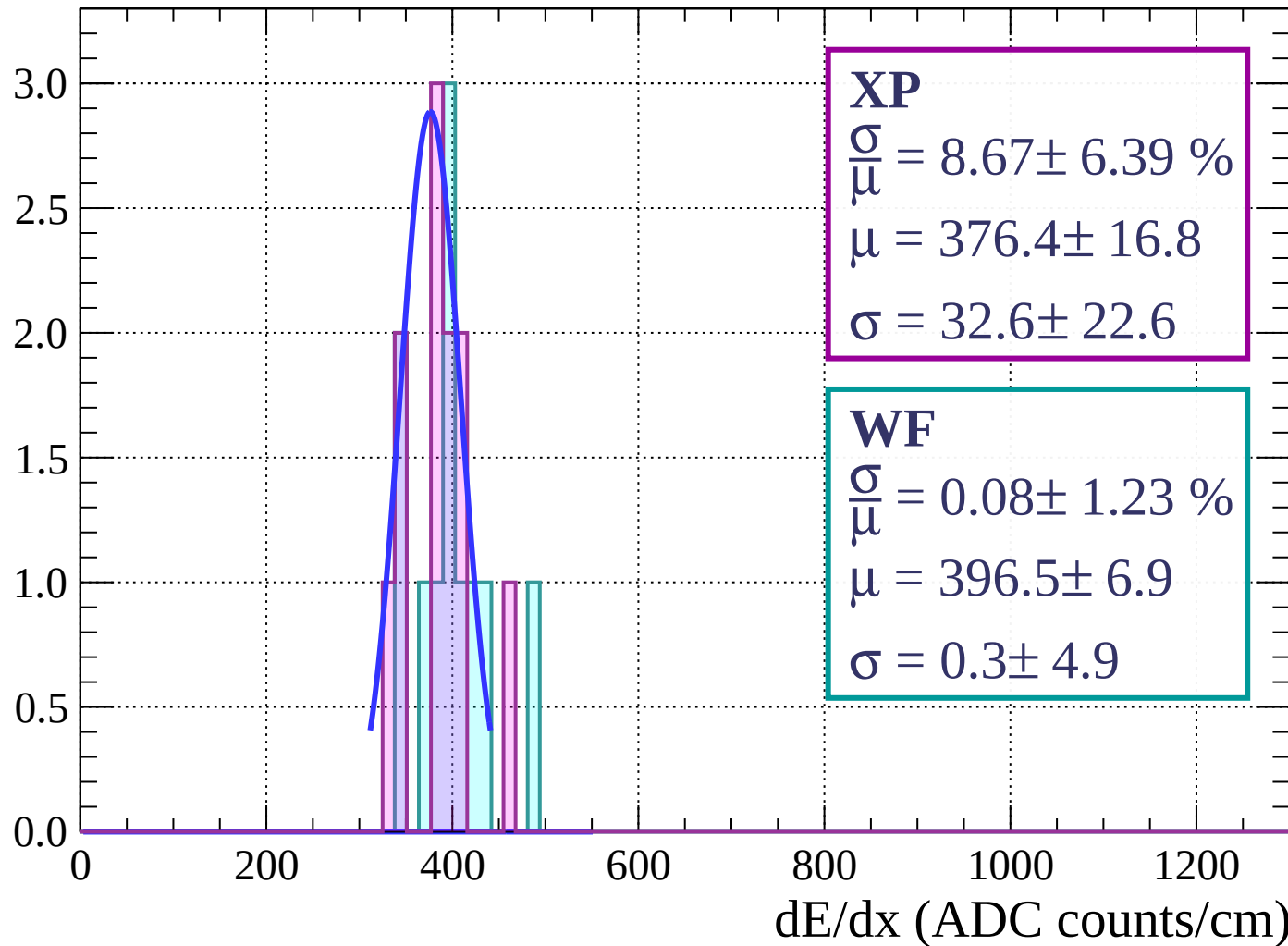
Energy loss | $75 < \phi < 78$

Count



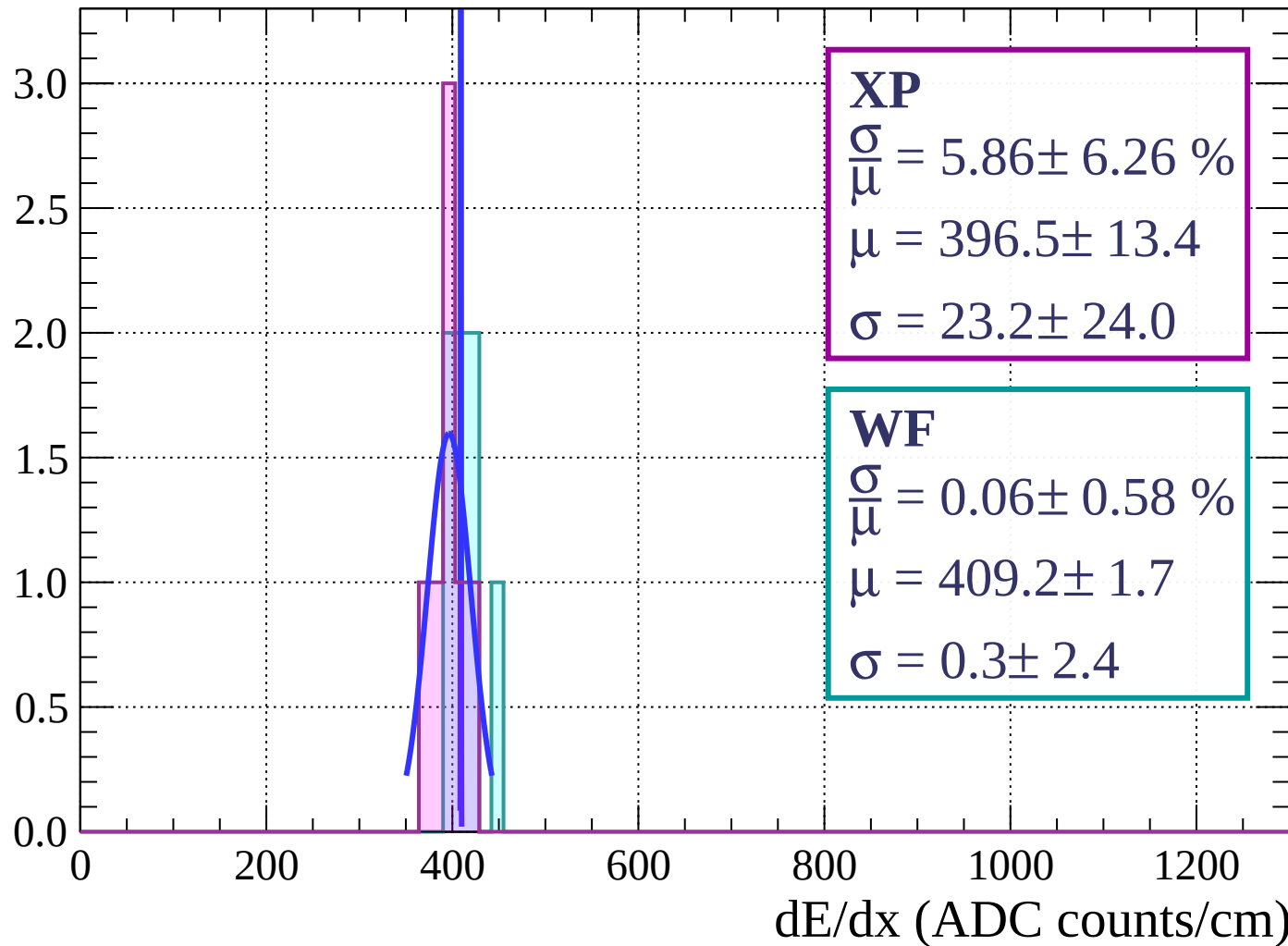
Energy loss | $78 < \phi < 81$

Count



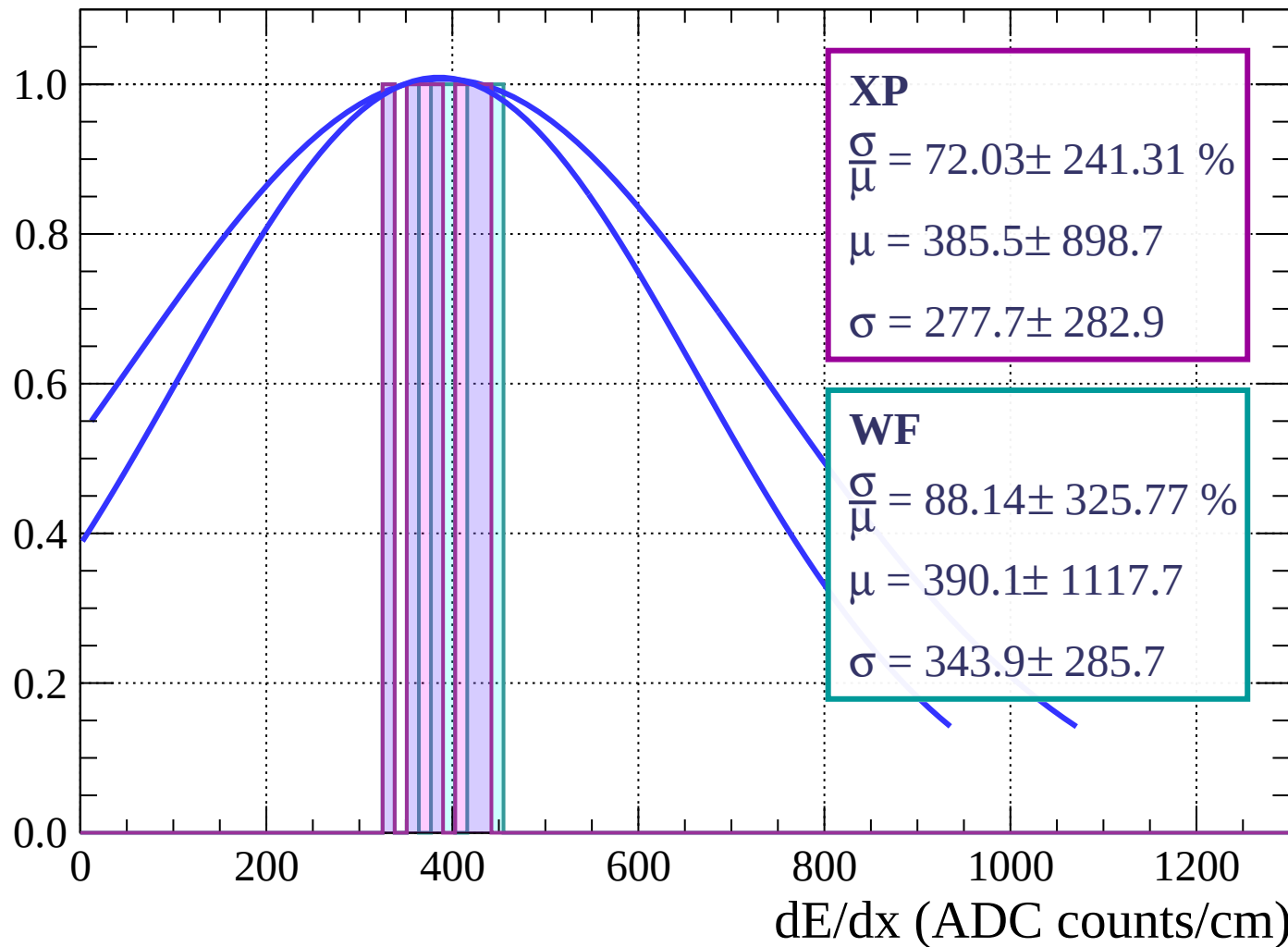
Energy loss | $81 < \phi < 84$

Count



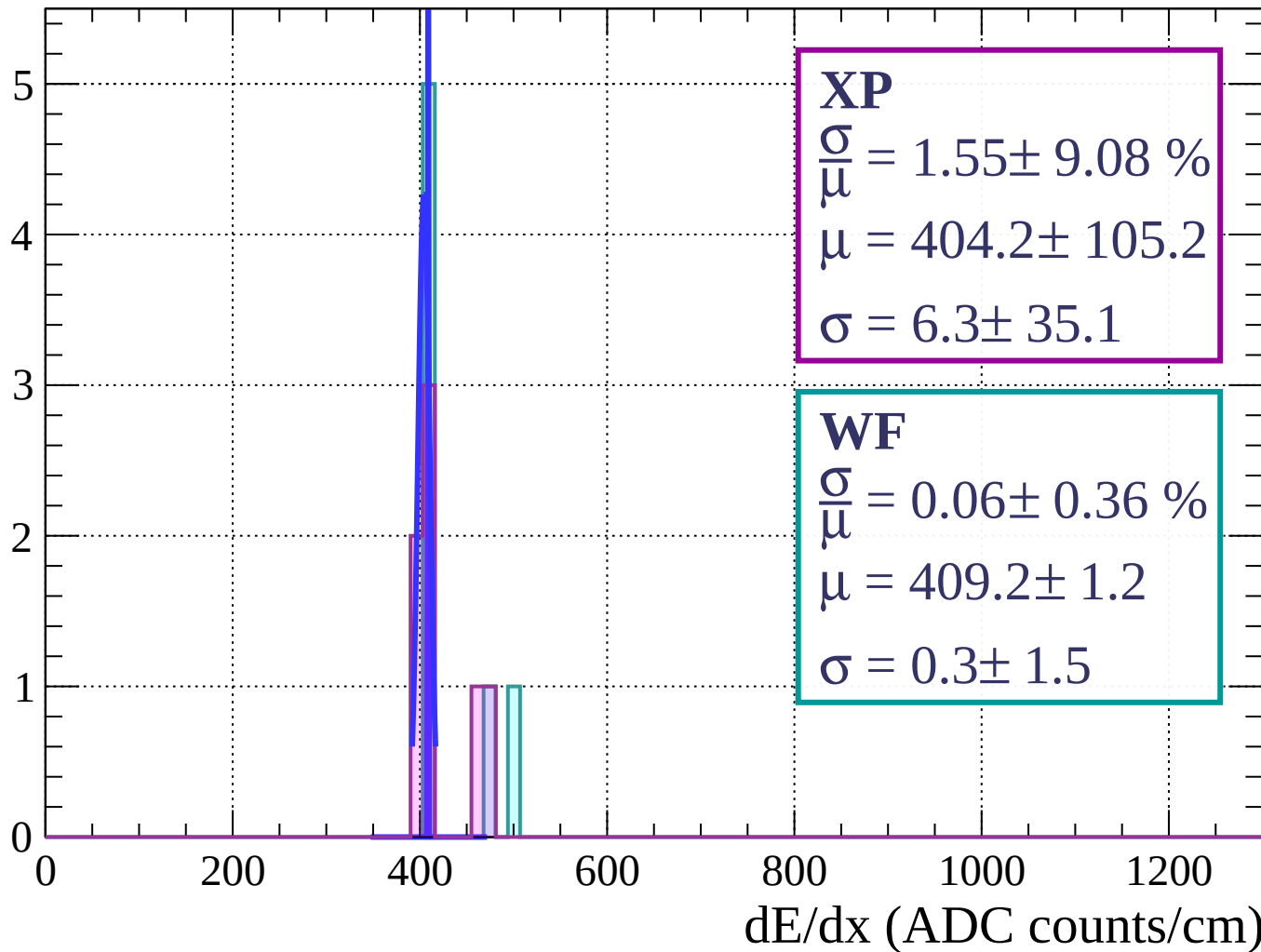
Energy loss | $84 < \phi < 87$

Count



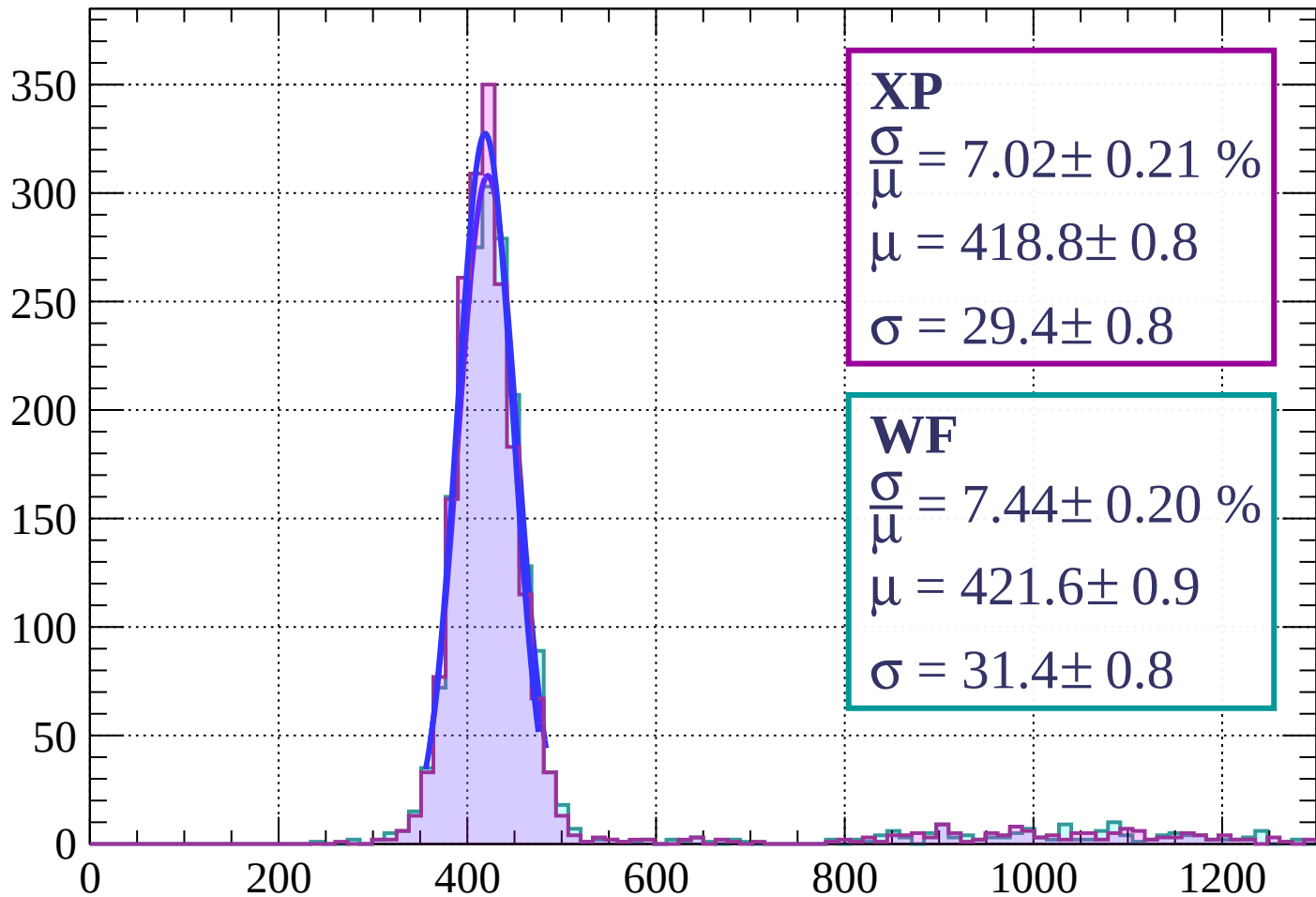
Energy loss | $87 < \phi < 90$

Count



Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 7.02 \pm 0.21 \%$$

$$\mu = 418.8 \pm 0.8$$

$$\sigma = 29.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.44 \pm 0.20 \%$$

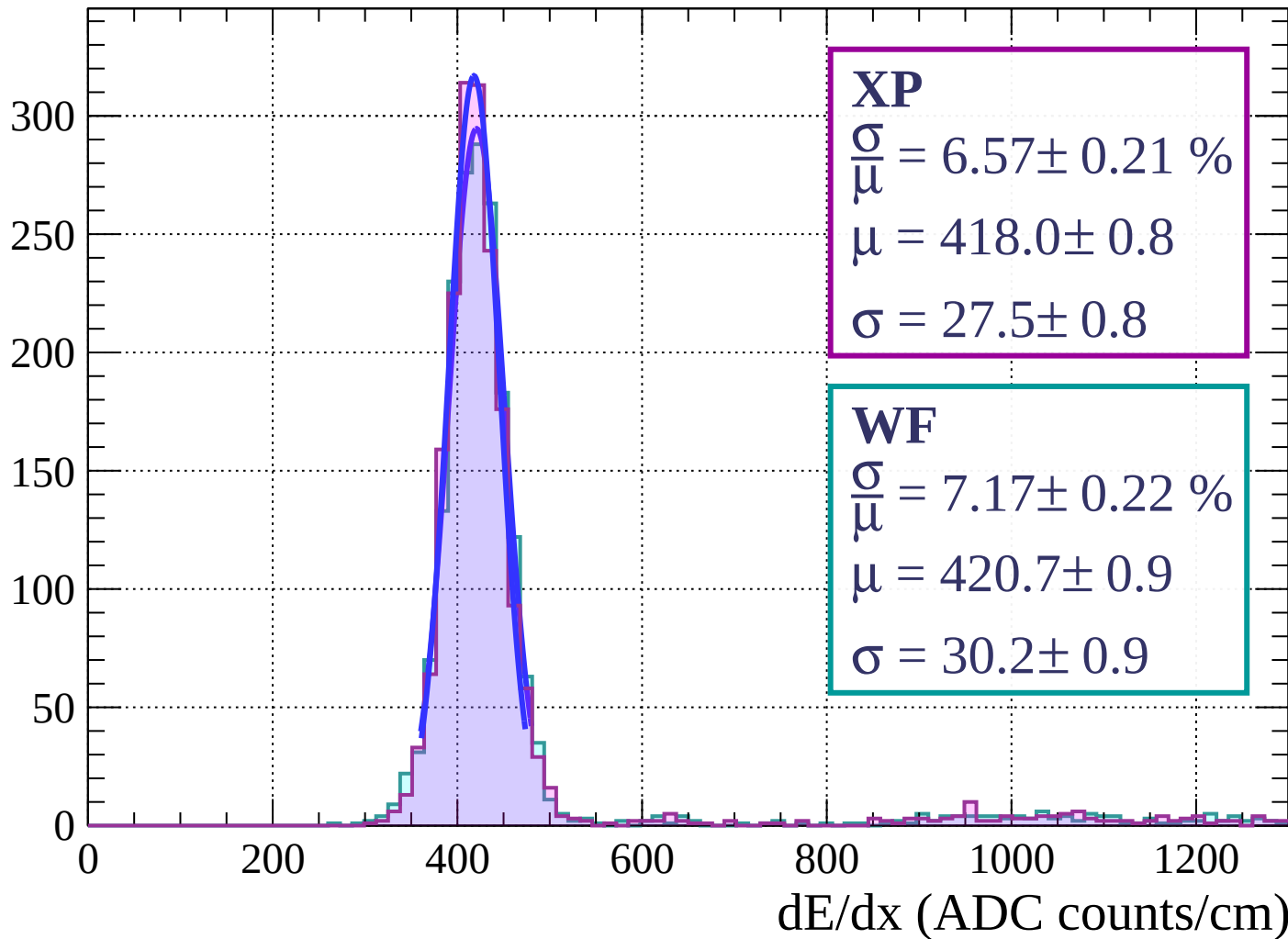
$$\mu = 421.6 \pm 0.9$$

$$\sigma = 31.4 \pm 0.8$$

dE/dx (ADC counts/cm)

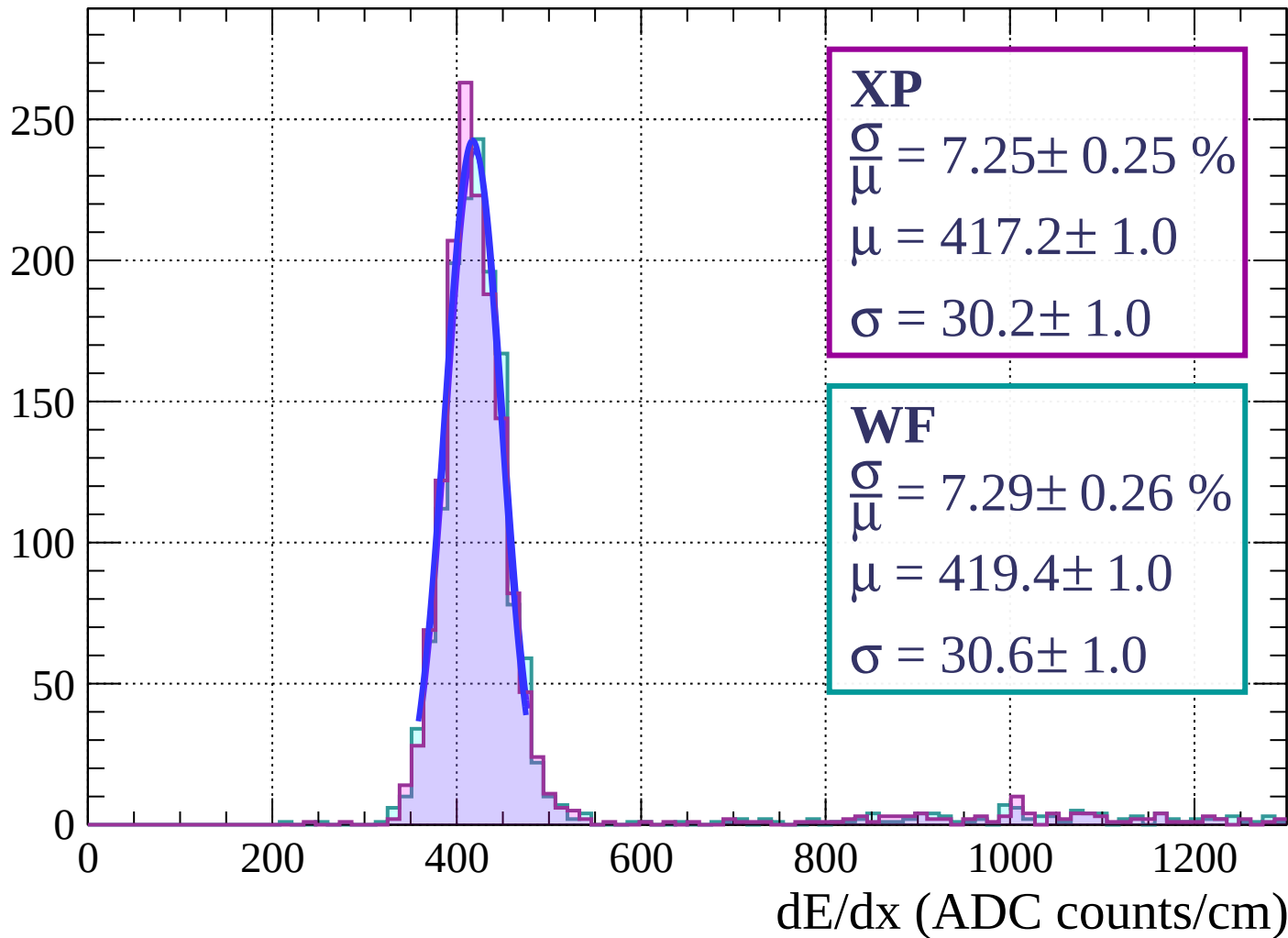
Energy loss | $3 < \theta < 6$

Count



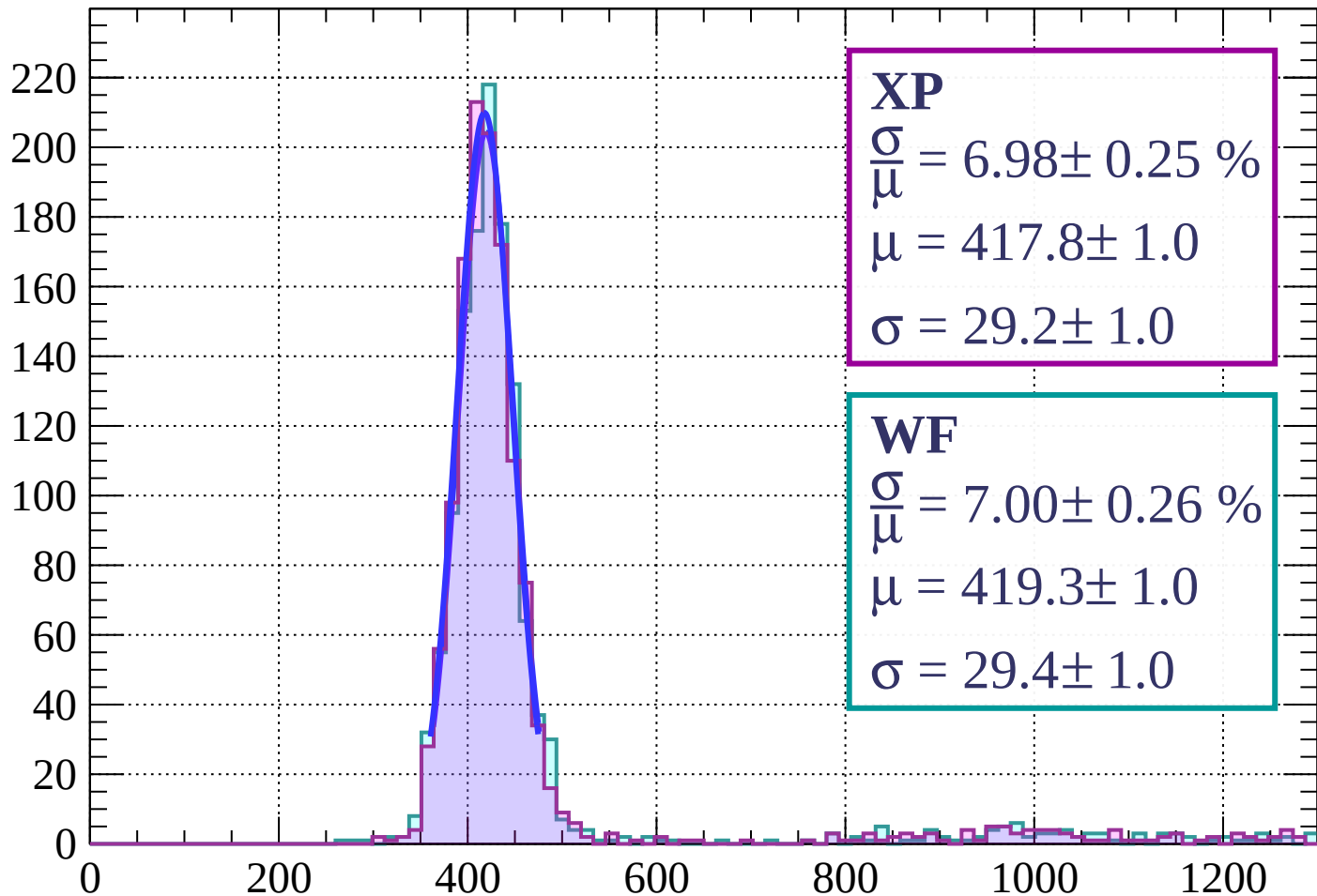
Energy loss | $6 < \theta < 9$

Count



Energy loss | $9 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 6.98 \pm 0.25 \%$$

$$\mu = 417.8 \pm 1.0$$

$$\sigma = 29.2 \pm 1.0$$

WF

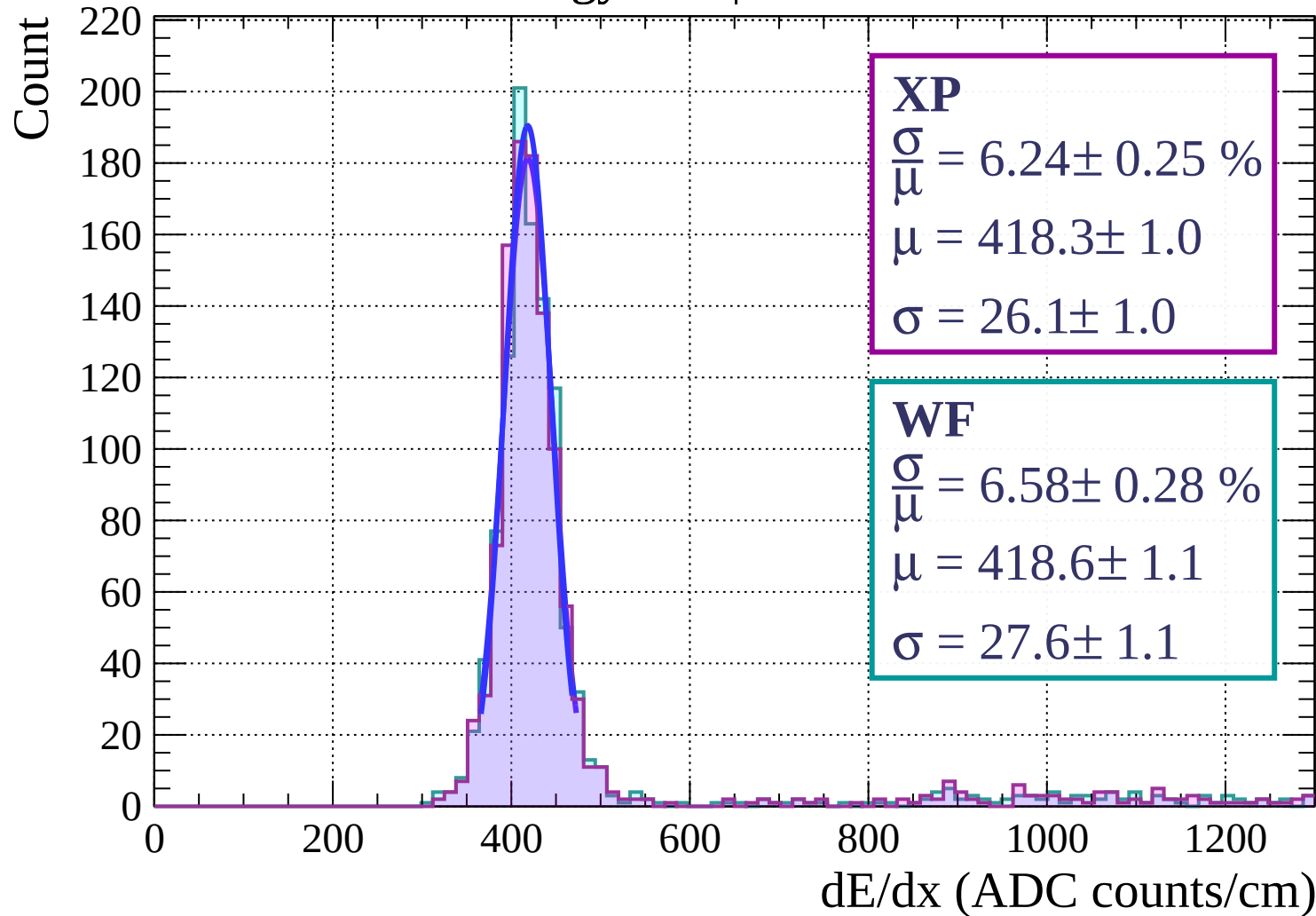
$$\frac{\sigma}{\mu} = 7.00 \pm 0.26 \%$$

$$\mu = 419.3 \pm 1.0$$

$$\sigma = 29.4 \pm 1.0$$

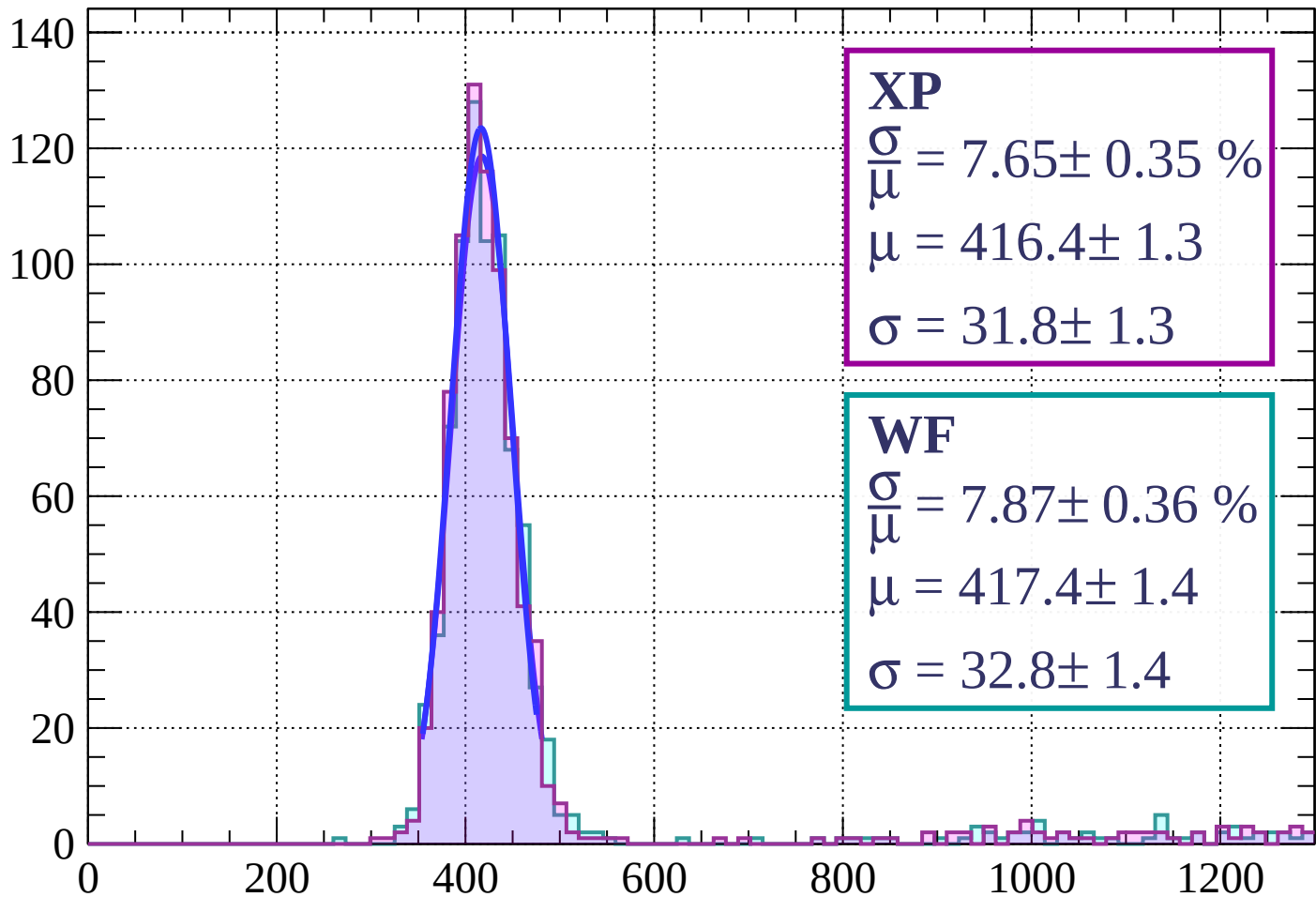
dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 15$



Energy loss | $15 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 7.65 \pm 0.35 \%$$

$$\mu = 416.4 \pm 1.3$$

$$\sigma = 31.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.87 \pm 0.36 \%$$

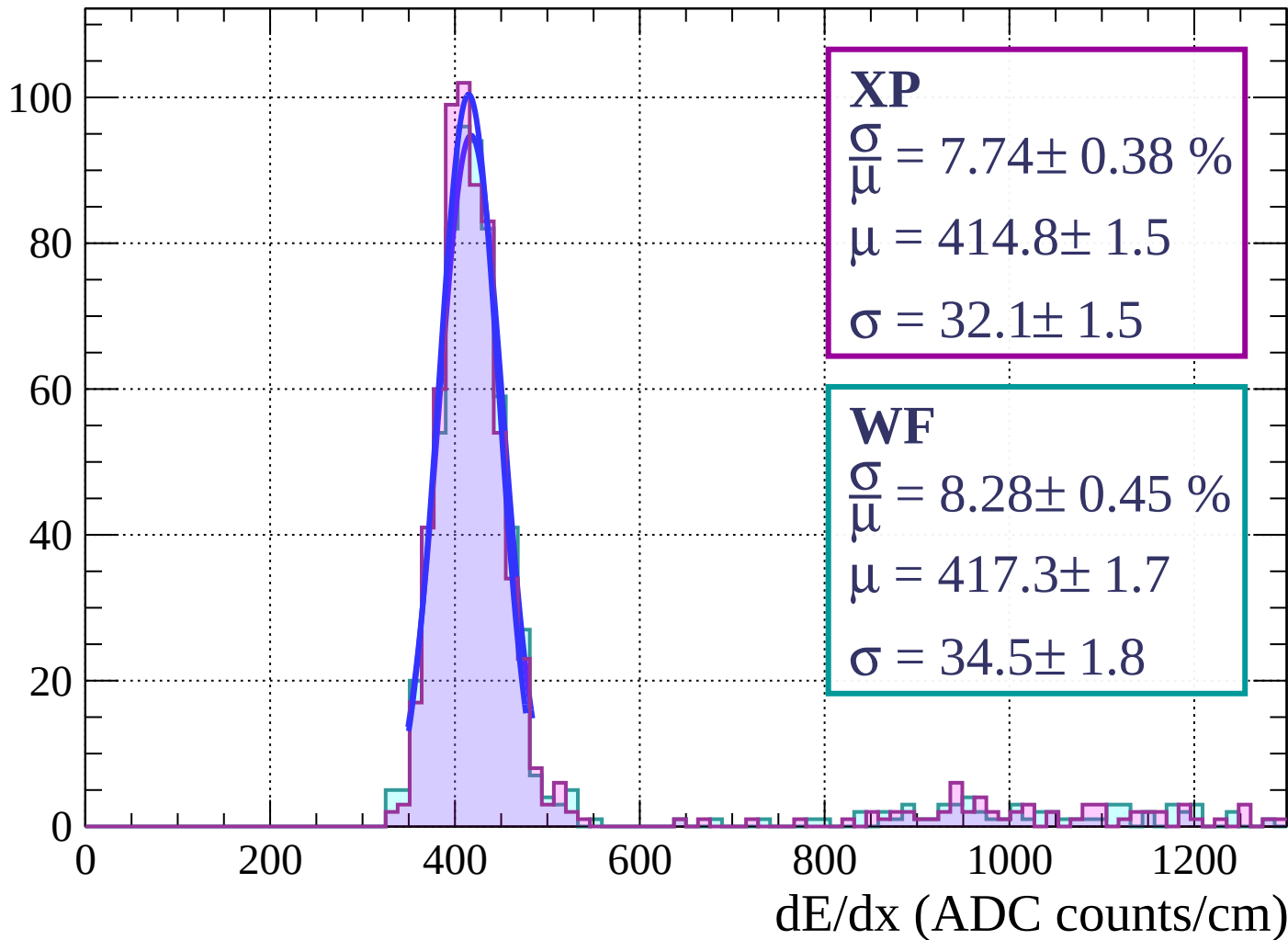
$$\mu = 417.4 \pm 1.4$$

$$\sigma = 32.8 \pm 1.4$$

dE/dx (ADC counts/cm)

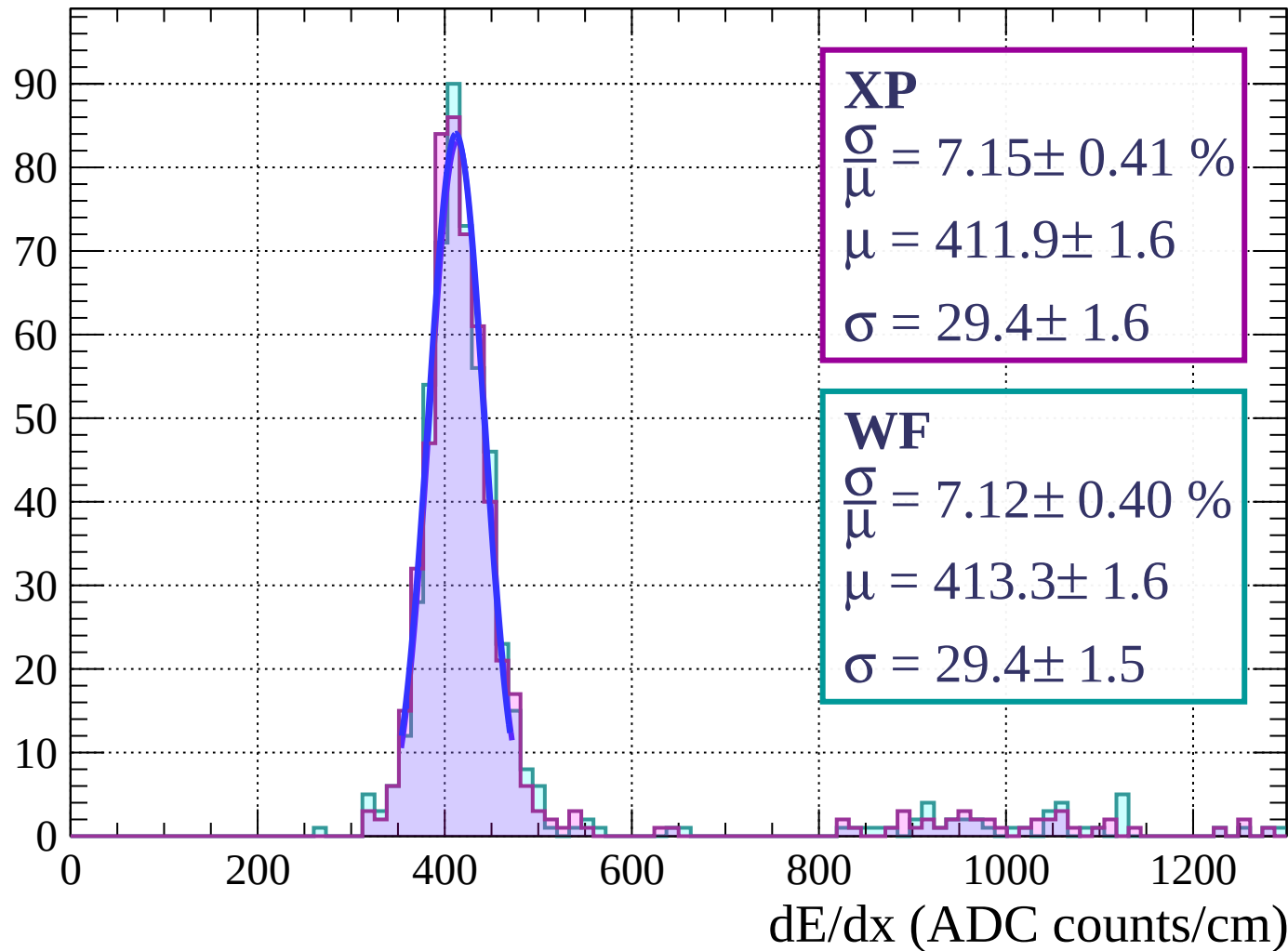
Energy loss | $18 < \theta < 21$

Count



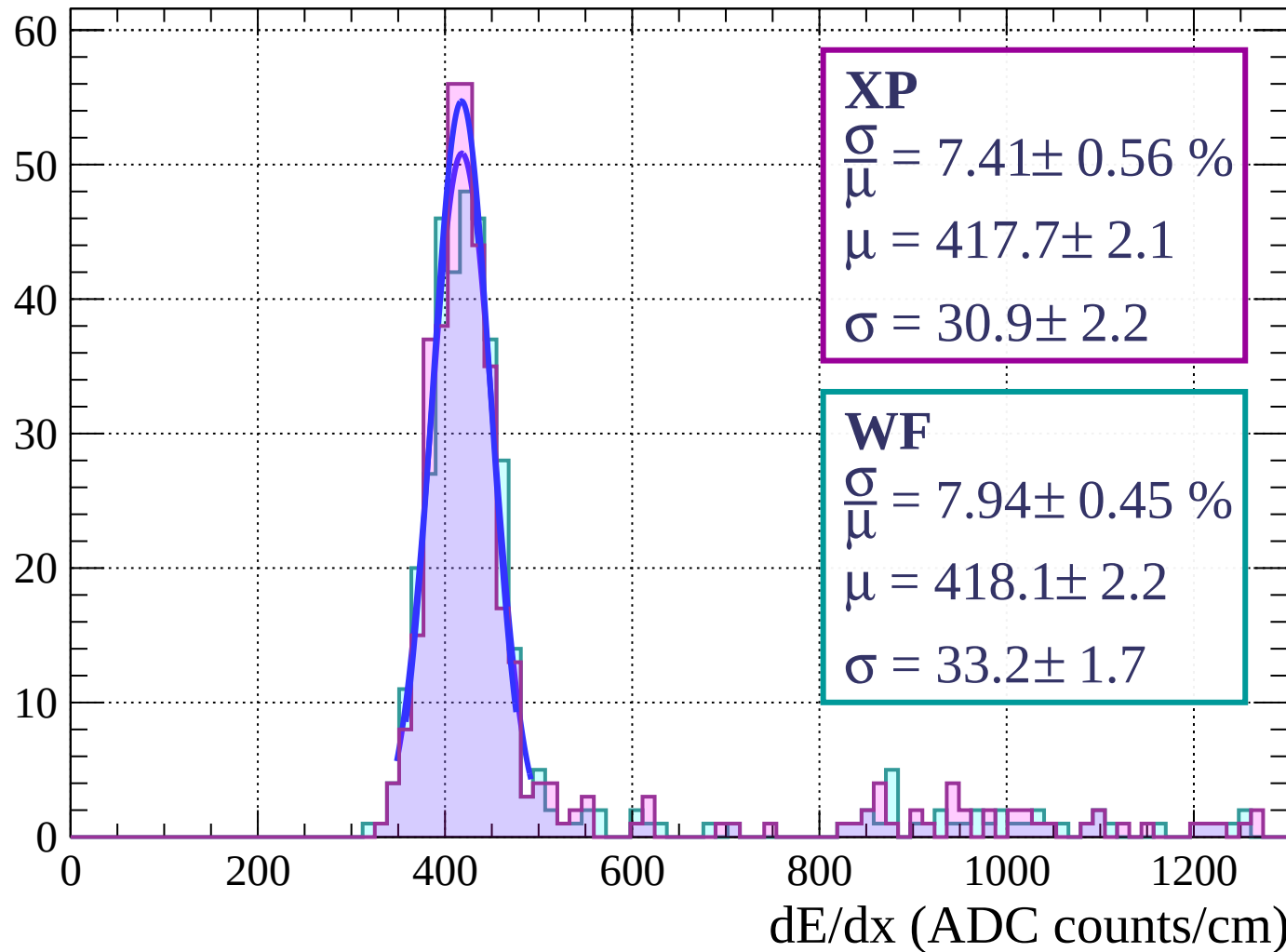
Energy loss | $21 < \theta < 24$

Count



Energy loss | $24 < \theta < 27$

Count



Energy loss | $27 < \theta < 30$

Count

XP

$$\frac{\sigma}{\mu} = 6.94 \pm 0.73 \%$$

$$\mu = 415.6 \pm 2.4$$

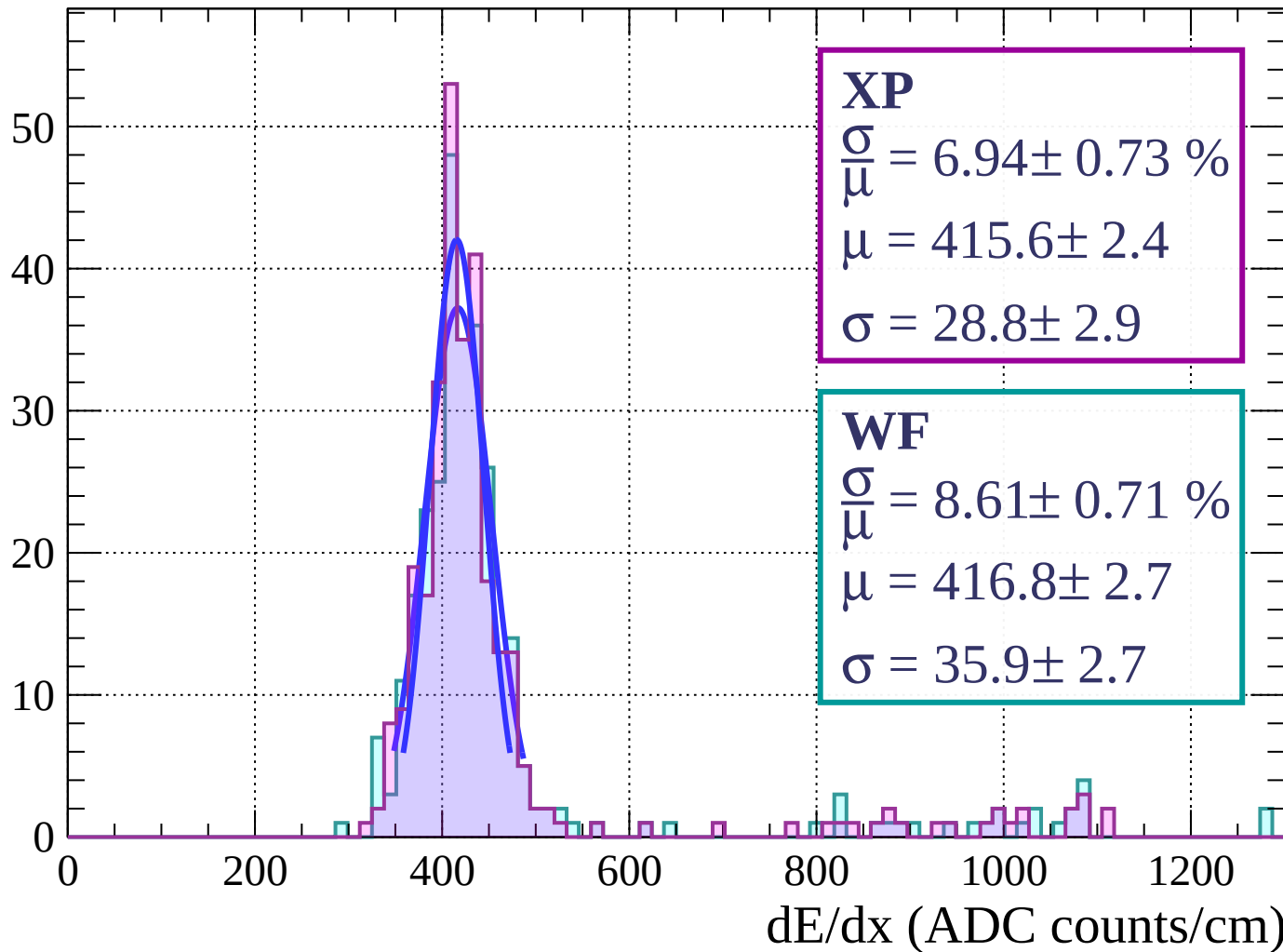
$$\sigma = 28.8 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 8.61 \pm 0.71 \%$$

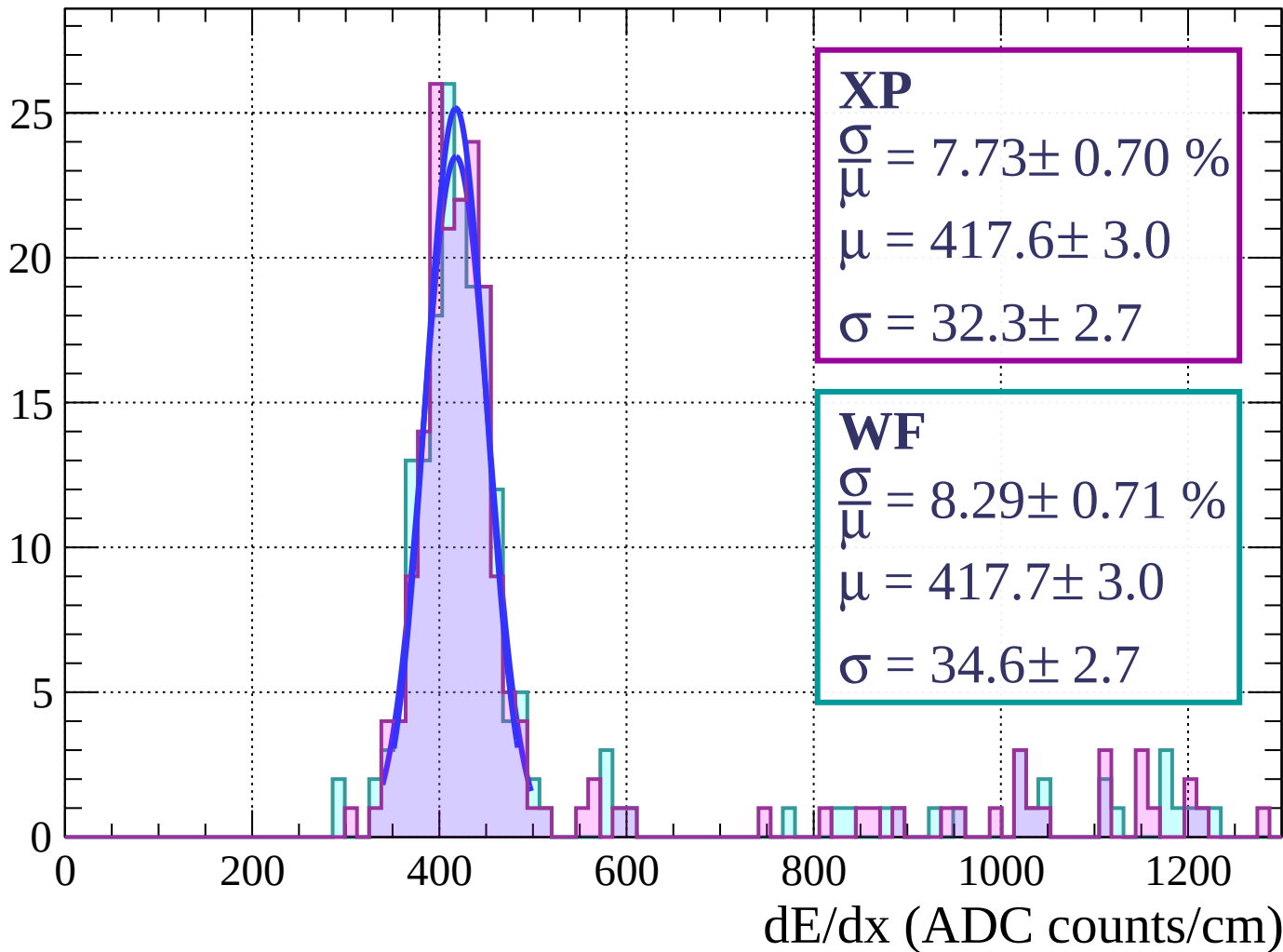
$$\mu = 416.8 \pm 2.7$$

$$\sigma = 35.9 \pm 2.7$$



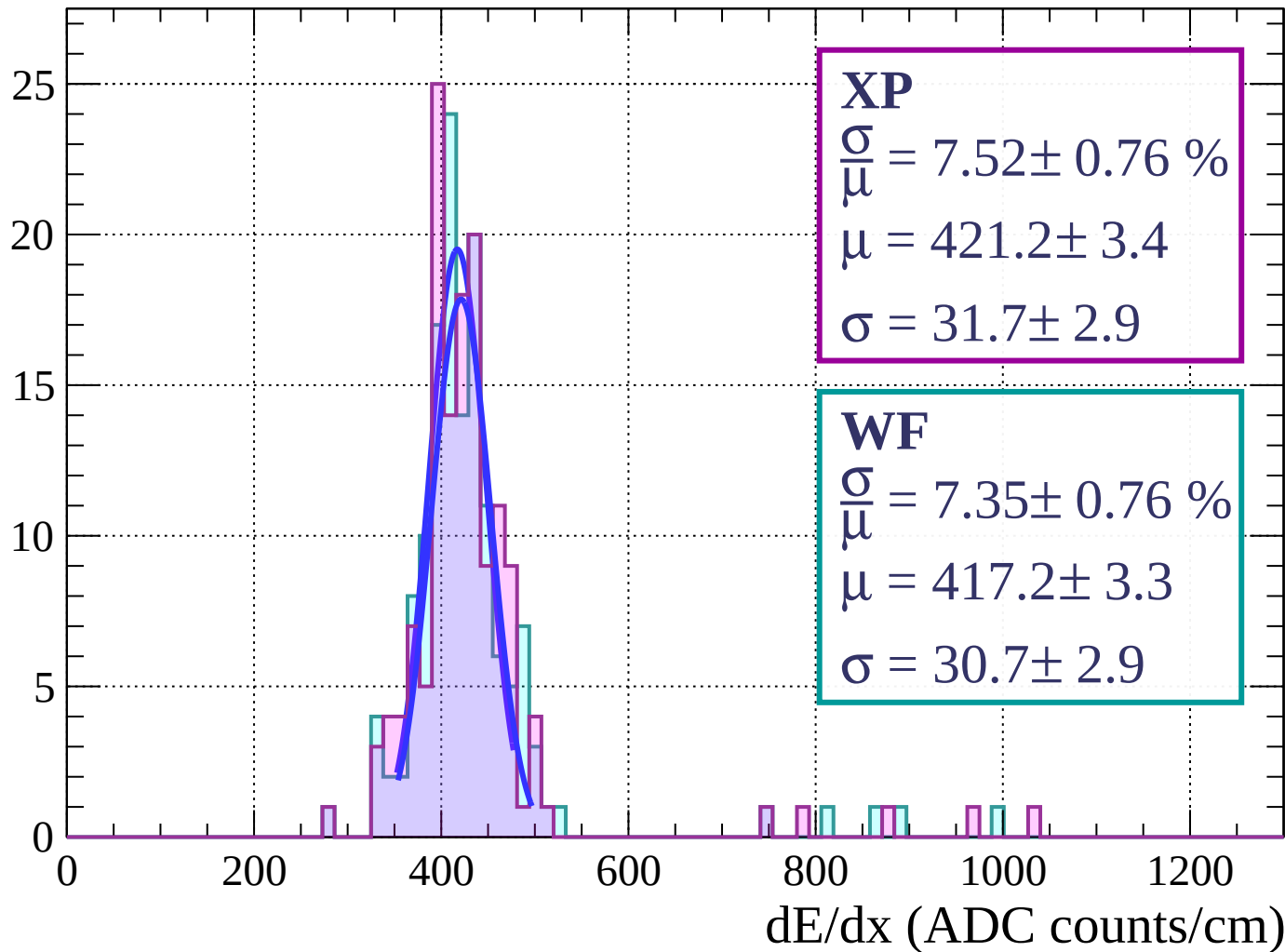
Energy loss | $30 < \theta < 33$

Count



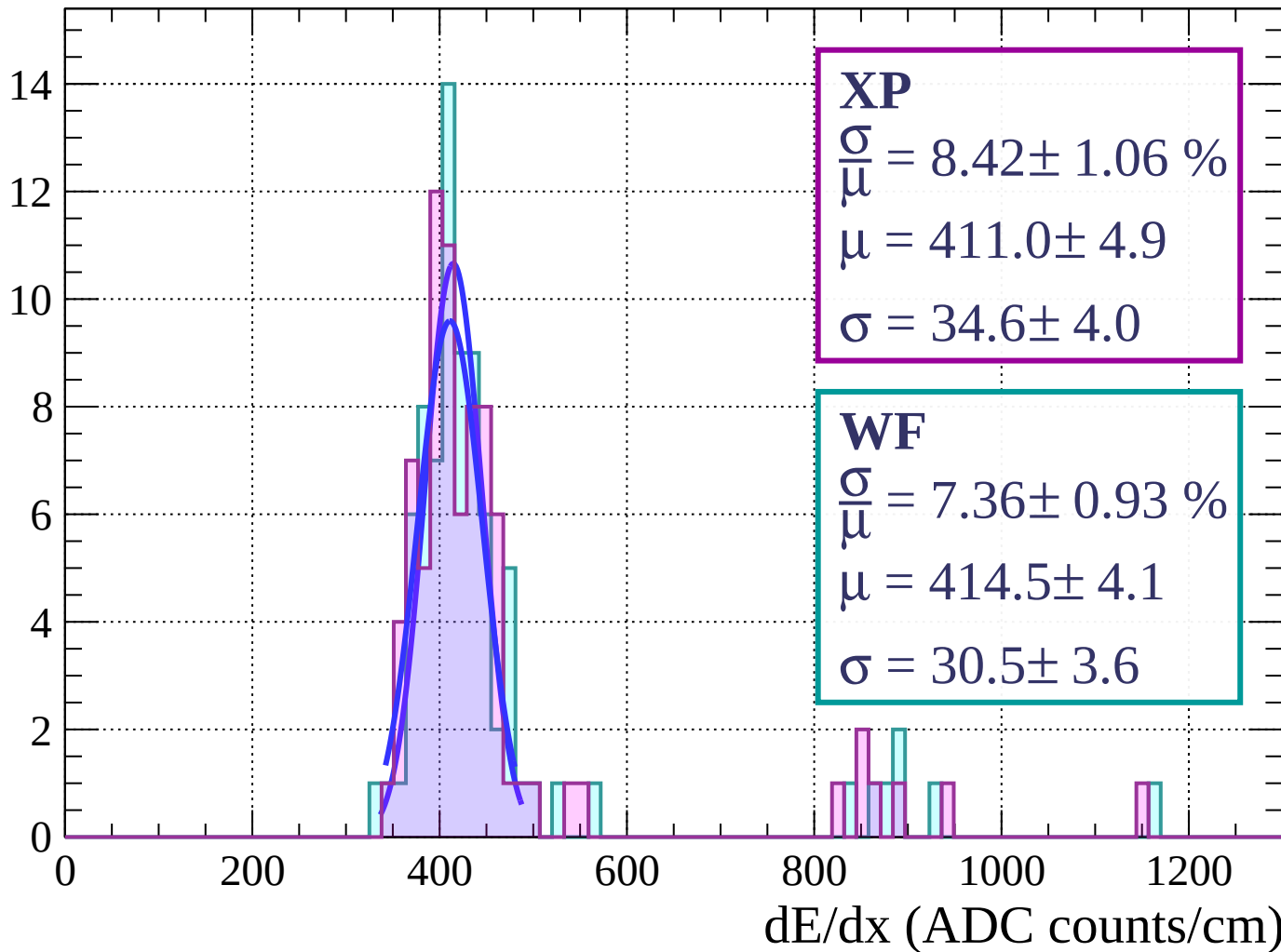
Energy loss | $33 < \theta < 36$

Count



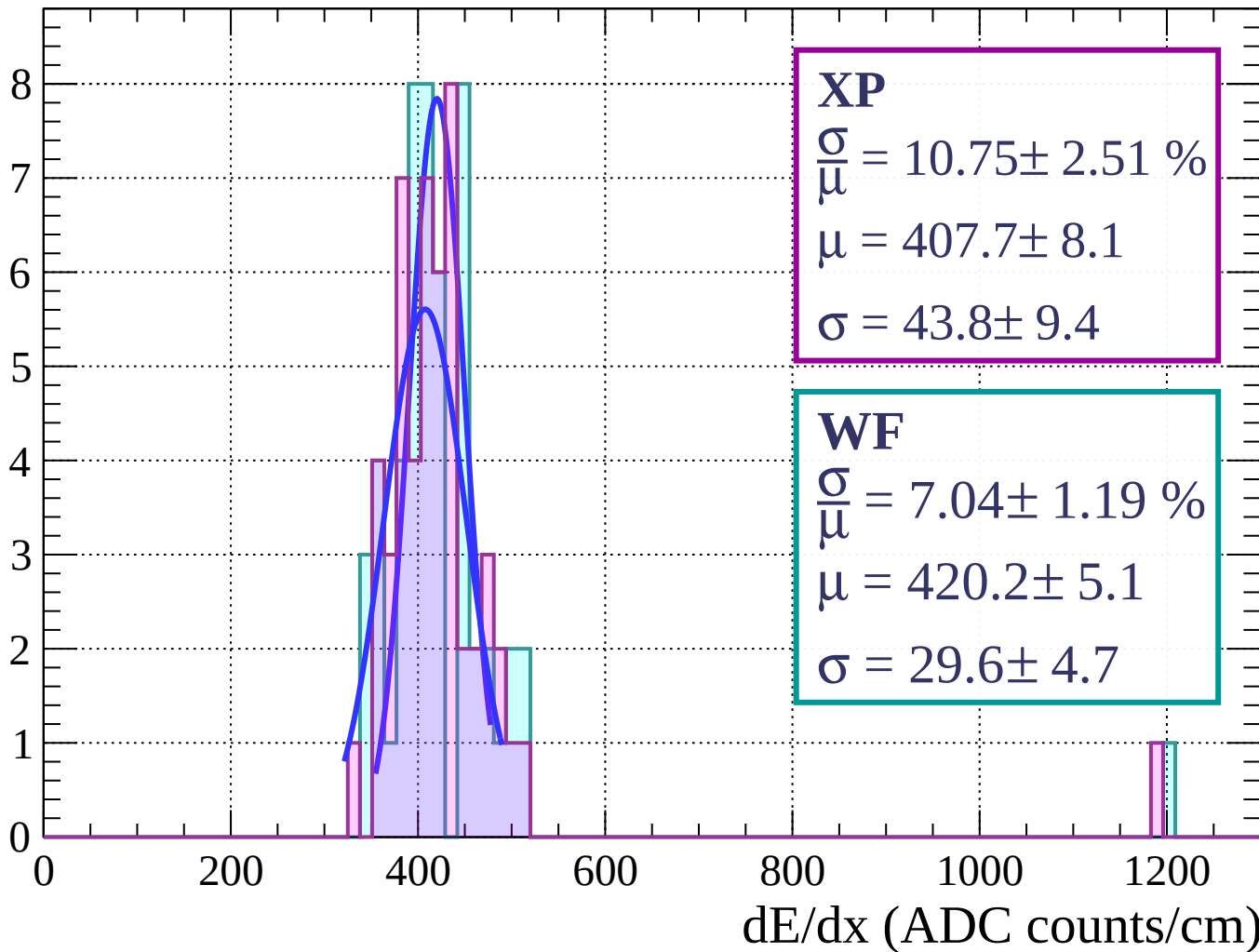
Energy loss | $36 < \theta < 39$

Count



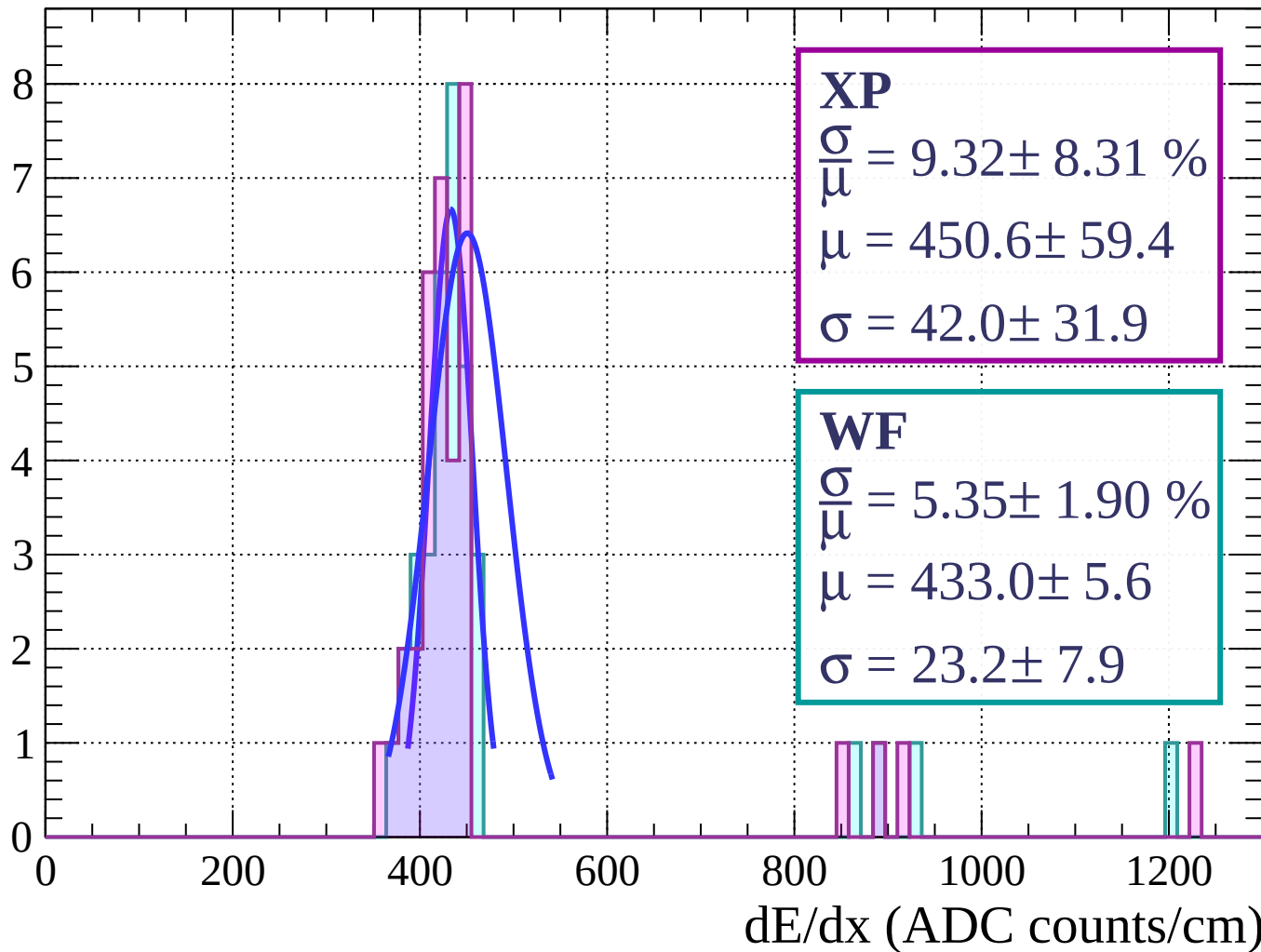
Energy loss | $39 < \theta < 42$

Count



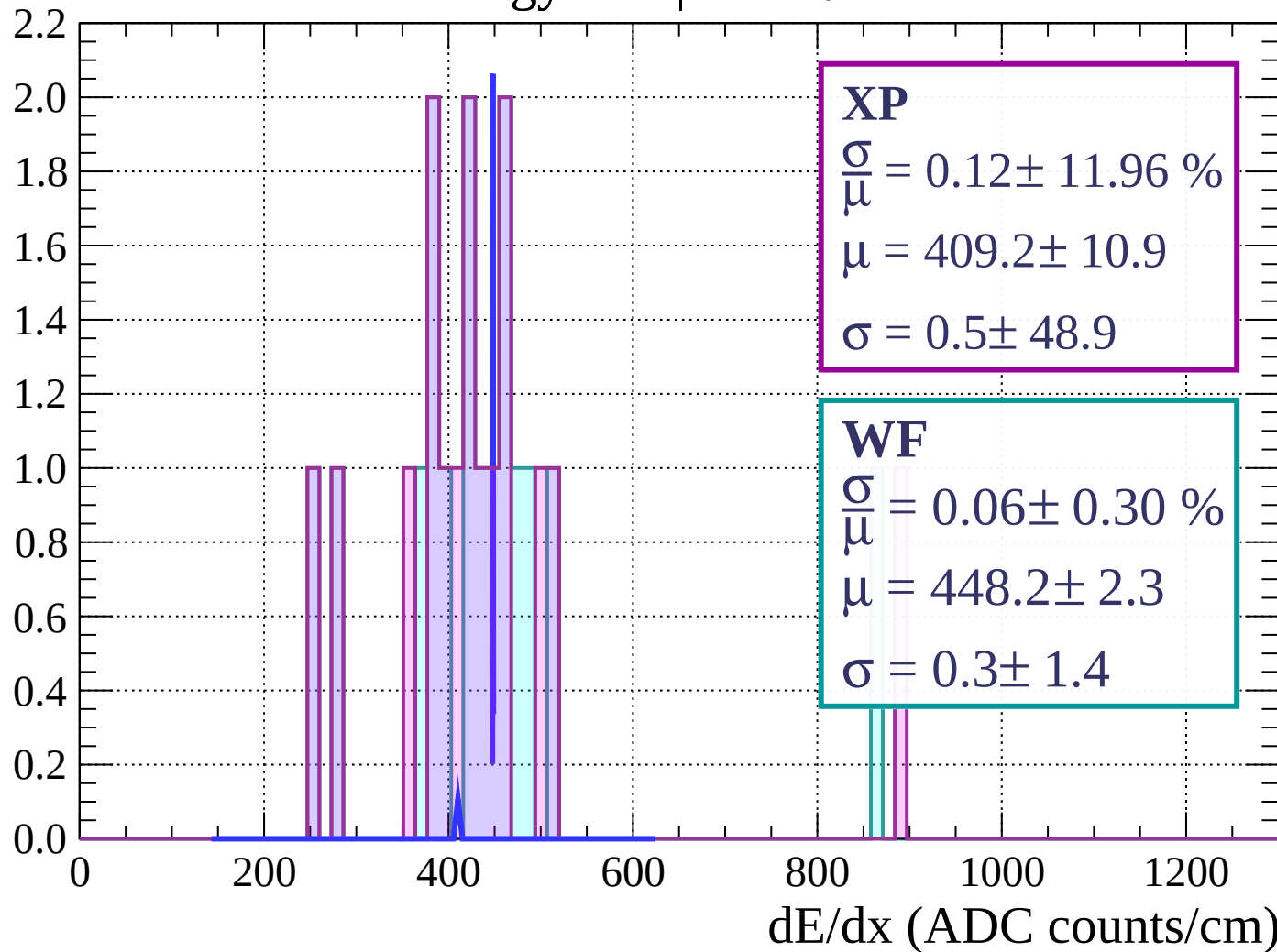
Energy loss | $42 < \theta < 45$

Count



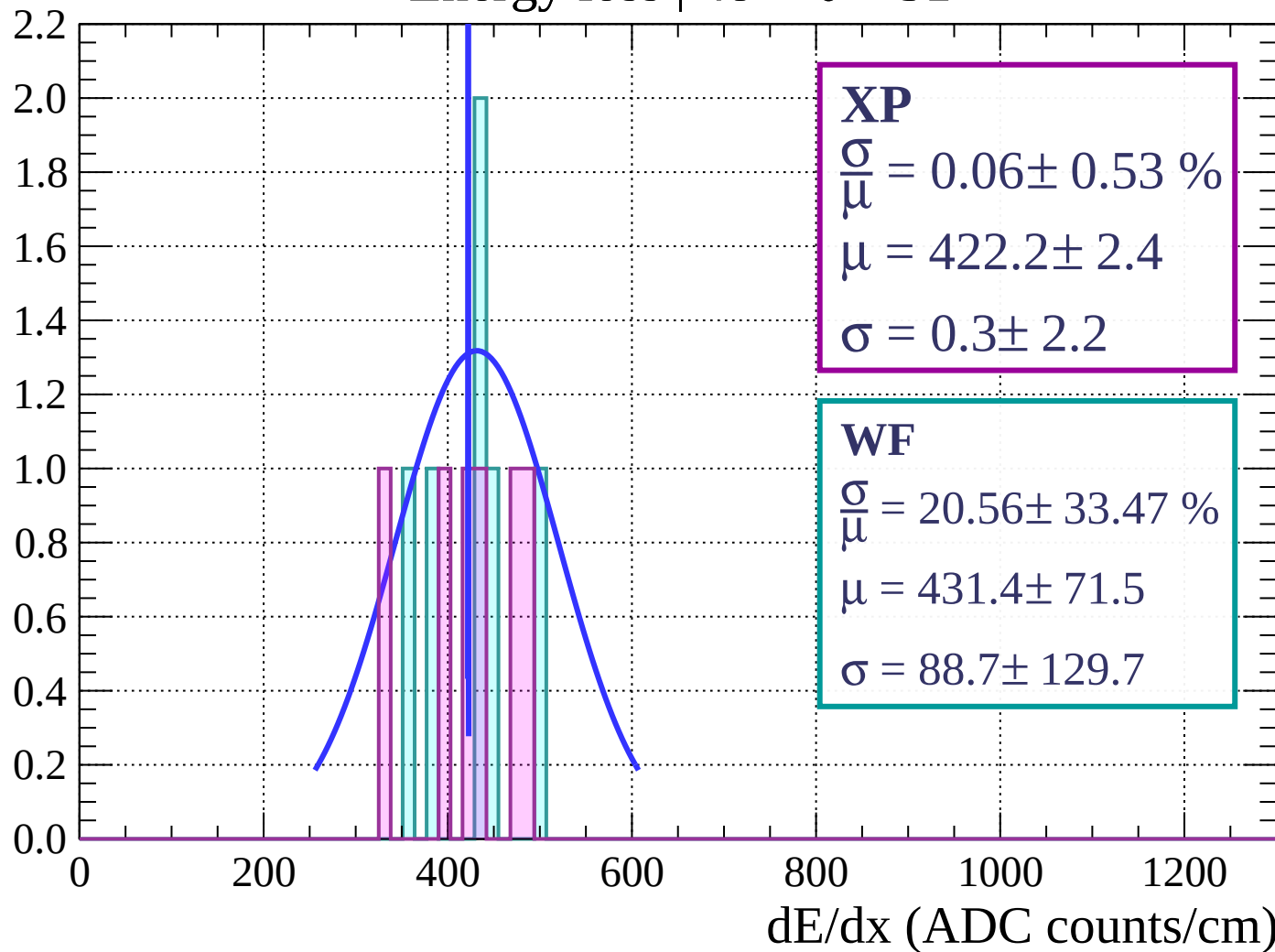
Energy loss | $45 < \theta < 48$

Count



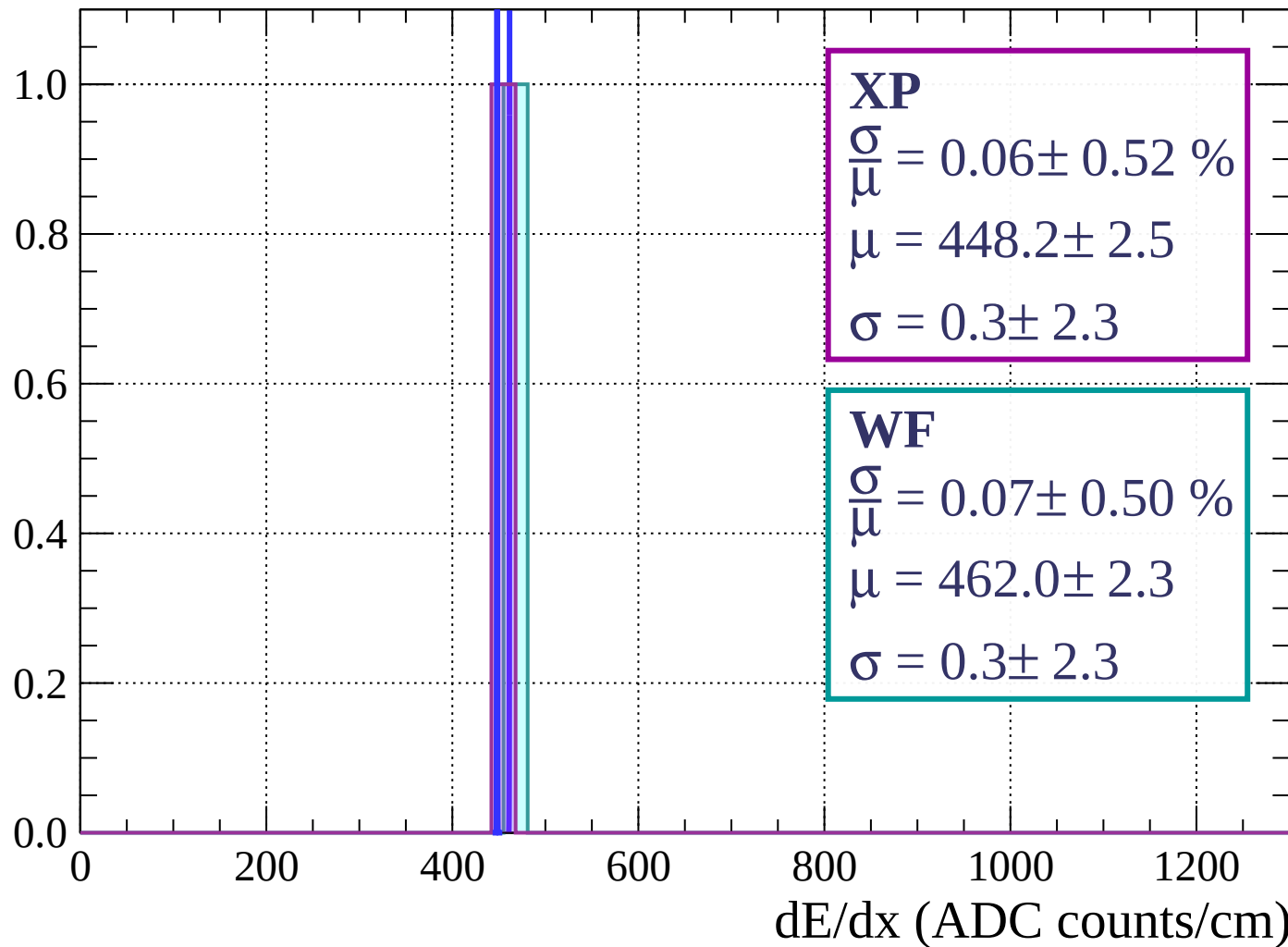
Energy loss | $48 < \theta < 51$

Count



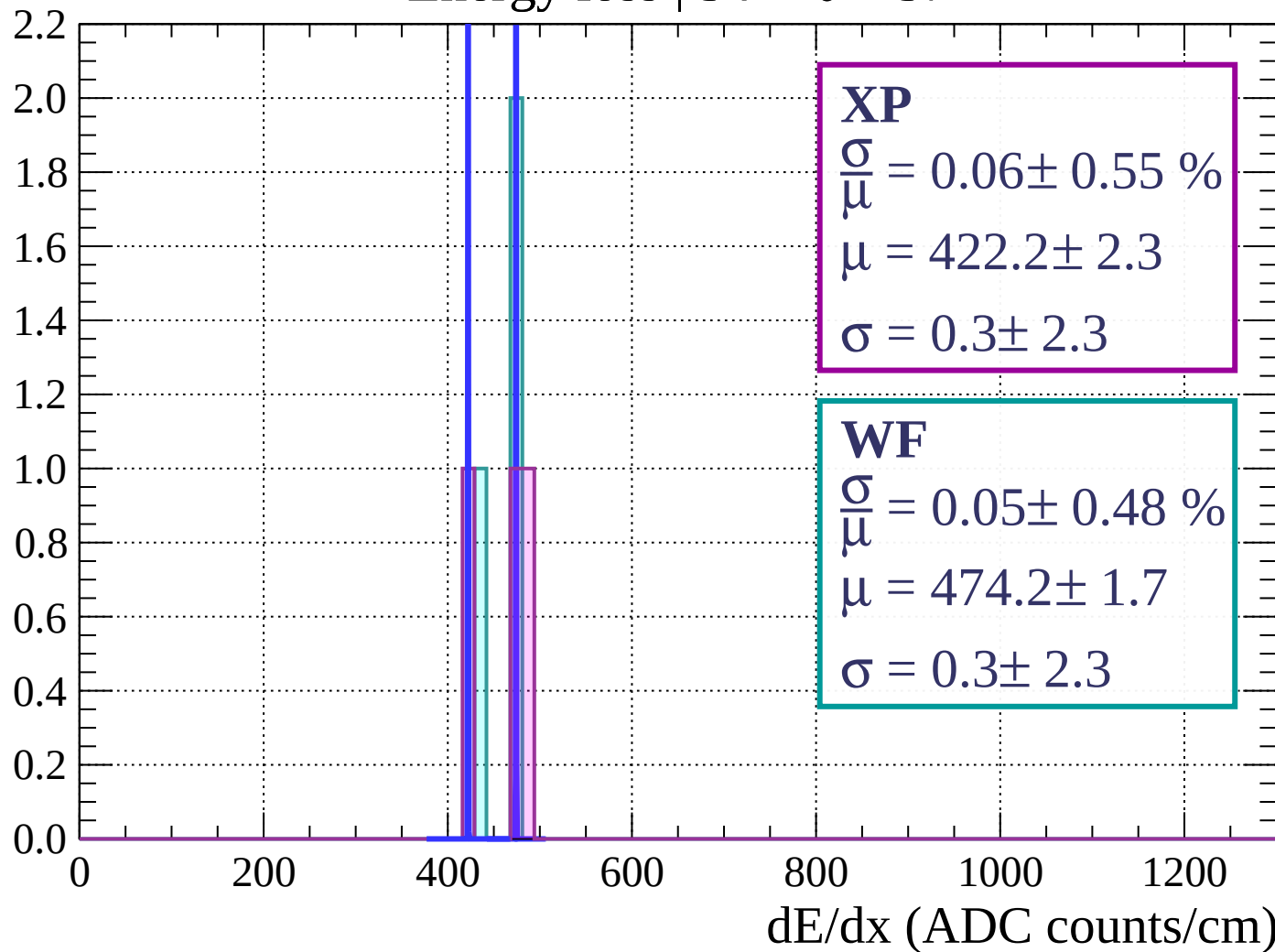
Energy loss | $51 < \theta < 54$

Count



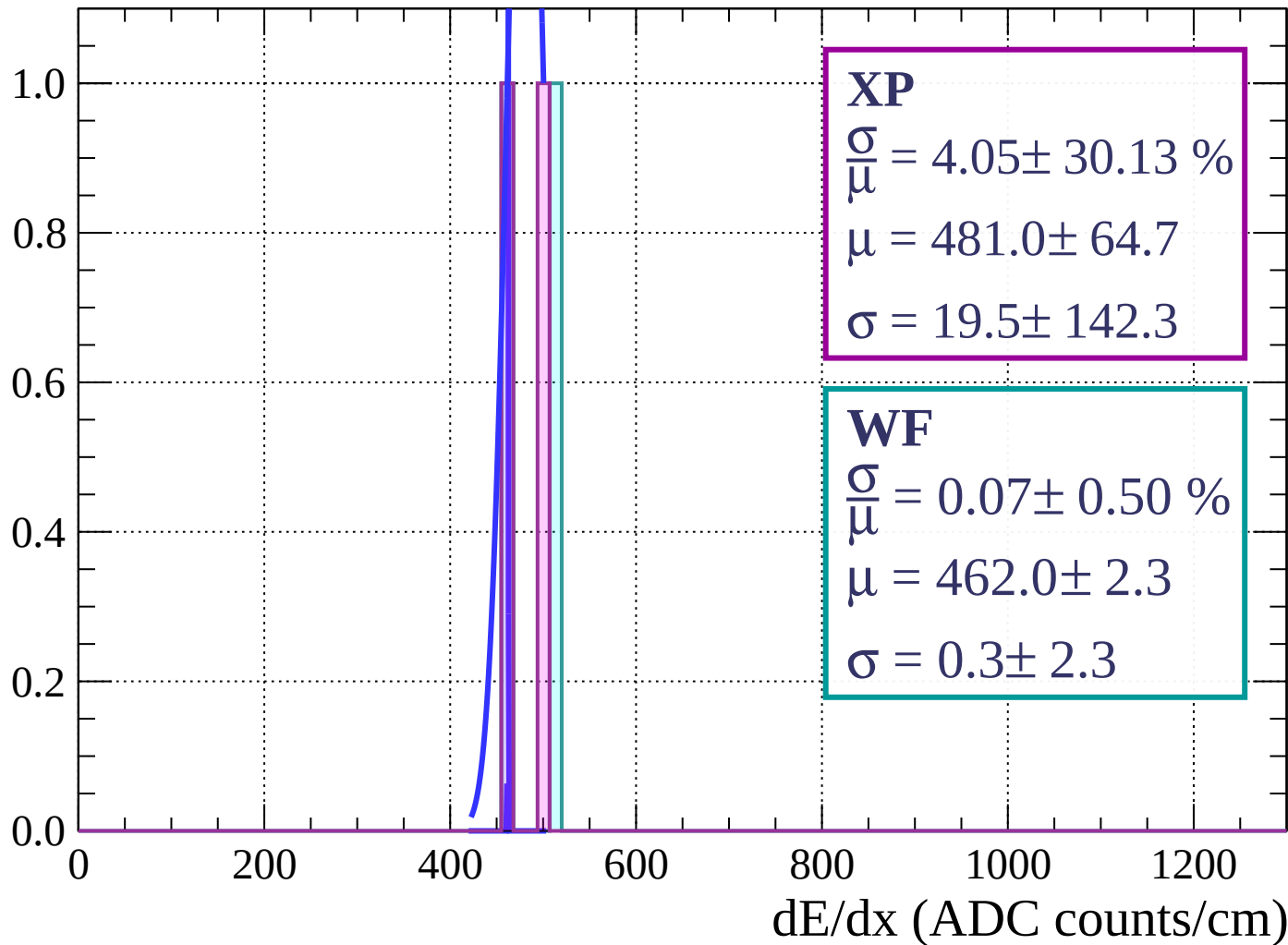
Energy loss | $54 < \theta < 57$

Count



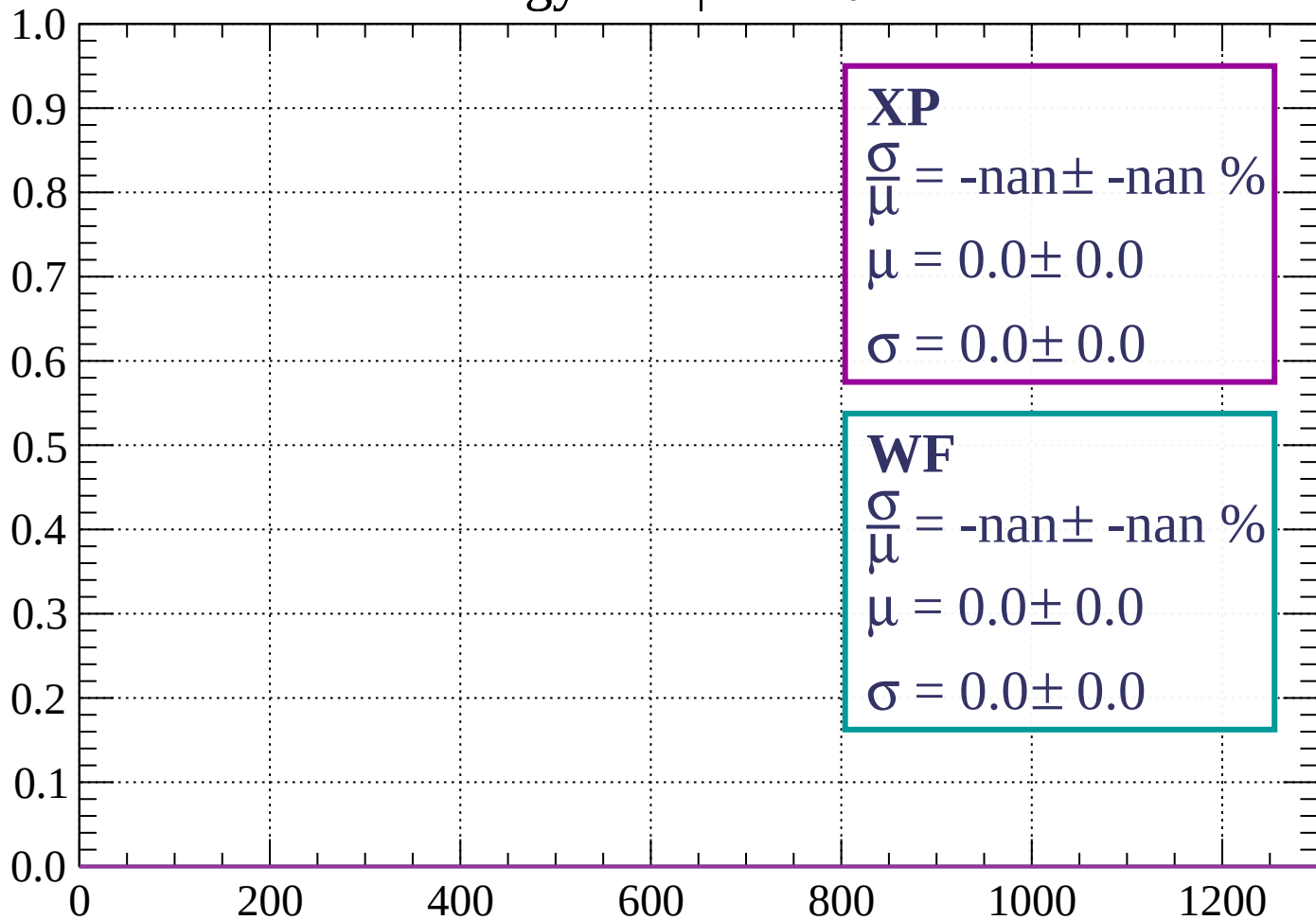
Energy loss | $57 < \theta < 60$

Count



Energy loss | $60 < \theta < 63$

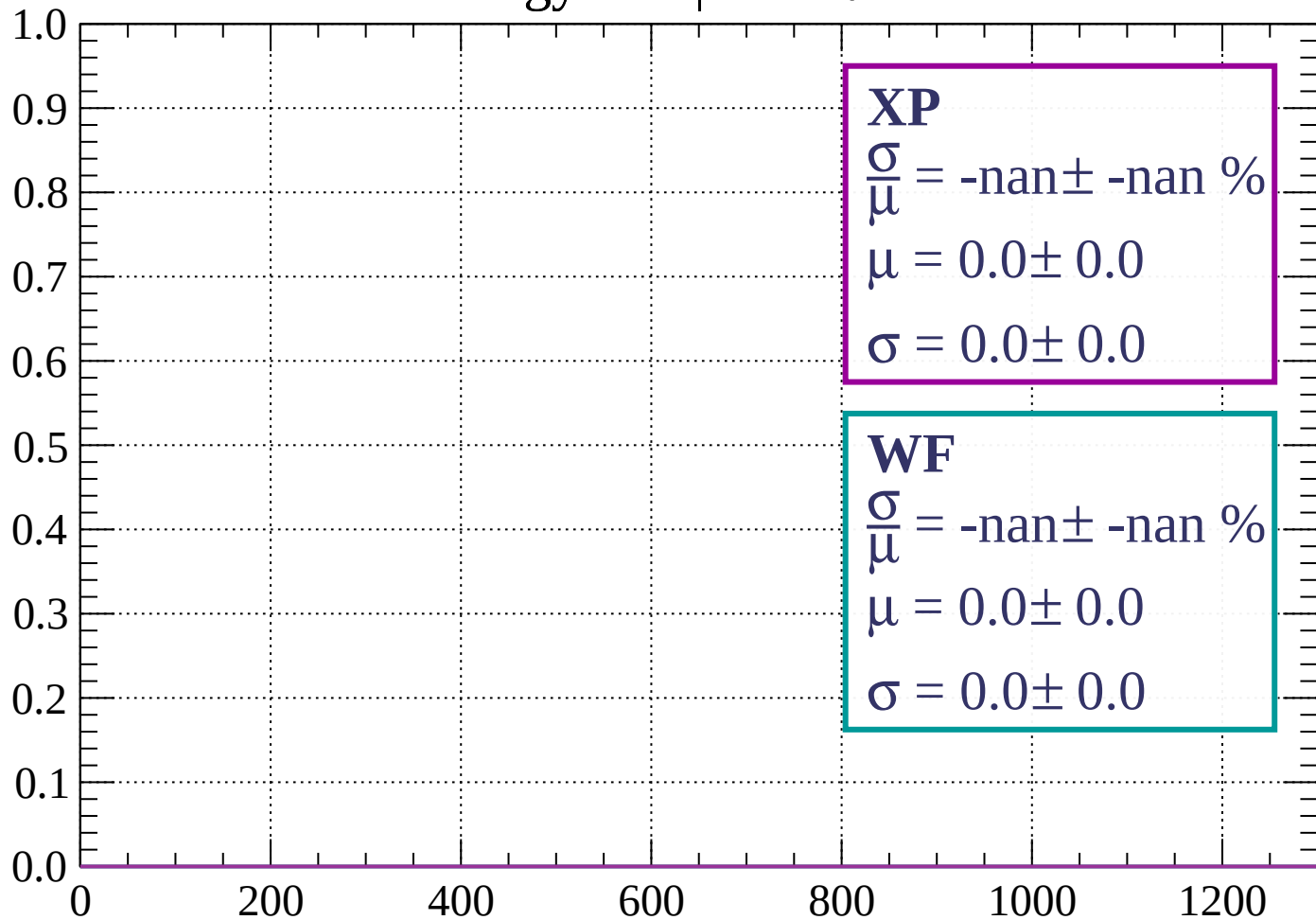
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \theta < 66$

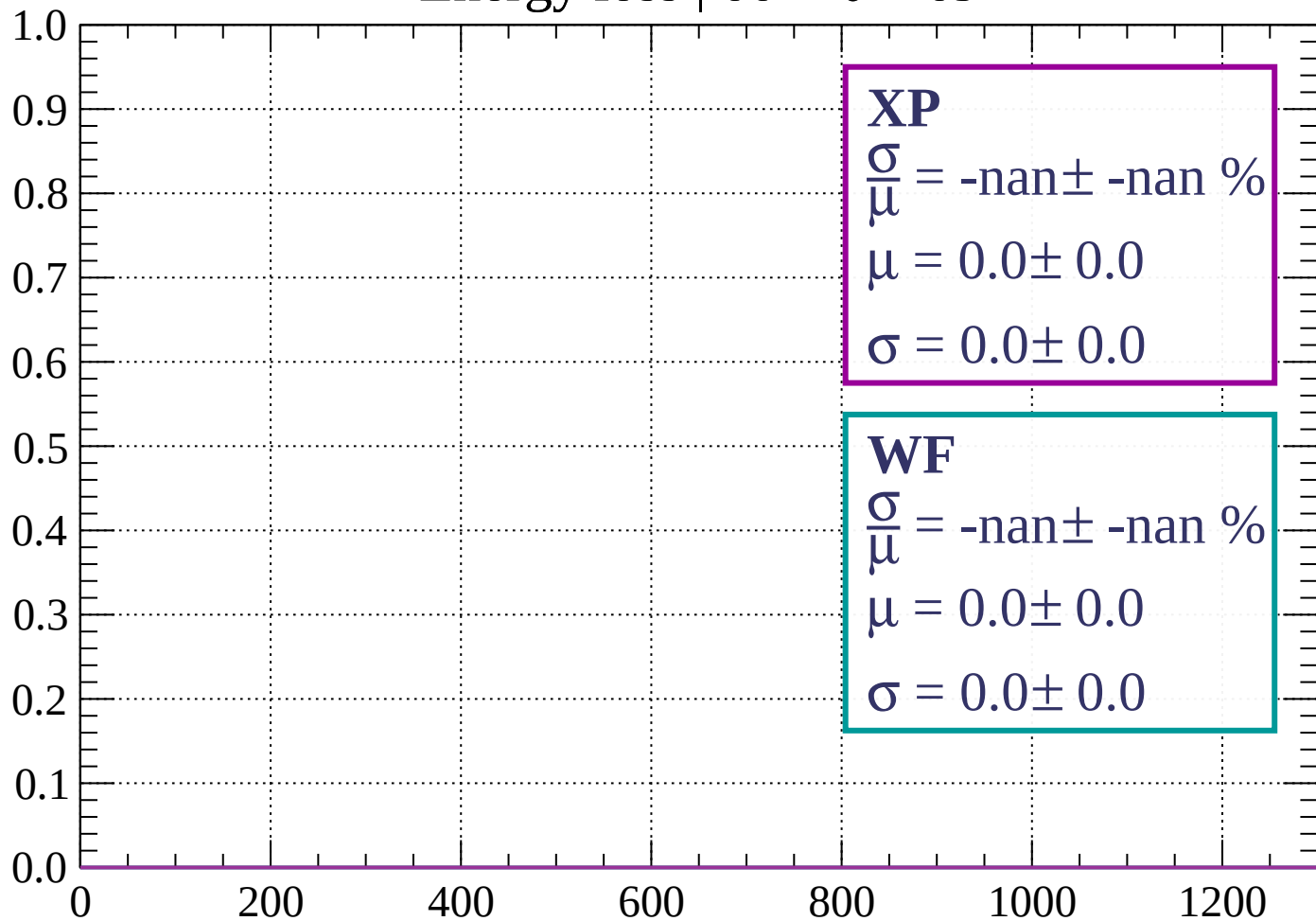
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

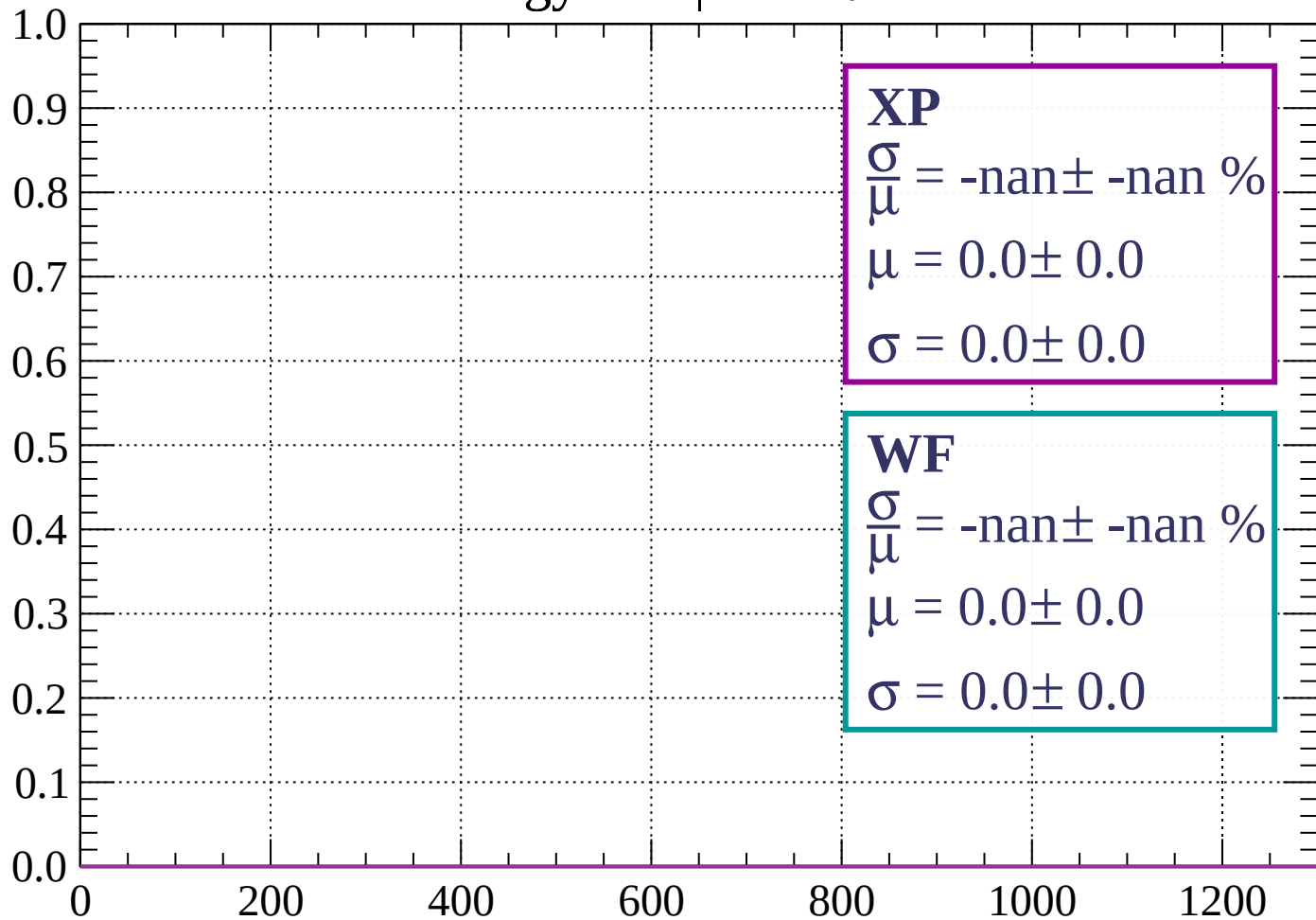
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

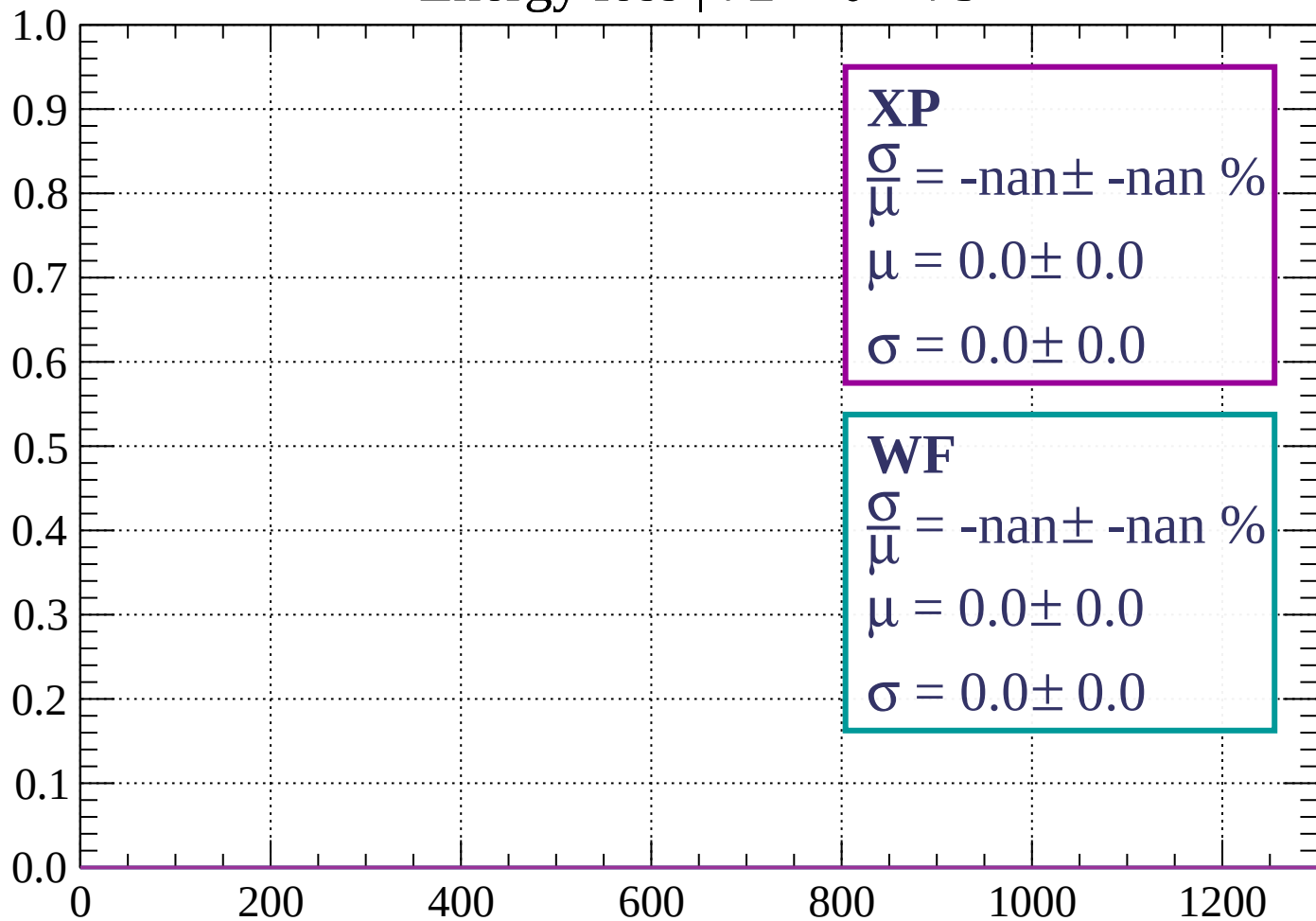
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

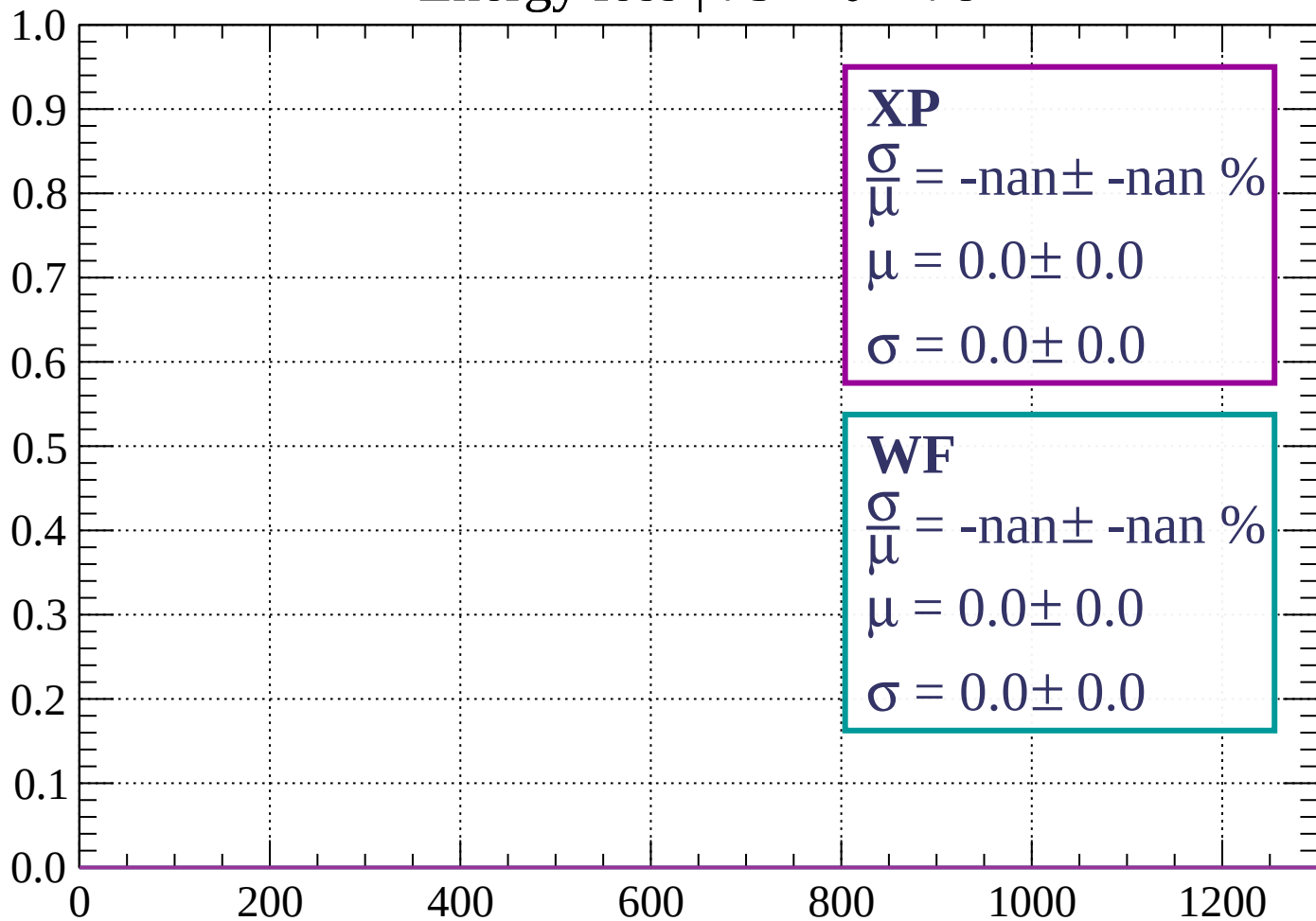
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

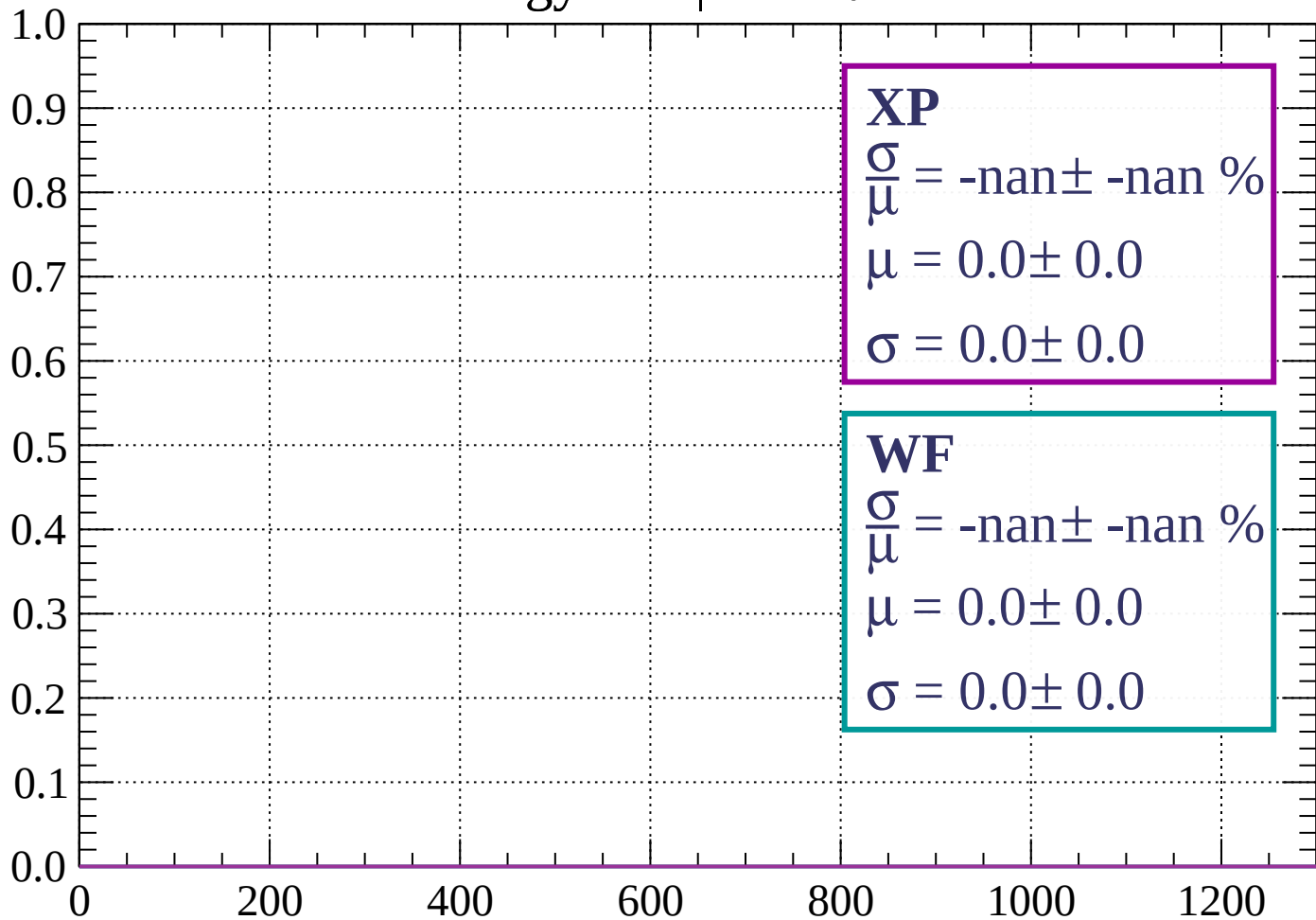
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

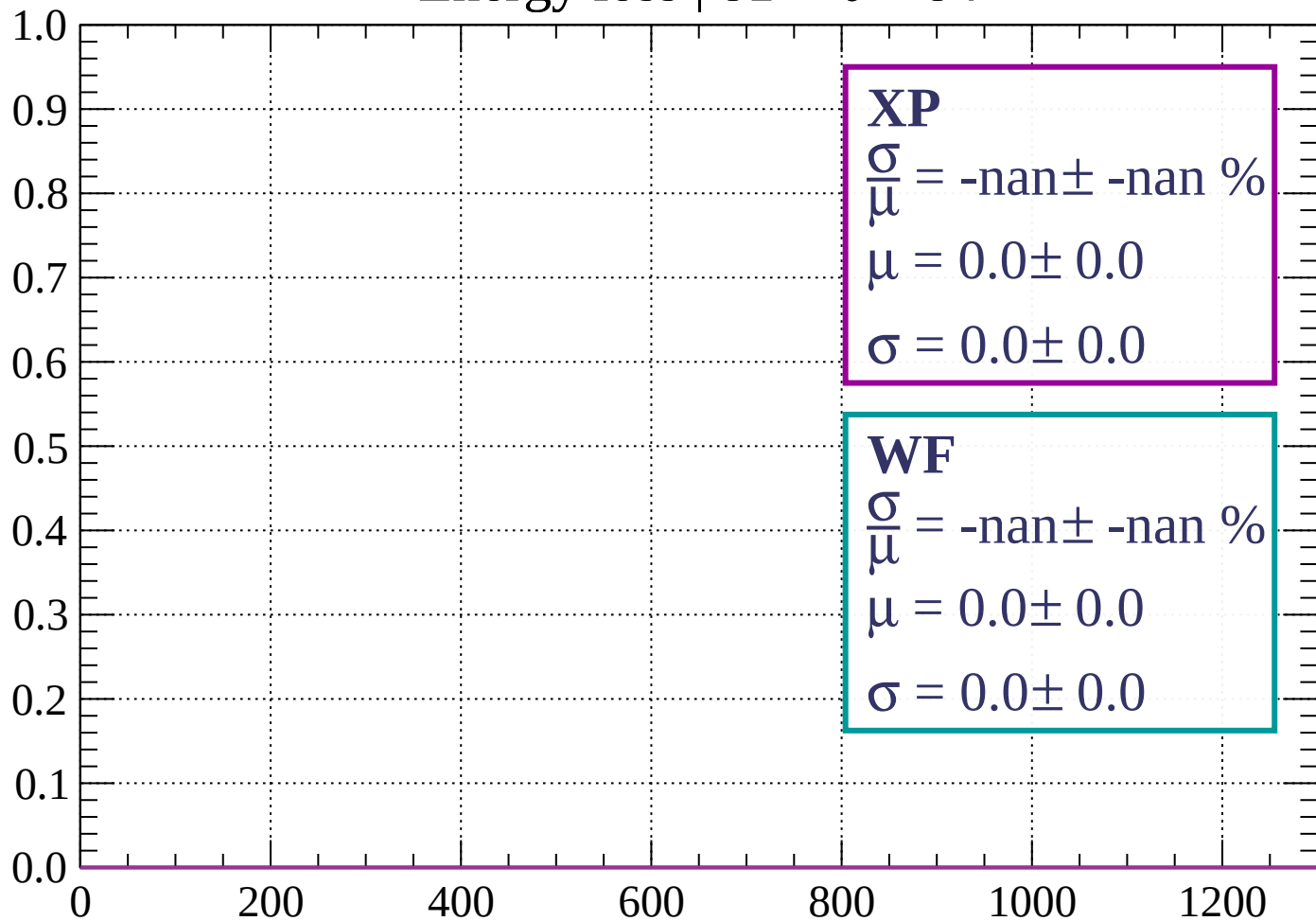
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

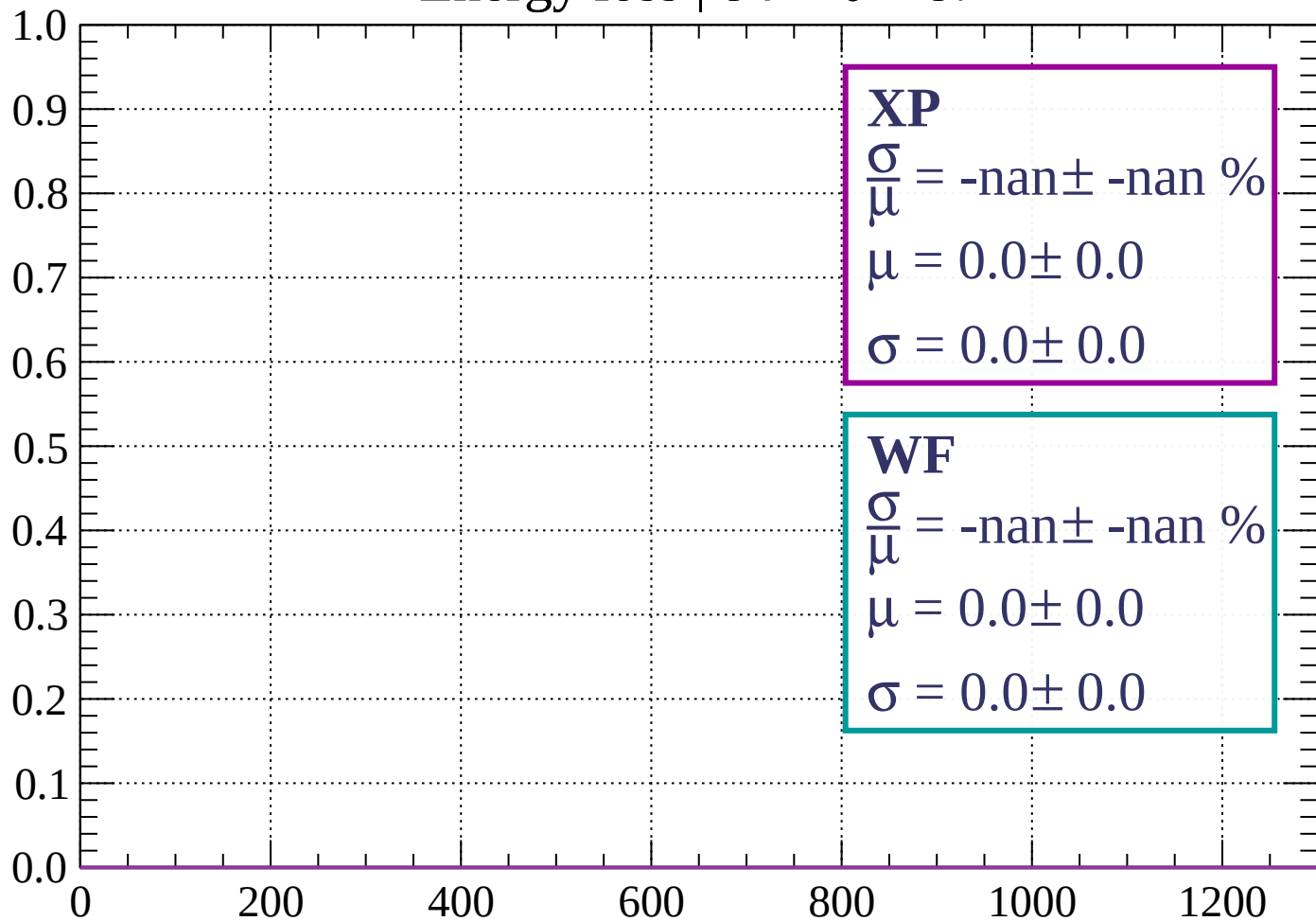
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

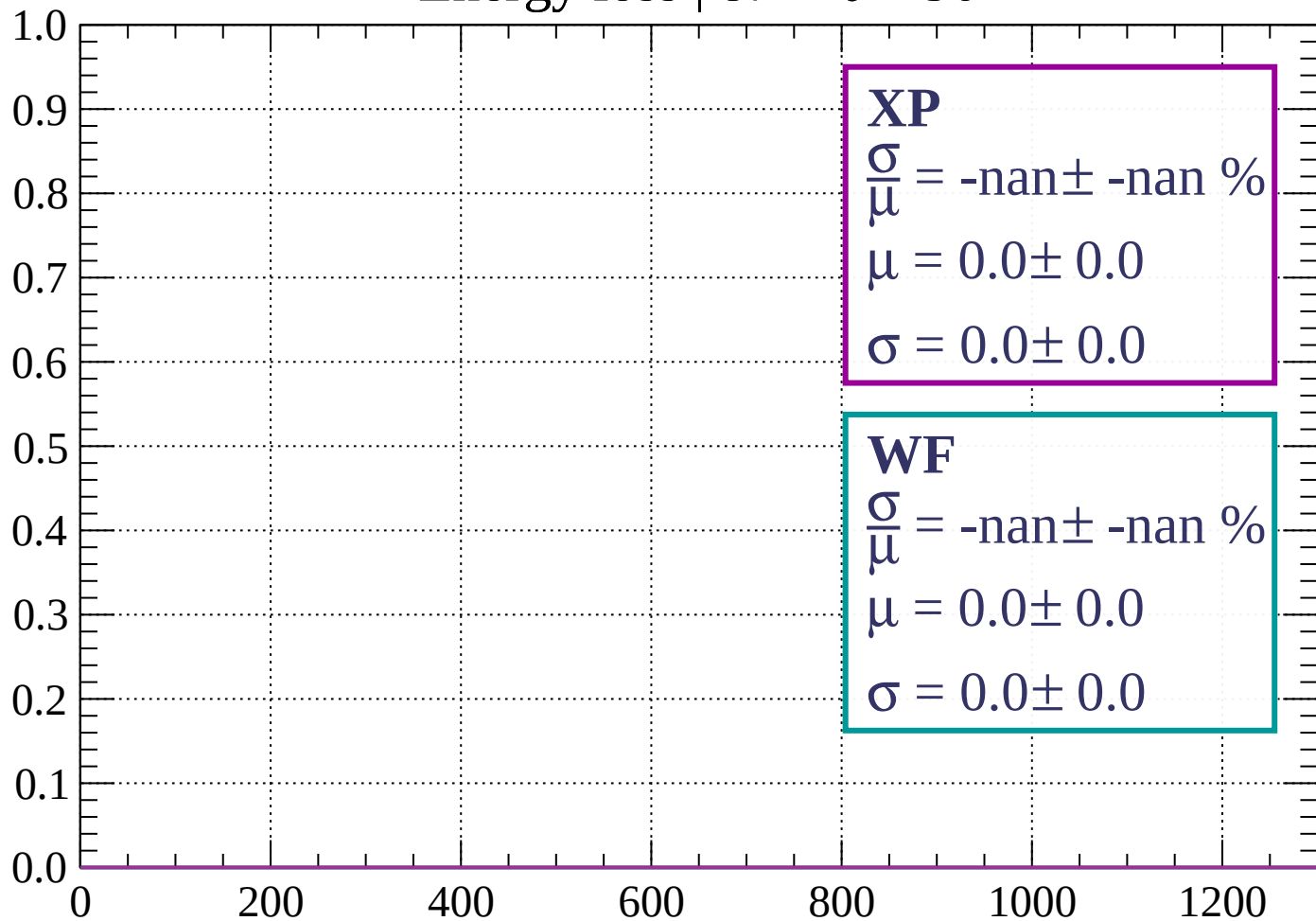
Count



dE/dx (ADC counts/cm)

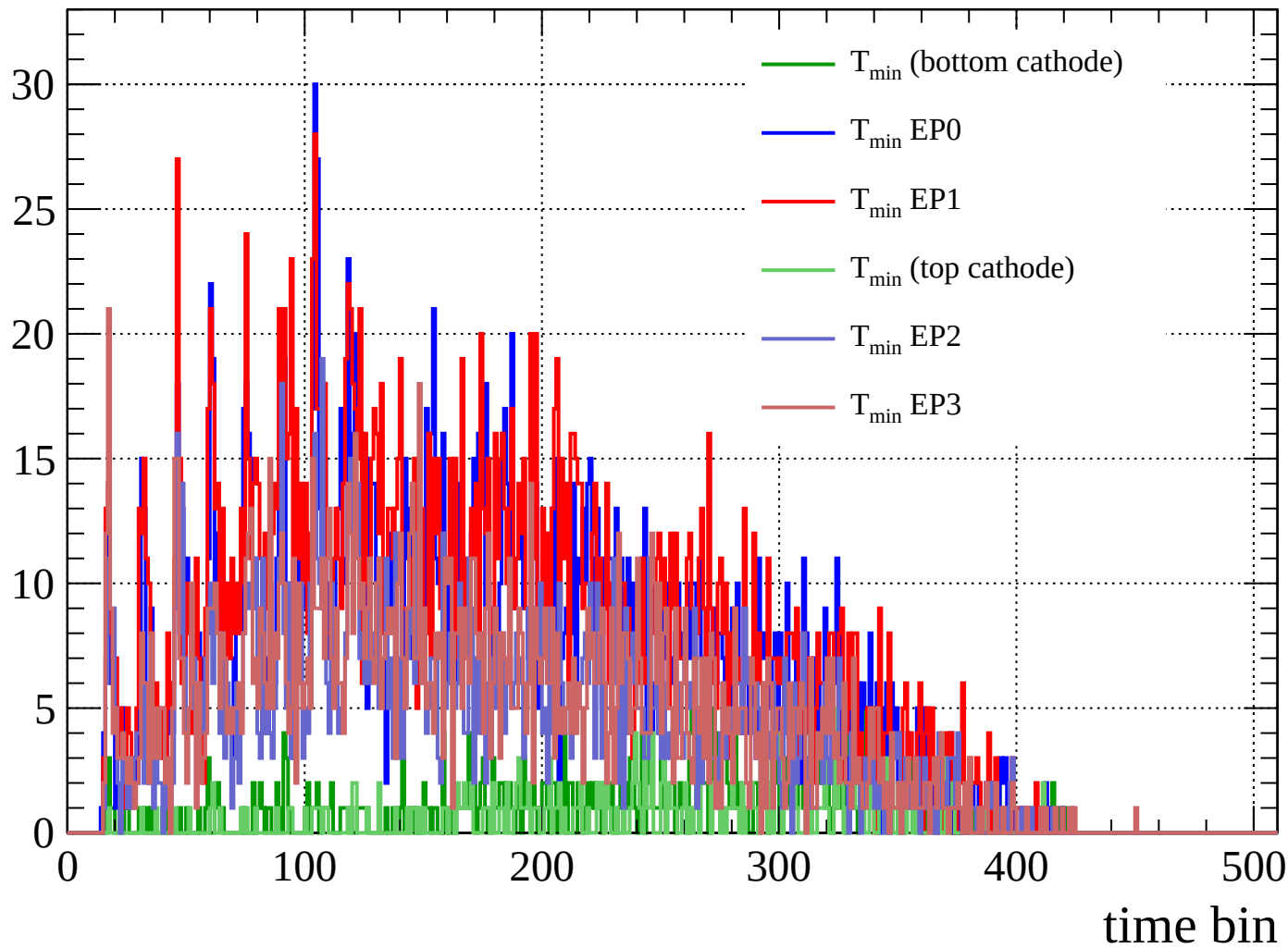
Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

Count



Count

